7.0 INSTRUMENTATION AND CONTROLS

7.1 INTRODUCTION

This chapter presents the various plant instrumentation and control systems by relating the functional performance requirements, design bases, system descriptions, design evaluations, and tests and inspections for each. The information provided in this chapter emphasizes the instruments and associated equipment that constitute the protection system as defined in Institute of Electrical and Electronics Engineers (IEEE) Standard 279-1971, Criteria for Protection Systems for Nuclear Power Generating Stations.

The standard Westinghouse solid state protection system design, which incorporates signal multiplexing for the control board and the plant computer, is applicable to VEGP.

The primary purpose of the instrumentation and control systems is to provide automatic protection and exercise proper control against unsafe and improper reactor operation during steady-state and transient power operations (American Nuclear Society (ANS) Conditions I, II, and III) and to provide initiating signals to mitigate the consequences of faulted conditions (ANS Condition IV). ANS conditions are discussed in chapter 15. Consequently, the information presented in this chapter emphasizes the instrumentation and control systems that are central to ensuring that the reactor can be operated to produce power in a manner that ensures no undue risk to the health and safety of the public.

It is shown that the applicable criteria and codes, such as general design criteria (GDC) and IEEE Standards, concerned with the safe generation of nuclear power are met by these systems. (See table 7.1.1-1 for a listing of applicable criteria as applied to instrumentation and control systems.)

A. Definitions

Terminology used in this chapter is based on the definitions given in IEEE Standard 279-1971, which is listed in subsection 7.1.2. In addition, the following definitions apply:

1. Degree of redundancy - The difference between the number of channels monitoring a variable and the minimum number of channels which, when tripped, would cause an automatic system trip.
2. Minimum degree of redundancy - The degree of redundancy below which operation is prohibited or otherwise restricted by the Technical Specifications.
3. Cold shutdown condition - When the reactor is subcritical by at least 1 percent $\Delta k/k$ and T_{avg} is $\leq 200°F$.
4. Hot shutdown condition - When the reactor is subcritical by an amount greater than or equal to the margin specified in the applicable Technical Specification and T_{avg} is greater than or equal to the temperature specified in the applicable Technical Specification.
5. Phase A containment isolation (CIA) - Closure of all purging ducts and nonessential process lines which penetrate containment initiated by the safety injection signal. (See subsection 6.2.4.)
6. Phase B containment isolation (CIB) - Not applicable.

7. Containment ventilation isolation - Closure of containment ventilation penetrations due to high radiation conditions existing inside the containment.

B. System Response Times

1. Reactor trip system response time - The time delays are defined as the time required for the reactor trip (i.e., the time the rods are free and begin to fall) to be initiated following a step change in the variable being monitored from 5 percent below to 5 percent above the trip setpoint.

2. Engineered safety features actuation system (ESFAS) response time - The interval required for the engineered safety features (ESF) sequence to be initiated subsequent to the point in time that the appropriate variable(s) exceed setpoints. The response time includes sensor/process (analog) and logic (digital) delay.

3. Reproducibility - This definition is taken from Scientific Apparatus Manufacturers Association (SAMA) Standard PMC-20.1-1973, Process Measurement and Control Terminology: "the closeness of agreement among repeated measurements of the output for the same value of input, under normal operating conditions over a period of time, approaching from both directions." It includes drift due to environmental effects, hysteresis, long-term drift, and repeatability. Long-term drift (aging of components, etc.) is not an important factor in accuracy requirements since, in general, the drift is not significant with respect to the time elapsed between testing. Therefore, long-term drift may be eliminated from this definition. In most cases reproducibility is a part of the definition of accuracy.

4. Accuracy - This definition is derived from SAMA Standard PMC-20.1-1973, Process Measurement and Control Terminology. An accuracy statement for a device falls under note 2 of the SAMA definition of accuracy, which means reference accuracy or the accuracy of that device at reference operating conditions: "reference accuracy includes conformity, hysteresis, and repeatability." To adequately define the accuracy of a system, the term "reproducibility" is useful as it covers normal operating conditions. The following terms, "trip accuracy" and "indicated accuracy," etc., include conformity and reproducibility under normal operating conditions. Where the final result does not have to conform to an actual process variable but is related to another value established by testing, conformity may be eliminated, and the term "reproducibility" may be substituted for accuracy.

5. Normal operating conditions - For this document, these conditions cover all normal process temperature and pressure changes. Also included are ambient temperature changes around the transmitter and racks.

6. Readout devices - The final device of a complete channel is considered a readout device. This includes indicators, recorders, isolators (nonadjustable), and controllers.

7. Channel accuracy - This definition includes accuracy of primary element, transmitter, and rack modules. It does not include readout devices or rack environmental effects but does include process and environmental effects on field-mounted hardware. Rack environmental effects are

included in the next two definitions to avoid duplication resulting from dual inputs.

8. Indicated and/or recorded accuracy - This definition includes channel accuracy, accuracy of readout devices, and rack environmental effects.

9. Trip accuracy - This definition includes comparator accuracy, channel accuracy for each input, and rack environmental effects. This is the tolerance expressed in process terms (or percent of span) within which the complete channel must perform its intended trip function. This includes all instrument errors but no process effects such as streaming. The term "actuation accuracy" may be used where the word "trip" might cause confusion, e.g., when starting pumps and other equipment.

10. Control accuracy - This definition includes channel accuracy, accuracy of readout devices (isolator and controller), and rack environmental effects. Where an isolator separates control and protection signals, the isolator accuracy is added to the channel accuracy to determine control accuracy, but credit is taken for tuning beyond this point; i.e., the accuracy of these modules (excluding controllers) is included in the original channel accuracy. The control accuracy is defined as the accuracy of the control signal in percent of the span of that signal. This includes gain changes where the control span is different from the span of the measured variable. Where controllers are involved, the control span is the input span of the controller. No error is included for the time the system is in a nonsteady-state condition.

7.1.1 IDENTIFICATION OF SAFETY-RELATED SYSTEMS

Safety-related instrumentation and control systems and their supporting systems are those systems required to ensure:

A. The integrity of the reactor coolant pressure boundary.

B. The capability to shut down the reactor and maintain it in a safe shutdown condition.

C. The capability to prevent or mitigate the consequences of accidents which could result in potential offsite exposures comparable to the guideline exposures of 10 CFR 100.

The definitions provided below are used to classify the instrumentation systems into the categories listed in chapter 7.0 of Regulatory Guide 1.70.

7.1.1.1 Reactor Protection System

The reactor protection system consists of the reactor trip system, the ESFAS, and the instrumentation and control power supply system.

7.1.1.1.1 Reactor Trip System

The reactor trip system is described in section 7.2. Design bases for the reactor trip system are given in paragraph 7.1.2.1. Figure 7.1.1-1 is a schematic diagram of this system.

7.1.1.1.2 Engineered Safety Features Actuation System

The ESFAS is a functionally defined system described in section 7.3. The equipment which provides the actuation functions is identified and discussed in section 7.3. Design bases for the ESFAS are given in paragraph 7.1.2.1.

The ESFAS are those instrumentation systems that are needed to actuate the equipment and systems required to mitigate the consequences of postulated design basis accidents. As discussed in section 7.3 the ESF requiring actuation are:

A. Emergency core cooling (section 6.3).

B. Main steam line and feedwater isolation (subsection 6.2.4).

C. Containment isolation (subsection 6.2.4).

D. Containment heat removal (subsection 6.2.2).

E. Containment combustible gas control (subsection 6.2.5).

F. Containment ventilation isolation (subsection 6.2.4).

G. Fuel building exhaust isolation (subsection 9.4.2).

H. Control room ventilation isolation (subsection 9.4.1).

I. Auxiliary feedwater supply (subsection 10.4.9).

7.1.1.1.3 Instrumentation and Control Power Supply System

Design bases for the instrumentation and control power supply system are given in paragraph 7.1.2.1. Further description of this system is provided in subsection 7.6.1 and in chapter 8.

7.1.1.2 Other Instrumentation Systems Required for Safety

7.1.1.2.1 Information Systems Important to Safety

Information systems important to safety provide information for the operator to manually perform reactor trip, ESF actuation, post-accident monitoring, or safe shutdown functions.

Identification of the equipment and information systems important to safety is provided in section 7.5. Descriptions of other indicating systems that provide information for monitoring equipment and processes are also provided in section 7.5.

Section 7.5 also summarizes information systems required to maintain the plant in a hot shutdown condition or to proceed to cold shutdown.

7.1.1.2.2 Interlock Systems Important to Safety and Mode Switchover Instrumentation

These safety-related instrumentation systems are the systems and components that have a preventive role in reducing the effects of accidents. Single failures in these systems do not inhibit reactor trip, ESF actuation, or functions required for safe shutdown. Other interlock systems important to safety consist of the following:

A. Residual heat removal isolation valve interlocks.

B. Refueling interlocks.

C. Accumulator motor-operated valve interlocks.

D. Emergency core cooling system switchover from injection mode to recirculation mode.

E. Interlocks for RCS pressure control during low temperature operation.

F. Isolation of nonsafety-related systems for safety-related systems.

Item B above is described in subsection 9.1.4. Item D is discussed in section 6.3. The remaining items are described in subsection 7.6.5.

7.1.1.3 Systems Required for Safe Shutdown

Systems required for safe shutdown are defined as those essential for pressure and reactivity control, coolant inventory makeup, and removal of residual heat once the reactor has been brought to a subcritical condition.

Identification of the equipment and systems required for safe shutdown is provided in section 7.4. Additional information regarding provisions for cold shutdown from outside the control room is also provided in section 7.4.

7.1.1.4 Control Systems Not Required for Safety

Control systems not required for safety are the automatic and manual systems with the primary purpose of normal load control, startup, and shutdown of the main power generating system. As shown in section 7.7, malfunctions in these systems do not result in unsafe conditions.

7.1.1.5 Comparison with Other Plants

The systems discussed in chapter 7 are compared with the systems of other plants of similar design in section 1.3.

7.1.2 IDENTIFICATION OF SAFETY CRITERIA

Paragraph 7.1.2.1 gives design bases for the systems identified in subsection 7.1.1. Design bases for nonsafety-related systems are provided in the sections that describe the systems. Considerations for instrument errors are included in the accident analyses presented in chapter 15. Functional requirements developed on the basis of the results of the accident analyses that have utilized conservative assumptions and parameters are used in designing these systems, and a preoperational testing program verifies the adequacy of the design. Accuracies are given in sections 7.2, 7.3, and 7.5.

The criteria listed in table 7.1.1-1 are considered in the design of the systems given in subsection 7.l.l. In general, the scope of these documents is given in the document itself. This determines the systems or parts of systems to which the document is applicable. A discussion of compliance with each document for systems in its scope is provided in the referenced sections given in table 7.1.1-1 for each criterion. Because some documents were issued after

design and testing had been completed, the equipment documentation may not meet the format requirements of some standards. Justification for any exceptions taken to each document for systems in its scope is provided in the referenced sections.

7.1.2.1 Design Bases

7.1.2.1.1 Reactor Trip System

The reactor trip system acts to limit the consequences of Condition II events (faults of moderate frequency such as loss of feedwater flow) by, at most, a shutdown of the reactor and turbine. The plant is capable of returning to operation after corrective action. The reactor trip system limits plant operation to ensure that the reactor safety limits are not exceeded during Condition II events and that these events can be accommodated without developing into more severe conditions. Reactor trip setpoints are given in the Technical Specifications.

The design requirements for the reactor trip system are derived by analyses of plant operating and fault conditions where automatic rapid control rod insertion is necessary in order to prevent or limit core or reactor coolant boundary damage. The design bases addressed in IEEE Standard 279-1971 are discussed in subsection 7.2.1. The design limits specified by Westinghouse for the reactor trip system are:

A. Minimum departure from nucleate boiling ratio (DNBR) shall not be less than the design basis limit as a result of any anticipated transient or malfunction (Condition II events).

B. Power density shall not exceed the rated linear power density for Condition II events. Refer to chapter 4 for fuel design limits.

C. The stress limit of the reactor coolant system for the various conditions shall be as specified in chapter 5.

D. Release of radioactive material shall be limited so as not to interrupt or restrict public use of areas beyond the exclusion radius as a result of any Condition III event.

E. For any Condition IV event, release of radioactive material shall not result in an undue risk to public health and safety.

7.1.2.1.2 Engineered Safety Features Actuation System

The engineered safety features actuation system (ESFAS) acts to limit the consequences of Condition III events (infrequent faults such as primary coolant spillage from a small rupture which exceeds normal charging system makeup and requires actuation of the safety injection system). The ESFAS acts to mitigate Condition IV events (limiting faults which include the potential for significant release of radioactive material).

The design bases for the ESFAS are derived from the design bases given in chapter 6 for the engineered safety features (ESF). Design bases requirements of IEEE Standard 279-1971 are addressed in paragraph 7.3.1.2. General design requirements are given below.

A. Automatic Actuation Requirements

The primary requirement of the ESFAS is to receive input signals (information) from the various ongoing processes within the reactor plant and containment and to automatically provide, as output, timely and effective signals to actuate the various components and subsystems comprising the ESF system.

B. Manual Actuation Requirements

The ESFAS must have provisions in the control room for manually initiating the functions of the ESF system.

7.1.2.1.3 Instrumentation and Control Power Supply System

The instrumentation and control power supply system provides continuous, reliable, regulated single phase ac power to all instrumentation and control equipment required for plant safety. Details of this system are provided in sections 7.6 and 8.3. The design bases are given below:

A. The inverter shall have the capacity and regulation required for the ac output for proper operation of the equipment supplied.

B. Redundant loads shall be assigned to different distribution panels which are supplied from different inverters.

C. Auxiliary devices that are required to operate dependent equipment shall be supplied from the same distribution panel to prevent the loss of electric power in one protection set from causing the loss of equipment in another protection set. No single failure shall cause a loss of power supply to more than one distribution panel.

D. Each of the distribution panels shall have access only to its respective inverter supply and a standby power supply.

E. The system shall comply with IEEE Standard 308-1974, section 5.4.

7.1.2.1.4 Emergency Power

Design bases and system description for the emergency power supply is provided in chapter 8.

7.1.2.1.5 Interlocks

Interlocks are discussed in sections 7.2, 7.3, 7.6, and 7.7. The protection (P) interlocks are given in tables 7.2.1-2 and 7.3.1-3. The safety analyses demonstrate that, even under conservative critical conditions for either postulated or hypothetical accidents, the protective systems ensure that the nuclear steam supply system (NSSS) is put into and maintained in a safe state following an ANS Condition II, III, or IV accident commensurate with applicable technical specifications and pertinent ANS criteria. The protective systems are designed to meet IEEE Standard 279-1971 and are entirely redundant and separate, including all permissives and blocks.

All blocks of a protective function are automatically cleared whenever the protective function is required to function in accordance with GDC 20, 21, and 22 and sections 4.11, 4.12, and 4.l3 of IEEE Standard 279-1971. Control interlocks (C) are identified in table 7.7-1. Because control interlocks are not safety related, they are not specifically designed to meet the requirements of IEEE protection system standards.

7.1.2.1.6 Bypasses

Bypasses are designed to meet the requirements of IEEE Standard 279-1971, sections 4.11, 4.12, 4.13, and 4.14. A discussion of bypasses provided is given in sections 7.2, 7.3, and 7.5.

A method has been developed to enable testing of the reactor trip system (RTS) and the engineered safety features actuation system (ESFAS) channels in the bypass condition as opposed to the tripped condition. At VEGP, bypass testing is provided for the 7300 process protection system, the nuclear instrumentation system, and various inputs to the solid state protection system.

The bypass test instrumentation (BTI) at VEGP will conform to applicable regulatory criteria including IEEE 279-1971 and Regulatory Guide 1.47 as well as prior regulatory guidance concerning tests in bypass. With implementation of the BTI, routine testing of analog RTS and ESFAS channels will be performed in a bypassed condition instead of a tripped condition. The Technical Specifications allow for the ability to test in the bypassed condition and govern the time that a channel can be in the bypassed condition for either test or maintenance. Reference 4 provides additional information concerning tests in bypass.

7.1.2.1.7 Equipment Protection

The criteria for equipment protection are given in chapter 3. Equipment related to safe operation of the plant is designed, constructed, and installed to protect it from damage. This is accomplished by working to accepted standards and criteria aimed at providing reliable instrumentation which is available under varying conditions. As an example, certain equipment is seismically qualified in accordance with IEEE Standard 344-1975. During construction, independence and separation are achieved, as required by IEEE Standard 279-1971, IEEE Standard 384-1981, and Regulatory Guide 1.75, either by barriers, physical separation, or demonstration test. This serves to protect against complete destruction of a system by fires, missiles, or other natural hazards.

7.1.2.1.8 Diversity

Functional diversity as discussed in reference 1 is designed into the system. The extent of diverse system variables is evaluated for a wide variety of postulated accidents. Generally, two or more diverse protection functions automatically terminate an accident before unacceptable consequences occur.

For example, there are automatic reactor trips based upon neutron flux measurements, reactor coolant loop temperature measurements, pressurizer pressure and level measurements, and reactor coolant pump underfrequency and undervoltage measurements. The system may also be activated manually and by initiation of a safety injection signal.

Regarding the ESFAS for a loss-of-coolant accident, a safety injection signal can be obtained manually or by automatic initiation from two diverse parameter measurements:

A. Low pressurizer pressure.

B. High containment pressure (high-1).

For a steam line break accident, safety injection signal actuation is provided by:

A. Low steam line pressure (lead-lag compensated).

B. Low pressurizer pressure.

For a steam line break inside containment, high containment pressure (high-1) provides an additional parameter for generation of the signal.

All of the above sets of signals are redundant and physically separated and meet the requirements of IEEE Standard 279-1971.

7.1.2.1.9 Bistable Trip Setpoints

Three values applicable to reactor trip and ESF actuation have been specified; they are safety limit, limiting value, and nominal value.

The safety limit is the value assumed in the accident analysis and is the least conservative value.

The limiting value is the technical specification value and is obtained by subtracting a safety margin from the safety limit. The safety margin accounts for instrument error, process uncertainties such as flow stratification and transport factor effects, etc.

The nominal value is the value set into the equipment and is obtained by subtracting allowances for instrument drift from the limiting value. The nominal value allows for the normal expected instrument setpoint drifts such that the technical specification limits are not exceeded under normal operation.

The setpoints that require trip action are given in the Technical Specifications. A further discussion on setpoints is found in paragraph 7.2.2.2.1.

The trip setpoint is determined by factors other than the most accurate portion of the instrument's range. The safety limit is determined only by the accident analysis.

As described above, allowance is then made for process uncertainties, instrument error, instrument drift, and calibration uncertainty to obtain the nominal value which is actually set into the equipment. The only requirement on the instrument's accuracy value is that over the instrument span, the error must always be less than or equal to the error value allowed in the accident analysis. The instrument does not need to be the most accurate at the setpoint value as long as it meets the minimum accuracy requirement. The accident analysis accounts for the expected errors at the actual setpoint.

Range selection for the instrumentation covers the expected range of the process variable being monitored consistent with its application. The design of the reactor protection and ESF systems is such that the bistable trip setpoints do not require process transmitters to operate within 5 percent of the high and low end of their calibrated span or range. Functional requirements established for every channel in the reactor protection and ESF systems stipulate the maximum allowable errors on accuracy, linearity, and reproducibility. The protection channels have the capability for and are tested to ascertain that the characteristics throughout the entire span in all aspects are acceptable and meet functional requirement specifications. As a result, no protection channel operates normally within 5 percent of the limits of its specified span.

In this regard, it should be noted that the specific functional requirements for response time, setpoint, and operating span are finalized contingent on the results and evaluation of safety studies to be carried out using data pertinent to the plant. Emphasis is placed on establishing adequate performance requirements under both normal and faulted conditions. This includes consideration of process transmitter margins such that, even under a highly improbable situation of full-power operation at the limits of the operating map (as defined by the high- and low-pressure reactor trip, ΔT overpower and overtemperature trip lines (DNB protection), and

the steam generator safety valve pressure setpoint), adequate instrument response is available to ensure plant safety.

7.1.2.1.10 Engineered Safety Features Motor Specifications

The voltage for the residual heat removal (RHR) pump motor and ESF auxiliary system pump motors rated 4 kV (and above) and 460 V is 75 percent of rated voltage at the motor terminals to start and accelerate the driven equipment. (For the boric acid transfer pump, 80 percent of rated voltage is required.) The motors are capable of accelerating the driven equipment from rest to operating speed within 4 s.

The minimum margin of motor torque over the pump full-load torque (as defined by the pump speed/torque curve) is sufficient to accelerate all the driven equipment as necessary and with 75 percent of rated voltage at the motor terminals from standstill to operating speed. (For the boric acid transfer pump and boron injection recirculation pump motors, 80 percent of rated voltage is required.)

Verification of the ESF pump motors capability to operate within design temperature ratings, including the National Electrical Manufacturers Association (NEMA) Test Specification MG1-20.43 ("number of starts"), is based on the design tests of the prototype motor that are performed at the manufacturer's test facilities, rather than by means of initial or periodic tests in the field.

Six stator resistance type temperature detectors embedded in two slots of each phase between top and bottom coil sides are provided on 4-kV motors, except for the RHR pump motors. Abnormalities in the motor windings may be monitored with this instrumentation. For conditions where the motor stalls or fails to start, the influence of these conditions is best monitored from the current versus time characteristics, and equipment protection for this is provided by the circuit breaker trip function.

The design of 4-kV ESF pump motors does not preclude the surveillance of the hot spots on the rotor side of the motor. Should there be justification for making use of it, the procedure for surveillance would include removal of the motor from the system and return to the manufacturer's test facilities for evaluation, by means of reverification of prototype test results. Westinghouse does not believe there are maintenance benefits over the method of evaluation of the effects of motor overloads by means of the six stator resistance type temperature detectors and the conventional method of equipment protection by means of a circuit breaker trip function coordinated with the motor thermal overload characteristic.

7.1.2.2 Independence of Redundant Safety-Related Systems

The safety-related systems described in subsection 7.1.1 are designed to meet the independence and separation requirements of General Design Criterion 22 and section 4.6 of IEEE Standard 279-1971. Conformance with the specific provisions of Regulatory Guide 1.75 is discussed in chapter 8 and paragraph 7.1.2.2.1.

The electrical power supply, instrumentation, and controls for redundant circuits of a nuclear plant have physical separation to preserve redundancy and to ensure that no single credible event will prevent operation of the associated function resulting from electrical conductor damage. Critical circuits and functions include power, control, and analog instrumentation associated with the operation of the reactor trip system or ESFAS. Credible events include, but are not limited to, the effects of short circuits, pipe rupture, missiles, fire, etc., and are considered in the basic plant design. Control board details are given in chapter 18. In the

control board, separation of redundant circuits is maintained as described in paragraph 7.1.2.2.2.

7.1.2.2.1 General

The physical separation criteria for redundant safety-related system sensors, sensing lines, wireways, cables, and components on racks meet recommendations contained in Regulatory Guide 1.75 with the following comments for NSSS equipment:

A. The design of the protection system relies on the provisions of IEEE Standard 384-1981 relative to overcurrent devices to prevent malfunctions in one circuit from causing unacceptable influences on the functioning of the protection system. The protection system uses redundant instrumentation channels and actuation trains and incorporates physical and electrical separation to prevent faults in one channel from degrading any other protection channel.

B. Separation recommendations for redundant instrumentation racks are not the same as those given in Regulatory Position C.16 of Regulatory Guide 1.75 for the control boards because of different functional requirements. Main control boards contain redundant circuits which are required to be physically separated from each other. However, since there are no redundant circuits which share a single compartment of an NSSS protection instrumentation rack and since these redundant protection instrumentation racks are physically separated from each other, the physical separation requirements specified for the main control board do not apply.

   However, redundant isolated control signal cables leaving the protection racks are brought into close proximity elsewhere in the plant, such as the control board. It could be postulated that electrical faults or interference at these locations might be propagated into all redundant racks and might degrade protection circuits because of the close proximity of protection and control wiring within each rack. Regulatory Guide 1.75 (Regulatory Position C.4) and IEEE Standard 384-1974 (section 4.5(3)) provide the option to demonstrate by tests that the absence of physical separation could not significantly reduce the availability of Class 1E circuits.

   Westinghouse test programs have demonstrated that Class 1E protection systems (nuclear instrumentation system, solid-state protection system, and 7300 process control system) are not degraded by non-Class 1E circuits sharing the same enclosure. Conformance to the requirements of IEEE Standard 279-1971 and Regulatory Guide 1.75 has been established and accepted by the Nuclear Regulatory Commission (NRC) based on the following, which is applicable to these systems at the VEGP.

   Tests conducted on the as-built designs of the nuclear instrumentation system and solid-state protection system were reported and accepted by the NRC in support of the Diablo Canyon application (Docket Nos. 50-275 and 50-323). Westinghouse considers these programs as applicable to all plants, including VEGP. Westinghouse tests on the 7300 process control system were covered in a report entitled, "7300 Series Process Control System Noise Tests," subsequently reissued as reference 2. In a letter dated April 20, 1977,[(3)] the NRC accepted the report in which the applicability of the VEGP is established.

C. The physical separation criteria for instrument cabinets within Westinghouse NSSS scope and the Westinghouse-supplied 7300 series for balance of plant scope meet the recommendations contained in section 6.7 of IEEE Standard 384-1981.

7.1.2.2.2 Specific Systems

Independence is maintained throughout the system, extending from the sensor to the devices actuating the protective function. Physical separation is used to achieve separation of redundant transmitters. Separation of wiring is achieved using separate wireways, cable trays, conduit runs, and containment penetrations for each redundant protection channel set. Redundant analog equipment is separated by locating modules in different protection rack sets. Each redundant channel set is energized from a separate ac power feed.

There are four separate process analog sets. Separation of redundant analog channels begins at the process sensors and is maintained in the field wiring, containment penetrations, and analog protection cabinets to the redundant trains in the logic racks.

In the nuclear instrumentation system, process systems, and the solid-state protection system input cabinets where redundant channel instrumentation are physically adjacent, there are no wireways or cable penetrations which would permit; for example, a fire resulting from electrical failure in one channel to propagate into redundant channels in the logic racks. Redundant analog channels are separated by locating modules in different cabinets. Since all equipment within any cabinet is associated with a single protection set, there is no requirement for separation of wiring and components within the cabinet.

Two reactor trip breakers are actuated by two separate logic matrices to interrupt power to the control rod drive mechanisms. The breaker main contacts are connected in series with the power supply so that opening either breaker interrupts power to all control rod drive mechanisms, permitting the rods to free-fall into the core.

A. Reactor Trip System

1. Separate routing is maintained for the four basic reactor trip system channel sets analog sensing signals, bistable output signals, and power supplies for such systems. The separation of these four channel sets is maintained from sensors to instrument cabinets to logic system input cabinets.

2. Separate routing of the redundant reactor trip signals from the redundant logic system cabinets is maintained. In addition, they are separated (by spatial separation, by provision of barriers, or by separate cable trays or wireways) from the four analog channel sets.

B. Engineered Safety Features Actuation System

1. Separate routing is maintained for the four basic sets of ESFAS analog sensing signals, bistable output signals, and power supplies for such systems. The separation of these four channel sets is maintained from sensors to instrument cabinets to logic system input cabinets.

2. Separate routing of the ESF actuation signals from the redundant logic system cabinets are maintained. In addition, they are separated by spatial separation, by provisions of barriers, or by separate cable trays or wireways from the four analog channel sets.

3. Separate routing of control and power circuits associated with the operation of ESF equipment is required to retain redundancies provided in the system design and power supplies.

C. Instrumentation and Control Power Supply System

The separation criteria presented also apply to the power supplies for the load centers and buses distributing power to redundant components and to the control of these power supplies.

Reactor trip system and ESFAS analog circuits may be routed in the same wireways, provided circuits have the same power supply and channel set identified (I, II, III, or IV).

In order to maintain separation between wiring on the main control board associated with different trains, mutually redundant safety train wiring is not terminated on a single device. Backup manual actuation switches link the separate trains by mechanical means to provide greater reliability of operator action for the manual reactor trip function and manual ESF actuations. The linked switches are themselves redundant so that operation of either set of linked switches will actuate safety trains A and B simultaneously. This is shown in figure 7.2.1-2. The design of the manual reactor trip function and manual ESF actuations conform with Regulatory Guide 1.62. (See also subsection 7.3.1.)

7.1.2.2.3 Fire Protection

For electrical equipment within the NSSS scope of supply, including the balance of plant Westinghouse-supplied 7300 series cabinets, Westinghouse specifies noncombustible or fire retardant material and conducts vendor-supplied specification reviews of this equipment, which includes assurance that materials are not used which may ignite or explode from an electrical spark, flame, or from heating or will independently support combustion. These reviews also include assurance of conservative current carrying capacities of all instrument cabinet wiring, which precludes electrical fires resulting from excessive overcurrent (I^2R) losses. For example, wiring used for instrument cabinet construction has Teflon or Tefzel insulation and is adequately sized based on current carrying capacities set forth by the National Electric Code. In addition, fire retardant paint is used on protection rack or cabinet construction to retard fire or heat propagation from rack to rack. Braided sheathed material used in the cables is noncombustible. For in-field wiring, cables in power trays are sized using derating factors listed in Insulated Cable Engineers Association (ICEA) Publication P-46-426 or Publication P-54-440. Paragraph 8.3.1.4.2 provides details regarding cable derating and cable tray fill.

For early warning protection against propagation of electrical fires, smoke or other detectors are provided for fire detection and alarm in remote wireways or other unattended areas where large concentrations of cables are installed.

The criteria and bases for the independence of electrical cable including routing, marking, and cable derating are covered in section 8.3. Fire detection and protection in the areas where wiring is installed is covered in subsection 9.5.1.

7.1.2.3 Physical Identification of Safety-Related Equipment

There are four separate protection sets identifiable with process equipment associated with the reactor trip and engineered safeguards actuation systems. A protection set may be comprised of more than a single process equipment cabinet. The color coding of each process equipment rack nameplate coincides with the color code established for the protection set of which it is a

part. Redundant channels are separated by locating them in different equipment cabinets. Separation of redundant channels begins at the process sensors and is maintained in the field wiring, containment penetrations, and equipment cabinets to the redundant trains in the logic racks. The solid-state protection system input cabinets are divided into four isolated compartments, each serving one of the four redundant input channels. Horizontal 1/8-in.-thick solid steel barriers coated with fire retardant paint separate the compartments. Four 1/8-in.-thick solid steel wireways coated with fire retardant paint enter the input cabinets vertically in their own quadrant. The wireway for a particular compartment is open only into that compartment so that flame could not propagate to affect other channels. At the logic racks the protection set color coding for redundant channels is clearly maintained until the channel loses its identity in the redundant logic trains. The color-coded nameplates described in subsection 8.3.1 provide identification of equipment associated with protective functions and their channel set association.

All noncabinet-mounted protective equipment and components are provided with an identification tag or nameplate. Small electrical components such as relays have nameplates on the enclosure which houses them. All cables are numbered with identification tags. Cable trays and conduits are identified using permanent markings which identify the associated separation group. The purpose of such markings is to facilitate cable routing identification for future modification or additions. Positive permanent identification of cables and/or conductors is made at all terminal points. There are also identification nameplates on the input panels of the solid-state logic protection system. See section 8.3 for further details of physical identification of balance of plant safety-related equipment.

7.1.2.4 Conformance to Criteria

A listing of applicable criteria and the sections where conformance is discussed is given in table 7.1.1-1. An additional discussion of Westinghouse conformance to Regulatory Guide 1.22 and IEEE Standards 338-1975 and 379-1972 is given in the following paragraph.

7.1.2.5 Conformance to Regulatory Guide 1.22

Periodic testing of the reactor trip and ESFAS, as described in section 1.9 and subsections 7.2.2 and 7.3.1, conforms with Regulatory Guide 1.22, Periodic Testing of Protection System Actuation Functions.

Where the ability of a system to respond to a bona fide accident signal is intentionally bypassed for the purpose of performing a test during reactor operation, each bypass condition is automatically indicated to the reactor operator in the main control room by a separate annunciator for the train in test. Test circuitry does not allow two trains of the SSPS to be tested at the same time so that extension of the bypass condition to the redundant system is prevented. Administrative and procedural controls are used to prevent simultaneous testing of more than one protection set of the analog circuitry.

The actuation logic for the reactor trip and ESFAS is tested as described in sections 7.2 and 7.3. As recommended by Regulatory Guide 1.22, where actuated equipment is not tested during reactor operation it has been determined that:

A. There is no practicable system design that would permit operation of the equipment without adversely affecting the safety or operability of the plant.

B. The probability that the protection system will fail to initiate the operation of the equipment is, and can be maintained, acceptably low without testing the equipment during reactor operation.

C. The equipment can routinely be tested when the reactor is shut down.

The list of reactor trip ESFAS equipment that cannot be tested at full power so as not to damage equipment or upset plant operation is:

A. Manual actuation switches for reactor trip system and ESFAS.

B. Turbine trip system.

C. Main steam line isolation valves (close).

D. Main feedwater and feedwater bypass isolation valves (close).

E. Feedwater regulating valves (close).

F. Main feedwater pump trip solenoids.

G. Reactor coolant pump breakers.

H. Reactor coolant pump seal water return valves (close).

I. Pressurizer power operated relief valves (PORVs) (open).

J. Instrument air containment isolation valves (close).

The justifications for not testing the above items at full power are discussed below:

A. Manual Actuation Switches for Reactor Trip System and ESFAS

These switches would cause initiation of their protection system function at power causing plant upset and/or reactor trip. It should be noted that the reactor trip function that is derived from the automatic safety injection signal is tested at power.

The analog signals, from which the automatic safety injection signal is derived, are tested at power in the same manner as the other analog signals and as described in paragraph 7.2.2.2.3. The processing of these signals in the solid-state protection system wherein their channel orientation converts to a logic train orientation is tested at power by the built-in semiautomatic test provisions of the solid-state protection system. The reactor trip breakers are tested at power as discussed in paragraph 7.2.2.2.3.

B. Turbine Trip System

Testing of the main turbine trip function under power operation is discussed in subsections 10.2.2 and 10.2.5.

C. Closing the Main Steam Line Isolation Valves

Main steam line isolation valves are routinely tested during refueling outages. Testing of the main steam line isolation valves to closure at power is not practical. As the plant power is increased, the coolant average temperature is programmed to increase. If the valves are closed under these elevated temperature conditions, the steam pressure transient would unnecessarily operate the steam generator relief valves and possibly the steam generator safety valves. The steam pressure transient produced would cause shrinkage in the steam generator level, which would cause the reactor to trip on low-low

steam generator water level. Testing during operation will decrease the operating life of the valve.

Based on the above-identified problems incurred with periodic testing of the main steam line isolation valves at power and since:

1. No practical system design permits operation of the valves without adversely affecting the safety or operability of the plant.
2. The probability that the protection system will fail to initiate the actuated equipment is acceptably low due to testing up to final actuation.
3. These valves are routinely tested during refueling outages.

The proposed resolution meets the guidelines of Regulatory Position D.4 of Regulatory Guide 1.22.

The main steam isolation valve actuator can be exercized periodically at power to approximately 90 percent of full open. Separate control switches and position lights are provided to test each redundant actuator hydraulic circuit on each valve.

The main steam isolation bypass valves can be closed at power, since the effect of the resulting change in steam flow is insufficient to upset the operation of the reactor average coolant temperature regulation system.

D. Closing the Main Feedwater and Feedwater Bypass Isolation Valves

The main feedwater and feedwater bypass isolation valves are routinely tested during refueling outages. Periodic testing of these isolation valves by closing them completely at power would induce steam generator water level transients and oscillations which would trip the reactor. These transient conditions would be caused by perturbing the feedwater flow and pressure conditions necessary for proper operation of the variable speed feedwater pump control system and the steam generator water level control system. Any operation which induces perturbations in the main feedwater flow, whether deliberate or otherwise, generally leads to a reactor trip and should be avoided.

Based on these identified problems incurred with periodic testing of these isolation valves at power and since:

1. No practical system design permits operation of these valves without adversely affecting the safety or operability of the plant.
2. The probability that the protection system will fail to initiate the activated equipment is acceptably low due to testing up to final actuation.
3. These valves are routinely tested during refueling outages.

The proposed resolution meets the guidelines of Regulatory Position D.4 of Regulatory Guide 1.22.

The main feedwater isolation valve actuator can be exercized periodically at power to approximately 90 percent of full open. Separate control switches and position lights are provided to test each redundant actuator hydraulic circuit on each valve. The feedwater bypass isolation valve cannot be similarly tested.

E. Closing the Feedwater Regulating Valves

These valves are routinely tested during refueling outages. To close them at power would adversely affect the operability of the plant. The verification of

operability of feedwater regulating valves at power is ensured by confirmation of proper operation of the steam generator water level control system. The actuation function of the solenoids, which provide the closing function, is periodically tested at power as discussed in paragraph 7.3.2.2.5. The operability of the slave relay which actuates the solenoid, which is the actuating device, is verified during this test. Although the closing of these regulating valves is blocked when the slave relay is tested, all functions are tested to ensure that no electrical malfunctions have occurred which could defeat the protective function. It is noted that the solenoids work on the deenergize-to-actuate principle, so that the feedwater regulating valves close upon either the loss of electrical power to the solenoids or loss of air pressure.

Based on the above, the testing of the isolating function of feedwater regulating valves meets the guidelines of Regulatory Position D.4 of Regulatory Guide 1.22.

At low power operation the bypass feedwater regulating valves are opened and the main feedwater regulating valves are closed. Testing of the bypass feedwater regulating valves under this condition is not permitted since it could cause unacceptable flow perturbation. The bypass feedwater regulating valves can be tested closed at full power since they are normally closed under these conditions. This is done by first opening and then closing the bypass feedwater regulating valves.

F. Main Feedwater Pump Trip Solenoids

Since no credit is taken for automatic tripping of the feedwater pumps, the main feedwater pump trip solenoids do not require periodic testing.

G. Reactor Coolant Pump Breakers

No credit is taken in the accident analyses for a reactor coolant pump breaker opening causing a direct reactor trip. Testing at power would result in a plant trip. Hence, these breakers are tested during scheduled refueling outages.

H. Reactor Coolant Pump Seal Water Return Valves (Close)

Seal water return line isolation valves are routinely tested during refueling outages. Closure of these valves during operation would cause the seal water system relief valve to lift, with the possibility of valve chatter. Valve chatter could damage this relief valve. Testing of these valves at power could cause equipment damage. Therefore, these valves are tested during scheduled refueling outages. As above, additional containment penetrations and containment isolation valves introduce additional unnecessary potential pathways for radioactive release following a postulated accident. Thus, the guidelines of Regulatory Position D.4 of Regulatory Guide 1.22 are met.

I. Pressurizer Power Operated Relief Valves (PORVs)

Testing of the pressurizer power relief valves to open at power is not practical. Opening of these valves at power would cause an unwarranted depressurization of the reactor coolant system. The valves should be routinely tested during plant shutdown with the block valve (corresponding to the pressurizer power relief valve being tested) in the open position, the relief valve solenoid is energized. The status of the relief valve is then verified to be in the open position via the limit switch indication. This solenoid should then be immediately deenergized and the position of the relief valve verified to be in the close position. This process should be repeated for each relief valve.

J. Instrument Air Containment Isolation Valve

The instrument air containment isolation valve is routinely tested during refueling outages. Testing of the instrument air containment isolation valve to close at power is not practical. Periodic testing of this valve by closing it completely would induce the change of the normal position of several pneumatic valves inside the containment (e.g., CVCS letdown line isolation valve, RCP seal no. 3 supply valves, steam generator blowdown isolation valves, containment drain sump isolation valve, and containment normal purge isolation valves).

The actuation function of the solenoids, which provide the closing function, is periodically tested at power as discussed in paragraph 7.3.1.2.2.5. The operability of the slave relay which actuates the solenoid, which is the actuating device, is verified during this test. Although the closing of this valve is blocked when the slave relay is tested, all functions are tested to ensure that no electrical malfunctions have occurred which could defeat the protective function. It is noted that the solenoids work on the deenergize-to- actuate principle, so that the valve closes upon either the loss of electrical power to the solenoids or loss of air pressure.

Based on the above, the testing of the isolating function of the valve meets the guidelines of Regulatory Position D.4 of Regulatory Guide 1.22.

7.1.2.6 Conformance to Regulatory Guide 1.53 and IEEE Standard 379-1972

The principles described in IEEE Standard 379-l972 are used in the design of both the Westinghouse protection system and the balance of plant ESFAS. These systems conform with the intent of this standard and the additional guidance of Regulatory Guide 1.53.

For the Westinghouse systems, the formal analyses are not documented exactly as outlined. Westinghouse has gone beyond the required analyses and has performed a fault tree analysis.(1)

The referenced report provides details of the analyses of the protection systems previously made to show conformance with the single-failure criterion set forth in section 4.2 of IEEE Standard 279-1971. The interpretation of the single-failure criterion provided by IEEE Standard 379-1972 does not indicate substantial differences with the Westinghouse interpretation of the criterion except in the methods used to confirm design reliability. Established design criteria in conjunction with sound engineering practices form the bases for the Westinghouse protection systems. The reactor trip and engineered safeguards actuation systems are each redundant safety systems. The required periodic testing of these systems discloses any failures or loss of redundancy which have occurred in the interval between tests, thus ensuring the availability of these systems.

7.1.2.7 Conformance to IEEE Standard 338-1975

The periodic testing of the reactor trip system and ESFAS conforms to the requirements of IEEE Standard 338-1975 with the following comments:

A. The surveillance requirements of the Technical Specifications for a protection system ensure that the system's functional operability is maintained comparable to the original design standards. Periodic tests at frequent intervals demonstrate this capability for the system, excluding sensors.

Overall protection systems response times are demonstrated by test. Sensors within the Westinghouse scope are demonstrated to be adequate for this design by vendor testing, by onsite tests in operating plants with appropriately similar design, or by suitable type testing. The nuclear instrumentation system detectors are excluded, since they exhibit response-time characteristics such that delays attributable to them are negligible in the overall channel response time required for safety.

Response time may be verified by actual response time tests in any series of sequential, overlapping, or total channel measurements, or by the summation of allocated sensor, signal processing, and actuation logic response times with actual response time tests on the remainder of the channel. Allocations for sensor response times may be obtained from: (1) historical records based on acceptable response time tests (hydraulic, noise, or power interrupt tests), (2) in place, onsite, or offsite (e.g., vendor) test measurements, or (3) vendor engineering specifications. Reference 5 provides the basis and methodology for using allocated sensor response times in the overall verification of the channel response time for specific sensors. Response time verification for other sensor types must be demonstrated by test.

Reference 6 provides the basis and methodology for using allocated signal processing and actuation logic response times in the overall verification of the protection system channel response time. The allocations for sensor, signal conditioning, and actuation logic response times must be verified prior to placing the component in operational service and reverified following maintenance that may adversely affect response time. In general, electrical repair work does not impact response time provided the parts used for repair are of the same type and value. Specific components may be replaced without verification testing. One example where response time could be affected is replacing the sensing assembly of a transmitter.

Each test shall include at least one logic train such that both logic trains are tested at least once per 36 months and one channel per function such that all channels are tested at least once every N times 18 months, where N is the total number of redundant channels in a specific protective function.

The measurement of response time provides assurance that the protective and ESF action function associated with each channel is completed within the time limit assumed in the accident analyses.

B. The reliability goals specified in section 4.2 of IEEE Standard 338-1975 are being developed, and adequacy of time intervals will be demonstrated at a later date.

C. The periodic time interval discussed in section 4.3 of IEEE Standard 338-1975 and specified in the Technical Specifications is conservatively selected to ensure that equipment associated with protection functions has not drifted beyond its minimum performance requirements. If any protection channel appears to be marginal or requires more frequent adjustments due to plant condition changes, the time interval is decreased to accommodate the situation until the marginal performance is resolved.

D. The test interval discussed in section 5.2 of IEEE Standard 338-1975 is developed primarily on past operating experience and modified if necessary to ensure that system and subsystem protection is reliably provided. Analytical methods for determining reliability are not used to determine test interval.

Based on the scope definition given in IEEE Standard 338-1975, no other systems described in chapter 7 are required to comply with this standard.

7.1.2.8 References

1. T. W. T. Burnett, "Reactor Protection System Diversity in Westinghouse Pressurized Water Reactors," WCAP-7306, April 1969.
2. Marasco, F. W., and Siroky, R. M., "Westinghouse 7300 Series Process Control System Noise Tests," WCAP-8892-A, June 1977.
3. Letter dated April 20, 1977, R. L. Tedesco (NRC) to C. Eicheldinger (Westinghouse).
4. Mermigos, J. F., "Bypass Test Instrumentation for the Vogtle Electric Generating Plant, Units 1 and 2," WCAP-13376, Revision 2, September 1992.
5. Howard, R. C., “Elimination of Pressure Sensor Response Time Testing Requirements,” WCAP-13632-P-A, Revision 2, January 1996.
6. Morgan, C. E., “Elimination of Periodic Protection Channel Response Time Tests,” WCAP-14036-P-A, Revision 1, October 1998.

TABLE 7.1.1-1 (SHEET 1 OF 7)

LISTING OF CRITERIA AS APPLIED TO
INSTRUMENTATION AND CONTROL SYSTEMS

Criteria	Title	Conformance Discussed In
1. General Design Criteria (GDC), Appendix A to 10 CFR 50		
GDC 1	Quality Standards and Records	3.1, 7.2.2
GDC 2	Design Bases for Protection Against Natural Phenomena	3.1, 7.2.1
GDC 3	Fire Protection	3.1, 9.5.1, 7.1.2
GDC 4	Environmental and Missile Design Bases	3.1, 7.2.1
GDC 5	Sharing of Structures, Systems, and Components	3.1
GDC 10	Reactor Design	3.1, 7.2.2
GDC 12	Suppression of Reactor Power Oscillations	3.1
GDC 13	Instrumentation and Control	3.1, 7.3.1, 7.3.2
GDC 15	Reactor Coolant System Design	3.1, 7.2.2
GDC 17	Electric Power Systems	3.1, 8.3, 8.2
GDC 19	Control Room	3.1
GDC 20	Protection System Functions	3.1, 7.1.2, 7.2.2, 7.3.1, 7.3.2
GDC 21	Protection System Reliability and Testability	3.1, 7.1.2, 7.2.2, 7.3.1, 7.3.2

TABLE 7.1.1-1 (SHEET 2 OF 7)

Criteria	Title	Conformance Discussed In
GDC 22	Protection System Independence	3.1, 7.1.2, 7.2.2, 7.3.1, 7.3.2
GDC 23	Protection System Failure Modes	3.1, 7.2.2, 7.3.1, 7.3.2
GDC 24	Separation of Protection and Control Systems	3.1, 7.2.2, 7.3.1, 7.3.2
GDC 25	Protection System Requirements for Reactivity Control Malfunctions	3.1, 7.3.2
GDC 26	Reactivity Control System Redundancy and Capability	3.1
GDC 27	Combined Reactivity Control Systems Capability	3.1, 7.3.1, 7.3.2
GDC 28	Reactivity Limits	3.1, 7.3.1, 7.3.2
GDC 29	Protection Against Anticipated Operational Occurrences	3.1, 7.2.2
GDC 33	Reactor Coolant Makeup	3.1
GDC 34	Residual Heat Removal	3.1
GDC 35	Emergency Core Cooling	3.1, 7.3.1, 7.3.2
GDC 37	Testing of Emergency Core Cooling System	3.1, 7.3.2
GDC 38	Containment Heat Removal	3.1, 7.3.1, 7.3.2
GDC 40	Testing of Containment Heat Removal System	3.1, 7.3.2
GDC 41	Containment Atmosphere Cleanup	3.1, 7.3.2
GDC 43	Testing of Containment Atmosphere Cleanup Systems	3.1, 7.3.2
GDC 44	Cooling Water	3.1

TABLE 7.1.1-1 (SHEET 3 OF 7)

Criteria	Title	Conformance Discussed In
GDC 46	Testing of Cooling Water System	3.1, 7.3.2
GDC 50	Containment Design Basis	3.1
GDC 54	Piping Systems Penetrating Containment	3.1, 6.2.4
GDC 55	Reactor Coolant Pressure Boundary Penetrating Containment	3.1, 6.2.4
GDC 56	Primary Containment Isolation	3.1, 6.2.4, .3.1
GDC 57	Closed System Isolation Valves	3.1, 6.2.4
2. IEEE Standards		
IEEE Std 279-1971 (ANSI N42.7-1972)	Criteria for Protection Systems for Nuclear Power Generating Stations	7.1, 7.2, 7.3, 7.6, 7.7
IEEE Std 308-1974	Criteria for Class 1E Electric Systems for Nuclear Power Generating Stations	7.6, 7.1, 8.1, 8.3
IEEE Std 317-1976	Electric Penetration Assemblies in Containment Structures for Nuclear Power Generating Stations	7.1.2, 8.1, 8.3
IEEE Std 323-1974	Qualifying Class 1E Equipment for Nuclear Power Generating Stations	3.11, 1.9, 8.1, 8.3 (RG 1.89)]
[(HISTORICAL) IEEE Std 336-1971 (ANSI N45.2.4-1972)	*Installation, Inspection and Testing Requirements for Instrumentation and Electric Equipment During the Construction of Nuclear Power Generating Stations*	*7.1.2, 8.0]*
IEEE Std 336-1985 (ASME NQA-1-1994)	Installation, Inspection, and Testing Requirements for Power, Instrumentation, and Control Equipment at Nuclear Facilities	7.1.2, 8.0
IEEE Std 338-1975	Trial-Use Criteria for the Periodic Testing of Nuclear Power Generating Station Protection Systems	7.1.2, 8.0
IEEE Std 344-1975 (ANSI N41.7)	Recommended Practices for Seismic Qualification of Class 1E Equipment for	3.10, 7.1.2, 8.1, 8.3

TABLE 7.1.1-1 (SHEET 4 OF 7)

Criteria	Title	Conformance Discussed In
	Nuclear Power Generating Stations	
IEEE Std 379-1972 (ANSI N41.2)	Trial-Use Guide for the Application of the Single Failure Criterion to Nuclear Power Generating Station Protection Systems	7.1.2, 8.0
IEEE Std 383-1974	Standard for Type Test of Class 1E Electric Cables, Field Splices, and Connectors for Nuclear Power Generating Stations	1.9.131, 8.1.4, 8.3
IEEE Std 384-1981	Standard Criteria for Independence of Class 1E Equipment and Circuits	7.1.2, 7.1.2, 8.1, 8.3
3. Regulatory Guides (RG)		
RG 1.6	Independence Between Redundant Standby (Onsite)Power Sources and Between Their Distribution Systems	1.9, 7.6, 8.0
RG 1.11	Instrument Lines Penetrating Primary Reactor Containment	1.9, 7.3.1
RG 1.12	Instrument for Earthquakes	1.9, 3.7.4
RG 1.22	Periodic Testing of Protection System Actuation Functions	1.9, 7.1.2, 7.3.2
RG 1.29	Seismic Design Classification	1.9
RG 1.30	Quality Assurance Requirements for the Installation, Inspection, and Testing of Instrumentation and Electric Equipment	1.9, 17.0
RG 1.32	Criteria for Safety-Related Electric Power Systems for Nuclear Power Plants	1.9
RG 1.45	Reactor Coolant Pressure Boundary Leakage Detection System	1.9, 5.2.5
RG 1.47	Bypassed and Inoperable Status Indication for Nuclear Power Plant Safety Systems	1.9, 7.5.5
RG 1.53	Application of the Single Failure Criterion to Nuclear Power Plant Protection Systems	1.9, 7.1.2

TABLE 7.1.1-1 (SHEET 5 OF 7)

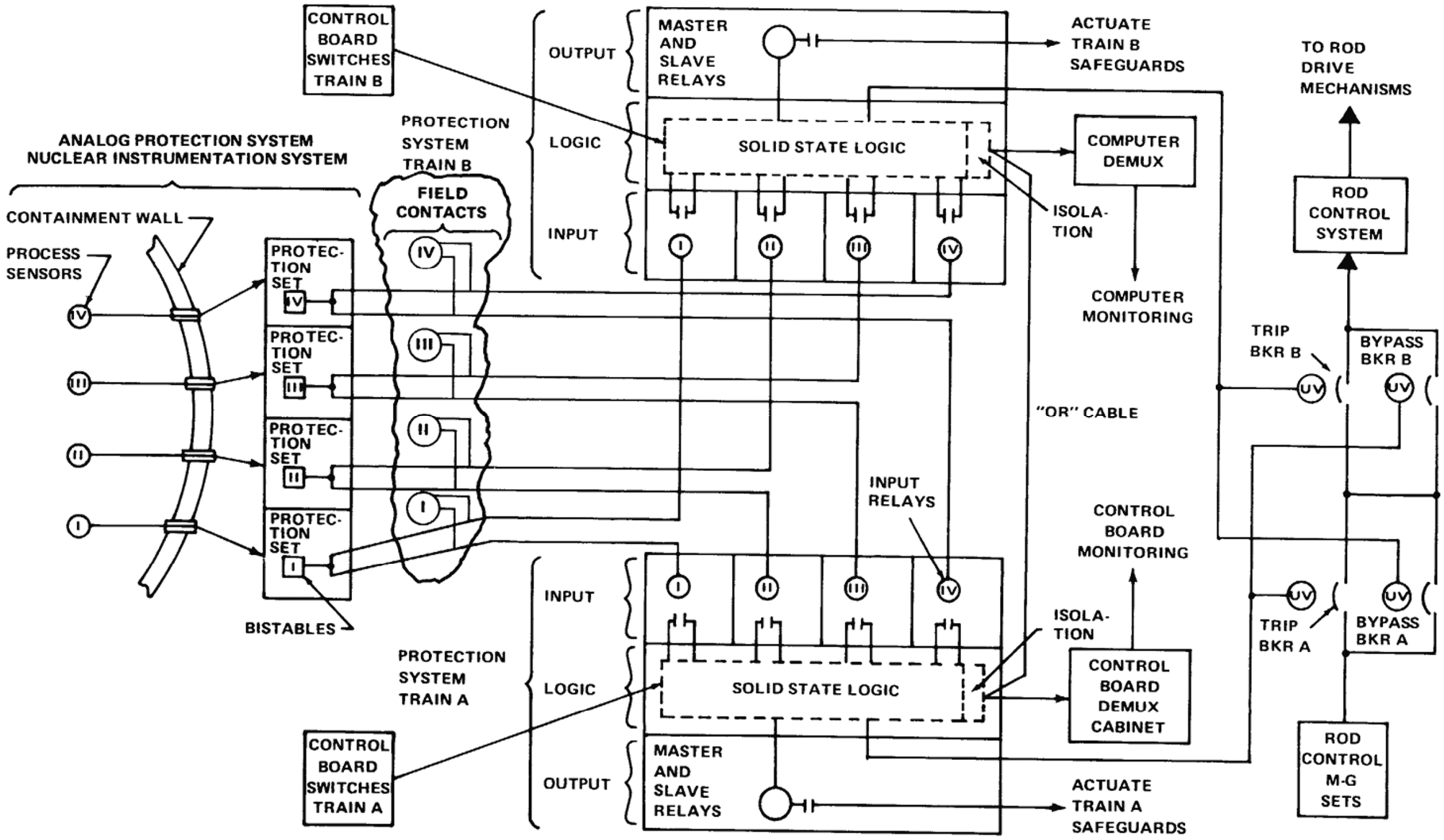
Criteria	Title	Conformance Discussed In
RG 1.62	Manual Initiation of Protection Actions	1.9, 7.3.1
RG 1.63	Electric Penetration Assemblies in Containment Structures for Light Water-Cooled Nuclear Power Plants	1.9, 8.1, 8.3
RG 1.67	Installation of Overpressure Protection Devices	3.9.B.3
RG 1.68	Initial Test Programs for Water-Cooled Nuclear Power Plants	1.9, 14.0
RG 1.70	Standard Format and Content of Safety Analysis Reports for Nuclear Power Plants LWR Edition	1.9
RG 1.75	Physical Independence of Electric Systems	1.9, 7.1.2
RG 1.80	Preoperational Testing of Instrumentation Air Systems	1.9, 9.3.1, 14.2.7
RG 1.89	Qualification of Class 1E Equipment for Nuclear Power Plants	1.9, 3.11
RG 1.95	Protection of Nuclear Power Plant Control Room Operators Against an Accidental Chlorine Release	1.9
RG 1.97	Instrumentation for Light Water-Cooled Nuclear Power Plants to Assess Plant Conditions During and Following an Accident	1.9
RG 1.100	Seismic Qualification of Electric Equipment for Nuclear Power Plants	1.9, 3.10
RG 1.105	Instrument Setpoints	7.1.2
RG 1.118	Periodic Testing of Electric Power and Protection Systems	1.9
RG 1.120	Fire Protection Guidelines for Nuclear Power Plants	1.9
4. Branch Technical Positions (BTP) ICSB		

TABLE 7.1.1-1 (SHEET 6 OF 7)

Criteria	Title	Conformance Discussed In
BTP ICSB 3	Isolation of Low-Pressure Systems from the High-Pressure Reactor Coolant System	7.6.2
BTP ICSB 4	Requirements on Motor-Operated Valves in the ECCS Accumulator Lines	7.6.4
BTP ICSB 5	Scram Breaker Test Requirements – Technical Specifications	7.2.2, Technical Specifications
BTP ICSB 9	Definition and Use of "Channel Calibration" - Technical Specifications	Technical Specifications
BTP ICSB 10	Electrical and Mechanical Equipment Seismic Qualification Program	3.10
BTP ICSB 12	Protection System Trip Point Changes for Operation with Reactor Coolant Pumps Out of Service	7.2.2, Technical Specifications
BTP ICSB 13	Design Criteria for Auxiliary Feedwater Systems	7.3.2
BTP ICSB 14	Spurious Withdrawals of Single Control Rods in Pressurized Water Reactors	7.7.2 15.2.1, 15.2.2, 15.3.6
BTP ICSB 15	Reactor Coolant Pump Breaker Qualification	3.10, 7.1.2, 7.2.1
BTP ICSB 18	Application of the Single Failure Criteria to Manually-Controlled Electrically-Operated Valves	Technical Specifications
BTP ICSB 20	Design of Instrumentation and Controls Provided to Accomplish Changeover from Injection to Recirculation Mode	7.6.5, 6.3.2
BTP ICSB 21	Guidance for Application of Regulatory Guide 1.47	7.5.5
BTP ICSB 22	Guidance for Application of Regulatory Guide 1.22	7.1.2

TABLE 7.1.1-1 (SHEET 7 OF 7)

Criteria	Title	Conformance Discussed In
BTP ICSB 26	Requirements for Reactor Protection System Anticipatory Trips	7.2.1



REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PROTECTION SYSTEM BLOCK DIAGRAM

FIGURE 7.1.1–1

7.2 REACTOR TRIP SYSTEM

7.2.1 DESCRIPTION

7.2.1.1 System Description

The reactor trip system automatically keeps the reactor operating within a safe region by shutting down the reactor whenever the limits of the region are approached. The safe operating region is defined by several considerations, such as mechanical/hydraulic limitations on equipment and heat transfer phenomena. Therefore, the reactor trip system keeps surveillance on process variables that are directly related to equipment mechanical limitations, such as pressure and pressurizer water level (to prevent water discharge through safety valves and uncovering heaters), and also on variables that directly affect the heat transfer capability of the reactor (e.g., flow and reactor coolant temperatures). Still other parameters utilized in the reactor trip system are calculated from various process variables. In any event, whenever a direct process or calculated variable exceeds a setpoint, the reactor will be shut down in order to protect against either gross damage to fuel cladding or loss of system integrity which could lead to release of radioactive fission products into the containment.

The following systems make up the reactor trip system. (See references 1, 2, 3, 5, 6, 7, and 8 for additional background information.)

- Process instrumentation and control system.
- Nuclear instrumentation system.
- Solid-state logic protection system.
- Reactor trip switchgear.
- Manual actuation circuit.

The reactor trip system consists of sensors, which monitor various plant parameters when connected with analog circuitry consisting of two to four redundant channels, and of digital circuitry, consisting of two redundant logic trains, which receives inputs from the analog protection channels to complete the logic necessary to automatically open the reactor trip breakers.

Either of the two trains, A or B, is capable of opening a separate and independent reactor trip breaker, RTA and RTB, respectively. The two trip breakers, in series, connect three-phase ac power from the rod drive motor generator sets to the rod drive power cabinets, as shown in drawing 1X6AA02-226. During plant power operation a dc undervoltage coil on each reactor trip breaker holds a trip plunger out against its spring, allowing the power to be available at the rod control power supply cabinets. For reactor trip, a loss of dc voltage to the undervoltage coil as well as energization of the shunt trip coils trips open the breaker. When either of the trip breakers opens, power is interrupted to the rod drive power supply, and the control and shutdown rods fall into the core. The rods cannot be withdrawn until the trip breakers are manually reset. The trip breakers cannot be reset until the abnormal condition which initiated the trip is corrected. Bypass breakers BYA and BYB are provided to permit testing of the trip breakers, as discussed in paragraph 7.2.2.2.3.

7.2.1.1.1 Functional Performance Requirements

The reactor trip system automatically initiates reactor trip:

A. Whenever necessary to prevent fuel damage for an anticipated operational transient (Condition II).

B. To limit core damage for infrequent faults (Condition III).

C. So that the energy generated in the core is compatible with the design provisions to protect the reactor coolant pressure boundary for limiting fault conditions (Condition IV).

The reactor trip system initiates a turbine trip signal whenever reactor trip is initiated. This prevents the reactivity insertion that would otherwise result from excessive reactor system cooldown to avoid unnecessary actuation of the engineered safety features actuation system.

The reactor trip system provides for manual initiation of reactor trip by operator action.

7.2.1.1.2 Reactor Trips

The various reactor trip circuits automatically open the reactor trip breakers whenever a condition monitored by the reactor trip system reaches a preset level. To ensure a reliable system, high-quality design, components, manufacturing, quality control, and testing are used. In addition to redundant channels and trains, the design approach provides a reactor trip system which monitors numerous system variables, thereby providing protection system functional diversity. The extent of this diversity has been evaluated for a wide variety of postulated accidents.

Table 7.2.1-1 provides a list of reactor trips which are described below. Table 7.2.1-2 provides a listing of the protection system interlocks and their P designations.

A. Nuclear Overpower Trips

The specific trip functions generated are described below.

1. Power Range High Neutron Flux Trip

The power range high neutron flux trip circuit trips the reactor when two out of the four power range channels exceed the trip setpoint.

There are two bistables in each channel, each with its own trip setting used for a high-and low-range trip setting. The high trip setting provides protection during normal power operation and is always active. The low trip setting, which provides protection during startup, can be manually bypassed when two out of the four power range channels read above approximately 10-percent power (P-10). Three out of the four channels below 10 percent automatically reinstate the trip function.

2. Intermediate Range High Neutron Flux Trip

The intermediate range high neutron flux trip circuit trips the reactor when one out of the two intermediate range channels exceeds the trip setpoint. This trip, which provides protection during reactor startup, can be manually blocked if two out of four power range channels are above approximately 10-percent power (P-10). Three out of the four power range channels below this value automatically reinstate the intermediate

range high neutron flux trip. The intermediate range channels (including detectors) are separate from the power range channels. The intermediate range channels can be individually bypassed at the nuclear instrumentation racks to permit channel testing during plant shutdown or prior to startup. This bypass action is monitored and annunciated on the control board.

3. Source Range High Neutron Flux Trip

The source range high neutron flux trip circuit trips the reactor when one out of the two source range channels exceeds the trip setpoint. This trip, which provides protection during reactor startup and plant shutdown, can be manually bypassed when one out of the two intermediate range channels reads above the P-6 setpoint value, and the trip is automatically reinstated when both intermediate range channels decrease below the P-6 setpoint value. This trip is automatically bypassed by two out of four logic from the power range protection interlock (P-10). This trip function can also be reinstated below P-10 by an administrative action requiring manual actuation of two control board-mounted switches. Each switch will reinstate the trip function in one of the two protection logic trains. The source range trip point is set between the P-6 setpoint (source range block power level) and the maximum source range power level. The channels can be individually bypassed at the nuclear instrumentation racks to permit channel testing during plant shutdown or prior to startup. This bypass action is monitored and annunciated on the control board.

4. Power Range High Positive Neutron Flux Rate Trip

This circuit trips the reactor when a sudden abnormal increase in nuclear power occurs in two out of four power range channels. This trip provides protection against rod ejection accidents of low worth from midpower and is always active. (See subsection 15.4.8.) This trip also provides protection when there is a rapid power increase (e.g., an uncontrolled rod cluster assembly withdrawal at power; see subsection 15.4.2).

Drawings 1X6AA02-227 and 1X6AA02-228 shows the logic for all of the nuclear overpower and rate trips.

B. Core Thermal Overpower Trips

The specific trip functions generated are described below.

1. Overtemperature ΔT Trip

This trip protects the core against low DNBR and trips the reactor on coincidence, as listed in table 7.2.1-1, with one set of temperature measurements per loop; where $\frac{\Delta t\ (1 + \tau_1 s)}{(1 + \tau_2 s)\ \ (1 + \tau_3 s)} \leq \Delta T$ setpoint.

The ΔT setpoint for this trip is continuously calculated by analog circuitry for each loop by solving the following expression:

$$\Delta T_0 \left[K_1 - K_2 \frac{(1+\tau_4 s)}{1+\tau_5 s} \left[(T_{avg} \frac{(1)}{1+\tau_6 s} - T^o_{avg}) \right] + K_3 (P - P_0) - f(\Delta\phi) \right]$$

where:

ΔT = measured ΔT.

ΔT_0 = indicated ΔT at rated thermal power.

T_{avg} = average reactor coolant temperature (°F).

$T°_{avg}$ = nominal T_{avg} at rated thermal power (°F).

P = pressurizer pressure (psig).

K_1 = preset bias.

K_2 = preset gain which compensates for effects of temperature on the DNB limits.

K_3 = preset gain which compensate for effect of pressure on the DNB limits.

τ_1, τ_2 = preset constants utilized in lead-lag compensator for ΔT.

P_0 = nominal RCS operating pressure (psig).

τ_3 = preset constant utilized in lag compensator for ΔT.

τ_4,τ_5 = preset constants which compensate for instrument time delay(s).

τ_6 = preset constant utilized for measured T_{avg} response compensation.

s = LaPlace transform operator.

f ($\Delta\phi$) = function of the neutron flux difference between upper and lower long ion chambers. (Refer to figure 7.2.1-1.)

A separate long ion chamber unit supplies the flux signal for each overtemperature ΔT channel.

Increases in $\Delta\phi$ beyond a predefined deadband results in a decrease in trip setpoint. (Refer to figure 7.2.1-1.)

For $T_{avg} < T°_{avg}$, the value of ($T°_{avg}$-T_{avg}) is clamped to limit the increase in the setpoint during cooldown transients.

The required one pressurizer pressure parameter per loop is obtained from separate sensors connected to three pressure taps at the top of the pressurizer. Four pressurizer pressure signals are obtained from the three taps by connecting one of the taps to two pressure transmitters. Refer to paragraph 7.2.2.3.3 for an analysis of this arrangement.

Drawing 1X6AA02-229 shows the logic for overtemperature ΔT trip function.

2. Overpower ΔT Trip

This trip protects against excessive power (fuel rod rating protection) and trips the reactor on coincidence as listed in table 7.2.1-1, with one set of temperature measurements per loop, where

$$\Delta T \frac{(1 + \tau_1 s)}{(1 + \tau_2 s)(1 + \tau_3 s)} \leq \Delta T \text{ setpoint.}$$

The ΔT setpoint for this trip is continuously calculated by analog circuitry for each loop using the following expression:

$$\Delta T_0 \left[K_4 - K_5 \frac{\tau_7 s}{1 + \tau_7 s} \frac{1}{1 + \tau_6 s} T_{avg} - K_5 \left[T_{avg} \left(\frac{1}{1 + \tau_6 s} \right) - T_{avg}^{\infty} \right] - f(\Delta\phi) \right]$$

where:

ΔT	=	measured ΔT.
ΔT_0	=	indicated ΔT rated thermal power.
$f(\Delta\phi)$	=	a function of the neutron flux difference between upper and lower long ion chamber section.
K_4	=	a preset bias.
K_5	=	a constant which compensates for instrument time delay.
K_6	=	a constant which compensates for the change in density flow and heat capacity of the water with temperature.
T_{avg}^{∞}	=	indicated T_{avg} at rated thermal power (°F).
T_{avg}	=	average reactor coolant temperature (°F).
τ_1, τ_2	=	preset constants utilized in lead-lag compensator for ΔT.

τ_3 = preset constant utilized in lag compensator for ΔT.

τ_7 = preset time constant (s).

s = LaPlace transform operator.

τ_6 = preset constant utilized for measured T_{avg} response compensation.

The source of temperature and flux information is identical to that of the overtemperature ΔT trip, and the resultant ΔT setpoint is compared to the same ΔT. Drawing 1X6AA02-229 shows the logic for this trip function.

C. Reactor Coolant System Pressurizer Pressure and Water Level Trips

The specific trip functions generated are described below.

1. Pressurizer Low-Pressure Trip

The purpose of this trip is to protect against low pressure which could lead to DNB. The parameter being sensed is reactor coolant pressure, as measured in the pressurizer. Above P-7 the reactor is tripped when the pressurizer pressure measurements (compensated for rate of change) fall below preset limits. This trip is blocked below P-7 to permit startup. The trip logic and interlocks are given in table 7.2.1-1.

The trip logic is shown in drawing 1X6AA02-230.

2. Pressurizer High-Pressure Trip

The purpose of this trip is to protect the reactor coolant system against system overpressure. The same sensors and transmitters used for the pressurizer low-pressure trip are used for the high-pressure trip, except that separate bistables are used for trip. These bistables trip when uncompensated pressurizer pressure signals exceed preset limits on coincidence, as listed in table 7.2.1-1. There are no interlocks or permissives associated with this trip function.

The logic for this trip is shown in drawing 1X6AA02-230.

3. Pressurizer High Water Level Trip

This trip is provided as a backup to the high pressurizer pressure trip and serves to prevent water relief through the pressurizer safety valves. This trip is blocked below P-7 to permit startup. The coincidence logic and interlocks of pressurizer high water level signals are given in table 7.2.1 1.

The trip logic for this function is shown in drawing 1X6AA02-230.

D. Reactor Coolant System Low Flow Trips

These trips protect the core from DNB in the event of a loss of coolant flow situation. Drawings 1X6AA02-228 and 1X6AA02-229 shows the logic for these trips. The means of sensing the loss of coolant flow are described below.

1. Low Reactor Coolant Flow

   The parameter sensed is reactor coolant flow. Four elbow taps in each coolant loop are used as a flow device that indicates the status of reactor coolant flow. The basic function of this device is to provide information as to whether or not a reduction in flow has occurred. An output signal from two out of the three bistables in a loop would indicate a low flow in that loop.

   The coincidence logic and interlocks are given in table 7.2.1-1

2. Reactor Coolant Pump Undervoltage Trip

   This trip is provided to protect against low flow which can result from loss of voltage to more than one reactor coolant pump motor (e.g., from plant blackout or reactor coolant pump breakers opening).

   Two undervoltage relays sense the voltage on the motor side of each reactor coolant pump breaker. These relays provide an output signal when the pump motor power bus voltage drops below approximately 70 percent of rated voltage. Signals from these relays are time delayed to prevent spurious trips caused by short-term voltage perturbations. The coincidence logic and interlocks are given in table 7.2.1-1.

3. Reactor Coolant Pump Underfrequency Trip

   This trip protects against low flow resulting from pump underfrequency, for example, a major power grid frequency disturbance. The function of this trip is to trip the reactor for an underfrequency condition greater than approximately 2.4 Hz.

   Two underfrequency relays sense the underfrequency on the motorside of each reactor coolant pump breaker. Signals from these relays are time delayed to prevent spurious trips caused by short-term frequency perturbations. The coincidence logic and interlocks are given in table 7.2.1-1.

E. Steam Generator Trip

The specific trip function generated is low-low steam generator water level trip.

This trip protects the reactor from loss of heat sink. This trip is actuated on two out of four low-low water level signals occurring in any steam generator.

The logic is shown in drawing 1X6AA02-231.

F. Reactor Trip on a Turbine Trip (Anticipatory)

The reactor trip on a turbine trip is actuated by two- out-of-three logic from emergency trip fluid pressure signals or by all closed signals from the turbine steam stop valves. A turbine trip causes a direct reactor trip above P-9. Although not credited in any safe`ty analysis, the reactor trip on turbine trip provides additional protection and conservatism beyond that required for the health and safety of the public. This trip is included as part of good engineering practice and prudent design and satisfies the requirement of TMI Action Items II.K.3.10 and II.K.3.12.

The turbine provides anticipatory trips to the reactor protection system from contacts which change position when the turbine stop valves close or when the turbine emergency trip fluid pressure goes below its setpoint.

Components specified for use as sensors for input signals to the reactor protection system for "emergency trip oil pressure low" and "turbine stop valves close" conform to the requirements of Institute of Electrical and Electronics Engineers (IEEE) 279-1971 and are environmentally qualified. However, pipe whip, jet impingement, and seismic criteria are not included in qualification, regarding mounting and location for that portion of the trip system located within non-Seismic Category 1 structures. (These criteria are also applicable to the steam dump solenoid valves and turbine impulse chamber pressure transmitters.)

Loss of signal from equipment located within non-Seismic Category 1 structures will result in a trip input to the reactor protection system.

In addition, the following measures will be taken to ensure the integrity of the cabling to the solid-state protection system (SSPS):

1. Inputs from the turbine steam stop valves will originate from four separate limit switches (one per valve), each of which is dedicated to providing an input to one channel of the SSPS. Cables carrying these signals will be routed in individual conduits. The four circuits will be separated from one another and from non-Class 1E circuits and identified according to the criteria imposed on Class 1E circuits, from their source up to their terminations within the SSPS cabinets.

   Additionally, fuses have been added in each turbine stop valve limit switch circuit before the circuit enters the turbine building. In the event of multiple ground faults, the fuses will isolate the affected channels and provide a trip signal input to the SSPS.

2. Input from the emergency trip oil pressure originates from three separate pressure transmitters powered from the balance of plant safety-related process instrumentation cabinet. The cables for these transmitters are routed in individual conduits within the turbine building, according to the criteria imposed on Class 1E circuits.

The logic for this trip is shown in drawings 1X6AA02-228 and 1X6AA02-240.

G. Safety Injection Signal Actuation Trip

A reactor trip occurs when the safety injection system is actuated. The means of actuating the safety injection system are described in section 7.3. This trip protects the core against a loss of reactor coolant or a steam line rupture.

Drawings 1X6AA02-232 and 1X6AA02-519 show the logic for this trip.

H. Manual Trip

The manual trip consists of two switches with two-train outputs on each switch on the main control board, and two single-train switches, one on each of the shutdown panels. One of the two-train outputs is used to actuate the train A reactor trip breaker, and the other output actuates the train B reactor trip breaker. Operating a manual trip switch removes the voltage from the undervoltage trip coil and energizes the shunt trip coil of each breaker.

There are no interlocks which can block this trip. Drawing 1X6AA02-227 shows the manual trip logic. The design conforms to Regulatory Guide 1.62, as shown in figure 7.2.1-2.

I. Solid State Protection System General Warning Alarm Reactor Trip

General warning alarm reactor trip is discussed in paragraph 7.2.2.2.3.

7.2.1.1.3 Reactor Trip System Interlocks

A. Power Escalation Permissives

The overpower protection provided by the out of core nuclear instrumentation consists of three discrete, but overlapping, ranges. Continuation of startup operation or power increase requires a permissive signal from the higher range instrumentation channels before the lower range level trips can be manually blocked by the operator.

A one-out-of-two intermediate range permissive signal (P-6) is required prior to source range trip blocking. Source range trips are automatically reactivated when both intermediate range channels are below the permissive (P-6) setpoint. There are two manual reset switches for administratively reactivating the source range level trip, if required, when it is between the permissive P-6 and P-10 setpoints. Source range level trip block is always maintained when above the permissive P-10 setpoint.

The intermediate range level trip and power range (low setpoint) trip can only be blocked after satisfactory operation and permissive information are obtained from two out of four power range channels. Four individual blocking switches are provided so that the low range power range trip and intermediate range trip can be independently blocked (one switch for each train).

These trips are automatically reactivated when three out of the four power range channels are below the permissive (P-10) setpoint, thus ensuring automatic activation of more restrictive trip protection.

The development of permissives P-6 and P-10 is shown in drawing 1X6AA02-228. All of the permissives are digital. They are derived from analog signals in the nuclear power range and intermediate range channels.

B. Blocks of Reactor Trips at Low Power

Interlock P-7 blocks a reactor trip at low power (below approximately 10 percent of full power) on a low reactor coolant flow in more than one loop, reactor coolant pump undervoltage, reactor coolant pump underfrequency, pressurizer low pressure, or pressurizer high water level. See drawings 1X6AA02-229 and 1X6AA02-230 for permissive applications. The low power signal is derived from three out of four power range neutron flux signals below the setpoint in coincidence with two out of two turbine impulse chamber pressure signals below the setpoint (low plant load). Turbine impulse chamber pressure transmitters and circuits in the turbine building are designed to criteria similar to the reactor trip on turbine trip circuits as described in paragraph 7.2.1.1.2.F. See drawings 1X6AA02-228 and 1X6AA02-240 for the derivation of P-7.

The P-8 interlock blocks a reactor trip when the plant is below approximately 48 percent of full power, on a low reactor coolant flow in any one loop. The block action (absence of the P-8 interlock signal) occurs when three out of four neutron flux power range signals are below the setpoint. Thus, below the P-8 setpoint, the reactor will be allowed to operate with one inactive loop, and trip will not occur until two loops are indicated as low flow. See drawing 1X6AA02-228 for derivation of P-8 and drawing 1X6AA02-229 for applicable logic.

7.2.1.1.4 Coolant Temperature Sensor Arrangement

The hot and cold leg temperature signals required for input to the protection and control functions are obtained using thermowell-mounted RTDs installed in each reactor coolant loop. The hot leg temperature measurement in each loop is accomplished using three fast-response, dual-element, narrow-range RTDs mounted in thermowells. The three thermowells in each loop are located within hot leg scoops 120 degrees apart in the cross-sectional plane of the piping, with one located at the top of the pipe, to obtain a representative temperature sample. The scoops have a flow hole machined into the end to facilitate the flow of water through holes in the leading edge of the scoop, past the thermowell, and back into the flow stream.

The temperatures measured by the three thermowell-mounted RTDs are different due to hot leg temperature streaming and vary as a function of thermal power. Therefore, these signals are averaged using electronic weighting to generate a hot leg average temperature. Provisions are incorporated into the process electronics to allow for operation with only two RTDs in service. The two RTD measurements can be biased to compensate for the loss of the third RTD as described in reference 4.

The cold leg temperature measurement in each loop is accomplished by one fast-response, dual-element, narrow-range RTD mounted in a thermowell.

7.2.1.1.5 Pressurizer Water Level Reference Leg Arrangement

The design of the pressurizer water level instrumentation employs the usual tank level measuring arrangement, using differential pressure between an upper and a lower tap on a column of water. A reference leg connected to the upper tap is kept full of water by condensation of steam at the top of the leg.

7.2.1.1.6 Analog System

The analog system consists of two instrumentation systems, the process instrumentation system, and the nuclear instrumentation system.

Process instrumentation includes those devices (and their interconnection into systems) which measure temperature, pressure, fluid flow, fluid level in tanks or vessels, and, occasionally, physicochemical parameters such as fluid conductivity or chemical concentration. Process instrumentation specifically excludes nuclear and radiation measurements. The process instrumentation includes the process measuring devices, power supplies, indicator, recorders, alarm actuating devices, controllers, signal conditioning devices, etc., which are necessary for day-to-day operation of the nuclear steam supply system, as well as for monitoring the plant and providing initiation of protective functions.

The primary function of nuclear instrumentation is to protect the reactor by monitoring the neutron flux and generating appropriate trips and alarms for various phases of reactor operating and shutdown conditions. It also provides a secondary control function and indicates reactor status during startup and power operation. The nuclear instrumentation system uses information from three separate types of instrumentation channels to provide three discrete protection levels. Each range of instrumentation (source, intermediate, and power) provides the necessary overpower reactor trip protection required during operation in that range. The overlap of instrument ranges provides reliable continuous protection, beginning with source level through the intermediate and low power level. As the reactor power increases, the overpower protection level is increased by administrative control and in-plant procedures after satisfactory higher range instrumentation operation is obtained. Automatic reset to more restrictive trip protection is provided when reducing power.

Various types of neutron detectors, with appropriate solid-state electronic circuitry, are used to monitor the leakage neutron flux from a completely shutdown condition to 200 percent of full power. The neutron flux covers a wide range between these extremes. Therefore, monitoring with several ranges of instrumentation is necessary.

The lowest range (source range) covers 7 decades of leakage neutron flux. The lowest observed count rate depends on the strength of the neutron sources in the core and the core multiplication associated with the shutdown reactivity. This is generally greater than two counts per second. The intermediate range covers 8 decades to 200 percent full power. Detectors and instrumentation are chosen to provide overlap between the higher portion of the source range and the lower portion of the intermediate range. The power range covers approximately 2 decades of the total instrumentation range to 120 percent full power. This is a linear range that overlaps with the higher portion of the intermediate range.

The system described above provides control room indication and recording of signals proportional to reactor neutron flux during core loading, shutdown, startup, and power operation, as well as during subsequent refueling. Startup rate indication for the source and intermediate range channels is provided at the control board. Reactor trip, rod stop, control, and alarm signals are transmitted to the reactor control and protection system for automatic plant control. Equipment failures and test status information are annunciated in the control room.

See references 1 and 2 for additional background information on the process and nuclear instrumentation.

7.2.1.1.7 Solid-State Logic Protection System

The solid-state logic protection system takes binary inputs (voltage/no voltage) from the process and nuclear instrument channels (nuclear steam supply system/balance of plant) and from field

instrument channels corresponding to conditions (normal/abnormal) of plant parameters. The system combines these signals in the required logic combination and generates a trip signal (no voltage) to the undervoltage trip attachment and shunt trip auxiliary relay coils of the reactor trip circuit breakers when the necessary combination of signals occur. The system also provides annunciator, status light, and computer input signals which indicate the condition of bistable input signals, partial trip, and full trip functions and the status of the various blocking, permissive, and actuation functions. In addition, the system includes means for semiautomatic testing of the logic circuits. See reference 3, 5, 6, 7, and 8 for additional background information.

7.2.1.1.8 Isolation Amplifiers

In certain applications, control signals are derived from individual protection channels through isolation amplifiers contained in the protection channel, as permitted by IEEE 279-1971.

In all of these cases, analog signals derived from protection channels for nonprotective functions are obtained through isolation amplifiers located in the analog protection racks. By definition, nonprotective functions include those signals used for control, remote process indication, and computer monitoring. Refer to paragraph 7.1.2.2.1 for a discussion of electrical separation of control and protection functions.

7.2.1.1.9 Power Supply and Environmental Variations

The power supply for the reactor trip system is described in section 7.6 and chapter 8. The environmental variations throughout which the system will perform are given in section 3.11.

7.2.1.1.10 Setpoints

The setpoints that require trip action are given in the Technical Specifications. A detailed discussion on setpoints is found in paragraph 7.1.2.1.9.

7.2.1.1.11 Seismic Design

The seismic design considerations for the reactor trip system are given in section 3.10. This design meets the requirements of General Design Criterion 2.

7.2.1.2 Design Bases Information

The information given below presents the design bases information requested by section 3 of IEEE 279-1971. Functional diagrams are presented in drawings 1X6AA02-225, 1X6AA02-226, 1X6AA02-227, 1X6AA02-228, 1X6AA02-229, 1X6AA02-230, 1X6AA02-231, 1X6AA02-232, 1X6AA02-233, 1X6AA02-234, 1X6AA02-235, 1X6AA02-236, 1X6AA02-237, 1X6AA02-238, 1X6AA02-239, 1X6AA02-240, 1X6AA02-494, 1X6AA02-495, 1X6AA02-496 and 1X6AA02-519.

7.2.1.2.1 Generating Station Conditions

The following are the plant conditions requiring reactor trip.

- DNBR approaching the design basis limit.
- Power density (kW/ft) approaching rated value for Condition II events. (See chapter 4 for fuel design limits.)

- Reactor coolant system overpressure creating stresses approaching the limits specified in chapter 5.

7.2.1.2.2 Generating Station Variables

The following are the variables required to be monitored in order to provide reactor trips. (See table 7.2.1-1.)

- Neutron flux.
- Reactor coolant temperature.
- Reactor coolant system pressure (pressurizer pressure).
- Pressurizer water level.
- Reactor coolant flow.
- Reactor coolant pump operational status (voltage and frequency).
- Steam generator water level.
- Turbine-generator operational status (trip fluid pressure and stop valve position).
- Safety injection signal.

7.2.1.2.3 Spatially Dependent Variables

The reactor coolant temperature is spatially dependent. (See subsection 7.3.1 for a discussion of this variable spatial dependence.)

7.2.1.2.4 Limits, Margins, and Setpoints

The parameter values that will require reactor trip are given in the Technical Specifications and in chapter 15. The accident analyses of chapter 15 demonstrate that the setpoints used in the Technical Specifications are conservative.

The setpoints for the various functions in the reactor trip system have been analytically determined such that the operational limits so prescribed will prevent fuel rod clad damage and loss of integrity of the reactor coolant system as a result of any American Nuclear Society (ANS) Condition II event (anticipated malfunction). As such, during any ANS Condition II event, the reactor trip system limits the following parameters to:

- Minimum DNBR = the design basis limit.
- Maximum system pressure = 2735 psig.
- Fuel rod maximum linear power for determination of protection setpoints = 22.4 kW/ft.

The accident analyses described in section 15.4 demonstrate that the functional requirements, as specified for the reactor trip system, are adequate to meet the above considerations, even assuming, for conservatism, adverse combinations of instrument errors. (Refer to table 15.3.1-

1.) A discussion of the safety limits associated with the reactor core and reactor coolant system, plus the limiting safety system setpoints, are presented in the Technical Specifications.

7.2.1.2.5 Abnormal Events

The malfunctions, accidents, or other unusual events which could physically damage reactor trip system components or could cause environmental changes are as follows:

- Earthquakes. (See chapters 2 and 3.)
- Fire. (See subsection 9.5.1.)
- Explosion - hydrogen buildup inside containment. (See section 6.2.)
- Missiles. (See section 3.5.)
- Flood. (See chapters 2 and 3.)
- Wind and tornadoes. (See section 3.3.)

The reactor trip system fulfills the requirements of IEEE 279-1971 to provide automatic protection and to provide initiating signals to mitigate the consequences of faulted conditions. The reactor trip system is protected against destruction of the system from fires, explosions, floods, wind, and tornadoes. (See each item above.)

7.2.1.2.6 Minimum Performance Requirements

A. Reactor Trip System Response Times

The response time of each reactor trip function shown in Technical Specification Table 3.3.1-1 is shown in FSAR table 7.2.1-4. Response time verification for selected components may use the predetermined allocation values provided in FSAR table 7.2.1-5.

Reactor trip system response time is defined in section 7.1. Typical maximum allowable time delays in generating the reactor trip signal are tabulated in table 7.2.1-3. See paragraph 7.1.2.7 A for a discussion of periodic response time verification.

B. Reactor Trip Accuracies

Accuracy is defined in section 7.1. Reactor trip accuracies are tabulated in table 7.2.1-3. An additional discussion on accuracy is found in subsection 7.1.2.

C. Protection System Ranges

Typical protection system ranges are tabulated in table 7.2.1-3. Range selection for the instrumentation covers the expected range of the process variable being monitored during power operation. Limiting setpoints are at least 5 percent from the end of the instrument span.

7.2.1.3 Final System Drawings

Functional block diagrams, electrical elementaries, and other drawings required to perform a safety review are listed in the safety-related drawing package. (See section 1.7.)

7.2.1.4 References

1. Reid, J. B., "Process Instrumentation for Westinghouse Nuclear Steam Supply Systems," WCAP-7913, January 1973. (Additional background information only.)
2. Lipchak, J. B., "Nuclear Instrumentation System," WCAP-8255, January 1974. (Additional background information only.)
3. Katz, D. N., "Solid State Logic Protection System Description," WCAP-7488-L (Proprietary), March 1971, and WCAP-7672 (Nonproprietary), May 1971. (Additional background information only.)
4. DiTommaso, S. M., Sterrett, C. R., "RTD Bypass Elimination Licensing Report for Vogtle Electric Generating Plant." WCAP-12788 (Proprietary), March 1, 1991. (Additional background information only.)
5. WCAP-16769-P Revision 1, “Westinghouse SSPS Universal Logic Board Replacement Summary Report 6D30225G01/G02/G03/G04.”
6. WCAP-16770-P Revision 0, “Westinghouse SSPS Safeguards Driver Board Replacement Summary Report 6D30252G01/G02.”
7. WCAP-16771-P Revision 0, “Westinghouse SSPS Undervoltage Driver Board Replacement Summary Report 6D30350G01/G02.”
8. WCAP-16772-P Revision 0, “Westinghouse SSPS Semi-Automatic Tester Board Replacement Summary Report 6D30520G01/G02/G03/G04/G05.”

7.2.2 ANALYSES

7.2.2.1 Failure Modes and Effects Analyses

An analysis of the reactor trip system has been performed. Results of this study and a fault tree analysis are presented in reference 1.

7.2.2.2 Evaluation of Design Limits

While most setpoints used in the reactor protection system are fixed, there are variable setpoints, most notably the over-temperature ΔT and overpower ΔT setpoints. All setpoints in the reactor trip system have been selected on the basis of engineering design or safety studies. The capability of the reactor trip system to prevent loss of integrity of the fuel cladding and/or reactor coolant system (RCS) pressure boundary during Condition II and III transients is demonstrated in chapter 15. These accident analyses are carried out using those setpoints determined from results of the engineering design studies. Setpoint limits are presented in the Technical Specifications. A discussion of the purpose of each of the various reactor trips and the accident analyses (where appropriate) which utilize this trip are presented below. It should be noted that the selected trip setpoints provide for a margin before protective action is actually required to allow for uncertainties and instrument errors. The design meets the requirements of General Design Criteria (GDC) 10 and 20.

7.2.2.2.1 Trip Setpoint Discussion

The departure from nucleate boiling ratio (DNBR) existing at any point in the core, for a given core design, can be determined as a function of the core inlet temperature, power output, operating pressure, and flow. Below the DNBR design basis limit there is likely to be significant local fuel cladding failure. Consequently, core safety limits, in terms of a DNBR equal to the design basis limit for the hot channel, can be developed as a function of core ΔT, T_{avg}, and pressure for specified flow as illustrated by the solid lines in figure 15.0.6-1. The dashed lines indicate the maximum permissible setpoint (ΔT) as a function of T_{avg} and pressure for the overtemperature and overpower reactor trip. Actual setpoint constants in the equation representing the dashed lines are as given in the Technical Specifications.

These values are conservative to allow for instrument errors. The design meets the requirements of GDC 10, 15, 20, and 29.

The DNBR is not a directly measurable quantity. However, the process variables that determine DNBR are sensed and evaluated. Small isolated changes in various process variables may not, individually, result in violation of a core safety limit. However, the combined variations, over sufficient time, may cause the overpower or overtemperature safety limit to be exceeded. The reactor trip system provides reactor trips associated with individual process variables, in addition to the overpower/overtemperature safety limit trips. Process variable trips prevent reactor operation whenever a change in the monitored value is such that a core or system safety limit could potentially be exceeded, should operation continue. Basically, the high-pressure, low-pressure, and overpower/over-temperature ΔT trips provide sufficient protection for slow transients, as opposed to such trips as low flow or high flux, which will trip the reactor for rapid changes in flow or flux, respectively, that would result in fuel damage before actuation of the slower responding ΔT trips could be effected.

Therefore, the reactor trip system has been designed to provide protection for fuel cladding and RCS pressure boundary integrity where:

A. A rapid change in a single variable or factor will quickly result in exceeding a core or a system safety limit.

B. A slow change in one or more variables will have an integrated effect which will cause safety limits to be exceeded.

Overall, the reactor trip system offers diverse and comprehensive protection against fuel cladding failure and/or loss of RCS integrity for Condition II and III accidents. This is demonstrated by table 7.2.2-1 which lists the various trips of the reactor trip system, the corresponding technical specification on safety limits and safety system settings, and the appropriate accident discussed in the safety analyses in which the trip could be utilized.

In accordance with Branch Technical Position I CSB 12 the reactor trip system automatically provides core protection during nonstandard operating configuration; i.e., operation with a loop out of service. Although operating with a loop out of service over an extended time is considered to be an unlikely event, no protection system setpoints need to be reset. This is because the nominal value of the power (P-8) interlock setpoint restricts the power, such that DNBRs less than the design basis limit will not be realized during any Condition II transients occurring during this mode of operation. This restricted power is considerably below the boundary of permissible values, as defined by the core safety limits for operation with a loop out of service. Thus, the P-8 interlock acts, essentially, as a high nuclear power reactor trip when operating with one loop not in service. By first resetting the coefficient setpoints in the overtemperature ΔT function to more restrictive values, as listed in the Technical Specifications, the P-8 setpoint can then be increased to the maximum value consistent with maintaining DNBR

above the design basis limit for Condition II transients in the one-loop shutdown mode. The resetting of the overtemperature ΔT trip and P-8 will be carried out under administrative control and the direction of authorized supervision and with the plant conditions prescribed in the Technical Specifications.

The design meets the requirements of GDC 21.

Preoperational testing is performed on reactor trip system components and systems to determine equipment readiness for startup. This testing serves as a further evaluation of the system design.

Analyses of the results of Condition I, II, III, and IV events, including considerations of instrumentation installed to mitigate their consequences, are presented in chapter 15. The instrumentation installed to mitigate the consequences of load rejection and turbine trip is given in section 7.4.

7.2.2.2.2 Reactor Coolant Flow Measurement

The elbow taps used on each loop in the primary coolant system are instrument devices that indicate the status of the reactor coolant flow. The basic function of devices is to provide information as to whether or not a reduction in flow has occurred. The correlation between flow and elbow tap signal is given by the following equation:

$$\frac{\Delta P}{\Delta P_o} = \left(\frac{W}{W_o}\right)^2$$

where ΔP_o is the pressure differential at the reference flow W_o, and ΔP is the pressure differential at the corresponding flow, W. The full-flow reference point is established during initial plant startup. The low-flow trip point is then established by extrapolating along the correlation curve. The expected absolute accuracy of the channel is within ±10 percent of full flow, and field results have shown the repeatability of the trip point to be within ±1 percent.

7.2.2.2.3 Evaluation of Compliance to Applicable Codes and Standards

The reactor trip system meets the criteria of the GDC, as indicated. The reactor trip system meets the requirements of section 4 of Institute of Electrical and Electronics Engineers (IEEE) 279-1971, as indicated below.

A. General Functional Requirement

The protection system automatically initiates appropriate protective action whenever a condition monitored by the system reaches a preset level. Functional performance requirements are given in paragraph 7.2.1.1.1. Paragraph 7.2.1.2.4 presents a discussion of limits, margins, and levels; paragraph 7.2.1.2.5 discusses unusual (abnormal) events; and paragraph 7.2.1.2.6 presents minimum performance requirements.

B. Single Failure Criterion

The protection system is designed to provide two, three, or four instrumentation channels for each protective function and two logic train circuits. These redundant channels and trains are electrically isolated and physically separated. Thus, any single failure within a channel or train will not prevent protective action at the system level, when required. Loss of input power to a channel or logic

train, the most likely mode of failure, will result in a signal calling for a trip. This design meets the requirements of GDC 23.

To prevent the occurrence of common mode failures, such additional measures as functional diversity, physical separation, and testing as well as administrative control during design, production, installation, and operation are employed, as discussed in reference 1. The design meets the requirements of GDC 21 and 22.

C. Quality of Components and Modules

For a discussion on the quality of the components and modules used in the reactor trip system, refer to chapter 17. The quality assurance applied conforms to GDC 1.

D. Equipment Qualification

For a discussion of the type of tests made to verify the performance requirements, refer to section 3.11. The test results demonstrate that the design meets the requirements of GDC 4.

E. Channel Integrity

Protection system channels required to operate in accident conditions maintain necessary functional capability under extremes of conditions relating to environment, power supply, malfunctions, and accidents. The power supply for the reactor trip system is described in chapter 8. The environmental variations, throughout which the system will perform, are given in section 3.11.

F. Independence

Channel independence is carried throughout the system, extending from the sensor to the devices actuating the protective function. Physical separation is used to achieve separation of redundant transmitters. Separation of wiring is achieved using separate wireways, cable trays, conduit runs, and containment penetrations for each redundant channel. Redundant analog equipment is separated by locating modules in different protection cabinets. Each redundant protection channel set is energized from a separate ac power feed. This design meets the requirements of GDC 21.

Two reactor trip breakers, which are actuated by two separate logic matrices, interrupt power to the control rod drive mechanisms. The breaker main contacts are connected in series with the power supply, so that opening either breaker interrupts power to all control rod drive mechanisms, permitting the rods to free fall into the core. (See figure 7.1.1-1.)

The design philosophy is to make maximum use of a wide variety of measurements. The protection system continuously monitors numerous diverse system variables. Generally, two or more diverse protection functions would terminate an accident before intolerable consequences could occur. This design meets the requirements of GDC 22.

G. Control and Protection System Interaction

The protection system is designed to be independent of the control system. In certain applications, the control signals and other nonprotective functions are derived from individual protective channels through isolation amplifiers. The isolation amplifiers are classified as part of the protection system and are located in the analog protection racks. Nonprotective functions include those signals used for control, remote process indication, and computer monitoring. The isolation amplifiers are designed, such that a short circuit, open circuit, or the application of credible fault voltages from within the cabinets on the isolated output portion of the circuit (i.e., the nonprotective side of the circuit) will not affect the input (protective) side of the circuit. The signals obtained through the isolation amplifiers are never returned to the protection racks. This design meets the requirements of GDC 24 and section 4.7 of IEEE 279-1971.

The results of applying various malfunction conditions on the output portion of the isolation amplifiers show that no significant disturbance to the isolation amplifier input signal occurred.

H. Derivation of System Inputs

To the extent feasible and practical, protection system inputs are derived from signals which are direct measures of the desired variables. Variables monitored for the various reactor trips are listed in paragraph 7.2.1.2.2.

I. Capability for Sensor Checks

The operational availability of each system input sensor during reactor operation is accomplished by cross-checking between channels that bear a known relationship to each other and that have readouts available. Channel checks are discussed in the Technical Specifications.

J. Capability for Testing

The reactor trip system is capable of being tested during power operation. Where only parts of the system are tested at any one time, the testing sequence provides the necessary overlap between the parts to ensure complete system operation. The testing capabilities are in conformance with Regulatory Guide 1.22, as discussed in paragraph 7.1.2.5.

The protection system is designed to permit periodic testing of the analog channel portion of the reactor trip system during reactor power operation without initiating a protective action, unless a trip condition actually exists. This is because of the coincidence logic required for reactor trip. These tests may be performed at any plant power, from cold shutdown to full power. Before starting any of these tests with the plant at power, all redundant reactor trip channels associated with the function to be tested must be in the normal (untripped) mode in order to avoid spurious trips. Setpoints are referenced in the precautions, limitations, and setpoints portion of the plant technical manual.

1. Analog Channel Tests

Analog channel testing of the process channels which produce the two-out-of-four or two-out-of-three protection logic is performed at the analog

instrumentation rack set by individually testing each instrumentation channel. Testing is accomplished through the use of a bypass testing instrumentation test panel installed in each of the 7300 protection channel sets. Use of this panel will prevent the initiation of an unwarranted protective action from that channel during the short period that it is undergoing test. Located on the test panel is a keylock switch which controls the use/operation of the panel during testing and normal operation. Activation of this keylock switch will provide an automatic and continuous indication (alarm and annunciator) in the control room to alert the operators that a 7300 process channel is being tested in the bypass condition. Individual toggle switches are also provided on the test panel for each 7300 bistable. The use of these switches will allow the primary field signal power to be replaced with an imposed test signal power to prevent disruption of the 26-V dc source provided from the protection system bistables to the SSPS input relays. These switches also isolate the 7300 outputs from the BTI panel. The keylock switch provided on the BTI test panel has two operable positions:

- NORMAL - The BTI test panel is disabled, along with all of the toggle switches on that test panel.
- BYPASS ENABLE - The BTI test panel has the capability through the use of the individual toggle switches to place a channel in bypass. When in this mode, automatic and continuous indication of a bypass condition or the potential for a bypass condition is provided to the control room.

When in the bypass enable keylock switch mode of operation, the individual toggle switches on the test panel have two operable positions:

- NORMAL - Live field signal power supplied to SSPS.
- BYPASS - Test signal power supplied to SSPS when an individual process channel toggle switch is placed in the bypass condition. To alert the test technician of this state of operation, a local status light is provided on the BTI test panel, one for each bistable to be tested, to indicate which channel is in test.

Reference 5 provides additional information on this subject.

The following analog channels will be tested as described above:

- T_{avg} and ΔT protection channel testing.
- Pressurizer pressure protection channel testing.
- Pressurizer water level protection channel testing.
- Steam generator water level protection channel testing.
- Reactor coolant low flow.
- Steam pressure protection channels.
- Containment pressure.
- Turbine (anticipatory trip) and trip fluid pressure channel testing.

The underfrequency and undervoltage protection channels are not equipped with a bypass capability for testing. These channels are tested by individually introducing dummy input signals into the instrumentation channels and observing the tripping of the appropriate output bistables. Process analog output to the logic circuitry is interrupted during individual channel testing by a test switch which deenergizes the associated logic input and inserts a proving lamp in the bistable output. Interruption of the bistable output to the logic circuitry for any cause (test, maintenance purposes, or removal from service) will cause that portion of the logic to be actuated (partial trip), accompanied by a partial trip alarm and channel status light actuation in the control room. Each channel contains those switches, test points, etc., necessary to test the channel. See references 2 and 3 for additional background information.

2. Nuclear Instrumentation Channel Tests

The nuclear instrumentation system (NIS) channels which produce a rod stop, permissive, or a reactor trip on one-out-of-two, one-out-of-four, two-out-of-four, or three-out-of four protection logic are provided with a bypass function to prevent the initiation of an unwarranted protective action from that channel during the short period that it is undergoing test. To permit testing of an NIS channel in the bypass mode, a BTI test panel is installed in each of the four NIS protection channel sets (racks). Located on the test panel is a keylock switch which controls the use/operation of the panel during testing and normal operation. Activation of this keylock switch will provide automatic and continuous indication (alarm and annunciator) in the control room to alert the operators that an NIS channel is being tested in the bypass condition. Individual make-before-break toggle switches are also provided on the test panel for each bistable associated with the protection channel set. The make-before-break switch is located on the NIS BTI panels only. Use of these switches will allow the primary field signal power to be replaced by an imposed test signal power to prevent disruption of the 118 V-ac from the protection system bistables to the SSPS input relays. These switches also isolate the NIS drawer outputs from the BTI panel.

The keylock switch provided on the BTI test panel has two operable positions:

- NORMAL - The BTI test panel is disabled, along with all of the toggle switches on that test panel.
- BYPASS ENABLE - The BTI test panel has the capability through the use of the individual toggle switches to place a channel in bypass. When in this mode, automatic and continuous indication of a bypass condition or the potential for a bypass condition is provided to the control room.

When in the bypass enable keylock switch mode of operation, the individual make-before-break toggle switches on the test panel have two operable positions and a transition position:

- NORMAL - Live field signal power supplied to SSPS.

- MID-POSITION (MAKE-BEFORE-BREAK) - Live field signal power and test signal power is supplied to the SSPS.
- BYPASS - Test signal power supplied to SSPS when an individual process channel toggle switch is placed in the bypass condition. To alert the test technician of this state of operation, a local status light is provided on the BTI test panel, one for each bistable to be tested, to indicate which channel is in test.

Since the power provided to the NIS is ac, prior to placing a nuclear instrumentation system channel in bypass, the live signal power signal and the test signal power sources must be in phase to prevent an unwarranted protective action. Alignment of phase is adjusted and verified at the test points provided in the BTI test panel per installation instructions. Once the sources are in phase, the make-before-break switch will be in a position to provide the 118 V-ac bypass voltage. Individual bypass status lights located on the bypass test panels are provided to indicate the bypassed condition of these bistable outputs to the SSPS.

Reference 5 provides additional information on this subject.

It should be noted that a valid trip signal would cause the channel under test to trip at a lower actual reactor power level. A reactor trip would occur when a second bistable trips. No provision has been made in the channel test circuit for reducing the channel signal level below that signal being received from the nuclear instrumentation system detector.

A nuclear instrumentation system channel which can cause a reactor trip through one of two protection logic (source or intermediate range) is provided with a bypass function which prevents the initiation of a reactor trip from that particular channel during the short period that it is undergoing test. Indication of these bypasses are provided locally via status lights on the NIS bypass test panel or remotely via a main control room annunciator.

Periodic tests of the nuclear instrumentation system are performed in accordance with the plant Technical Specifications.

Any deviations noted during the performance of these tests are investigated and corrected in accordance with the established calibration and trouble shooting procedures provided in the plant technical manual for the nuclear instrumentation system. Control and protection trip settings are indicated in the plant technical manual under precautions, limitations, and setpoints.

For additional background information on the nuclear instrumentation system, see reference 3.

3. Solid-State Logic Testing

The reactor logic trains of the reactor trip system are designed to be capable of complete testing at power. After the individual channel analog

testing is complete, the logic matrices are tested from the train A and train B logic rack test panels. This step provides overlap between the analog and logic portions of the test program. During this test, all of the logic inputs are actuated automatically in all combinations of trip and nontrip logic. Trip logic is not maintained sufficiently long to permit opening of the reactor trip breakers.

The reactor trip undervoltage coils are "pulsed" in order to check continuity. During logic testing of one train, the other train can initiate any required protective functions. Annunciation is provided in the control room to indicate when a train is in test (train output bypassed) and when a reactor trip breaker is bypassed. Logic testing can be performed in less than 30 min.

A direct reactor trip resulting from undervoltage or underfrequency on the reactor coolant pump buses is provided, as discussed in paragraph 7.2.1.1.2.D and shown in drawings 1X6AA02-228 and 1X6AA02-229. The logic for these trips is capable of being tested during power operation. When parts of the trip are being tested, an overlap is provided between parts so that a complete logic test is provided. Thus, complete testing of protection system equipment is possible.

This design complies with the testing requirements of IEEE 279-1971 and 338-1975, as discussed in subsection 7.1.2.

The permissive and block interlocks associated with the reactor trip system and engineered safety features actuation system are given in tables 7.2.1-2 and 7.2.1-3 and designated protection or "P" interlocks. As a part of the protection system, these interlocks are designed to meet the testing requirements of IEEE 279-1971 and 338-1975.

Testing of all protection system interlocks is provided by the logic testing and semiautomatic testing capabilities of the solid state protection system. In the solid state protection system the undervoltage trip attachment and shunt trip auxiliary relay coils (reactor trip) and master relays (engineered safeguards actuation) are pulsed for all combinations of trip or actuation logic, with and without the interlock signals. For example, reactor trip on low flow (two out of four loops showing two out of three low flow) is tested to verify operability of the trip above P-8 and nontrip below P-7. (See drawing 1X6AA02-229). Interlock testing may be performed at power.

Testing of the logic trains of the reactor trip system includes a check of the input relays and a logic matrix check. The following sequence is used to test the system:

- Check of Input Relays

  During testing of the process instrumentation system and nuclear instrumentation system channels, each channel bistable is placed in a trip mode, causing one input relay in train A and one in train B to deenergize. A contact of each relay is connected to a universal logic

printed circuit card. This card performs both the reactor trip and monitoring functions. Each reactor trip input relay contact causes a status lamp and an annunciator on the control board to operate. Either the train A or train B input relay operation will light the status lamp and annunciator.

Each train contains a multiplexing test switch. At the start of a process or nuclear instrumentation system test, this switch (in either train) is placed in the A + B position. The A + B position alternately allows information to be transmitted from the two trains to the control board. A steady status lamp and annunciator indicates that input relays in both trains have been deenergized. A flashing lamp means that both the input relays in the two trains did not deenergize. Contact inputs to the logic protection system, such as reactor coolant pump bus underfrequency relays, operate input relays which are tested by operating the remote contacts, as described above, and use the same type of indications as those provided for bistable input relays.

Actuation of the input relays provides the overlap between the testing of the logic protection system and the testing of those systems supplying the inputs to the logic protection system. Test indications are status lamps and annunciators on the control board. Inputs to the logic protection system are checked one channel at a time, leaving the other channels in service. For example, a function that trips the reactor when two out of four channels trip becomes a one out of three trip when one channel is placed in the trip mode. Both trains of the logic protection system remain in service during this portion of the test.

- Check of Logic Matrices

  Logic matrices are checked, one train at a time. Input relays are not operated during this portion of the test. Reactor trips from the train being tested are inhibited with the use of the input error inhibit switch on the semiautomatic test panel in the train. At the completion of the logic matrix tests, one bistable in each channel of process instrumentation or nuclear instrumentation is tripped to check closure of the input error inhibit switch contacts.

  The logic test scheme uses pulse techniques to check the coincidence logic. All possible trip and nontrip combinations are checked. Pulses from the tester are applied to the inputs of the universal logic card at the same terminals that connect to the input relay contacts. Thus, there is an overlap between the input relay check and the logic matrix check. Pulses are fed back from the reactor trip breaker undervoltage trip attachment and shunt trip auxiliary relay coils to the tester. The pulses are of such short duration that the reactor trip breaker undervoltage coil armature cannot respond mechanically.

  Periodic testing of the solid state protection system includes testing of the master and slave relays from the system's relay test panel.

Test indications that are provided are an annunciator in the control room, indicating that reactor trips from the train have been blocked and that the train is being tested, and green and red lamps on the semiautomatic tester to indicate a good or bad logic matrix test. Protection capability provided during this portion of the test is from the train not being tested.

The testing capability meets the requirements of GDC 21.

- General Warning Alarm Reactor Trip

  Each of the two trains of the solid state protection system is continuously monitored by the general warning alarm reactor trip subsystem. The warning circuits are actuated if undesirable train conditions are set up by improper alignment of testing systems, circuit malfunction or failure, etc., as listed below. A trouble condition in a logic train is indicated in the control room. However, if any one of the conditions exists in train A at the same time any one of the conditions exists in train B, the general warning alarm circuits will automatically trip the reactor.

  - Loss of either of two 48-V dc or either of two 15-V dc power supplies.
  - Printed circuit card improperly inserted.
  - Input error inhibit switch in the inhibit position.
  - Slave relay tester mode selector in test position.
  - Multiplexing selector switch in inhibit position.
  - Train bypass breaker racked in and closed.
  - Permissive or memory test switch not in off position.
  - Logic function test switch not in off position.
  - Loss of power to slave relay output cabinet.

4. Testing of Reactor Trip Breakers

Normally, reactor trip breakers 52/RTA and 52/RTB are in service, and bypass breakers 52/BYA and 52/BYB are withdrawn (out of service). In testing the protection logic, pulse techniques are used to avoid tripping the reactor trip breakers, thereby eliminating the need to bypass them during this testing (drawing 1X6AA02-226). The following procedure describes the method used for testing the trip breakers:

- With bypass breaker 52/BYA racked out, manually close and trip it to verify its operation.
- Rack in and close 52/BYA. Manually trip 52/RTA through a protection system logic matrix while at the same time operating the "Auto Shunt Trip Block" pushbutton on the automatic shunt trip panel. This verifies operation of the undervoltage trip attachment (UVTA) when the breaker trips. After reclosing RTA, trip it again by operation of the

"Auto Shunt Trip Test" pushbutton on the automatic shunt trip panel. This is to verify tripping of the breaker through the shunt trip device.

- Reset 52/RTA.
- Trip and rack out 52/BYA.
- Repeat the above steps to test trip breaker 52/RTB using bypass breaker 52/BYB.

Auxiliary contacts of the bypass breakers are connected into the alarm system of their respective trains such that, if either train is placed in test while the bypass breaker of the other train is closed, both reactor trip breakers and both bypass breakers will automatically trip.

Auxiliary contacts of the bypass breakers are also connected in such a way that if an attempt is made to close the bypass breaker in one train while the bypass breaker of the other train is already closed, both bypass breakers will automatically trip.

Test panels are provided near the reactor trip breakers for verifying auxiliary and cell switch contacts used in the P-4 SSPS and turbine trip signals. In addition, a voltmeter and selector switch are available on the front of each reactor trip switchgear that may be used to determine auxiliary contact position for input to SSPS.

The train A and train B alarm systems operate separate annunciators in the control room. The two bypass breakers also operate an annunciator in the control room. Bypassing of a protection train with either the bypass breaker or with the test switches will result in audible and visual indications.

The complete reactor trip system is normally required to be in service. However, to permit online testing of the various protection channels or to permit continued operation in the event of a subsystem instrumentation channel failure, the Technical Specifications define the minimum number of operable channels. The Technical Specifications also define the required restriction to operation in the event that the channel operability requirements cannot be met.

K. Channel Bypass or Removal from Operation

The protection system is designed to permit periodic testing of the analog channel portion of the reactor trip system during reactor power operation, without initiating a protective action, unless a trip condition actually exists.

L. Operating Bypasses

Where operating requirements necessitate automatic or manual bypass of a protective function, the design is such that the bypass is removed automatically whenever permissive conditions are not met. Devices used to achieve automatic removal of the bypass of a protective function are considered part of the protective system and are designed in accordance with the criteria of this section.

Indication is provided in the control room if some part of the system has been administratively bypassed or taken out of service.

M. Indication of Bypasses

Bypass indication is discussed in paragraph 7.5.5 and section 1.9.

N. Access to Means for Bypassing

The design provides for administrative control of access to the means for manually bypassing channels or protective functions.

O. Multiple Setpoints

For monitoring neutron flux, multiple setpoints are used. When a more restrictive trip setting becomes necessary to provide adequate protection for a particular mode of operation or set of operating conditions, the protective system circuits are designed to provide positive means or administrative control to ensure that the more restrictive trip setpoint is used. The devices used to prevent improper use of less restrictive trip settings are considered part of the protective system and are designed in accordance with the criteria of this section.

P. Completion of Protective Action

The protection system is so designed that, once initiated, a protective action goes to completion. Return to normal operation requires action by the operator.

Q. Manual Initiation

Switches are provided on the control board for manual initiation of protective action. Failure in the automatic system does not prevent the manual actuation of the protective functions. Manual actuation relies on the operation of a minimum of equipment.

R. Access

The design provides for administrative control of access to all setpoint adjustments, module calibration adjustments, and test points.

S. Identification of Protective Actions

Protective channel identification is discussed in paragraph 7.1.2. Indication is discussed in item T below.

T. Information Readout

The protective system provides the operator with complete information pertinent to system status and safety. All transmitted signals (flow, pressure, temperature, etc.) which can cause a reactor trip will either be indicated or recorded for every channel, including all neutron flux power range currents (top detector, bottom detector, algebraic difference, and average of bottom and top detector currents).

Any reactor trip will actuate an alarm and an annunciator. Such protective actions are indicated and identified down to the channel level.

Alarms and annunciators are also used to alert the operator of deviations from normal operating conditions, so that he may take appropriate corrective action to avoid a reactor trip. Actuation of any rod stop or trip of any reactor trip channel will actuate an alarm.

U. System Repair

The system is designed to facilitate the recognition, location, replacement, and repair of malfunctioning components or modules. Refer to the discussion in item J above.

7.2.2.3 Specific Control and Protection Interactions

7.2.2.3.1 Neutron Flux

Four power range neutron flux channels are provided for overpower protection. An isolated auctioneered high signal is derived by auctioneering of the four channels for automatic rod control. If any channel fails in such a way as to produce a low output, that channel is incapable of proper overpower protection but will not cause control rod movement because of the auctioneer. Two out of four overpower trip logic will ensure an overpower trip if needed, even with an independent failure in another channel.

In addition, channel deviation signals in the control system will give an alarm if any neutron flux channel deviates significantly from the average of the flux signals. Also, the control system will respond only to rapid changes in indicated neutron flux; slow changes or drifts are compensated by the temperature control signals. Finally, an overpower signal from any nuclear power range channel will block manual rod withdrawal. Automatic rod withdrawal capability of the rod control system has been disabled. The setpoint for this rod stop is below the reactor trip setpoint.

7.2.2.3.2 Coolant Temperature

The accuracy of the resistance temperature detector (RTD) loop temperature measurements is demonstrated during plant preoperational tests by comparing temperature measurements from all loop RTDs with one another, as well as with the temperature measurements obtained from the wide range RTDs located in the hot leg and cold leg piping of each loop. The comparisons are done with the RCS in an isothermal condition. The linearity of the ΔT measurements obtained from the hot leg and cold leg loop RTDs, as a function of plant power, is also checked during plant startup tests. The absolute value of ΔT versus plant power is not important, per se, as far as reactor protection is concerned. Reactor trip system setpoints are based upon percentages of the indicated ΔT at nominal full power rather than on absolute values of ΔT. This is done to account for loop differences which are inherent. Therefore, the percent ΔT scheme is relative, not absolute, and therefore provides better protective action without the expense of accuracy. For this reason, the linearity of the ΔT signals as a function of power is of importance rather than the absolute values of the ΔT. As part of the plant preoperational tests, the loop RTD signals will be compared with the core exit thermocouple signals.

Reactor control is based upon signals derived from protection system channels after isolation by isolation amplifiers, such that no feedback effect can perturb the protection channels.

Since control is based on the average temperature of the loop with the highest temperature, the control rods are always moved based upon the most pessimistic temperature measurement with respect to margins to departure from nucleate boiling. A spurious low average temperature measurement from any loop temperature control channel will cause no control action. A spurious high average temperature measurement will cause rod insertion (safe direction).

Channel deviation signals in the control system will give an alarm if any temperature channel deviates significantly from the auctioneered (highest) value. Manual rod withdrawal blocks and

turbine runback (power demand reduction) will also occur if any two out of the four overtemperature or overpower ΔT channels indicate an adverse condition.

7.2.2.3.3 Pressurizer Pressure

The pressurizer pressure protection channel signals are used for high- and low-pressure protection and as inputs to the overtemperature ΔT trip protection function. Isolated output signals from these channels are used for pressure control. These are used to control pressurizer spray and heaters and power-operated relief valves. Pressurizer pressure is sensed by fast response pressure transmitters.

A spurious high-pressure signal from one channel can cause decreasing pressure by actuation of either spray or relief valves. Additional redundancy is provided in the low pressurizer pressure reactor trip and in the logic for safety injection to ensure low pressure protection.

Overpressure protection is based upon the positive surge of the reactor coolant produced as a result of turbine trip under full load, assuming the core continues to produce full power. The self-actuated safety valves are sized on the basis of steam flow from the pressurizer to accommodate this surge at a setpoint of 2485 psig and an accumulation of 3 percent. Note that no credit is taken for the relief capability provided by the power-operated relief valves during this surge.

In addition, operation of any one of the power-operated relief valves can maintain pressure below the high-pressure trip point for most transients. The rate of pressure rise achievable with heaters is slow, and ample time and pressure alarms are available to alert the operator of the need for appropriate action.

Redundancy is not compromised by having a shared tap (paragraph 7.2.1.1.2), since the logic for this trip is two out of four. If the shared tap is plugged, the affected channels will remain static. If the impulse line bursts, the indicated pressure will drop to zero. In either case the fault is easily detectable, and the protective function remains operable.

7.2.2.3.4 Pressurizer Water Level

Three pressurizer water level channels are used for reactor trip. Isolated signals from these channels are used for pressurizer water level control. A failure in the level control system could fill or empty the pressurizer at a slow rate (on the order of 1/2 h or more).

The high water level trip setpoint provides sufficient margin, such that the undesirable condition of discharging liquid coolant through the safety valves is avoided. Even at full- power conditions, which would produce the worst thermal expansion rates, a failure of the water level control would not lead to any liquid discharge through the safety valves. This is due to the automatic high pressurizer pressure reactor trip actuating at a pressure sufficiently below the safety valve setpoint.

For control failures which tend to empty the pressurizer, two out of four logic for safety injection action on low pressure ensures that the protection system can withstand an independent failure in another channel. In addition, ample time and alarms exist to alert the operator of the need for appropriate action.

7.2.2.3.5 Steam Generator Water Level

The basic function of the reactor protection circuits associated with low-low steam generator water level is to preserve the steam generator heat sink for removal of long-term residual heat. Should a complete loss of feedwater occur, the reactor would be tripped on low-low steam

generator water level. In addition, redundant auxiliary feedwater pumps are provided to supply feedwater in order to maintain residual heat removal after trip. This reactor trip acts before the steam generators are dry. This reduces the required capacity, increases the time interval before auxiliary feedwater pumps are required, and minimizes the thermal transient on the reactor coolant system and steam generators. Therefore, a low-low steam generator water level reactor trip circuit is provided for each steam generator to ensure that sufficient initial thermal capacity is available in the steam generator at the start of the transient. Two out of four low-low steam generator water level trip logic ensures a reactor trip if needed, even with an independent failure in another channel used for control, and when degraded by an additional second postulated random failure.

A spurious low signal from the feedwater flow channel being used for control would cause an increase in feedwater flow. The mismatch between steam flow and feedwater flow produced by the spurious signal would actuate alarms to alert the operator of the situation in time for manual correction. If the condition continues, a two out of four high-high steam generator water level signal in any loop, independent of the indicated feedwater flow, will cause feedwater isolation and trip the turbine. The turbine trip will result in a subsequent reactor trip if power is above the P-9 setpoint. The high-high steam generator water level trip is an equipment protective trip, preventing excessive moisture carryover which could damage the turbine blading.

In addition, the three-element feedwater controller incorporates reset action on the level error signal, such that with expected controller settings, a rapid increase or decrease in the flow signal would cause only a small change in level, before the controller would compensate for the level error. A slow change in the feedwater signal would have no effect at all. A spurious low or high steam flow signal would have the same effect as high or low feedwater signal, as discussed above.

A spurious high steam generator water level signal from the protection channel used for control will tend to close the feedwater valve. A spurious low steam generator water level signal will tend to open the feedwater valve. Before a reactor trip would occur, two out of four channels in a loop would have to indicate a low-low water level. Any slow drift in the water level signal will permit the operator to respond to the level alarms and to take corrective action.

Automatic protection is provided in case the spurious high level reduces feedwater flow sufficiently to cause low-low level in the steam generator. Automatic protection is also provided in case the spurious low-level signal increases feedwater flow sufficiently to cause high level in the steam generator. A turbine trip and feedwater isolation would occur on two out of four high-high steam generator water level in any loop.

7.2.2.4 Additional Postulated Accidents

Loss of plant instrument air or loss of component cooling water is discussed in subsection 7.3.1. Load rejection and turbine trip are discussed in further detail in section 7.7.

The control interlocks, called rod stops, that are provided to prevent abnormal power conditions which could result from excessive control rod withdrawal are discussed in paragraph 7.7.1.4 and listed on table 7.7.1-1. Excessively high-power operation (which is prevented by blocking of manual rod withdrawal), if allowed to continue, might lead to a safety limit (as given in the Technical Specifications) being reached. The automatic rod withdrawal capability of the rod control system has been disabled.

Before such a limit is reached, protection will be available from the reactor trip system. At the power levels of the rod block setpoints, safety limits have not been reached. Therefore, these rod withdrawal stops do not come under the scope of safety-related systems and are considered control systems.

7.2.2.5 Tests and Inspections

The reactor trip system meets the testing requirements of IEEE 338-1975, as discussed in paragraph 7.1.2.7. The testability of the system is discussed in paragraph 7.2.2.2.3. The initial and subsequent test intervals are specified in the Technical Specifications. Written test procedures and documentation, conforming to the requirements of IEEE 338-1975, will be available for audit by responsible personnel. Periodic testing conforms with Regulatory Guide 1.22, as discussed in subsections 7.1.2 and 7.2.2.

7.2.2.6 References

1. Gangloff, W. C., and Loftus, W. D., "An Evaluation of Solid State Logic Reactor Protection in Anticipated Transients," WCAP-7706-L (Proprietary) and WCAP-7706 (Nonproprietary), February 1971.
2. Reid, J. B., "Process Instrumentation for Westinghouse Nuclear Steam Supply Systems," WCAP-7913, January 1973. (Additional background information only.)
3. Lipchak, J. B., "Nuclear Instrumentation System," WCAP-8255, January 1974. (Additional background information only.)
4. Katz, D. N., "Solid State Logic Protection System Description," WCAP-7488-L (Proprietary), March 1971, and WCAP-7672 (Nonproprietary), May 1971. (Additional background information only.)
5. Mermigos, J. F., "Bypass Test Instrumentation for the Vogtle Electric Generating Plant, Units 1 and 2," WCAP-13376, Revision 2, September 1992.
6. WCAP-16769-P Revision 1, “Westinghouse SSPS Universal Logic Board Replacement Summary Report 6D30225G01/G02/G03/G04.”
7. WCAP-16770-P Revision 0, “Westinghouse SSPS Safeguards Driver Board Replacement Summary Report 6D30252G01/G02.”
8. WCAP-16771-P Revision 0, “Westinghouse SSPS Undervoltage Driver Board Replacement Summary Report 6D30350G01/G02.”
9. WCAP-16772-P Revision 0, “Westinghouse SSPS Semi-Automatic Tester Board Replacement Summary Report 6D30520G01/G02/G03/G04/G05.”

TABLE 7.2.1-1 (SHEET 1 OF 2)

LIST OF REACTOR TRIPS

Reactor Trips	Coincidence Logic	Interlocks	Comments
Power range high neutron flux	2/4	Manual block of low setting permitted by P-10	High and low setting; manual block and automatic reset of low setting by P-10
Intermediate range high neutron flux	1/2	Manual block permitted by P-10	Manual block and automatic reset
Source range high neutron flux	1/2	Manual block permitted by P-6; interlocked with P-10	Manual block and automatic reset; automatic block above P-10
Power range high positive neutron flux rate	2/4	No interlocks	-
Overtemperature ΔT_{avg}	2/4	No interlocks	-
Overpower ΔT_{avg}	2/4	No interlocks	-
Pressurizer low pressure	2/4	Interlocked with P-7	Blocked below P-7
Pressurizer high pressure	2/4	No interlocks	-
Pressurizer high water level	2/3	Interlocked with P-7	Blocked below P-7
Low reactor coolant flow	2/3 in any loop	Interlocked with P-7	Low flow in one loop and P-8 will cause a reactor trip when above P-8, and a low flow in two loops will cause a reactor trip when above P-7; blocked below P-7
Reactor coolant pump bus undervoltage	Low voltage sensed for pumps 1 or 2 and 3 or 4.	Interlocked with P-7	Blocked below P-7

TABLE 7.2.1-1 (SHEET 2 OF 2)

Reactor Trips	Coincidence Logic	Interlocks	Comments
Reactor coolant pump bus underfrequency	Underfrequency sensed for pumps 1 or 2 and 3 or 4	Interlocked with P-7	Blocked below P-7
Solid state protection system general warning alarm	Both trains	No interlocks	-
Low-low steam generator water level	2/4 per loop	No interlocks	-
Turbine-generator		Interlocked with P-9	Blocked below P-9
a. Low auto stop oil pressure b. Turbine stop valve close	2/3 4/4		
Safety injection signal	coincident with actuation of safety injection	No interlocks	See section 7.3 for engineering safety features actuation conditions
Manual	1/2 per train	No interlocks	

TABLE 7.2.1-2 (SHEET 1 OF 2)

PROTECTION SYSTEM INTERLOCKS

Designation	Derivation	Function
Power Escalation Permissives		
P-6	Presence of P-6: 1/2 neutron flux (intermediate range) above approximately 2.0 x 10^{-5} % rated thermal power	Allows manual block of source range reactor trip
	Absence of P-6: 2/2 neutron flux (intermediate range) below setpoint	Defeats the block of source range reactor trip
P-10	Presence of P-10: 2/4 neutron flux (power range) above setpoint	Allows manual block of power range (low setpoint) reactor trip
		Allows manual block of intermediate range reactor trip and intermediate range rod stops (C-1)
		Blocks source range reactor trip (backup for P-6)
	Absence of P-10: ¾ neutron flux (power range) below setpoint	Defeats the block of power range (low setpoint) reactor trip
		Defeats the block of intermediate range reactor trip and intermediate range rod stops (C-1)
		Inputs to P-7

TABLE 7.2.1-2 (SHEET 2 OF 2)

Designation	Derivation	Function
Blocks of Reactor Trips		
P-11	2/3 pressurizer pressure below setpoint	Allows manual block of safety injection actuation on low pressurizer pressure signal and low steam line pressure signal (lead/lag compensated)
	2/3 pressurizer pressure above setpoint	Defeats manual block of safety injection actuation
P-7	Absence of P-7: ¾ neutron flux (power range) below setpoint (from P-10) and 2/2 turbine impulse chamber pressure below setpoint (from P-13)	Blocks reactor trip on low reactor coolant flow in more than one loop, undervoltage, underfrequency, pressurizer low pressure, and pressurizer high level
P-8	Absence of P-8: ¾ neutron flux (power range) below setpoint	Blocks reactor trip on low reactor coolant flow in a single loop
P-9	Absence of P-9: ¾ neutron flux (power range) below 40 percent power	Blocks reactor trip on turbine trip
P-13	Absence of P-13: 2/2 turbine impulse chamber pressure below 10 percent of full load	Input to P-7

TABLE 7.2.1-3 (SHEET 1 OF 3)

REACTOR TRIP SYSTEM INSTRUMENTATION

Reactor Trip Signal	Typical Range	Typical Trip Accuracy	Typical Time Response (s)
Power range high neutron flux	1 to 120% of full power	1% of full power	≤ 0.5[(a)]
Intermediate range high neutron flux	8 decades of neutron flux overlapping source range by 2 decades	± 5% of full scale; ± 1% of full scale from 10^{-4} to 50% full power	N/A
Source range high neutron flux	6 decades of neutron flux (1 to 10^6 counts/s)	± 5% of full scale	≤ 0.5[(a)]
Power range high positive neutron flux rate	+ 15% of full power	± 5%	≤ 0.65[(a)]
Overtemperature ΔT	T_{hot} 530° to 630°F T_{cold} 530° to 630°F T_{avg} 530° to 630°F P_{PRZR} 1700 to 2500 psig F(ΔΦ) -50 to +50 $\Delta T_{setpoint}$ 0° to 100°F	± 3.2°F	[(a) (c)]
Overpower ΔT	T_{hot} 530° to 650°F T_{cold} 530° to 630°F T_{avg} 530° to 630°F $\Delta T_{setpoint}$ 0° to 100°F	± 2.7°F	[(a) (c)]
Pressurizer low pressure	1700 to 2500 psig	± 18 psi (compensated signal)	≤ 2
Pressurizer high pressure	1700 to 2500 psig	± 18 psi (noncompensated signal)	≤ 2
Pressurizer high water level	Entire cylindrical portion of pressurizer (distance between taps)	± 2.3% of full range ΔP between taps at design temperature and pressure	N/A

TABLE 7.2.1-3 (SHEET 2 OF 3)

Reactor Trip Signal	Typical Range	Typical Trip Accuracy	Typical Time Response (s)
Low reactor coolant flow	0 to 120% of rated flow	± 2.5% of full flow within range of 70 to 100% of full flow[a] ± 1%	a. Single loop (above P-8) b. Two loops (above P-7
Reactor coolant pump undervoltage	0 to 100% rated voltage		≤ 1.5
Reactor coolant pump underfrequency	3 to 80 Hz	± 0.1 Hz	≤ 0.6
Low-low steam generator water level[b]	±~6 ft from nominal full load water level	± 2.3% of ΔP signal overpressure range of 700 to 1200 psig	≤ 2
Turbine trip oil pressure	0 to 2000 psig		N/A

a. Neutron detectors are exempt from response time testing. Response time of the neutron flux signal portion of the channel shall be measured from detector output or input of first electronic component in channel. (This provision is not applicable to construction permits docketed after January 1, 1978. See Regulatory Guide 1.118, June 1978.)

b. See also Technical Specification 3.3.3.

c. RTD time constants are verified by measurement. The following channel response times are calculated for narrow range RTD time constants ≤ 5.5 seconds. In both conditions, sensor (RTD) response times have been mathematically removed such that only electronic delays are included. The choice between"without dynamics" and "with dynamics" depends on the method chosen to verify response time. When using allocation times in table 7.2.1-5, the withoutdynamics values are utilized. When using actual measurements, the values chosen must match the test conditions. Dynamics refers to functions usually performed by NLL cards in the 7300 Process Control System and include lead-lag, rate-lag, and lag functions.

	Function	Without Dynamics	With Dynamics
1	Overtemperature ΔT, T_{avg} input	2.000 s	2.469 s
2	Overtemperature ΔT, pressurizer pressure input	8.000 s	8.000 s
3	Overtemperature ΔT, nuclear flux input	8.000 s	8.000 s
4	Overtemperature ΔT, ΔT input	2.000 s	6.159 s
5	Overpower ΔT, T_{avg} input	2.000 s	2.341 s
6	Overpower ΔT, ΔT input	2.000 s	6.159 s

TABLE 7.2.1-3 (SHEET 3 OF 3)

For measured RTD time constants (plus 10% uncertainty) of more than 5.5 seconds, adjust the channel response times as follows:

	Function	Reduction
1	Overtemperature ΔT, T_{avg} input	Reduce by amount RTD time constant plus 10% uncertainty exceeds 5.5 s.
2	Overtemperature ΔT, pressurizer pressure input	No adjustment.
3	Overtemperature ΔT, nuclear flux input	No adjustment.
4	Overtemperature ΔT, ΔT input	Reduce by amount RTD time constant plus 10% uncertainty exceeds 5.5 s.
5	Overpower ΔT, T_{avg} input	Reduce by amount RTD time constant plus 10% uncertainty exceeds 5.5 s.
6	Overpower ΔT, ΔT input	Reduce by amount RTD time constant plus 10% uncertainty exceeds 5.5 s.

TABLE 7.2.1-4 (SHEET 1 OF 3)

REACTOR TRIP SYSTEM INSTRUMENTATION RESPONSE TIMES

FUNCTIONAL UNIT	RESPONSE TIME
1. Manual Reactor Trip	N/A
2. Power Range, Neutron Flux (N-0041, N-0042, N-0043, N-0044)	≤ 0.5 s[(a)]
3. Power Range, Neutron Flux, High Positive Rate (N-0041, N-0042, N-0043, N-0044)	≤ 0.65 s[(a)]
5. Intermediate Range, Neutron Flux (N-0035, N-0036)	N/A
6. Source Range, Neutron Flux (N-0031, N-0032)	N/A
7. Overtemperature ΔT (TE-0411, TE-0421, TE-0431, TE-0441)	(a)(c)
8. Overpower ΔT (TE-0411, TE-0421, TE-0431, TE-0441)	(a)(c)
9. Pressurizer Pressure--Low (PI-0455, PI-0456, PI-0457, PI-0458)	≤ 2 s
10. Pressurizer Pressure--High (PI-0455, PI-0456, PI-0457, PI-0458)	≤ 2 s
11. Pressurizer Water Level--High (LI-0459, LI-0460, LI-0461)	N/A
12. Reactor Coolant Flow--Low	

Loop 1	Loop 2	Loop 3	Loop 4	
FI-0414	FI-0424	FI-0434	FI-0444	
FI-0415	FI-0425	FI-0435	FI-0445	
FI-0416	FI-0426	FI-0436	FI-0446	
a. Single Loop (Above P-8)				≤ 1 s
b. Two Loops (Above P-7 and below P-8)				≤ 1 s

TABLE 7.2.1-4 (SHEET 2 OF 3)

FUNCTIONAL UNIT	RESPONSE TIME
13. Steam Generator Water Level--Low-Low [b]	≤ 2 s

Loop 1	Loop 2	Loop 3	Loop 4
LI-0519	LI-0529	LI-0539	LI-0549
LI-0518	LI-0528	LI-0538	LI-0548
LI-0517	LI-0527	LI-0537	LI-0547
LI-0551	LI-0552	LI-0553	LI-0554

14.	Undervoltage - Reactor Coolant Pumps	≤ 1.5 s
15.	Underfrequency - Reactor Coolant Pumps	≤ 0.6 s
16.	Turbine Trip	
	a. Low Fluid Oil Pressure (PI-6161, PI-6162, PI-6163)	N/A
	b. Turbine Stop Valve Closure	N/A
17.	Safety Injection Input from ESF	N/A
18.	Reactor Trip System Interlocks	N/A
19.	Reactor Trip Breakers	N/A
20.	Automatic Trip and Interlock Logic	N/A

a. Neutron detectors are exempt from response time testing. Response time of the neutron flux signal portion of the channel shall be measured from detector output or input of first electronic component in channel. (This provision is not applicable to construction permits docketed after January 1, 1978. See Regulatory Guide 1.118, June 1978.)

b. See also Technical Specification 3.3.3.

c. RTD time constants are verified by measurement. The following channel response times are calculated for narrow range RTD time constants ≤ 5.5 seconds. In both conditions, sensor (RTD) response times were mathematically removed such that only electronic delays are included. The choice between “without dynamics” and “with dynamics” depends on the method chosen to verify response time. When using allocation times in table 16.3-3a, the “without dynamics” values are utilized. When using actual measurements, the values chosen must match the test conditions. Dynamics refers to functions usually performed by NLL cards in the 7300 Process Control System and include lead-lag, rate-lag, and lag functions.

TABLE 7.2.1-4 (SHEET 3 OF 3)

Function	Without Dynamics	With Dynamics
Overtemperature ΔT, T_{avg} input	2.000 s	2.469 s
Overtemperature ΔT, pressurizer pressure input	8.000 s	8.000 s
Overtemperature ΔT, nuclear flux input	8.000 s	8.000 s
Overtemperature ΔT, ΔT input	2.000 s	6.159 s
Overpower ΔT, T_{avg} input	2.000 s	2.341 s
Overpower ΔT, ΔT input	2.000 s	6.159 s

For measured RTD time constants (plus 10% uncertainty) of more than 5.5 seconds, adjust the channel response times as follows:

Function	Reduction
Overtemperature ΔT, T_{avg} input	Reduce by amount RTD time constant plus 10% uncertainty exceeds 5.5 s.
Overtemperature ΔT, pressurizer pressure input	No adjustment.
Overtemperature ΔT, nuclear flux input	No adjustment.
Overtemperature ΔT, ΔT input	Reduce by amount RTD time constant plus 10% uncertainty exceeds 5.5 s.
Overpower ΔT, T_{avg} input	Reduce amount RTD time constant plus 10% uncertainty exceeds 5.5 s.
Overpower ΔT, ΔT input	Reduce by amount RTD time constant plus 10% uncertainty exceeds 5.5 s.

TABLE 7.2.1-5 (SHEET 1 OF 2)

REACTOR TRIP ALLOCATION TIMES

Function	Sensor	Time	7300/NIS String	Time	SSPS Relays	Time
PZR PRESS HI	Tobar 32PG	200 ms	NLP + NAL	65 ms	Input	20 ms
	Veritrak 76PH	200 ms				
	Rosemount 1154SH9	200 ms				
PZR PRESS LO	Tobar 32PG	200 ms	NLP + NAL	65 ms	Input	20 ms
	Veritrak 76PH	200 ms				
	Rosemount 1154SH9	200 ms				
SG LEVEL LO-LO	Tobar 32DP	400 ms	NLP + NAL	65 ms	Input	20 ms
	Veritrak 76DP	400 ms				
	Rosemount 1154DH5	200 ms				
RCS FLOW LO	Tobar 32DP	400 ms	NLP + NAL	65 ms	Input	20 ms
	Veritrak 76DP	400 ms				
	Rosemount 1153HB5	200 ms				
OPDT (Vary T_{avg})	Weed N9004E-2B	(1)	NRA+NSA+NSA+NSA+NSA+NAL	368 ms	Input	20 ms
OPDT (Vary ΔT)	Weed N9004E-2B	(1)	NRA+NSA+NSA+NAL	293 ms	Input	20 ms
OTDT (Vary T_{avg})	Weed N9004E-2B	(1)	NRA+NSA+NSA+NSA+NSA+NAL	368 ms	Input	20 ms
OTDT (Vary ΔT)	Weed N9004E-2B	(1)	NRA+NSA+NSA+NAL	293 ms	Input	20 ms
OTDT (Vary Press)	Tobar 32PG	200 ms	NLP+NSA+NSA+NAL	140 ms	Input	20 ms
	Veritrak 76PH	200 ms				
	Rosemount 1154SH9	200 ms				

TABLE 7.2.1-5 (SHEET 2 OF 2)

Function	Sensor	Time	7300/NIS String	Time	SSPS Relays	Time
OTDT (Vary Flux)	Detectors Exempt	N/A	NIS (1 ms)+NSA+NCH+ NSA+NAL	148.5 ms	Input	20 ms
RCP VOLTAGE LO	GE NGV/SAM	(1)	N/A	N/A	Input	20 ms
RCP FREQ LO	ABB	(1)	N/A	N/A	Input	20 ms
NIS LEVEL HI	Detectors Exempt	N/A	NIS FEMA	65 ms	Input	20 ms
NIS RATE HI	Detectors Exempt	N/A	NIS FEMA	200 ms	Input	20 ms
CNMT PRESS REACTOR TRIP FROM SI	Barton 764/351	1.0 s	NLP+NAL	65 ms	Input	20 ms
SEAMLINE PRESS REACTOR TRIP FROM SI	Tobar 32PA	200 ms	NLP+NAL	65 ms	Input	20 ms
	Veritrak 76PG	200 ms				
	Rosemount 1154SH9	200 ms				
	Rosemount 1153GB9	200 ms				

Note 1: Allocated sensor times are not used for these variables. These components will continue to be tested as required.

Allocated sensor times are derived from method (3), section (9), WCAP-13632, revision 2 (Vendor Engineering Specifications). Tobar, Veritrak, and Barton times were provided in table 9-1. Rosemount times are from Rosemount manuals 4302 and 4631 The Rosemount response time specifications may also be found in NUREG/CR-5383. Transmitter FMEAs are based upon EPRI report NP-7243 revision 1.

Values for 7300 cards are from tables 4-7 through 4-12 of WCAP-14036, revision 1. Cards installed are 4NCH, 4NRA, 6NLP, 4NSA, and 9NAL or older artwork levels. NIS components installed are summing and level Amp (3359C48G01), isolation Amp (6065D75G01), rate circuit assembly (3359C41G01), and bistable relay driver assembly (3359C39G01). These were evaluated per NIS FMEA schematic diagram 6065D99.

SSPS input and master relays are Potter & Brumfield KH series relays. SSPS slave relays are Potter & Brumfield MDR relays. Values are tabulated from section 4.8, Westinghouse SSPS FMEA.

TABLE 7.2.2-1 (SHEET 1 OF 5)

REACTOR TRIP CORRELATION

Trip[a]		Accident[b]	Technical Specification
A.	NUCLEAR OVERPOWER TRIPS		
1.	Power range high neutron flux trip (low setpoint)	Uncontrolled rod cluster control assembly bank withdrawal from a subcritical or low-power startup condition (15.4.1) Feedwater system malfunctions that result in a decrease in feedwater temperature (15.1.1) Spectrum of rod cluster control assembly ejection accidents (15.4.8)	(c)
2.	Intermediate range high neutron flux trip	Uncontrolled rod cluster control assembly bank withdrawal from a subcritical or low-power startup condition (15.4.1)	(c)
3.	Source range high neutron flux trip	Uncontrolled rod cluster control assembly bank withdrawal from a subcritical or low-power startup condition (15.4.1)	(c)
4.	Power range high positive neutron flux rate trip	Spectrum of rod cluster control assembly ejection accidents (15.4.8) Uncontrolled rod cluster control assembly bank withdrawal at power (RCS overpressure event only) (15.4.2)	(c)
5.	Power range high neutron flux trip (high setpoint)	Uncontrolled rod cluster control assembly bank withdrawal from a subcritical or low-power startup condition (15.4.1) Uncontrolled rod cluster control assembly bank withdrawal at power (15.4.2)	(c)

TABLE 7.2.2-1 (SHEET 2 OF 5)

Trip[a]	Accident[b]	Technical Specification
	Startup of an inactive reactor coolant pump at an incorrect temperature (15.4.4)	
	Feedwater system malfunctions that result in a decrease in feedwater temperature (15.1.1)	
	Excessive increase in secondary steam flow (15.1.3)	
	Inadvertant opening of a steam generator relief or safety valve (15.1.4)	
	Spectrum of steam system piping failures inside and outside of containment in a PWR (15.1.5) Spectrum of rod cluster control assembly ejection accidents (15.4.8)	
B. CORE THERMAL OVERPOWER TRIPS		
1. Overtemperature ΔT trip	Uncontrolled rod cluster control assembly bank withdrawal at power (15.4.2)	(c)
	Chemical and volume control system malfunction that results in a decrease in the boron concentration in the reactor coolant(15.4.6)	
	Loss of external electrical load (15.2.2)	
	Turbine trip (15.2.3)	
	Feedwater system malfunctions that result in a decrease in feedwater temperature (15.1.1)	
	Excessive increase in secondary steam flow (15.1.3)	
	Inadvertent opening of a pressurizer safety or relief valve (15.6.1)	
	Inadvertent opening of a steam generator relief or safety valve (15.1.4)	

TABLE 7.2.2-1 (SHEET 3 OF 5)

	Trip[a]	Accident[b]	Technical Specification
		Loss-of-coolant accidents resulting from the spectrum of postulated piping breaks within the reactor coolant pressure boundary (15.6.5)	
2.	Overpower ΔT trip	Uncontrolled rod cluster control assembly bank withdrawal at power (15.4.2)	(c)
		Feedwater system malfunctions that result in a decrease in feedwater temperature (15.1.1)	
		Excessive increase in secondary steam flow (15.1.3)	
		Inadvertent opening of a steam generator relief or safety valve (15.1.4)	
		Spectrum of steam system piping failures inside and outside of containment in a PWR (15.1.5)	
C.	REACTOR COOLANT SYSTEM PRESSURIZER PRESSURE AND WATER LEVEL TRIPS		
1.	Pressurizer low pressure trip	Inadvertent opening of a pressurizer safety or relief valve (15.6.1)	(c)
		Loss-of-coolant accidents resulting from the spectrum of postulated piping breaks within the reactor coolant pressure boundary (15.6.5)	
		Steam generator tube failure (15.6.3)	
2.	Pressurizer high pressure trip	Uncontrolled rod cluster control assembly bank withdrawal at power (15.4.2)	(c)
		Loss of external electrical load (15.2.2)	
		Turbine trip (15.2.3)	
3.	Pressurizer high water level trip	Uncontrolled rod cluster control assembly bank withdrawal at power (15.4.2)	(c)

TABLE 7.2.2-1 (SHEET 4 OF 5)

Trip[a]		Accident[b]	Technical Specification
		Loss of external electrical load (15.2.2)	
		Turbine trip (15.2.3)	
D.	REACTOR COOLANT SYSTEM LOW FLOW TRIPS		
1.	Low reactor coolant flow	Partial loss of forced reactor coolant flow (15.3.1)	(c)
		Loss of nonemergency ac power to the station auxiliaries (15.2.6)	
		Complete loss of forced reactor coolant flow (15.3.1)	
2.	Reactor coolant pump undervoltage trip	Complete loss of forced reactor coolant flow (15.3.1)	(c)
3.	Reactor coolant pump underfrequency trip	Complete loss of forced reactor coolant flow (15.3.1)	(c)
E.	STEAM GENERATOR TRIP		
	Low-low steam generator water level trip	Loss of normal feedwater flow (15.2.7)	(c)
		Feedwater System Malfunction (15.1.2)	
F.	REACTOR TRIP ON A TURBINE TRIP		
	Reactor trip on turbine trip	Loss of external electrical load (15.2.2)	(c)
		Turbine trip (15.2.3)	
		Loss of nonemergency ac power to the station auxiliaries (15.2.6)	(c)
G.	SAFETY INJECTION SIGNAL ACTUATION TRIP		
	Safety injection signal actuation trip	Inadvertent opening of a steam generator relief or safety valve (15.1.4)	(c)

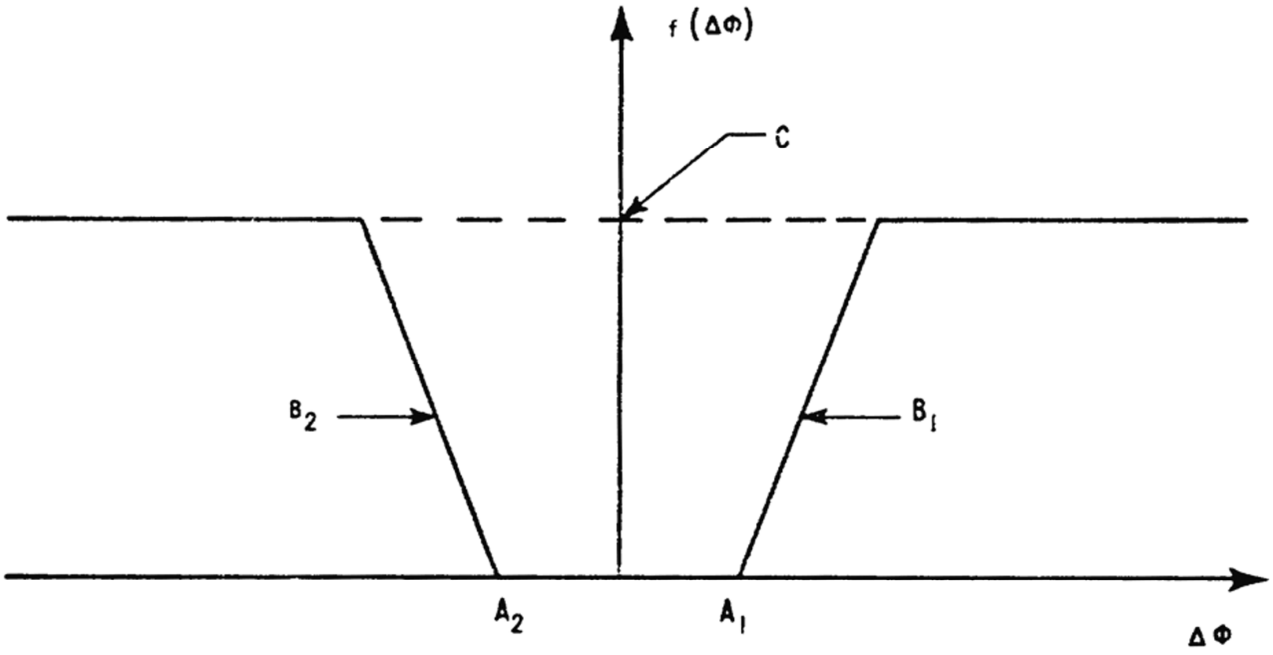
TABLE 7.2.2-1 (SHEET 5 OF 5)

Trip[a]		Accident[b]	Technical Specification
H.	MANUAL TRIP		
	Manual trip	Available for all accidents (chapter 15.)	(c)

a. Trips are listed in order of discussion in section 7.2.

b. References refer to accident analyses presented in chapter 15.

c. Trip safety settings will be incorporated in the Technical Specifications.

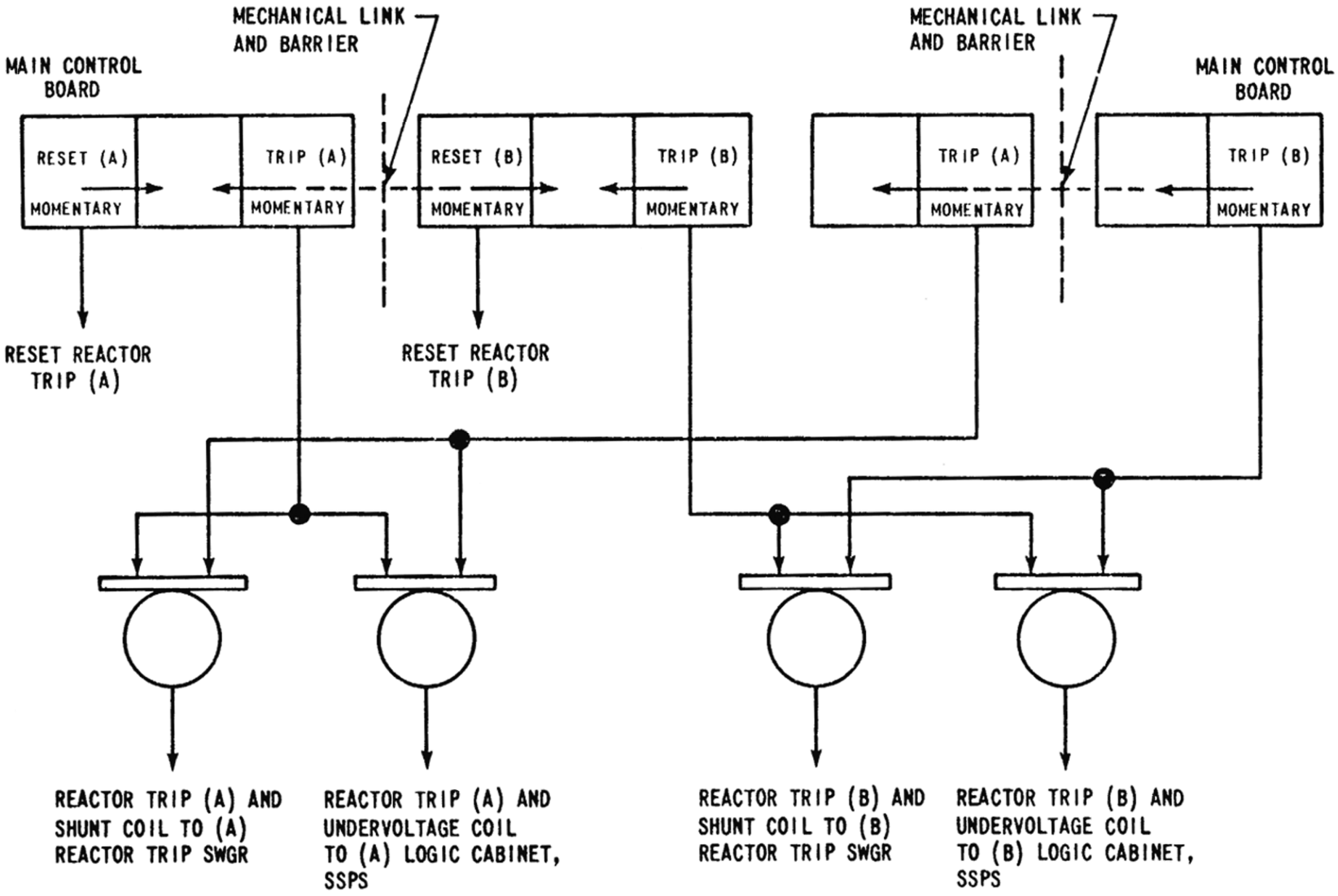


ΔΦ - NEUTRON FLUX DIFFERENCE BETWEEN UPPER AND LOWER LONG ION CHAMBERS

A_1, A_2 - LIMIT OF F (ΔΦ) DEADBAND

B_1, B_2 - SLOPE OF RAMP: DETERMINES RATE AT WHICH FUNCTION REACHES IT'S MAXIMUM VALUE ONCE DEADBAND IS EXCEEDED

C - MAGNITUDE OF MAXIMUM VALUE THE FUNCTION MAY ATTAIN



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REACTOR TRIP/ENGINEERED
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ACTUATION MECHANICAL LINKAGE

FIGURE 7.2.1–2

7.3 ENGINEERED SAFETY FEATURES SYSTEMS

In addition to the requirements for a reactor trip for anticipated abnormal transients, the facility is provided with adequate instrumentation and controls to sense accident situations and initiate the operation of necessary engineered safety features (ESF). The occurrence of a limiting fault, such as a loss-of-coolant accident (LOCA) or a steam line break, requires a reactor trip plus actuation of one or more of the ESF in order to prevent or mitigate damage to the core and reactor coolant system (RCS) components and to ensure containment integrity.

To accomplish these design objectives the ESF system has proper and timely initiating signals which are to be supplied by the sensors, transmitters, and logic components making up the various instrumentation channels of the engineered safety features actuation system (ESFAS).

7.3.1 NUCLEAR STEAM SUPPLY SYSTEM ESFAS

7.3.1.1 Introduction

The ESFAS uses selected plant parameters, determines whether or not predetermined safety limits are being exceeded, and, if they are, combines the signals into logic matrix combinations indicative of primary or secondary system boundary ruptures (Condition III or IV events). Once the required logic combination is completed, the system sends actuation signals to the appropriate ESF components. The ESFAS meets the requirements of General Design Criteria (GDC) 13, 20, 27, 28, and 38.

7.3.1.1.1 System Description

The ESFAS is a functionally defined system described in this section. The equipment which provides the actuation functions identified in paragraph 7.3.1.1.1.1 is listed below and discussed in this section. (For additional background information refer to references 1, 2, 3, 6, 7, 8, and 9).

A. Process instrumentation and control system.(1)

B. Solid-state logic protection system.(2)

C. ESF test cabinet.(3)

D. Manual actuation circuits.

The ESFAS consists of two discrete portions of circuitry as follows:

A. An analog portion consisting of three to four redundant channels per parameter or variable to monitor various plant parameters such as the RCS and steam system pressures, temperatures, and flows and containment pressures.

B. A digital portion consisting of two redundant logic trains which receive inputs from the analog protection channels and perform the logic needed to actuate the ESF.

Each digital train is capable of actuating the ESF equipment required. The intent is that any single failure within the ESFAS shall not prevent system action when required.

7.3.1.1.1.1 Function Initiation. The specific functions which rely on the ESFAS for initiation are:

A. A reactor trip, provided one has not already been generated by the reactor trip system.

B. Cold leg injection isolation valves which are opened for injection of borated water by centrifugal charging pumps into the cold legs of the RCS.

C. Charging pumps, SI pumps, residual heat removal pumps, and associated valving which provide emergency makeup water to the cold legs of the RCS following a LOCA.

D. Containment air cooling units which cool the containment and limit the potential for release of fission products from the containment by reducing the pressure following an accident.

E. Those pumps which serve as part of the heat sink for containment cooling, e.g., nuclear service cooling water and component cooling water pumps.

F. Motor-driven auxiliary feedwater pumps and steam generator blowdown line isolation valve.

G. Phase A containment isolation which prevents fission product release (isolation of all lines not essential to reactor protection).

H. Steam line isolation to prevent the continuous, uncontrolled blowdown of more than one steam generator and thereby uncontrolled RCS cooldown.

I. Main feedwater line isolation as required to prevent or mitigate the effect of excessive cooldown.

J. Start of the emergency diesels to ensure backup supply of power to emergency and supporting systems components.

K. Isolation of the control room intake ducts and normal heating, ventilation, and air-conditioning (HVAC) units and actuation of the control room emergency HVAC system to meet control room occupancy requirements following a LOCA.

L. Containment spray actuation which initiates containment spray to reduce containment pressure and temperature following a LOCA or steam line break accident inside of containment.

M. Reactor cavity post-accident purge units.

N. Containment purge isolation.

O. Actuation of the control building ESF safety feature electrical equipment room.

P. Isolation of the auxiliary building normal HVAC system and actuation of the auxiliary building emergency ventilation system.

Q. ESF-chilled water pumps and chillers.

R. Auxiliary feedwater pumphouse ESF HVAC systems.

7.3.1.1.1.1.1 Analog Initiating Circuitry. The process analog sensors and racks for the ESFAS are discussed in reference 1. Discussed in this report are the parameters to be measured including pressures, flows, tank and vessel water levels, and temperatures, as well as the measurement and signal transmission considerations. These latter considerations include the transmitter, orifices and flow elements, and resistance temperature detectors, as well as automatic calculations, signal conditioning, and location and mounting of the devices.

The sensors monitoring the primary system are located as shown on the piping flow diagrams in chapter 5. The secondary system sensor locations are shown on the steam system flow diagrams given in chapter 10.

Containment pressure is sensed by four physically separated differential pressure transmitters mounted by strong supports outside of the containment. These are connected to the containment atmosphere by a filled and sealed hydraulic transmission system. The distance from penetration to transmitter is kept to a minimum, and separation is maintained. This arrangement and the pressure sensors external to the containment form a double barrier and conform to GDC 56 and Regulatory Guide 1.11.

7.3.1.1.1.1.2 Digital Initiating Circuitry. The ESF logic racks are discussed in detail in references 2, 6, 7, 8, and 9. The description includes the considerations and provisions for physical and electrical separation as well as details of the circuitry. Reference 2 also covers certain aspects of online test provisions, provisions for test points, considerations for the instrument power source, and considerations for accomplishing physical separation. The outputs from the analog channels are combined into actuation logic as shown in drawings 1X6AA02-228, 1X6AA02-229, 1X6AA02-230, 1X6AA02-231, 1X6AA02-232, and 1X6AA02-519.

To facilitate ESF actuation testing, two-bay cabinets (one per train) are provided which enable operation, to the maximum practical extent, of safety feature loads on a group-by-group basis until actuation of all devices has been checked. Final actuation testing is discussed in detail in this section.

7.3.1.1.1.2 Logic. The outputs from the analog channels are combined into actuation logic as shown in drawings 1X6AA02-229, 1X6AA02-230, 1X6AA02-231, 1X6AA02-232, and 1X6AA02-519. Sensing of the variables by the analog circuitry is discussed in reference 1 and in section 7.2. Tables 7.3.1-1 and 7.3.1-2 give additional information pertaining to logic and functions.

7.3.1.1.1.3 Bypasses, Interlocks, and Sequencing. The interlocks associated with the ESFAS are outlined in table 7.3.1-3. These interlocks satisfy the functional requirements, including those for operational bypasses (Refer to P-11 in table 7.3.1-3.) discussed in subsection 7.1.2. The functions of sequencing electrical equipment are not part of the ESFAS.

7.3.1.1.1.4 Redundancy and Diversity. The redundant concept is applied to both the analog and logic portions of the system. Separation of redundant analog channels begins at the process sensors and is maintained in the field wiring, containment building penetrations, and analog protection racks terminating at the redundant safeguards logic racks. The design meets the requirements of GDC 20, 21, 22, 23, and 24.

7.3.1.1.1.5 Final Actuation Circuitry. The outputs of the solid-state logic protection system (the slave relays) are energized to actuate, as are most final actuators and actuated devices. Examples of these devices are:

A. Safety injection (identified also as emergency core cooling) system pump and valve actuators. See chapter 6 for flow diagrams and additional information.

B. Containment isolation phase A (CIA) signal isolates all nonessential process lines on receipt of SI signal. For further information see subsection 6.2.4.

C. Emergency fan coolers, air handling units, and water chillers. (See section 6.2.)

D. Nuclear service cooling water pump and valve actuators. (See section 9.2.)

E. Auxiliary feedwater pumps start. (See section 10.4.)

F. Emergency diesel generator start. (See section 8.3.)

G. Main feedwater isolation. (See section 10.4.)

H. Ventilation isolation valve and damper actuators. (See section 6.4.)

I. Steam line isolation valve actuators. (See section 10.3.)

J. Containment spray pump and valve actuators. (See section 6.2.)

If an accident is assumed to occur coincident with a loss of offsite power, the ESF loads are sequenced onto the diesel generators to prevent overloading them. This sequence is discussed in chapter 8. The design meets the requirements of GDC 35.

7.3.1.1.2 Design Bases Information

The functional diagrams 1X6AA02-228, 1X6AA02-229, 1X6AA02-230, 1X6AA02-231, 1X6AA02-232, and 1X6AA02-519 provide a graphic outline of the functional logic associated with the ESFAS. Requirements for the ESF system are given in chapter 6. Given below is the design bases information required in Institute of Electrical and Electronics Engineers (IEEE) 279-1971.[4]

7.3.1.1.2.1 Plant Conditions. The following is a summary of those plant conditions requiring protective action:

A. Primary System

1. Rupture in small pipes or cracks in large pipes.
2. Rupture of an RCS (LOCA).
3. Steam generator tube rupture.

B. Secondary System

1. Minor secondary system pipe breaks resulting in steam release rates equivalent to the opening of a single dump, relief, or safety valve.
2. Rupture of a major steam pipe.

7.3.1.1.2.2 Plant Variables. The following list summarizes the plant variables required to be monitored for the automatic initiation of SI during each accident identified in the preceding section. Post-accident monitoring requirements are given in table 7.5.2-1.

A. Primary System Accidents

1. Pressurizer pressure.
2. Containment pressure (not required for steam generator tube rupture).

B. Secondary System Accidents

1. Pressurizer pressure.
2. Steam line pressures.
3. Containment pressure.

7.3.1.1.2.3 Spatially Dependent Variables. The only variable sensed by the ESFAS which has spatial dependence is reactor coolant temperature. The effect on the measurement is neutralized by electronic averaging.

7.3.1.1.2.4 Limits, Margins, and Levels. Prudent operational limits, available margins, and setpoints before onset of unsafe conditions requiring protective action are discussed in chapter 15 and the Technical Specifications.

7.3.1.1.2.5 Abnormal Events. The malfunctions, accidents, or other unusual events which could physically damage protection system components or could cause environmental changes are as follows:

A. LOCA. (See chapter 15.)

B. Steam breaks. (See chapter 15.)

C. Earthquakes. (See chapters 2 and 3.)

D. Fire. (See subsection 9.5.1.)

E. Explosion-hydrogen buildup inside containment. (See subsection 6.2.5.)

F. Missiles. (See section 3.5.)

G. Flood. (See chapters 2 and 3.)

7.3.1.1.2.6 Minimum Performance Requirements. Minimum performance requirements are as follows:

A. System Response Times

The response time of each ESFAS function shown in Technical Specification Table 3.3.2-1 is shown in FSAR table 7.3.1-6. Response time verification for selected components may use the predetermined allocation values provided in FSAR table 7.3.1-7.

See paragraph 7.1.2.7A for a discussion of periodic response time verification.

The ESFAS response time is defined as the interval required for the ESF sequence to be initiated subsequent to the time that the appropriate variable(s) exceed the setpoint(s). The ESF sequence is initiated by the output of the ESFAS. This is brought about by the operation of the dry contacts of the slave relays (600 and 700 series relays) in the output cabinets of the solid-state protection system. The response times include the time interval between the time the parameter sensed by the sensor exceeds the safety setpoint and the time the solid-state protection system slave relay dry contacts are operated. These values are maximum allowable values consistent with the safety analyses and the Technical Specifications and are systematically verified during plant preoperational startup tests. These maximum delay times include all compensation and therefore require that any such network be aligned and operating during verification testing.

The ESFAS is always capable of having response time tests performed using the same methods as those tests performed during the preoperational test program or following significant component changes.

Maximum allowable time delays in generating the actuation signal for loss-of-coolant protection are given in table 7.3.1-4.

Maximum allowable time delays in generating the actuation signal for secondary system protection are given in table 7.3.1-5.

B. System Accuracies

Typical accuracies required for generating the required actuation signals for loss-of-coolant protection are given in table 7.3.1-4.

Typical accuracies required in generating the required actuation signals for secondary system protection are given in table 7.3.1-5.

C. Ranges of Sensed Variables

Typical ranges of sensed variables to be accommodated until conclusion of protective action is ensured are given in table 7.3.1-4 for loss-of-coolant protection.

Typical ranges of sensed variables to be accommodated until conclusion of protective action is ensured are given in table 7.3.1-5 for secondary system protection.

7.3.1.1.3 Final System Drawings

The schematic diagrams for the system discussed in this section are identified in section 1.7.

7.3.1.2 Analysis

7.3.1.2.1 Failure Modes and Effects Analyses

Failure modes and effects analyses have been performed on ESF systems equipment within the Westinghouse scope of supply.(5) The balance of plant system interfacing with the ESF system equipment meets the failure modes and effects analyses interface requirements in WCAP-8760.(5) The VEGP balance of plant ESF systems, although not identical, have been designed to equivalent safety design criteria. Other FMEA are included in the pertinent sections of this report.

7.3.1.2.2 Compliance with Standards and Design Criteria

Discussions of GDC are provided in various sections of chapter 7 where a particular GDC is applicable. Conformance with certain IEEE standards is presented in subsection 7.1.2. Conformance with Regulatory Guide 1.22 is discussed in paragraph 7.1.2.5. The discussion given below shows that the ESFAS conforms with IEEE 279-1971.(4)

7.3.1.2.2.1 Single Failure Criteria. The discussion presented in paragraph 7.2.2.2.3 is applicable to ESFAS, with the following exception.

In the ESF, a loss of instrument power calls for actuation of ESF equipment controlled by the specific bistable that lost power (containment spray excepted). The actuated equipment must have power to comply. The power supply for the protection systems is discussed in sections 7.6 and 8.3. For containment spray, the final bistables are energized to trip to avoid spurious actuation. In addition, manual containment spray requires a simultaneous actuation of two

manual controls. This is considered acceptable because spray actuation on high-3 containment pressure signal provides automatic initiation of the system via protection channels meeting the criteria of reference 2. Moreover, two sets (two switches per set) of containment spray manual initiation switches are provided to meet the requirements of IEEE 279-1971. Also, it is possible for all ESF equipment (e.g., valves and pumps) to be individually, manually actuated from the control board. Hence, a third mode of containment spray initiation is available. The design meets the requirements of GDC 21 and 23.

7.3.1.2.2.2 Equipment Qualification. Equipment qualifications are discussed in sections 3.10 and 3.11.

7.3.1.2.2.3 Channel Independence. The discussion presented in paragraph 7.2.2.2.3 is applicable. The ESF slave relay outputs from the solid-state logic protection cabinets are redundant, and the actuations associated with each train are energized up to and including the final actuators by the separate ac power supplies which power one logic train each.

7.3.1.2.2.4 Control and Protection System Interaction. The discussions presented in paragraph 7.2.2.2.3 are applicable.

7.3.1.2.2.5 Capability for Sensor Checks and Equipment Test and Calibration. The discussions of system testability in paragraph 7.2.2.2.3 are applicable to the sensor, analog circuitry, and logic trains of the ESFAS.

The following discussions cover those areas in which the testing provisions differ from those for the reactor trip system.

7.3.1.2.2.5.1 Testing of ESFAS. The ESF systems are tested to provide assurance that the systems operate as designed and are available to function properly in the event of an accident. The testing program meets the requirements of GDC 21 and Regulatory Guide 1.22 as discussed in paragraph 7.1.2.5. The tests described herein and further discussed in subsection 6.3.4 meet the requirements on testing of the emergency core cooling system (ECCS) as stated in GDC 37, except for the operation of those components that will cause an actual safety injection. The test, as described, demonstrates the performance of the full operational sequence that brings the system into operation, the transfer between normal and emergency power sources, and the operation of associated cooling water systems. The safety injection and residual heat removal pumps are started and operated and their performance verified in a separate test discussed in subsection 6.3.4. When the pump tests are considered in conjunction with the ECCS test, the requirements of GDC 37 on testing of the ECCS are met as closely as possible without causing an actual safety injection.

Testing, as described in subsections 6.3.4 and 7.2.2 and herein, provides periodic testability during reactor operation of all logic and components associated with the ECCS. This design meets the requirements of Regulatory Guide 1.22 as discussed in the above sections. The program is as follows:

A. Prior to initial plant operations, ESF system tests are conducted.

B. Subsequent to initial startup, ESF system tests are conducted on one train ON A STAGGERED TEST BASIS during each regularly scheduled refueling outage.

C. During online operation of the reactor, all of the ESF analog and digital circuitry are fully tested. In addition, essentially all of the ESF final actuators are fully tested. The

remaining few final actuators whose operation is not compatible with continued online plant operation is checked by means of continuity testing.

D. During normal operation, the operability of testable final actuation devices of the ESF systems is tested by manual initiation from the control room.

7.3.1.2.2.5.2 Performance Test Acceptability Standard for the "SI" (Safety Injection Signal) and for the "CS" (Containment Spray Actuation) Actuation Signals Generation. During reactor operation the basis for ESFAS acceptability is the successful completion of the overlapping tests performed on the initiating system and the ESFAS. (See figure 7.3.1-1.) Checks of process indications verify operability of the sensors. Analog checks and tests verify the operability of the analog circuitry from the input of the analog circuits up to and including the logic input relays except for the input relays associated with the containment spray function which are tested during the solid-state logic testing. Solid-state logic testing also checks the digital signal path from and including logic input relay contacts through the logic matrices and master relays and performs continuity tests on the coils of the output slave relays; final actuator testing operates the output slave relays and verifies operability of those devices which require safeguards actuation and which can be tested without causing plant upset. A continuity check is performed on the actuators of the untestable devices. Operation of the final devices is confirmed by control board indication and by visual observation that the appropriate pump motor breakers close and automatic valves have completed their travel.

The basis for acceptability for the ESF interlocks is control board indication of proper receipt of the signal upon introducing the required input at the appropriate setpoint. Maintenance checks (performed during regularly scheduled refueling outages), such as resistance to ground of signal cables in radiation environments are based on qualification test data which identifies what constitutes acceptable radiation degradation.

7.3.1.2.2.5.3 Frequency of Performance of ESFAS. During reactor operation, complete system testing (excluding sensors or those devices whose operation would cause plant upset) is performed periodically as specified in the Technical Specifications. Testing, including the sensors, is also performed during scheduled plant outages for refueling.

7.3.1.2.2.5.4 ESF Actuation Test Description. The following sections describe the testing circuitry and procedures for the online portion of the testing program. The guidelines used in developing the circuitry and procedures are:

A. The test procedures must not involve the potential for damage to any plant equipment.

B. The test procedures must minimize the potential for accidental tripping.

C. The provisions for online testing must minimize complication of ESF actuation circuits so that their reliability is not degraded.

7.3.1.2.2.5.5 Description of Initiation Circuitry. Several systems comprise the total ESF system, the majority of which may be initiated by different process conditions and be reset independently of each other.

The functions listed in paragraph 7.3.1.1.1.1 (excluding items H and L) are initiated by a common signal (Safety Injection) which in turn may be generated by different process conditions.

In addition, operation of all other vital auxiliary support systems, such as auxiliary feedwater, component cooling water, and nuclear service cooling water, is initiated by the safety injection signal.

Each function is actuated by a logic circuit which is duplicated for each of the two redundant trains of ESF initiation circuits.

The output of each of the initiation circuits consists of a master relay which drives slave relays for contact multiplication as required. The logic, master, and slave relays are mounted in two redundant and independent solid-state logic protection cabinets designated train A and train B. The master and slave relay circuits operate various pump and fan circuit breakers or motor starters, motor-operated valve starters, solenoid-operated valves, emergency generator starting equipment, and other ESF actuation devices.

7.3.1.2.2.5.6 Analog Testing. Analog testing is identical to that used for reactor trip circuitry and is described in paragraph 7.2.2.2.3.

An exception to this is containment spray, which is energized to actuate two out of four and reverts to two out of three when one channel is in test.

7.3.1.2.2.5.7 Solid-State Logic Testing. Except for containment spray channels, solid-state logic testing is the same as that discussed in paragraph 7.2.2.2.3. During logic testing of one train, the other train can initiate the required ESF function. For additional details, see references 2, 6, 7, 8, and 9.

7.3.1.2.2.5.8 Actuator Testing. At this point, testing of the initiation circuits through operation of the master relay and its contacts to the coils of the slave relays has been accomplished. The ESFAS logic slave relays in the solid-state protection system output cabinets are subjected to coil continuity tests by the output relay tester in the solid-state protection system cabinets. Slave relays (e.g., K601 and K602) do not operate because of reduced voltage applied to their coils by the mode selector switch (test/operate). A multiple position master relay selector switch selects the master relays and corresponding slave relays to which the coil continuity test voltage is applied. The master relay selector switch is returned to off before the mode selector switch is returned to off. The mode selector switch is placed back in the operate mode. However, failure to do so does not result in defeat of the protective function. The ESFAS slave relays are activated during the testing by the online test cabinet, so that overlap testing is maintained.

The ESFAS final actuation device or actuated equipment testing is performed from the engineered safeguards test cabinets. These cabinets are located near the solid-state logic protection system equipment. There is one set of test cabinets provided for each of the two protection trains A and B. Each set of cabinets contains individual test switches necessary to actuate the slave relays. To prevent accidental actuation, test switches are of the type that must be rotated and then depressed to operate the slave relays. Assignments of contacts of the slave relays for actuation of various final devices or actuators has been made such that groups of devices or actuated equipment can be operated individually during plant operation without causing plant upset or equipment damage. In the unlikely event that a safety injection signal is initiated during the test of the final device that is actuated by this test, the device will already be in its safeguards position.

During this last procedure, close communication between the main control room operator and the tester at the test cabinet is required. Prior to the energizing of a slave relay, the operator in the main control room ensures that plant conditions permit operation of the equipment that is actuated by the relay. After the tester has energized the slave relay, the main control room operator observes that all equipment has operated as desired using the appropriate indicating lamps, monitor lamps, and annunciators on the control board and records all operations. The operator then resets all devices and prepares for operation of the next slave relay actuated equipment.

By means of the procedure outlined above, all ESF devices actuated by ESFAS initiation circuits, with the exceptions noted in paragraph 7.1.2.5 under a discussion of Regulatory Guide 1.22, are operated by the automatic circuitry.

7.3.1.2.2.5.9 Actuator Blocking and Continuity Test Circuits. Those few final actuation devices that cannot be designed to be actuated during plant operation (discussed in paragraph 7.1.2.5) have been assigned to slave relays for which additional test circuitry has been provided to individually block actuation of a final device upon operation of the associated slave relay during testing. Operation of these slave relays, including contact operations and continuity of the electrical circuits associated with the final devices control, is checked in lieu of actual operation. The circuits provide for monitoring of the slave relay contacts, the devices control circuit cabling and control voltage, and the devices control actuation solenoids. Interlocking prevents blocking the output from more than one output relay in a protection train at a time. Interlocking between trains is also provided to prevent continuity testing in both trains simultaneously; therefore, the redundant device associated with the protection train not under test is available in the event protective action is required. If an accident occurs during testing, the automatic actuation circuitry overrides testing as noted above. One exception to this is if the accident occurs while testing a slave relay whose output must be blocked, those few final actuation devices associated with this slave relay are not actuated; however, the redundant devices in the other train are operational and perform the required safety function. Actuation devices to be blocked are identified in paragraph 7.1.2.5.

The continuity test circuits for these components that cannot be actuated online are verified by providing lights on the engineered safeguards test cabinets.

The typical schemes for blocking operation of selected protection function actuator circuits are shown in figure 7.3.1-2 as details A and B. The schemes operate as explained below and are duplicated for each safeguards train.

Detail A shows the circuit for contact closure for protection function actuation. Under normal plant operation with equipment not under test, the test lamps DS* for the various circuits are energized, verifying that the blocking functions are not in use. Typical circuit path is through the normally closed test relay contact K8* and through test lamp connections one to three. Coils X1 and X2 are capable of being energized for protection function actuation upon closure of solid-state logic output relay contacts K*. Coil X1 is typical for a motor control center starter coil, and X2 is typical for a breaker closing auxiliary coil, motor starter master coil, coil of a solenoid valve, auxiliary relay, etc. When the contacts K8* are opened to block energizing of coil X1 or X2, the white lamp is deenergized, and the slave relay K* may be energized to perform continuity testing. The continuity test is performed by depressing the test lamp assembly and observing that the test lamp lights. The circuit path is through test lamp connections two to one (contact K8* open), through relay contact K*, and finally through actuator coil X1 or X2. Sufficient current flows in the circuit to cause the test lamp to light, but the current is insufficient to cause coil X1 or X2 to operate. When the K* relay is reset, depressing the lamp assembly does not cause the lamp to light. After the K8* relay is reset, the test lamp lights, verifying that the blocking action is removed and the circuit is in its normal operable condition.

Detail B shows the circuit for contact opening for protection function actuation. Under normal plant operation with equipment not under test for 125 V-dc actuation devices, the white test lamps DS* for the various circuits are energized, and green test lamps DS* are deenergized.

Typical circuit path for white lamp DS* is through the normally closed solid-state logic output relay contact K*. Coil Y2 is typical for a solenoid valve coil, auxiliary relay, etc. When the contact K8* is closed to block deenergizing of coil Y2, the green test lamp is energized and the slave relay K* may be energized to verify operation (opening of its contacts). Opening of the K*

contact is verified by the white lamp DS* deenergizing. When the K* relay is reset, the white lamp DS* reenergizes, verifying that the K* relay contact has closed. After the K8* relay is reset, the green test lamp should be deenergized, which verifies that the circuit is now in its normal (i.e., operable) position.

7.3.1.2.2.5.10 Time Required for Testing. It is estimated that analog testing can be performed at a rate of several channels per hour. Logic testing of both trains A and B can be performed in less than 30 min. Testing of actuated components, including those which can only be partially tested, will be a function of control room operator availability. It is expected to require several shifts to accomplish these tests. Automatic actuation circuitry overrides testing, except for those few devices associated with a single slave relay whose outputs must be blocked and then only while blocked. It is anticipated that continuity testing associated with a blocked slave relay could take several minutes. During this time the redundant devices in the other trains would be functional.

7.3.1.2.2.5.11 Summary of Online Testing Capabilities. The procedures described provide capability for complete checking from the process signal to the logic cabinets and from there to the individual pump and fan circuit breakers or starters, valve starters, pilot solenoid valves, and other equipment including all field cabling actually used in the circuitry called upon to operate for an accident condition. For those few devices whose operation could adversely affect plant or equipment operation, the same procedure provides for checking from the process signal to the logic rack. To check the final actuation device, a continuity test of the individual control circuits is performed.

The procedures require testing at various locations:

A. Analog testing and verification of bistable setpoint are accomplished at process analog racks. Verification of bistable relay operation is done at the main control room status lights.

B. Logic testing through operation of the master relays and low voltage application to slave relays is done at the solid-state protection system logic rack test panel.

C. Testing of pumps, fans, and valves is done at the engineered safeguards cabinet test panel located near the solid-state protection system logic racks in combination with the control room operator.

D. Continuity testing for those circuits that cannot be operated is done at the same test cabinet mentioned in item C above.

The reactor coolant pump essential service isolation valves consist of the isolation valves for the auxiliary component cooling water return and the seal water return header.

The main reason for not testing these valves periodically is that the reactor coolant pumps may be damaged. Although pump damage from this type of test would not result in a situation which endangers the health and safety of the public, it could result in unnecessary shutdown of the reactor for an extended period of time while the reactor coolant pump or any of its components are replaced.

Containment spray system pump tests are performed periodically. The pump tests are performed with the isolation valves in the spray pump discharge lines closed. The valves are tested with the pump stopped.

7.3.1.2.2.5.12 Testing During Shutdown. ECCS tests are performed periodically in accordance with the Technical Specifications with the RCS isolated from the ECCS by closing

the appropriate valves. A test safety injection signal will then be applied to initiate operation of active components (pumps and valves) of the ECCS. This is in compliance with GDC 37.

7.3.1.2.2.5.13 Periodic Maintenance Inspections. The maintenance procedures which follow are accomplished in accordance with applicable plant procedures. The frequency depends on the operating conditions and requirements of the reactor power plant. If any degradation of equipment operation is noted, either mechanically or electrically, remedial action is taken to repair, replace, or readjust the equipment. Optimum operating performance must be achieved at all times.

Typical maintenance procedures include the following:

A. Check cleanliness of all exterior and interior surfaces.

B. Check all fuses for corrosion.

C. Inspect for loose or broken control knobs and burned out indicator lamps.

D. Inspect for moisture and check the condition of cables and wiring.

E. Mechanically check all connectors and terminal boards for looseness, poor connection, or corrosion.

F. Inspect the components of each assembly for signs of overheating or component deterioration.

G. Perform complete system operating check.

The balance of the requirements listed in reference 1 (sections 4.11 through 4.22) are discussed in paragraph 7.2.2.2.3. Section 4.20 receives special attention in section 7.5.

7.3.1.2.2.6 Manual Resets and Blocking Features. The manual reset feature is provided in the standard design of the Westinghouse solid-state protection system design for two basic purposes:

A. The feature permits the operator to start an interruption procedure in the event of false actuation.

B. Although system actuation is automatic, the reset feature enables the operator to start a manual takeover of the system to handle unexpected events which can be better dealt with by operator appraisal of changing conditions following an accident.

It is most important to note that manual control of the system does not occur once actuation has begun by just resetting the associated logic devices alone. Components seal in (latch) so that removal of the actuate signal, in itself, neither cancels nor prevents completion of protective action nor provides the operator with manual override of the automatic system by this single action. In order to take complete control of the system to interrupt its automatic operation, the operator must manually unlatch relays which have latched the initial actuation signals in the associated motor control centers, and trip the pump motor circuit breakers.

The manual reset feature therefore, does not perform a bypass function. It is merely the first of several manual operations required to take control of the automatic system should such an action be considered necessary.

In the event that the operator anticipates system actuation and erroneously concludes that it is undesirable or unnecessary and imposes a standing reset condition in one train by operating and holding the corresponding reset switch at the time the actuation signal is transmitted, the other train will automatically carry the protective action to completion. In the event that the reset condition is imposed simultaneously in both trains at the time the actuation signals are

generated, the automatic sequential completion of system action is interrupted and control is taken by the operator. Manual takeover will be maintained, even though the reset switches are released, if the original initiate signal exists. Should the initiate signal then clear and return again, automatic system actuation will repeat.

Note also that any time delays imposed on the system action are to be applied after the initiating signals are latched. Delay of actuation signals for fluid systems lineup, load sequencing, and other operations are not sufficient to allow the operator time to interrupt automatic completion with manual reset alone, as would be necessary if the time delay was imposed prior to latching of the initial actuation signal.

The manual block features associated with pressurizer and steam line safety injection signals provide the operator with the means to block initiation of safety injection during plant startup. These block features meet the requirements of section 4.12 of IEEE 279-1971, in that automatic removal of the block occurs when plant conditions require the protection system to be functional.

If a steam line rupture occurs while both of these safety injection actuation signals are blocked, steam line isolation will occur on high negative steam pressure rate. An alarm for steam line isolation will alert the operator of the accident.

For large loss-of-coolant accidents (LOCAs), sufficient mass and energy would be released to the containment to automatically actuate safety injection when the containment high pressure setpoint (high-l) is reached. Additionally, the operator would be alerted to the occurrence of a LOCA by the following safety-related indications:

A. Loss of pressurizer level (a low level alarm is provided).

B. Rapid decrease of reactor coolant system pressure.

C. Increase in containment pressure.

In addition to the above, the following indications are normally available to the operator at the control board:

A. Radiation alarms.

B. Increase in sump water level.

C. Decrease off scale of accumulator water levels and decrease in pressure (a low water level alarm and low pressure alarm is provided for each accumulator).

D. ECCS valve and pump position indication, status lights, and annunciators.

E. Flow from ECCS pumps.

For very small LOCAs (approximately less than 2-in. diameter) in which the containment high pressure setpoint may not be reached, the operator would observe the safety-related indications plus the first two normally available indications. In addition, a charging flow/letdown mismatch would provide the operator with another indication of leakage from the reactor coolant system.

Since the operator would observe the pressurizer level and receive additional indications that a LOCA occurred, a manual safety injection would be initiated immediately. As presented in WCAP-8356, the time to uncover the core following a small break is relatively long (e.g., greater than 10 min for a 2-in. break). The operator would, therefore, have sufficient time to manually initiate safety injection.

As part of WCAP-10599, ERG Validation Program Final Report, June 1984, a simulator response to a LOCA with safety injection blocked is included. Although this was a substantial sized LOCA, the operator actions for this LOCA are more limiting than those for a small-break

LOCA, and therefore bound the small-break LOCA. Sufficient operator action time was available to perform the necessary actions to mitigate the consequences of this event.

7.3.1.2.2.7 Manual Initiation of Protective Actions (Regulatory Guide 1.62). There are eight individual main steam stop valve momentary control switches (two per loop) mounted on the control board. Each switch, when actuated, will isolate one of the main steam lines. In addition, there are two system level switches. Each switch actuates all eight main steam line isolation valves and associated bypass valves at the system level.

Manual initiation of switchover to recirculation is in conformance with section 4.17 of IEEE 279-1971 with the following comment.

Manual initiation of containment isolation consists of two momentary control switches mounted on the control board. Each switch, when actuated, will provide for actuation of containment isolation (as well as containment ventilation isolation).

Manual initiation of containment spray consists of four momentary control switches mounted on the control board. Actuation of containment spray and resultant containment ventilation isolation will occur only if two associated control switches are operated simultaneously.

Manual initiation of either one of two redundant safety injection actuation main control board-mounted switches provides for actuation of the components required for reactor protection and mitigation of adverse consequences of the postulated, modify accident, including delayed actuation of sequence-started emergency electrical loads, as well as for the cold leg recirculation mode following a loss of primary coolant accident. Therefore, once safety injection is initiated, those components of the ECCS (see section 6.3) which are realigned as part of the semiautomatic switchover go to completion on refueling water storage tank low-low water level without any manual action. Manual operation of other components or manual verification of proper position as part of emergency procedures is not precluded nor otherwise in conflict with the above described conformance to section 4.17 of IEEE 279-1971 of the semiautomatic switchover circuits.

No exception to the requirements of IEEE 279-1971 has been taken in the manual initiation circuit of safety injection. Although section 4.17 of IEEE 279-1971 requires that a single failure within common portions of the protective system shall not defeat the protective action by manual or automatic means, the standard does not specifically preclude the sharing of initiated circuitry logic between automatic and manual functions. It is true that the manual safety injection actuation associated with one safety train (e.g., train A) shares portions of the automatic actuation circuitry of the same train; however, a single failure in shared functions does not defeat the protective action of the redundant actuation train (e.g., in this case train B). A single failure in shared functions does not defeat the protective action of the safety function. It is further noted that the sharing of the logic by manual and automatic actuation is consistent with the system level action requirements of section 4.17 of IEEE 279-1971 and with the minimization of complexity.

7.3.1.2.2.8 Further Considerations. In addition to the considerations given above, a loss of instrument air or loss of component cooling water to vital equipment has been considered. Neither the loss of instrument air nor the loss of component cooling water (assuming no other accident conditions) can cause safety limits given in the Technical Specification to be exceeded. Likewise, loss of either one of the two will not adversely affect the core or the reactor coolant system nor will it prevent an orderly shutdown if this is necessary. Furthermore, all pneumatically operated valves and controls will assume a safe operating position upon loss of

instrument air. It is also noted that, for conservatism during the accident analysis (chapter 15), credit is not taken for the instrument air system nor any control system being operable.

The design does not provide any circuitry which will directly trip the reactor coolant pumps on a loss of auxiliary component cooling water. Indication in the control room is provided whenever auxiliary component cooling water is lost. The reactor coolant pumps can run about 10 min after a loss of auxiliary component cooling water. This provides adequate time for the operator to correct the problem or trip the plant if necessary.

In regard to the auxiliary feedwater system refer to subsection 7.3.7.

7.3.1.2.3 Summary

The effectiveness of the ESFAS is evaluated in chapter 15, based on the ability of the system to contain the effects of condition III and IV events, including loss-of-coolant and steam break accidents. The ESFAS parameters are based upon the component performance specifications which are given by the manufacturer or verified by test for each component. Appropriate factors to account for uncertainties in the data are factored into the constants characterizing the system.

The ESFAS must detect Condition III and IV events and generate signals which actuate the ESF. The system must sense the accident condition and generate the signal actuating the protection function reliably and within a time determined by and consistent with the accident analyses in chapter 15.

The ESF actuating signals, once generated, are latched in the actuation logic output relays and remain active until the manual reset is performed by the operator. Such reset will not reverse the actuation of any ESF equipment, all of which will remain in its emergency mode until deenergized by the operator on an individual basis. For details see the logic diagrams referenced in section 1.7.

Much longer times are associated with the actuation of the mechanical and fluid system equipment associated with ESF. This includes the time required for switching and bringing pumps and other equipment to speed and the time required for them to take load.

Operating procedures require that the complete ESFAS normally be operable. However, redundancy of system components is such that the system operability assumed for the safety analyses can still be met with certain instrumentation channels out of service. Channels that are out of service are to be placed in the tripped mode or in the case of containment spray, in the bypass mode.

7.3.1.2.3.1 Loss-of-Coolant Protection. By the analysis of LOCA and in system tests, it has been verified that except for very small reactor coolant system breaks which can be protected against by the charging pumps followed by an orderly shutdown, the effects of various LOCAs are reliably detected by the low pressurizer pressure signal; the ECCS is actuated in time to prevent or limit core damage.

For large coolant system breaks the passive accumulators inject first, because of the rapid pressure drop. This protects the reactor during the unavoidable delay associated with actuating the active ECCS equipment.

High containment pressure also actuates the ECCS. Therefore, emergency core cooling actuation can be brought about by sensing this other direct consequence of a primary system break; that is, the ESFAS detects the leakage of the reactor coolant into the containment. Then

generation time of the actuation signal of about 1.5 s after detection of the consequences of the accident is adequate.

Containment spray will provide additional emergency cooling of containment and also limit fission product releases upon sensing elevated containment pressure (high-3) to mitigate the effects of a LOCA.

The delay time between detection of the accident condition and the generation of the actuation signal for the system is assumed to be about 1.0 s. However, this time is short as compared to that required for startup of the fluid systems.

The analyses in chapter 15 show that the diverse methods of detecting the accident condition and the time for generation of the signals by the protection systems are adequate to provide reliable and timely protection against the effects of loss-of-coolant.

7.3.1.2.3.2 Steam Line Break Protection. The ECCS is also actuated to protect against a steam line break. About 2.0 s elapse between sensing low steam line pressure and generation of the actuation signal. Analysis of steam line break accidents assuming this delay for signal generation shows that the ECCS is actuated for a steam line break in time to limit or prevent further core damage for steam line break cases.

Additional protection against the effects of steam line break is provided by feedwater isolation which occurs upon actuation of the ECCS. Feedwater line isolation is initiated to prevent excessive cooldown of the reactor vessel and thus protect the reactor coolant system.

Additional protection against a steam line break accident is provided by closure of all steam line isolation valves to prevent uncontrolled blowdown of all steam generators. The generation of the protection system signal (from high negative steam pressure rate) (about 2.0 s) is again short as compared to the time required to close the fast acting steam line isolation valves (approximately 5 s).

In addition to actuation of the ESF, the steam line break accident results in a reactor trip. The core reactivity is further reduced by borated water injected by the ECCS.

The analyses in chapter 15 of the steam line break accidents and an evaluation of the protection system instrumentation and channel design show that the ESFAS is effective in preventing or mitigating the effects of a steam line break accident.

7.3.1.3 References

1. Reid, J. B., "Process Instrumentation for Westinghouse Nuclear Steam Supply System (4 Loop Plant Using WCID 7300 Series Process Instrumentation)," WCAP-7913, March 1973.
2. Katz, D. N., "Solid State Logic Protection System Description," WCAP-7488-L (Proprietary) and WCAP-7672 (Nonproprietary), June 1971. (Additional background information only).
3. Swogger, J. W., "Testing of Engineered Safety Features Actuation System," WCAP-7705, Revision 2, January 1976. (Information only, i.e., not a generic topical WCAP.)
4. The Institute of Electrical and Electronics Engineers, Inc., "IEEE Standard: Criteria for Protection System for Nuclear Power Generating Stations," IEEE 279-1971.
5. Mesmeringer, J. C., "Failure Mode and Effects Analysis (FMEA) of the Engineered Safety Features Actuation System" WCAP-8584, Revision 1 (Proprietary), and WCAP-8760, Revision 1 (Nonproprietary), February 1980.

6. WCAP-16769-P Revision 1, “Westinghouse SSPS Universal Logic Board Replacement Summary Report 6D30225G01/G02/G03/G04.”
7. WCAP-16770-P Revision 0, “Westinghouse SSPS Safeguards Driver Board Replacement Summary Report 6D30252G01/G02.”
8. WCAP-16771-P Revision 0, “Westinghouse SSPS Undervoltage Driver Board Replacement Summary Report 6D30350G01/G02.”
9. WCAP-16772-P Revision 0, “Westinghouse SSPS Semi-Automatic Tester Board Replacement Summary Report 6D30520G01/G02/G03/G04/G05.”

7.3.2 EMERGENCY CORE COOLING SYSTEM

7.3.2.1 Description

7.3.2.1.1 System Description

An important engineered safety feature (ESF) is the emergency core cooling system which includes a collection of fluid system components described as the safety injection system (SIS). Refer to section 6.3 for a description and analysis of the system. Portions of the SIS which are actuated by the ESFAS include these components:

A. Residual heat removal/low-head safety injection (SI) pumps in both trains.

B. Charging pumps/high-head SI pumps in both trains.

C. Air-operated isolation valves. These include isolation valves for accumulators fill line, test.

D. Motor-operated isolation valves. These include 8808A, 8808B, 8808C, and 8808D for the accumulators.

E. A flow diagram description is shown in figure 6.3.2-1. The principal description and evaluation of this system is provided in section 6.3.

7.3.2.1.1.1 Initiating Circuits and Logic. The function of initiation of SI is described in paragraph 7.3.1.1.1 with specific functions identified in table 7.3.1-2. The logic for the initiation of SI is shown in drawings 1X6AA02-232 and 1X6AA02-519.

7.3.2.1.1.2 Bypass, Interlocks, and Sequencing. There are no operating or online testing bypasses provided for the SI pump motors or valve operators. The associated interlocks are described in section 7.6. The pump motors for high-head SI and low-head SI are sequenced as shown in drawings 1X3D-AA-K02A and 1X3D-AA-K02B.

7.3.2.1.1.3 Redundancy and Diversity. The system is composed of redundant trains A and B. The instrumentation and controls of the components and equipment in train A are physically and electrically separate and independent of the instrumentation and controls of the components and equipment in train B. The redundancy and independence provided between safety trains A and B are adequate to maintain equipment functional capabilities following design bases events.

7.3.2.1.1.4 Status Indication and Display. Pumps and valves which are an integral part of or associated with the engineered safeguards (used for injection, containment spray, and recirculation) have an operation/position status light.

ESF remote-operated valves have position indication on the control board in two places to show proper positioning of the valves. Red and green indicator lights are located next to the manual control station showing open and closed positions. The ESF (SI) positions of these valves are displayed by an energized light on the monitor light panels, which consist of an array of white lights which are off when the valves are in their normal or required positions for power operations. The monitor lights for automatically actuated valves are energized when the valve is in the automatically actuated position. For the centrifugal charging pump alternate minimum flow valves (HV-8508A and B), the monitor panel lights indicate that the valves are in the enabled mode; therefore, valve position is indicated only at the handswitches. These monitor lights thus enable the operator to quickly assess the status of the ESF systems. These indications are derived from contacts integral to the valve operators. The circuits for the ESF monitor lights are classified as associated circuits and have electrical and physical separation. In the cases of the accumulator isolation valves, redundancy of position indication is provided by valve stem-mounted limit switches which actuate annunciators on the control board when the valves are not correctly positioned for ESF actuation.

The stem-mounted switches for the accumulator isolation valves are independent of the limit switches in the motor operator.

7.3.2.1.1.5 Support Systems. The following systems are required for support of the ESF:

A. Nuclear service cooling water system. (See subsection 9.2.1.)

B. Component cooling water system. (See subsection 9.2.2.)

C. Electrical power distribution systems. (See chapter 8.)

7.3.2.1.2 Design Basis Information.

Refer to section 6.3.

7.3.2.1.3 Final System Drawings.

Refer to section 6.3.

7.3.2.2 Analysis

Refer to chapter 15 and section 6.3.

7.3.3 CONTAINMENT COMBUSTIBLE GAS CONTROL SYSTEM

7.3.3.1 Description

The concentration of hydrogen in the containment atmosphere is monitored by the hydrogen monitor system described in subsection 6.2.5. The containment combustible gas control equipment (described briefly below and more completely in subsection 6.2.5) maintains this hydrogen concentration below the minimum concentration capable of combustion.

7.3.3.1.1 System Description

A. Subsystems

1. Hydrogen monitors.
2. Hydrogen recombiners.
3. Post-loss-of-coolant accident (post-LOCA) purge exhaust system.
4. Post-LOCA cavity purge system.
5. Containment cooling system fans.

B. Initiating Circuits

The containment combustible gas control equipment (table 7.3.3-1) is operated manually from control switches located in the main control room or at local stations. It is not necessary for the monitor, recombiner, or purge equipment to be initiated automatically because it would take approximately 7 days for the hydrogen concentration to reach the control limit of 4-percent hydrogen by volume with no hydrogen reduction system in operation. The containment cooler fans start automatically and run at slow speed upon receipt of a safety injection signal (SIS). (See subsection 7.3.11.) The post-LOCA cavity purge system starts automatically on SIS.

C. Logic

The combustible gas control system is manually controlled, except for items under automatic start mentioned in paragraph 7.3.3.1.1.B above, as shown in drawings 1X5DN013-4, 1X5DN015-1, 1X5DN017-2, 1X5DN013-1, 1X5DN013-2 and 1X5DN013-4.

D. Bypass

Indication of system bypass is provided as described in section 7.5. The containment isolation system (CIS) isolates the purge exhaust lines which can manually be reopened when necessary.

E. Interlocks

There are no interlocks on these controls.

F. Sequencing

On SIS or loss of offsite power coincident with SIS, the containment fan coolers are sequenced on at low speed at the 30.5-s sequencer step. On loss of offsite power only, the fans are sequenced on at high speed at the same step.

G. Redundancy

Controls are provided on a one-to-one basis with the mechanical equipment so that the controls preserve the redundancy of the mechanical equipment.

H. Diversity

Diversity of control is provided in that the combustible gas control equipment may be

controlled from local controls at the power distribution equipment, as well as from the main control room panels.

I. Actuated Devices

Table 7.3.3-1 lists the actuated devices.

J. Supporting Systems

The supporting systems required for these controls are the Class 1E ac power system (described in section 8.3) and the containment atmosphere monitoring system (described in subsection 6.2.5).

7.3.3.1.2 Design Basis

Design bases for the containment combustible gas control system are such that operation is controlled manually from the main control room and no single failure prevents the containment combustible gas control system from functioning. In addition, the following conditions are considered for the control system components:

A. Range of Transient and Steady-State Conditions and Circumstances

The electrical power supply characteristics for the controls on this system are as described in section 8.3. The range of possible environmental conditions for these controls is as described in section 3.11.

B. Malfunctions, Accidents, or Other Unusual Events

1. Fire protection is discussed in subsection 9.5.1.
2. Missile protection is discussed in section 3.5.
3. Earthquake protection is discussed in sections 3.7.B and 3.7.N.

7.3.3.1.3 Drawings

There is no automatic actuation signal for this system, although the equipment controls include interfaces with sensors and with other devices. However, at the device level, the containment cooler fans and the post-LOCA cavity purge fans automatically start upon receipt of SIS, and the containment post-LOCA purge exhaust isolation valves automatically close on receipt of CVI. References to the drawings associated with this system are provided as described in the introductory material for this section.

Control logic diagrams for the individual devices are shown in drawings 1X5DN013-4, 1X5DN015-1, 1X5DN017-2, 1X5DN019-1, 1X5DN019-2, 1X5DN013-1, 1X5DN013-2 and 1X5DN013-4. These compare with the Preliminary Safety Analysis Report (PSAR) as follows:

A. Recombiner Controls

For recombiners, there is no functional change, but fault protection is added.

B. Mixing Fan Controls

Functionally the containment cooler fans operate as shown in drawings 1X5DN013-1, 1X5DN013-2 and 1X5DN013-4. Details of motor overload protection have been added since the PSAR. The containment cooler fans are loaded onto the diesel

generators as indicated in drawings 1X3D-AA-K02A, 2X3D-AA-K02A, 1X3D-AA-K02B and 2X3D-AA-K02B.

The electrical schematic diagrams listed in section 1.7 are in accordance with the control logic diagrams.

7.3.3.2 Analysis

A. Conformance to Nuclear Regulatory Commission (NRC) General Design Criteria

The applicable criteria are listed in table 7.1.1-1. No deviations or exceptions to those criteria are taken. (See section 3.1.)

B. Conformance to Regulatory Guide 1.7

Conformance is described in subsection 6.2.5 and summarized in section 1.9.

C. Conformance to Institute of Electrical and Electronics Engineers (IEEE) Standard 279-1971

The design of the control system is based on the applicable requirements of IEEE Standard 279-1971, as follows:

1. General Functional Requirement - Paragraph 4.1

   The containment cooler fans and the post-LOCA cavity purge fans are able to function automatically and reliably over the full range of transients for all plant conditions for which credit was taken in the analyses. The rest of the system functions for all of these plant conditions when manually initiated. The system response time and accuracy are as required in the accident analyses. The hydrogen sampling line is manually actuated.

2. Single Failure Criterion - Paragraph 4.2

   Through use of redundant, independent systems, as previously described, any single failure or multiple failures resulting from a single credible event will not prevent the system from performing its intended function when required.

3. Quality of Components and Modules - Paragraph 4.3

   Components and modules used in the construction of the system exhibit a quality consistent with the nuclear power plant design life objective, require minimum maintenance, and have low failure rates. The program for quality assurance is described in chapter 17.

4. Equipment Qualification - Paragraph 4.4

   The system is qualified to perform its intended functions under the environmental conditions specified in sections 3.10.B, 3.10.N, 3.11.B, and 3.11.N.

5. Channel Integrity - Paragraph 4.5

   All channels maintain functional capability under all conditions described in paragraph 7.3.3.1.2.

6. Channel Independence - Paragraph 4.6

   Discussions of the means used to ensure channel independence are given in paragraphs 7.1.2.2 and 8.3.1.4.

7. Control and Protection System Interaction - Paragraph 4.7

   No credible failure at the output of an isolation device will prevent the associated channel from performing its intended function. No single random failure in one channel will prevent the other channel from performing the intended function.

8. Derivation of System Outputs - Paragraph 4.8

   To the extent feasible, the system inputs are from direct measurement of the desired variable.

9. Capability of Sensor Checks - Paragraph 4.9

   Sufficient means have been provided to check the operational availability of the system.

10. Testing and Calibration - Paragraph 4.10

    The control system has the capability of testing the devices used to derive the final system output. No jumpers are used for testing.

11. Channel Bypass or Removal from Operation - Paragraph 4.11

    Testing of one channel can be accomplished during reactor operation without initiating a protective action at the system level.

12. Operating Bypasses - Paragraph 4.12

    There are no permissive conditions on bypasses. Bypass of one channel will not bypass the other channel. Bypass of one system will not bypass any other system.

13. Indication of Bypass - Paragraph 4.13

    If the protective action of any part of the system has been bypassed or deliberately rendered inoperative, the fact will be continuously indicated in the control room, as described in section 7.5.

14. Access to Means for Bypassing - Paragraph 4.14

    Appropriate administrative controls will be applied to ensure that access to the means for manually bypassing the system is adequately protected.

15. Multiple Setpoints - Paragraph 4.15

    The system is designed so that there are no multiple setpoints.

16. Completion of Protective Action Once It Is Initiated - Paragraph 4.16

    The system is designed so that once protective action is initiated, it is carried through to completion.

17. Manual Initiation - Paragraph 4.17

    Manual initiation of each function is provided in the control system with a minimum of equipment by direct control of power distribution equipment and solenoid valves from panel-mounted control switches. System level actuation of the safety function is not provided since the time required for operation of these functions allows the station operator to take individual action for each controlled device.

18. Access to Setpoint Adjustments, Calibration, and Test Points - Paragraph 4.18

    Appropriate administrative controls are applied to ensure that access to the means for adjusting, calibrating, and testing the system is adequately protected.

19. Identification of Protective Actions - Paragraph 4.19

    System protective actions are described and identified down to the channel level.

20. Information Readout - Paragraph 4.20

    Sufficient information is provided to allow the station operator to make a prompt decision regarding the system operating requirements. The indications required for these decisions are provided by device status lights, the systems status monitor panel, and supporting systems, as listed in the system description discussed in paragraph 7.3.3.1.1.J.

21. System Repair - Paragraph 4.21

    The system is designed to facilitate the recognition, location, replacement, repair, and adjustment of malfunctioning components or modules.

22. Identification - Paragraph 4.22

    Protection system components are identified, as described in paragraph 7.1.2.3.

D. Conformance to Nuclear Regulatory Commission (NRC) Regulatory Guides

   The applicability of regulatory guides is as shown in table 7.1.1-1 and summarized in section 1.9. References to the discussions of these regulatory guides are presented in paragraphs 7.1.2.5, 7.1.2.6, and 7.1.2.7.

E. Periodic Testing

   Periodic testing of the mechanical equipment associated with this system is discussed in subsection 6.2.5. There is no automatic actuation equipment for the entire system, but there is automatic device actuation, as described in paragraph 7.3.3.1.3. Provisions for periodic testing of the actuation system are discussed in the Technical Specifications.

F. Failure Modes and Effects Analysis

   See table 6.2.5-2.

7.3.4 CONTAINMENT PURGE ISOLATION SYSTEM

7.3.4.1 Description

The containment purge isolation system detects any abnormal amount of radioactivity in the containment and initiates appropriate action to ensure that any release of radioactivity to the environs is controlled. The containment purge systems are isolated by containment ventilation isolation (CVI) signals. A detailed description of those systems is given in subsection 6.2.4.

7.3.4.1.1 System Description

A. Initiating Circuits

Redundant area radiation monitors in the containment and an independent radiation monitor in the purge line consisting of gaseous, particulate, and iodine radiation monitors measure the radioactivity levels in the containment. These monitors, through their data processing modules, provide digital radioactivity signals to the engineered safety features actuation system (ESFAS) logic. The logic generates redundant CVI actuation signals.

B. Logic

Logic diagrams for the ESFAS are provided in drawings 1X6AA02-225, 1X6AA02-226, 1X6AA02-227, 1X6AA02-228, 1X6AA02-229, 1X6AA02-230, 1X6AA02-231, 1X6AA02-232, 1X6AA02-233, 1X6AA02-234, 1X6AA02-235, 1X6AA02-236, 1X6AA02-237, 1X6AA02-238, 1X6AA02-239, 1X6AA02-240, 1X6AA02-494, 1X6AA02-495, 1X6AA02-496, 1X6AA02-519, 1X5DN019-1 and 1X5DN019-2. These diagrams show the actuation systems and bypass interlock provisions. The logic for the containment purge isolation subsystem is included in these figures.

C. Bypass

Switches are provided to bypass defective monitors to preclude spurious actuation.

D. Interlocks

There are no interlocks on these controls.

E. Sequencing

The system is energized on the first step of load sequencing.

F. Redundancy

Controls are provided on a one-to-one basis with the mechanical equipment so that the controls preserve the redundancy of the mechanical equipment.

G. Diversity

Diversity of sensing is provided in that containment purge isolation can be actuated by the containment vent gaseous iodine, air particulate radiation monitors, or containment area radiation monitors.

H. Actuated Devices

Table 7.3.4-1 lists the actuated devices.

I. Supporting Systems

Supporting systems for the containment purge isolation are the four Class 1E 125-V dc power supplies, the Class 1E ac power system discussed in section 8.3, and the instrument air system described in section 9.3. The isolation function is fail-safe with respect to all of these support systems; that is to say, loss of any one of these support systems will not prevent isolation.

7.3.4.1.2 Design Bases

The design bases for the containment purge isolation system are described in paragraphs 6.2.4.1.1 and 7.3.3.1.2.

7.3.4.1.3 Drawings

The logic for the containment purge isolation system is shown in the ESFAS logic diagrams, drawings 1X6AA02-225, 1X6AA02-226, 1X6AA02-227, 1X6AA02-228, 1X6AA02-229, 1X6AA02-230, 1X6AA02-231, 1X6AA02-232, 1X6AA02-233, 1X6AA02-234, 1X6AA02-235, 1X6AA02-236, 1X6AA02-237, 1X6AA02-238, 1X6AA02-239, 1X6AA02-240, 1X6AA02-494, 1X6AA02-495, 1X6AA02-496, 1X6AA02-519, 1X5DN019-1 and 1X5DN019-2.

7.3.4.2 Analysis

A. Conformance to Nuclear Regulatory Commission (NRC) General Design Criteria

The applicable criteria are listed in table 7.1.1-1. No deviations or exceptions to those criteria are taken. Compliance is summarized in section 3.1.

B. Conformance to Institute of Electrical and Electronics Engineers (IEEE) Standard 279-1971

The design of the control system conforms to the applicable requirements of IEEE Standard 279-1971, as listed and discussed in paragraph 7.3.3.2C. The ranges and setpoints are given in the Technical Specifications.

C. Conformance to NRC Regulatory Guides

The applicability of the regulatory guides is as shown in table 7.1.1-1 and summarized in section 1.9. References to the discussions of these regulatory guides are presented in subsection 7.1.2.

D. Periodic Testing

Periodic testing of the mechanical equipment associated with this system is discussed in section 9.4. Periodic testing of the actuation system is discussed in the Technical Specifications.

7.3.5 FUEL HANDLING BUILDING VENTILATION ISOLATION

7.3.5.1 Description

A description of the entire fuel handling building ventilation system is given in subsection 9.4.2.

7.3.5.1.1 Initiating Circuits

A. Four redundant two-channel-oriented and train-oriented gaseous radioactivity monitors, together with their data processing modules, provide a digital signal to the balance of plant (BOP) safety actuation system when preset radiation levels are exceeded.

B. Two manual actuation switches also are wired into the BOP safety actuation system.

C. Upon receipt of the inputs from items A through C above, the BOP safety actuation system logic circuitry produces a fuel handling building isolation signal for both train A and train B (FHBI-A and FHBI-B). This signal in turn causes the post-accident filter units, train A and train B, fans to start. Starting these fans then causes the inlet and discharge dampers to open. Isolation dampers are closed automatically.

D. When radiation signals return to normal conditions, the post-accident heating, ventilation and air-conditioning (HVAC) systems continue to operate until reset manually.

E. Switches are provided to bypass defective monitors to preclude spurious actuation. The FHBI signal may be blocked using the normal channel test blocks. Channel bypass is indicated in the control room.

F. There are no interlocks on these controls.

G. The system is energized on the first step of load sequencing.

H. Controls are provided on a one-to-one basis with the mechanical equipment so that the controls preserve the redundancy of the mechanical equipment. There are two channels of actuation initiated by redundant radioactivity monitors, and redundant manual initiation switches.

I. Diversity of control is provided in that the fuel handling building ventilation isolation system can be actuated by either automatic signals or manual control.

J. Table 7.3.5-1 lists the actuated devices.

K. Supporting systems for the fuel handling building ventilation isolation system actuation are the two Class IE 125-V dc power supplies, the two Class-IE vital 120-V ac power systems discussed in section 8.3, and the instrument air system described in subsection 9.3.1. Loss of any one of these support systems will not prevent isolation.

7.3.5.2 Design Bases

The design bases for the fuel handling building ventilation isolation system are discussed in paragraph 9.4.2.2.1.1. Additionally, the design bases described in paragraph 7.3.1.1.2 are applicable for the control system components.

7.3.5.3 Drawings

The logic diagrams for the fuel handling building ventilation isolation actuation system are included in drawings AX5DN020-1, AX5DN020-2, AX5DN020-3, AX5DN027-1, AX5DN028-1, AX5DN029-1 and AX5DN029-3.

The control logic diagrams, the electrical schematic diagrams, the piping and instrument diagrams, and the physical location drawings for this system are included in the references in section 1.7.

7.3.5.4 Analysis

A. Conformance to Nuclear Regulatory Commission (NRC) General Design Criteria

   The applicable criteria are listed in table 7.1.1-1. No deviations or exceptions to those criteria are taken. Compliance is summarized in section 3.1.

B. Conformance to Institute of Electrical and Electronics Engineers (IEEE) Standard 279-1971

   The design of the control system conforms to the applicable requirements of IEEE Standard 279-1971, as listed and discussed in paragraph 7.3.3.2C.

C. Conformance to NRC Regulatory Guides

   The applicability of the regulatory guides is as shown in table 7.1.1-1 and summarized in section 1.9. References to the discussions of conformance to these regulatory guides are presented in paragraph 7.1.2.

D. Failure Mode and Effects Analysis

   See table 9.4.2-2.

E. Periodic Testing

   Periodic testing of the mechanical equipment associated with this system is discussed in subsection 9.4.2.

7.3.6 CONTROL ROOM VENTILATION ISOLATION

7.3.6.1 Description

Upon detection of high gaseous radioactivity levels in the control room outside air intake, the normal HVAC system is isolated as described in sections 6.4 and 9.4. The control room HVAC system switches to the emergency mode of operation where a small supply of outside air is provided to maintain a set positive pressure in the control room. This positive pressure will prevent the ingress of the local ambient atmosphere. Normal ventilation is restored only by manual operation by the plant operator and is maintained only if the local ambient atmosphere poses none of the monitored hazards.

7.3.6.1.1 System Description

A. Actuating Circuits

The gaseous radioactivity level of the air provided to the main control room from the local ambient atmosphere is monitored by four redundant monitors (two per each intake duct).

The signals from these monitors are transmitted to bistables in the engineered safety features actuation system. If acceptable levels are exceeded, the control room is isolated, as described above.

The sensitivities and response times of these monitors are listed in table 7.3.6-1.

In addition to the above, control room isolation is initiated manually.

B. Logic

The control room ventilation isolation actuation system logic is included in drawings AX5DN020-4, AX5DN020-5, AX5DN020-6, AX5DN020-8, AX5DN020-10, AX5DN031-1, AX5DN031-2, AX5DN031-4, AX5DN032-3, AX5DN032-1, AX5DN034-2 and AX5DN037-1. For emergency operation, both trains of the affected unit receive a start signal. However, a permissive is provided which does not allow the lag unit to start unless there is a low-flow condition in the lead unit. The actuation signal is transmitted to each actuated device and, subject to the provisions of bypass or override, causes each device to assume its safe state.

C. Bypass

Channel selector switches, with a test block feature, are provided in the instrumentation control circuit to enable testing of the instrument control circuit independently of the redundant control circuit. Channel bypass is indicated at the system level in the control room.

Manual override is available by means of pull-to-lock switches on the fans.

D. Interlocks

Operational interlocks are as shown in drawings AX5DN020-4, AX5DN020-5, AX5DN020-6, AX5DN020-8, AX5DN020-10, AX5DN031-1, AX5DN031-2, AX5DN031-4, AX5DN032-3, AX5DN032-1, AX5DN034-2 and AX5DN037-1 and as identified in section 6.4.

E. Sequencing

The control room ventilation isolation system is powered from the Class 1E power system and energized on the first (0.5 s) step of the load sequencing, except for the control room filter units which start automatically after the 30.5 s step.

F. Redundancy

Controls are provided on a one-to-one basis with the mechanical equipment so that the controls preserve the redundancy of the mechanical equipment. Redundancy is provided in the gaseous radioactivity monitors, the actuation signals, and manual actuation switches.

G. Diversity

Diversity of actuation is provided in that the control room ventilation system may be isolated by either an automatic system or by operator manual actuation. Diversity is provided by actuation from the gaseous radioactivity and manual switches.

H. Actuated Devices

Table 7.3.6-2 lists the actuated devices.

I. Supporting System

The supporting system required for the controls are the four Class 1E 125 V-dc power supplies, vital Class 1E ac system described in section 8.3, and instrument air system described in section 9.3.1.

7.3.6.1.2 Design Bases

The design bases for the control room ventilation isolation system are such that no single failure can prevent the isolation of the control room ventilation system. The trip points are provided in the Technical Specifications.

Additionally, the design bases described in subsection 6.4.1 are applicable to the control system components.

7.3.6.1.3 Drawings

The logic diagram for the control room ventilation isolation actuation system is included in drawings AX5DN020-4, AX5DN020-5, AX5DN020-6, AX5DN020-8, AX5DN020-10, AX5DN031-1, AX5DN031-2, AX5DN031-4, AX5DN032-3, AX5DN032-1, AX5DN034-2 and AX5DN037-1.

Other drawings pertaining to this system are included in the references in section 1.7.

7.3.6.2 Analysis

A. Conformance to Nuclear Regulatory Commission (NRC) General Design Criteria

The applicable criteria are listed in table 7.1.1-1. No deviations or exceptions to those criteria are taken. Compliance is summarized in section 3.1.

B. Conformance to Institute of Electrical and Electronics Engineers (IEEE) Standard 279-1971

The design of the control system conforms to the applicable requirements of IEEE Standard 279-1971, as listed and discussed in paragraph 7.3.3.2.C. The setpoints are provided in the Technical Specifications.

C. Conformance to NRC Regulatory Guides

The applicability of regulatory guides is as shown in table 7.1.1-1 and summarized in section 1.9. References to the discussions of these regulatory guides are presented in table 7.1.1-1.

D. Failure Mode and Effects Analysis

This analysis is given in table 6.4.4-1.

E. Periodic Testing

Periodic testing of the mechanical equipment associated with this system is discussed in subsection 9.4.1. Provisions for the periodic testing of the actuation system are discussed in the Technical Specifications.

7.3.7 AUXILIARY FEEDWATER SYSTEM

7.3.7.1 Description

The auxiliary feedwater system (AFWS) consists of two motor-driven pumps, one steam turbine-driven pump, and piping, valves, instruments, and controls, as shown in drawings 1X4DB161-2 and 1X4DB161-3. The pumps are started automatically on receipt of signals from the actuation logic, as shown in drawings 1X5DN117-1, 1X5DN117-2, 1X5DN117-3, 1X5DN120-1, 1X5DN120-2, 1X5DN120-3, 1X5DN120-5, 1X5DN120-6, 1X5DN121-1, 1X5DN121-2, 1X5DN122-1 and 1X5DN122-2. The two motor-driven pumps can also be started manually from control switches in the control room or at the remote shutdown control panel. The turbine-driven pump can also be started manually from the control room or at the local control panels located in the auxiliary feedwater pumphouse.

The preferred source of water for the AFWS is the condensate storage tank (CST). This tank is Seismic Category 1.

Each motor-driven pump feeds two steam generators through individual motor-operated flow control valves. AFWS flow can be regulated manually from the control room or from the remote shutdown panels.

The turbine-driven pump feeds all four steam generators through individual dc motor-operated control valves. AFWS valves can be operated manually from the control room or from the local control panels located in the auxiliary feedwater pumphouse.

AFWS flow indication is provided for each steam generator in the control room and at the remote shutdown control panel.

The AFWS pump turbine is supplied steam from two of the four main steam lines. Each of the steam supply lines to the turbine driver is equipped with a check valve and a normally open motor-operated gate valve. These steam lines join to form a header which leads to the turbine through a normally closed supply valve and normally open trip/throttle valve, both of which are dc motor-operated, and a normally open electro-hydraulically operated speed governing valve. Control of these valves, as well as manual speed control for the turbine-driven pump, is provided in the control room and at the local control panels located in the auxiliary feedwater pumphouse.

The status of the motor-driven pumps, the turbine-driven pump, the turbine steam supply valves, and the turbine stop valves is indicated in the control room.

The AFWS equipment is described in subsection 10.4.9.

In addition to initiating functions described above, the auxiliary feedwater actuation signal (AFWAS) closes the steam generator blowdown and sample isolation valves, when auxiliary feedwater is required by plant conditions. However, the steam generator sample isolation valves may be opened 30 seconds after closure due to an auxiliary feedwater auto-start signal to allow operators to obtain a sample. All remote manually operated valves in the normal suction from the CST and in the discharge to the steam generators are normally open.

7.3.7.1.1 System Description

A. Initiating Circuits

The AFWAS motor-driven (AFWAS-M) starting the motor-driven auxiliary feedwater pumps is generated on the occurrence of any one of the following signals:

1. Manual start.
2. Trip of both main feedwater pumps.
3. The two out of four low-low water level signals in any one steam generator.
4. Safety injection (SI).
5. Loss of offsite power.
6. Anticipated Transient Without SCRAM (ATWS) Mitigation System Actuation Circuitry (AMSAC).

All automatic actuations of the motor-driven pumps are subject to load sequencing. The AFWAS turbine-driven (AFWAS-T) starting the turbine-driven auxiliary feedwater pumps is generated on the occurrence of any one of the following signals:

1. Manual start.
2. The two out of four low-low water level signals on any two steam generators.
3. Loss of offsite power.
4. AMSAC.

The steam generator sample line isolation valves and the steam generator blowdown isolation valves are all automatically closed on the occurrence of a steam generator low-low water level, safety injection signal, a loss of offsite power signal, trip of both main feedwater pumps, or AMSAC actuation.

B. Logic

See drawings 1X5DN117-1, 1X5DN117-2, 1X5DN117-3, 1X5DN120-1, 1X5DN120-2, 1X5DN120-3, 1X5DN120-5, 1X5DN120-6, 1X5DN121-1, 1X5DN121-2, 1X5DN122-1 and 1X5DN122-2.

C. Bypass

Control switches in the control room to modulate the feedwater pump discharge valves have override features to maintain the required steam generator water levels. This also permits manual closure of the valves if necessary to isolate the flow to a faulted steam generator. (See FSAR paragraph 10.4.9.2.2.3 and drawings 1X5DN121-1 and 1X5DN121-2.)

D. Interlocks

There are no other interlocks other than those shown in drawings 1X5DN117-1, 1X5DN117-2, 1X5DN117-3, 1X5DN120-1, 1X5DN120-2, 1X5DN120-3, 1X5DN120-5, 1X5DN120-6, 1X5DN121-1, 1X5DN121-2, 1X5DN122-1 and 1X5DN122-2.

E. Redundancy

Sufficient actuation and control channels are provided throughout the AFWS to

ensure the required flow to at least two steam generators in the event of a single failure.

F. Diversity

The AFWS is diversified by utilizing a turbine-driven pump with dc motor-operated valves and two ac motor-driven pumps with ac motor-operated valves. Diversity in initiating signals can be seen in drawings 1X5DN117-1, 1X5DN117-2, 1X5DN117-3, 1X5DN120-1, 1X5DN120-2, 1X5DN120-3, 1X5DN120-5, 1X5DN120-6, 1X5DN121-1, 1X5DN121-2, 1X5DN122-1 and 1X5DN122-2.

G. Actuated Devices

1. Auxiliary feedwater pump turbine steam supply valves (two).
2. Auxiliary feedwater pump turbine stop valve.
3. Auxiliary feedwater motor-operated valves (eight).
4. Auxiliary feedwater pump electric motors (two).
5. Steam turbine-driven AFWS pump drain line to condenser HV-5178.
6. Vacuum degasifier isolation valve HV-5087.
7. Steam generator blowdown isolation valves (four).
8. Steam generator blowdown sample isolation valves (eight).
9. Auxiliary feedwater pump recirculation valves (two).

H. Supporting Systems

The Class 1E ac and dc power systems are required for auxiliary feedwater control.

I. Portion of System Not Required for Safety

Instrumentation provided for monitoring system performance is not required for safety, except for the instrumentation that is shown on table 7.5.2-1.

7.3.7.1.2 Design Bases

Auxiliary feedwater is required, as described in subsection 10.4.9. No single failure shall prevent this system from operating.

The system must provide full auxiliary feedwater flow within 1 min of the detection of any condition requiring auxiliary feedwater.

7.3.7.1.3 Drawings

The logic diagram for the AFWAS is included in drawings 1X5DN117-1, 1X5DN117-2, 1X5DN117-3, 1X5DN120-1, 1X5DN120-2, 1X5DN120-3, 1X5DN120-5, 1X5DN120-6, 1X5DN121-1, 1X5DN121-2, 1X5DN122-1 and 1X5DN122-2.

Other drawings pertaining to this system are referenced in section 1.7.

7.3.7.2 Analysis

A. Compliance to Nuclear Regulatory Commission (NRC) General Design Criteria (GDC)

Compliance is summarized in section 3.1.

1. GDC 13

   Instrumentation necessary to monitor station variables associated with hot shutdown is provided in the main control room and on the auxiliary shutdown control panel. Controls for the AFWS are provided at each location. A description of the surveillance instrumentation is provided in section 7.5.

2. GDC 19

   All controls and indications required for safe shutdown of the reactor are provided in the main control room. In the event that the main control room must be evacuated, adequate controls and indications are located outside the main control room to bring to and maintain the reactor in a hot standby condition and provide capability to achieve cold shutdown.

   The remote shutdown control panels, located outside the main control room, are described in section 7.4.

3. GDC 34

   The AFWS provides an adequate supply of feedwater to the steam generators to remove reactor decay heat following reactor trip. Two steam generators with auxiliary feedwater supply are sufficient to remove reactor decay heat without exceeding design conditions of the reactor coolant system.

4. Other GDC

   The remaining applicable general design criteria are listed in table 7.1.1-1 and subsection 10.4.9.

B. Conformance to Institute of Electrical and Electronics Engineers (IEEE) Standard 279-1971

   The design of the control system conforms to the applicable requirements of IEEE Standard 279-1971, as listed and discussed in paragraph 7.2.2.3 and subsection 7.3.2, except that this system is automatically actuated. The setpoints are provided in the Technical Specifications.

C. Conformance to NRC Regulatory Guides

   The applicability of regulatory guides is shown in table 7.1.1-1 and summarized in section 1.9. References to the discussions of these regulatory guides are presented in table 7.1.1-1.

D. Failure Modes and Effects Analysis

   See table 10.4.9-4.

E. Periodic Testing

   Periodic testing of the mechanical equipment associated with this system is discussed in paragraph 10.4.9.4. Provisions for the periodic testing of the actuation system are discussed in the Technical Specifications.

7.3.8 MAIN STEAM AND FEEDWATER ISOLATION

7.3.8.1 Description

The signals that initiate automatic closure of the main steam isolation, main steam isolation valve bypass, feedwater isolation, and feedwater isolation bypass valves are generated in the engineered safety features actuation system (ESFAS) described in subsection 7.3.1. The logic diagrams for the generation of these signals are shown in drawings 1X6AA02-225, 1X6AA02-226, 1X6AA02-227, 1X6AA02-228, 1X6AA02-229, 1X6AA02-230, 1X6AA02-231, 1X6AA02-232, 1X6AA02-233, 1X6AA02-234, 1X6AA02-235, 1X6AA02-236, 1X6AA02-237, 1X6AA02-238, 1X6AA02-239, 1X6AA02-240, 1X6AA02-494, 1X6AA02-495, 1X6AA02-496, and 1X6AA02-519. The remainder of this section concentrates on the non-Westinghouse portion of the main steam and feedwater isolation system.

The main steam and main feedwater isolation valves are operated by hydraulic actuators. The actuators are powered by compressed gas accumulators, which are controlled by electrically operated solenoid valves. Each main feedwater isolation valve has two actuators. Each actuator is controlled from a separate Class 1E electrical system, and each is capable of closing the valve independently of the other. Each main steam isolation valve has one separate Class 1E electrical system actuator.

The main steam isolation valve bypass valves are operated by a pneumatic diaphragm operator, each with one separate Class 1E electrical system actuator; the bypass feedwater isolation valves are pneumatic piston operator, each with two separate 1E electrical system actuators.

7.3.8.1.1 System Description

A. Initiating Circuits

The main steam isolation, main steam isolation valve bypass, feedwater isolation, and feedwater isolation bypass valves close automatically upon receipt of automatic close signals (steam line isolation signal for steam isolation and feedwater isolation signal for feedwater isolation) from the Westinghouse solid-state protection system.

The steam line isolation signal is generated by any of the following:

1. High steam pressure rate.
2. Low steam line pressure.
3. High containment pressure.

A feedwater isolation signal is generated by the following:

1. Steam generator high level (two out of four for each steam generator).
2. Safety injection.
3. Reactor trip coincident with low T_{avg}.

Manual operation is also provided.

B. Logic

Refer to drawings 1X6AA02-225, 1X6AA02-226, 1X6AA02-227, 1X6AA02-228, 1X6AA02-229, 1X6AA02-230, 1X6AA02-231, 1X6AA02-232, 1X6AA02-233, 1X6AA02-234, 1X6AA02-235, 1X6AA02-236, 1X6AA02-237, 1X6AA02-238,

1X6AA02-239, 1X6AA02-240, 1X6AA02-494, 1X6AA02-495, 1X6AA02-496, 1X6AA02-519, 1X5DN149-1, 1X5DN149-2, 1X5DN149-3, 1X5DN150-1, 1X5DN150-2, 1X5DN150-3, 1X5DN149-4 and 1X5DN150-4.

C. Bypass

See subsection 7.3.1.

D. Interlocks

See subsection 7.3.1.

E. Redundancy

Two isolation valves in series (train oriented) are provided, ensuring steam line isolation.

F. Diversity

See subsection 7.3.1 for a discussion of diversity with regard to the automatic actuation signal.

G. Actuated Devices

The actuated devices are the main steam and feedwater isolation valves. Refer to table 7.3.8-1.

H. Supporting Systems

The system makes use of the Class 1E dc power systems and of the compressed air system.

I. Portions of the System Not Required for Safety

The operator for each valve includes provisions for manually opening the valve. Instrumentation is provided for measuring the accumulator pressures. Neither of these provisions is required for safety.

7.3.8.1.2 Design Bases

The design bases for the main steam and feedwater isolation actuation system are provided in subsection 7.3.1. The design bases for the remainder of the main steam and feedwater isolation system are that the system isolates the main steam and feedwater when required and that no single failure can prevent isolation from occurring. See subsection 7.3.1 for additional discussion.

In addition, paragraph 7.3.3.1.2 is applicable to the control system components.

7.3.8.1.3 Drawings

See drawings 1X6AA02-225, 1X6AA02-226, 1X6AA02-227, 1X6AA02-228, 1X6AA02-229, 1X6AA02-230, 1X6AA02-231, 1X6AA02-232, 1X6AA02-233, 1X6AA02-234, 1X6AA02-235, 1X6AA02-236, 1X6AA02-237, 1X6AA02-238, 1X6AA02-239, 1X6AA02-240, 1X6AA02-494, 1X6AA02-495, 1X6AA02-496, 1X6AA02-519, 1X5DN149-1, 1X5DN149-2, 1X5DN149-3, 1X5DN150-1, 1X5DN150-2, 1X5DN150-3, 1X5DN149-4 and 1X5DN150-4.

7.3.8.2 Analysis

A. Compliance to Nuclear Regulatory Commission (NRC) General Design Criteria

Compliance is summarized in section 3.1.

B. Conformance to Institute of Electrical and Electronics Engineers (IEEE) Standard 279-1971. The design of the valve control system conforms to the applicable requirements of IEEE Standard 279-1971, as listed and discussed in subsection 7.3.2, except that the system is automatically actuated. The setpoints are provided in the Technical Specifications.

C. Conformance to NRC Regulatory Guides

The applicability of regulatory guides is shown in table 7.1.1-1 and summarized in section 1.9.

D. Periodic Testing

The main steam isolation valve control system includes provisions for verifying the proper operation of the electronic logic circuits, checking the accumulator pressure in each actuator, and for performing a 10-percent close test of each valve. The frequency of control system testing is provided in the Technical Specifications. The mechanical system testing provisions are given in subsection 10.3.4.

7.3.9 NUCLEAR SERVICE COOLING WATER

7.3.9.1 Description

The nuclear service cooling water (NSCW) consists of the ultimate heat sink and the NSCW pumps, piping, valves, exchangers, and other components. The NSCW system is described in subsection 9.2.1. The ultimate heat sink is described in subsection 9.2.5.

The referenced sections also state the safety design bases and the power generation design bases for their respective systems.

7.3.9.1.1 System Description

A. Initiating Circuits

For train A, two of the three NSCW cooling tower pumps are normally operating during power generation, with one spare. In the event of an accident requiring safety injection, the safety injection signal A ensures that two out of three remain in operation. If any one pump drops out, pump discharge manifold low pressure and the pump interlock circuitry starts the spare pump. The design is similar for the train B NSCW cooling tower pumps. For either train, the load sequencer must also be in operation. Manual initiation is also provided from the control room and remote shutdown panels.

Transfer pumps in each basin are used to transfer water between basins; they are powered by the same source as the train power source for the basin into which they pump. The operation of these pumps is manual only.

The first fan to start in each NSCW tower is interlocked to start when the tower's

spray valve opens and will stop when the spray valve closes. The spray valve begins to open when the NSCW return temperature is above 75°F and begins to close when the temperature falls below 65°F.

The other three fans in each NSCW tower are controlled by independent temperature switches that are dependent on the NSCW return header temperatures. These fans are set to start sequentially through a range of 79°F to 87°F. Automatic trip of the three tower fans on decreasing temperature is provided, with the fans set to trip sequentially through a range of 77°F to 71°F.

To protect against tower icing in the event of low ambient temperature, two motor-operated interlocked valves function to bypass the cooling spray headers and return water directly to the cooling tower basin. Manual initiation is also provided from the control room and remote shutdown panels.

B. Logic

Drawings 1X5DN086-1, 1X5DN087-1, 1X5DN087-2, 1X5DN087-3, 1X5DN087-4, 1X5DN089-1, 1X5DN089-2, 1X5DN089-3, 1X5DN090-1, 1X5DN090-2, 1X5DN090-3, 1X5DN087-5, and 1X5DN090-6 show the logic for NSCW engineered safety features.

C. System Bypass

System bypass, nonauto, power failure, or overload are indicated and alarmed at the systems status monitor panel (QBPS).

D. Interlocks

Interlocks are described in subsection 9.2.1 and 9.2.5 and are shown in drawings 1X5DN086-1, 1X5DN087-1, 1X5DN087-2, 1X5DN087-3, 1X5DN087-4, 1X5DN089-1, 1X5DN089-2, 1X5DN089-3, 1X5DN090-1, 1X5DN090-2, 1X5DN090-3, 1X5DN087-5, and 1X5DN090-6.

E. Redundancy

Redundancy is provided by trains A and B and controls on a one-to-one basis with the mechanical equipment, so that controls preserve the redundancy of the mechanical equipment.

F. Diversity

Diversity is provided by trains A and B, as well as by control from the control room and the remote safe shutdown panels.

G. Radiation Monitoring

Radiation monitoring is provided by a radiation monitor in the return line to each NSCW cooling tower.

H. Actuated Devices

Table 7.3.9-1 lists the actuated devices.

I. Supporting Systems

a. The Class 1E ac power system (described in chapter 8).

b. Makeup water wells.

c. Makeup from river.

7.3.9.1.2 Drawings

Drawings 1X4DB133-1, 1X4DB133-2, 1X4DB134, 1X4DB135-1, and 1X4DB135-2 show the NSCW system and the ultimate heat sink. Drawings 1X5DN086-1, 1X5DN087-1, 1X5DN087-2, 1X5DN087-3, 1X5DN087-4, 1X5DN089-1, 1X5DN089-2, 1X5DN089-3, 1X5DN090-1, 1X5DN090-2, 1X5DN090-3, 1X5DN087-5, and 1X5DN090-6 show the systems actuation logic.

7.3.9.2 Analysis

A. Compliance with Nuclear Regulatory Commission General Design Criteria

Compliance is summarized in section 3.1. See subsection 7.3.1.

B. Conformance to Institute of Electrical and Electronics Engineers (IEEE) Standard 279-1971

The design of the NSCW control system conforms to the applicable requirements of IEEE Standard 279-1971 as listed and discussed in subsection 7.3.2 and paragraph 7.3.7.2. The setpoints which result in Engineered Safety Features Actuation System (ESFAS) actuation of NSCW system are in the Technical Specifications.

C. Failure Modes and Effects Analysis

See table 9.2.1-2.

7.3.10 COMPONENT COOLING WATER SYSTEM

7.3.10.1 Description

The component cooling water system (CCWS) consists of three pumps in each of two trains (A and B), heat exchangers, surge tank, interconnecting pipes, valves, and fittings. The CCWS is described in subsection 9.2.2. Subsection 9.2.2 also addresses the safety design bases and the power generation design bases.

7.3.10.1.1 Engineered Safety Features Initiating Circuits

A. One or two out of three 50-percent capacity CCWS pumps are normally operated in one train. The appearance of a safety injection signal A or loss of offsite power starts train A pumps via the load sequencer.

B. A pump discharge header low pressure signal will also start any pump not running.

C. Pumps can also be started manually from the control room or the remote shutdown panels.

D. Should any pump drop out, the continued presence of the safety injection signal A automatically ensures that two pumps are operating.

E. The safety injection signal B starts train B similarly to train A.

F. Low-low level switches on the surge tank are provided to automatically stop the CCWS pumps before insufficient net positive suction head (NPSH) conditions occur in the pump suction. Each train has its own, separate surge tank, thus maintaining separation.

G. Radiation monitoring is provided in the return line to the CCWS pumps suctions, which would include any return to the surge tanks.

H. Power failure, bypass for test or maintenance, switches in local mode are monitored; alarms and system inoperable lights occur on system status monitoring panel (QBPS) in the control room.

I. Table 7.3.10-1 lists the actuated devices.

J. Supporting systems

1. The Class 1E ac power system.
2. Nuclear service cooling water system.
3. Nuclear steam supply system-balance of plant engineered safety features actuation system.
4. Reactor makeup water storage tanks.
5. Demineralized water storage tank.
6. CCWS surge tanks.

   Makeup to the component cooling water surge tanks is provided automatically from the Seismic Category 1 reactor makeup water storage tank or the demineralized water storage tank.

7.3.10.1.2 Design Bases

Subsection 9.2.2 covers the safety design bases and the power generation design bases.

7.3.10.1.3 Drawings

Drawings 1X5DN091-1, 1X5DN091-2, 1X5DN091-3, 1X5DN092-1, and 1X5DN092-2 show the logic diagrams for the CCWS.

7.3.10.2 Analysis

A. Conformance to Nuclear Regulatory Commission general design criteria is discussed in subsection 7.3.1 and section 3.1.

B. Conformance to Institute of Electrical and Electronics Engineers 279-1971 is listed and discussed in paragraphs 7.3.1.2, and 7.3.7.2.

C. A failure modes and effects analysis is given in table 9.2.2-3.

7.3.11 CONTAINMENT HEAT REMOVAL SYSTEM

7.3.11.1 Description

The containment heat removal system consists of eight train- oriented, fan-operated cooling units and the containment spray system. The cooling unit and the spray system are described in subsection 6.2.2.

7.3.11.1.1 Design Bases

A. Safety design bases for the cooling units are described in paragraph 6.2.2.1.1.1 and for the spray system in paragraph 6.2.2.2.1.1.1.

B. Power generation bases are described in paragraph 6.2.2.1.1.2 for the cooling units. The containment spray system has no power generation design bases.

7.3.11.1.2 Actuating Signals and Circuits

A. The containment safety-related cooling units start via the load sequencer on the receipt of a safety injection signal. All fans are energized on the 30.5 s sequencer step following the receipt of a safety injection signal. The cooling units can also be started and stopped from the control room and from the remote shutdown panels.

B. The containment spray system is initiated by the receipt of a high containment pressure signal (high-3).

C. The safety evaluation is covered in paragraph 6.2.2.1.3.

D. Actuated devices that are tested are listed in table 7.3.11-1.

7.3.11.2 Analysis

The analysis for the engineered safety features, including the systems covered by this section, is covered in paragraph 7.3.1.2. The failure modes and effects analysis of the containment heat removal system is given in table 6.2.2-3. The containment spray failure modes and effects analysis is given in table 6.2.2-5.

7.3.11.3 Summary

The summary is covered in paragraph 7.3.1.

7.3.11.4 Loss-of-Coolant Protection

This analysis is covered in paragraph 7.3.1.

7.3.11.5 Drawings

A. Drawings 1X5DN013-1, 1X5DN013-2, and 1X5DN013-4 show the logic for the containment coolers.

B. Drawings 1X3D-BD-J01A, 1X3D-BD-J01B, 1X3D-BD-J02A, 1X3D-BD-J02B, 1X3D-BD-J02C, and 1X3D-BD-J02D show the electrical elementary diagrams for the containment spray system. The containment spray system starts independently of the sequence to load the safety injection signal. This makes it possible to start within the required time independently of the safety injection signal status.

7.3.12 CONTROL BUILDING ENGINEERED SAFETY FEATURES HEATING, VENTILATION, AND AIR-CONDITIONING SYSTEM

7.3.12.1 Description

The control building engineered safety features (ESF) heating, ventilation, and air-conditioning (HVAC) system provides a proper environment and temperature for the Class 1E electrical equipment and personnel, both during normal operations and under postulated accident conditions. It also serves to reduce or limit the release of fission products to the control building following a postulated loss-of-coolant accident or fuel handling accident. Except for the control room essential HVAC system described in detail in section 6.4 and subsection 7.3.6, the other subsystems of the control building ESF HVAC system are as follows:

A. Control building safety feature electrical equipment rooms HVAC system.

B. Control building HVAC equipment rooms ESF ventilation system (level 3).

C. Control building auxiliary relay rooms ESF air-conditioning units.

A detailed description of these systems is given in subsection 9.4.5.

7.3.12.1.1 System Description

A. Actuating Circuits

1. Control building safety feature electrical equipment rooms HVAC system components (air-conditioning units, exhaust fans, and associated dampers) are actuated upon:
   a. Safety injection (signal A and signal B).
   b. Manual initiation.
2. Control building HVAC equipment rooms ESF ventilation system components (control room ESF chiller rooms exhaust fans) are actuated upon:
   a. Control room ESF chiller room high temperature.
   b. Manual actuation.
3. Control building auxiliary relay rooms ESF air-conditioning units are actuated upon:
   a. Safety injection (signal A and signal B).
   b. Manual initiation.

B. Logic

The control building ESF HVAC system logic is shown in drawings AX5DN008-1, AX5DN056-6, 1X5DN044-1, 1X5DN045-1, and AX5DN056-2. Logic is designed in such a manner that a momentary loss of the control power will not prevent or reverse the safety actuation of any equipment and the reset of the safety injection signal will not trip the actuated equipment without deliberate subsequent operator action.

C. Bypass

Bypass of each subsystem (except the auxiliary relay rooms ESF air-conditioning) comprising the control building ESF HVAC system is indicated in the control room. Such bypass may result from either control power failure, system component failure,

manual override at the component level, or transfer to local control. The bypass indication logic is shown in drawings AX5DN008-1, AX5DN056-6, 1X5DN044-1, 1X5DN045-1, and AX5DN056-2. The manual override capability is provided only for the control building control room ESF chiller rooms exhaust fans.

D. Interlocks

There are no interlocks other than those shown in drawings AX5DN008-1, AX5DN056-6, 1X5DN044-1, 1X5DN045-1, and AX5DN056-2.

E. Sequencing

All loads other than the supporting ESF chiller compressor motors (item J) are energized on the first step of load sequencing. The ESF chiller compressor motors start automatically after the 30.5-s sequencer step.

F. Redundancy

All equipment, instruments, and controls are fully redundant and arranged in two completely independent trains (A and B).

G. Seismic Qualification

All components comprising the control building ESF HVAC system are Seismic Category 1 and remain functional during and after a safe shutdown earthquake.

H. Diversity

Diversity of actuation is provided in that the control building ESF HVAC system can be operated either manually from any one of the two physically separated locations (i.e., main control room and the shutdown panels) or automatically.

I. Actuated Devices

Table 7.3.12-1 lists the actuated devices.

J. Supporting Systems

The following systems are required to be operational for proper functioning of the control building ESF HVAC system:

1. Class 1E 480-V ac system.
2. Class 1E 120-V ac system.
3. Class 1E 125-V dc system.
4. Class 1E 4160-V ac system.
5. Essential chilled water system.

Under emergency conditions (safety injection or loss of offsite power) the Class 1E electric power systems remain operational, as described in section 8.3. The essential chilled water system is described in subsection 9.2.9. It is automatically actuated by the safety injection signal.

7.3.12.1.2 Design Bases

The design bases for the control building ESF HVAC system are such that no single failure within that system nor any supporting system shall prevent it from performing its safety function. A detailed description of the system's design bases is provided in subsection 9.4.5.

7.3.12.1.3 Drawings

The drawings pertaining to the control building ESF HVAC system (including logic diagrams shown in drawings AX5DN008-1, AX5DN056-6, 1X5DN044-1, 1X5DN045-1, and AX5DN056-2) are included in the references in section 1.7.

7.3.12.2 Analysis

The analysis presented in subsection 7.3.1 for the ESF pertains also to the system discussed herein. The failure mode and effects analyses of the control building ESF HVAC systems are given in table 9.4.5-2.

7.3.12.3 Summary

The summary is covered in paragraph 7.3.1.

7.3.13 AUXILIARY BUILDING ENGINEERED SAFETY FEATURES HEATING, VENTILATION, AND AIR-CONDITIONING SYSTEM

7.3.13.1 Description

The auxiliary building engineered safety features (ESF) heating, ventilation, and air conditioning (HVAC) system performs the following safety functions:

- Maintains proper temperatures in safety-related switchgear, motor control center (MCC), pump and heat exchanger rooms during postulated accident conditions, station blackout, and manual conditions.

- Minimizes the release of airborne radioactivity to the outside atmosphere resulting from recirculation line and component leakage into the piping penetration area ECCS and ESF pump rooms during an accident condition.

The system maintains a negative pressure in the piping penetration area and ESF pump rooms and filters the exhaust from the negative pressure boundary. The auxiliary building ESF HVAC system is comprised of the following two systems:

- Auxiliary building ESF room coolers.

- Piping penetration area filtration and exhaust system.

Both systems are described in detail in subsection 9.4.3.

7.3.13.1.1 System Description

A. Activity Circuits

1. The ESF room coolers are actuated upon:

- Safety injection signal (signal A and signal B) or an automatic actuation signal generated by actuation of the corresponding equipment (pump or heat exchanger).
- Room temperature high signal.
- Manual actuation.

For details see drawings 1X5DN030-1, 1X5DN030-3, 1X5DN030-4, 1X5DN030-5, and 1X5DN065-1.

2. The piping penetration area filtration and exhaust unit motors and their associated dampers and heaters are actuated upon:

   - Containment ventilation isolation signal (signal A and signal B).
   - Manual actuation.

   Upon automatic actuation of the piping penetration area filtration exhaust system, the piping penetration area is automatically isolated from the auxiliary building normal HVAC system.

B. Logic

The auxiliary building ESF HVAC system logic is shown in drawings 1X5DN030-1, 1X5DN030-3, 1X5DN030-4, 1X5DN030-5, and 1X5DN065-1. Logic is designed in such a manner that a momentary loss of the control power will not prevent or reverse the safety actuation of any equipment, and it is designed such that the reset of the safety injection signal will not trip the actuated equipment without deliberate subsequent operator action.

C. Bypass

Bypass of either subsystem comprising the auxiliary building ESF HVAC system is indicated in the control room. Such bypass may result from control power failure, system component failure, manual override at the component level, or transfer to local control. The bypass indication logic is shown in drawings 1X5DN030-1, 1X5DN030-3, 1X5DN030-4, 1X5DN030-5, and 1X5DN065-1. The manual override capability is provided only for the piping penetration area filtration and exhaust units.

D. Interlocks

There are no interlocks other than those shown in drawings 1X5DN030-1, 1X5DN030-3, 1X5DN030-4, 1X5DN030-5, and 1X5DN065-1.

E. Sequencing

The piping penetration area filtration and exhaust units are permitted to start on the containment ventilation isolation signal during the 15.5-s sequencer step for (1-s) or after sequencing is complete (after 30.5 s). All other loads are energized on the first step of load sequencing. The supporting ESF chiller compressor motors start automatically after sequencing is complete. Heaters may be manually loaded after sequencing is completed.

F. Redundancy

All equipment, instruments, and controls are fully redundant and arranged in two completely independent trains (A and B).

G. Seismic Qualification

All components comprising the auxiliary building ESF HVAC system are Seismic Category 1 and remain operational during and after a safe shutdown earthquake.

H. Diversity

Diversity of actuation is provided in that the auxiliary building ESF room coolers can be operated either manually from any one of the two physically separated locations (i.e., main control room and the shutdown panels) or automatically. The automatic actuation occurs upon either high room temperature or safety injection; this also enhances system diversity. The penetration area filtration and exhaust units can be actuated either manually from the control room or automatically.

I. Actuated Devices

Table 7.3.13-1 lists the actuated devices.

J. Supporting Systems

The following systems are required to be operational for proper functioning of the auxiliary building ESF HVAC system:

- Class 1E 480-V ac system.
- Class 1E 120-V ac system.
- Class 1E 125-V dc system.
- Essential chilled water system.

Under emergency conditions (safety injection or loss of offsite power) the Class IE electric power system remains operational, as described in section 8.3. The essential chilled water system is described in subsection 9.2.9. It is automatically actuated by the safety injection signal.

7.3.13.1.2 Design Bases

The design bases for the auxiliary building ESF HVAC system are such that no single failure within that system nor any supporting system can prevent it from performing its safety function. A detailed description of system design bases is provided in subsection 9.4.3.

7.3.13.1.3 Drawings

The drawings pertaining to the auxiliary building ESF HVAC system (including logic diagrams shown in drawings 1X5DN030-1, 1X5DN030-3, 1X5DN030-4, 1X5DN030-5, and 1X5DN065-1) are included in the references in section 1.7.

7.3.13.2 Analysis

The analysis presented in subsection 7.3.1 for the ESF pertains also to the systems discussed herein. The failure modes and effects analysis of the auxiliary building ESF HVAC system is given in table 9.4.3-3.

7.3.13.3 Summary

The summary is covered in paragraph 7.3.1.

7.3.14 AUXILIARY FEEDWATER PUMPHOUSE ENGINEERED SAFETY FEATURES HEATING, VENTILATION, AND AIR-CONDITIONING SYSTEM

7.3.14.1 Description

The auxiliary feedwater pumphouse engineered safety features (ESF) heating, ventilation, and air-conditioning (HVAC) system provides a suitable environment for equipment and maintenance personnel within the auxiliary feedwater pump rooms. It consists of one wall-mounted air supply fan and damper in each of the two motor-driven auxiliary feedwater pump rooms and two dampers facilitating natural convection in the turbine-driven auxiliary feedwater pump room. A detailed description of this system is given in subsection 9.4.8.

7.3.14.1.1 System Description

A. Actuating Circuits

1. The motor-driven auxiliary feedwater pump room air supply fans and dampers are actuated upon:
   - Room temperature high signal.
   - Manual actuation.
2. The turbine-driven auxiliary feedwater pump room dampers are actuated upon:
   - Turbine-driven auxiliary feedwater pump automatic start signal.
   - Manual actuation.

B. Logic

The auxiliary feedwater pumphouse ESF HVAC system logic is shown in drawings 1X5DN068-1 and 1X5DN068-3. Logic is designed so that a momentary loss of control power can not prevent or reverse the safety actuation of any equipment.

C. Bypass

Bypassed/inoperable status for the motor-driven auxiliary feedwater pump room air supply fans and dampers is indicated in the control room.

The manual override capability is provided only for the motor-driven auxiliary feedwater pump room air supply fans.

D. Interlocks

There are no interlocks other than those shown in drawings 1X5DN068-1 and 1X5DN068-3.

E. Sequencing

All equipment comprising the auxiliary feedwater pumphouse ESF HVAC system is energized on the first (0.5-s) sequencer step.

F. Redundancy

There is no redundancy in the auxiliary feedwater pumphouse ESF HVAC system.

G. Seismic Qualification

All components comprising the auxiliary feedwater pumphouse ESF HVAC system are Seismic Category 1 and remain functional during and after a safe shutdown earthquake.

H. Diversity

Diversity of actuation is provided in that the equipment can be operated either automatically or manually from any one of the two physically separated locations.

I. Actuated Devices

Table 7.3.14-1 lists the actuated devices.

J. Supporting Systems

The following systems are required to be operational for proper functioning of the auxiliary feedwater pumphouse ESF HVAC system:

- Class 1E 480-V ac system.
- Class 1E 125-V dc system.

These systems remain operational under emergency conditions. (See section 8.3.)

7.3.14.1.2 Design Bases

The design bases for the auxiliary feedwater pumphouse ESF HVAC system are outlined in subsection 9.4.8.

7.3.14.1.3 Drawings

The drawings pertaining to the auxiliary feedwater pumphouse ESF HVAC system (including logic diagrams shown in drawings 1X5DN068-1 and 1X5DN068-3) are included in the references in section 1.7.

7.3.14.2 Analysis

The analysis presented in subsection 7.3.1 generally applies. Although the auxiliary feedwater pumphouse ESF HVAC system is not redundant, its malfunctioning poses no threat to safety functions of the auxiliary feedwater system, due to the excessive redundancy of the latter. A loss of either of the three auxiliary feedwater pumps that might potentially result from the malfunction of its respective ventilation equipment does not impair the auxiliary feedwater supply function. (See subsection 10.4.9.) The failure modes and effects analysis of the auxiliary feedwater pumphouse ESF HVAC system is given in table 9.4.8-2.

7.3.14.3 Summary

The summary covered in subsection 7.3.1 generally applies. All specific features of the auxiliary feedwater pumphouse ESF HVAC system are discussed above and in subsection 9.4.8.

7.3.15 DIESEL GENERATOR BUILDING ENGINEERED SAFETY FEATURES HEATING, VENTILATION, AND AIR-CONDITIONING SYSTEM

7.3.15.1 Description

The diesel generator building engineered safety features (ESF) heating, ventilation, and air-conditioning (HVAC) system is designed to remove the heat added to the building atmosphere by operating diesel generators, their associated equipment, and solar load. The system is comprised of two identical and completely independent trains, each serving one diesel generator.

Each such subsystem includes two 50-percent capacity ESF fan units connected in parallel to common ductwork. For a detailed description of the diesel generator building ESF HVAC system, see subsection 9.4.7.

7.3.15.1.1 System Description

A. Actuating Circuits

As noted above, each of the two diesel generator building ESF HVAC system trains includes two ESF fan units. The first ESF fan unit is actuated upon:

- Diesel generator running signal.
- Manual actuation.

The second (standby) unit is actuated upon:

- Diesel generator running and room temperature high signal.
- Manual actuation.

Following its actuation, the system maintains the air temperature within the recommended range by modulating appropriate dampers. For details see drawings 1X5DN058-1, 1X5DN058-3, 1X5DN058-4, and 1X5DN058-5.

B. Logic

The diesel generator building ESF HVAC system logic is shown in drawings 1X5DN058-1, 1X5DN058-3, 1X5DN058-4, and 1X5DN058-5. Logic is designed in such a manner that a momentary loss of the control power does not prevent or reverse the safety actuation of any equipment. Once actuated, the system operates until the actuating signals disappear. The first ESF fan is stopped manually. The second (standby) ESF fan stops when room temperature drops below setpoint.

The dampers then automatically return to normal position. Should the diesel generator restart, the ESF HVAC system actuates again without operator intervention.

C. Bypass

Bypass of the diesel generator building ESF HVAC system is indicated in the control room. Such bypass may result from either system control power loss, fan motor breaker inoperable position, component failure, transfer to local control, or manual override of the fan motor actuating signal. The bypass indication logic is shown in drawings 1X5DN058-1, 1X5DN058-3, 1X5DN058-4, and 1X5DN058-5.

D. Interlocks

There are no interlocks other than those shown in drawings 1X5DN058-1, 1X5DN058-3, 1X5DN058-4, and 1X5DN058-5.

E. Sequencing

The ESF fan motors may be energized after the 30.5-s sequencer step based on process requirements. All other loads are energized at the first of load sequencing.

F. Redundancy

The diesel generator building ESF HVAC system is not redundant, since each of its trains (A and B) serves the corresponding diesel generator. Nevertheless, such arrangement preserves redundancy at the diesel generator system level.

G. Seismic Qualification

All components of the diesel generator ESF HVAC system are Seismic Category 1 and remain operational during and after a safe shutdown earthquake.

H. Diversity

Diversity of actuation is provided in that the diesel generator building ESF HVAC system can be operated either manually from any one of the two physically separated locations (i.e., main control room and the shutdown panels) or automatically with each of the two fans in either train being actuated by different signals. (See item A.)

I. Actuated Devices

Table 7.3.15-1 lists the actuated devices.

J. Supporting Systems

The following systems are required to be operational for proper functioning of the diesel generator building ESF HVAC system:

- Class 1E 480-V ac system.
- Class 1E 120-V ac system.
- Class 1E 125-V dc system.

As described in chapter 8, all systems remain operational under emergency conditions.

7.3.15.1.2 Design Bases

The design bases for the diesel generator building ESF HVAC system are given in subsection 9.4.7.

7.3.15.1.3 Drawings

The drawings pertaining to the diesel generator building ESF HVAC system (including logic diagrams shown in drawings 1X5DN058-1, 1X5DN058-3, 1X5DN058-4, and 1X5DN058-5) are included in references in section 1.7.

7.3.15.2 Analysis

The analysis presented in subsection 7.3.1 generally applies. Although the diesel generator building ESF HVAC system is not redundant, its malfunction does not impair the performance and redundancy at the diesel generator system level. (See item F.) The failure modes and effects analysis of the diesel generator building ESF HVAC system is given in table 9.4.7-2.

7.3.15.3 Summary

The summary covered in subsection 7.3.1 generally applies. All specific features of the diesel generator building ESF HVAC system are discussed above and in subsection 9.4.7.

7.3.16 ELECTRICAL TUNNEL ENGINEERED SAFETY FEATURES HEATING, VENTILATION, AND AIR-CONDITIONING SYSTEM

7.3.16.1 Description

The safety function of the electrical tunnel engineered safety features (ESF) heating, ventilation, and air-conditioning (HVAC) system is to provide adequate environment for the Class 1E cables routed through electrical tunnels. The tunnels serviced by this system are as follows:

- Two diesel power cable tunnels (trains A and B).
- Two nuclear service cooling water (NSCW) tower cable tunnels (trains A and B).
- Turbine building and auxiliary building train A tunnel. (The corresponding train B tunnel is ventilated by convection only.)

Each of the above tunnels has a single fan unit. The electrical tunnel ESF HVAC system is described in detail in subsection 9.4.9.

7.3.16.1.1 System Description

A. Actuating Circuits

The fan motor in every fan unit is actuated upon:

1. Tunnel temperature high signal, with the exception of fans 1-1540-B7-005-000 and 2-1540-B7-005-000, the turbine building to auxiliary building train A tunnel ventilation fans. As described in subsection 9.4.9, this fan does not automatically start on high temperature.
2. Manual actuation.

For details see drawing 1X5DN069-1.

B. Logic

The electrical tunnel ESF HVAC system logic is shown in drawing 1X5DN069-1. The logic is designed in such a manner that a momentary loss of the control power does not prevent or reverse the safety actuation of any equipment.

C. Bypass

Bypass of either fan comprising the electrical tunnel ESF HVAC system is indicated in the control room. Such bypass may result from control power failure, system component failure, or manual override at the component level. Bypass indication for manual override of the turbine building to auxiliary building train A electrical tunnel HVAC system is not indicated in the control room. As described in subsection 9.4.9, the train A electrical tunnel ventilation fan is manually started. The manual override capability is provided on all five fan units.

D. Interlocks

There are no interlocks other than those shown in drawing 1X5DN069-1.

E. Sequencing

The electrical tunnel ESF HVAC fan motors are energized on the first (0.5-s) step of load sequencing, except for the train A tunnel fan which is manually actuated.

F. Redundancy

The electrical tunnel ESF HVAC system is comprised of five completely independent subsystems, each serving a different tunnel. Although these subsystems are not redundant, it does not impair the redundancy at the system level. (Each tunnel and its respective fan unit belong to the same train.)

G. Seismic Qualification

All components comparing the electrical tunnel ESF HVAC system are Seismic Category 1 and remain operational during and after a safe shutdown earthquake.

H. Diversity

Diversity of actuation is provided in that the electrical tunnel ESF fan units can be operated either manually or automatically with the exception of the turbine building to auxiliary building train A tunnel. The fan unit in that tunnel is started manually.

I. Actuated Devices

Table 7.3.16-1 lists the actuated devices.

J. Supporting Systems

The only supporting system that has to be operational for the electrical tunnel ESF HVAC system to function properly is the Class 1E 480-V ac power system. As described in chapter 8, this system remains operational under any postulated emergency conditions.

7.3.16.1.2 Design Bases

The design bases for the electrical tunnel ESF HVAC system are discussed in subsection 9.4.9.

7.3.16.1.3 Drawings

The drawings pertaining to the electrical tunnel ESF HVAC system (including logic diagrams shown in drawing 1X5DN069-1) are included in the references in section 1.7.

7.3.16.2 Analysis

The analysis presented in subsection 7.3.1 generally applies. As noted in item F above, redundancy of the vital diesel generator power system is not compromised by the lack of redundancy in the electrical tunnel ESF HVAC system. The failure modes and effects analysis of the latter system is given in table 9.4.9-3.

7.3.16.3 Summary

The summary covered in paragraph 7.3.1 generally applies. All specific features of the electrical tunnel ESF HVAC system are discussed above and in subsection 9.4.9.

7.3.17 DIESEL GENERATOR FUEL OIL SYSTEM

7.3.17.1 Description

A separate diesel generator fuel oil system provides sufficient and independent fuel oil supply for each diesel generator engine under all conditions and plant operating modes. Each such system consists of a diesel fuel oil storage tank, a diesel fuel oil day tank, two diesel fuel oil storage tank pumps, an engine-driven fuel oil pump, associated pipes, valves, filters, instrumentation, and controls. The diesel generator fuel oil system is described in detail in subsection 9.5.4.

7.3.17.1.1 System Description

A. Actuating Circuits

Each diesel generator has two 100-percent capacity diesel fuel oil storage tank pumps operating alternately for greater reliability. Each pump is actuated upon:

1. Day tank fuel oil level low signal.
2. Low pressure at the discharge of the other operating pump.
3. Manual actuation.

The details of the actuation logic are drawn in drawing 1X5DN107-1.

B. Logic

The diesel generator fuel oil system logic is shown in drawing 1X5DN107-1. The logic is designed in such a manner that a momentary loss of the control power does not prevent or reverse the actuation of the pumps.

C. Bypass

Bypass of each diesel fuel oil storage pump is indicated in the control room. Such bypass may result from either control power failure, system component failure, or manual override of the pump automatic actuation capability. The bypass indication logic is shown in drawing 1X5DN107-1.

D. Interlocks

There are no interlocks other than those shown in drawing 1X5DN107-1.

E. Sequencing

The diesel generator fuel oil storage pumps are energized on the first (0.5 s) step of load sequencing.

F. Redundancy

The fuel oil system for each diesel generator is partially redundant. The fuel oil storage pumps are fully redundant, as there are two of these for each diesel generator. All other portions of each diesel generator fuel oil system are not redundant. This, however, does not impair the redundancy of the diesel generator onsite power system, which consists of two completely independent and redundant trains A and B for each nuclear power generating unit.

G. Seismic Qualification

All components comprising the diesel generator fuel oil system are Seismic Category 1 and remain operational during and after a safe shutdown earthquake.

H. Diversity

Diversity of actuation is provided in that the diesel generator fuel oil system can be operated either automatically or manually.

I. Actuated Device

Actuated devices are listed in table 7.3.17-1.

J. Supporting Systems

The following systems are required to be operational for the diesel generator fuel oil system to function properly:

1. Class 1E 480-V ac system.
2. Class 1E 120-V ac system.

As described in chapter 8, all Class 1E power systems remain operational under emergency conditions.

7.3.17.1.2 Design Bases

The design bases for the diesel generator fuel oil system are given in subsection 9.5.4.

7.3.17.1.3 Drawings

The drawings pertaining to the diesel generator fuel oil system (including logic diagrams drawn in drawing 1X5DN107-1) are included in the references in section 1.7.

7.3.17.2 Analysis

The failure mode and effects analysis of the diesel generator fuel oil system is given in table 9.5.4-2.

7.3.17.3 Summary

The summary covered in paragraph 7.3.1 generally applies. All specifics of the diesel generator fuel oil system are discussed above in subsection 9.5.4.

TABLE 7.3.1-1

INSTRUMENTATION OPERATING CONDITIONS FOR ENGINEERED SAFETY FEATURES

No.	Functional Unit	No. of Channels	No. of Channels to Trip
1	Safety injection		
	Manual	2	1
	High containment pressure (high-1)	3	2
	Low steam line pressure lead-lag compensated	12 (3 per steam line)	2 in any one steam line
	Pressurizer low pressure [(a)]	4	2
2	Containment spray		
	Manual[(b)]	4	2
	Containment pressure (high-3)	4	2

a. Permissible bypass if reactor coolant pressure is less than 2000 psig.

b. Manual actuation of containment spray system requires the simultaneous operation of two separate switches. The requirements for the simultaneous operation of two switches is desirable to prevent inadvertent spray actuation.

TABLE 7.3.1-2 (SHEET 1 OF 2)

INSTRUMENTATION OPERATING CONDITIONS FOR ISOLATION FUNCTIONS

No.	No. of Functional Unit	No. of Channels	No. of Channels to Trip
1	Containment isolation		
	Automatic safety injection (phase A)	See items 1b through 1d of table 7.3.1-1	
	Manual (phase A)	2	1
2	Steam line isolation		
	High steam line negative pressure rate	12 (3 per steam line)	2 per steam line in any steam line
	Containment pressure (high-2)	3	2
	Low steam line pressure	12 (3 per steam line)	2 per steam line in any steam line
	Manual[(a)]	2 per steam line	1 per steam line

TABLE 7.3.1-2 (SHEET 2 OF 2)

No.	Functional Unit	No. of Channels	No. of Channels to Trip
3	Feedwater line isolation		
	Safety injection	See items 1a through 1d of table 7.3.1-1	
	Steam generator high-high level 2/4 on any steam generator	16 (4 per steam generator)	2 per steam generator
	Reactor trip coincident with low T_{avg}	2 (reactor trip) 4 (low T_{avg})	1 2
	Manual	1 per feedwater line	1 per feedwater line

a. Two tandem switches (one for train A and one for train B) will simultaneously close all main steam line and main steam bypass isolation valves at the system level.

TABLE 7.3.1-3 (SHEET 1 OF 2)

INTERLOCKS FOR ENGINEERED SAFETY FEATURES ACTUATION SYSTEM

Designation	Input	Function Performed
P-4	Reactor trip	Actuates turbine trip.
		Closes main and bypass feedwater valves on low T_{avg} below set point.
		Prevents opening of main and bypass feedwater valves which were closed by safety injection or high-high steam generator water level.
		Allows manual block of the automatic reactuation of safety injection.
		Transfers steam dump control from the load rejection controller to the plant trip controller.
	Reactor not tripped	Defeats the block preventing automatic reactuation of safety injection.
P-11	2/3 pressurizer pressure below setpoint	Allows manual block of safety injection actuation on low pressurizer pressure signal.
		Allows manual block of safety injection actuation and steam line isolation on low compensated steam line pressure signal and allows steam line isolation on high steam line negative pressure rate.

TABLE 7.3.1-3 (SHEET 2 OF 2)

Designation	Input	Function Performed
	2/3 pressurizer pressure above setpoint	Defeats manual block of safety injection actuation on low pressurizer pressure and safety injection and steam line isolation on low steam line pressure and defeats steam line isolation on high steam line negative pressure rate.
P-12[a]	2/4 low-low T_{avg} below setpoint	Blocks steam during except cooldown condenser dump valves. Allows manual bypass of steam dump block for the cooldown valves only.
	2/4 low-low T_{avg} above setpoint	Defeats the manual bypass of steam dump block.
P-14	2/4 steam generator high-high water level above setpoint on any steam generator	Closes all feedwater regulating valves and isolation valves. Trips all main feedwater pumps which close the pump discharge valves. Actuates turbine trip.

a. ESF interlock not applicable.

TABLE 7.3.1-4

PRIMARY SYSTEM ACCIDENTS AND REQUIRED INSTRUMENTATION
RUPTURES IN SMALL PIPES, CRACKS IN LARGE PIPES,
RUPTURES OF LARGE PIPES, AND STEAM GENERATOR TUBE RUPTURE

Channel	Response Time(s)[a]	Accuracy[a]	Range
Pressurizer pressure	(c)	±1.75 percent of span	1700 to 2500 psig
Containment pressure[b]	(d)	±1.75 percent of span	0 to 115 percent of containment design pressure

a. See section 7.1 for definitions of engineered safety features actuation system response time and accuracy.

b. Not required for steam generator tube rupture.

c. Total time from step change in pressurizer pressure until start of safety injection pumps is 27 s with offsite power available and 40 s with offsite power unavailable.

d. Total time from step change in containment pressure until full containment spray is obtained is 94 s (includes 29 s for diesel start and sequencing and 65 s for filling the spray header).

TABLE 7.3.1-5

SECONDARY SYSTEM ACCIDENTS AND REQUIRED INSTRUMENTATION,
MINOR SECONDARY SYSTEM PIPE BREAK, AND
MAJOR SECONDARY SYSTEM PIPE BREAK

Item	Channel	Response Time(s)[a]	Accuracy [a]	Range
1	Containment pressure[b]	(b)	±1.75 percent of full scale	0 to 115 percent of containment design pressure
2	Steam line pressure	10.0[c]	±2.25 percent of span	0 to 1300 psig
3	Steam line pressure rate	See item 2 for sensor characteristics		
4	T_{avg}	N/A	±2°F	530 to 630°F
5	Pressurizer pressure	(d)	±1.75 percent of span	1700 to 2500 psig

a. See section 7.1 for definitions of engineered safety features actuation system response time and accuracy.

b. Total time from step change in containment pressure until full containment spray is obtained is 94 s (includes 29 s for diesel start and sequencing and 65 s for filling the spray header).

c. Total time from step change in steam pressure until steam line isolation valves are fully closed.

d. Total time from step change in pressurizer pressure until start of safety injection pumps is 27 s with offsite power available and 40 s with offsite power unavailable.

TABLE 7.3.1-6 (SHEET 1 OF 6)

ENGINEERED SAFETY FEATURES RESPONSE ITEMS

INITIATION SIGNAL AND FUNCTION			RESPONSE TIME IN SECONDS
1.	Manual Initiation		
	a.	Safety Injection (ECCS)	N/A
		Feedwater Isolation	N/A
		Component Cooling Water	N/A
		Containment Cooling Fans	N/A
		Nuclear Service Cooling Water	N/A
		Containment Ventilation Isolation	N/A
	b.	Containment Spray	N/A
	c.	Phase "A" Isolation	N/A
	d.	Auxiliary Feedwater	N/A
	e.	Steam Line Isolation	N/A
	f.	Control Room Ventilation Emergency Mode Actuation	N/A
	g.	Reactor Trip	N/A
	h.	Start Diesel Generators	N/A
2.	Containment Pressure--High-1		
	a.	Safety Injection (ECCS)	$\leq 39^{(1)}/27^{(5)}$
	b.	Reactor Trip (from SI)	≤ 2
	c.	Feedwater Isolation	≤ 7
	d.	Phase "A" Isolation	$\leq 2^{(6)}$

TABLE 7.3.1-6 (SHEET 2 OF 6)

INITIATION SIGNAL AND FUNCTION			RESPONSE TIME IN SECONDS
	e.	Containment Ventilation Isolation	≤ 1.5[6]
	f.	Auxiliary Feedwater	≤ 60
	g.	Nuclear Service and Component Cooling Water	≤ 100[1]/88.5[2]
	h.	Containment Cooling Fans	≤ 48[1]/36.5[2]
	i.	Control Room Ventilation Emergency Mode Actuation	69.3[12]/99.3[13]
	j.	Start Diesel Generators	≤ 13.5[7]
3.	Pressurizer Pressure--Low		
	a.	Safety Injection (ECCS)	≤ 39[1]/27[5]
	b.	Reactor Trip (from SI)	≤ 2
	c.	Feedwater Isolation	≤ 7
	d.	Phase "A" Isolation	≤ 2[6]
	e.	Containment Ventilation Isolation	≤ 1.5[6]
	f.	Auxiliary Feedwater	≤ 60
	g.	Nuclear Service and Component Cooling Water	≤ 100[1]/88.5[2]
	h.	Containment Cooling Fans	≤ 48[1]/36.5[2]
	i.	Control Room Ventilation Emergency Mode Actuation	69.3[12]/99.3[13]
	j.	Start Diesel Generators	≤ 13.5[7]
4.	Steam Line Pressure--Low		
	a.	Safety Injection (ECCS)	≤ 39[1]/27[5]

TABLE 7.3.1-6 (SHEET 3 OF 6)

INITIATION SIGNAL AND FUNCTION			RESPONSE TIME IN SECONDS
	b.	Reactor Trip (from SI)	≤ 2
	c.	Feedwater Isolation	≤ 7
	d.	Phase "A" Isolation	≤ 2[6]
	e.	Containment Ventilation Isolation	≤ 1.5[6]
	f.	Auxiliary Feedwater	≤ 60
	g.	Nuclear Service and Component Cooling Water	≤ 100[1]/88.5[2]
	h.	Containment Cooling Fans	≤ 48[1]/36.5[2]
	i.	Control Room Ventilation Emergency Mode Actuation	69.3[12]/99.3[13]
	j.	Start Diesel Generators	≤ 13.5[7]
	k.	Steam Line Isolation	≤ 10[3]
5.	Containment Pressure--High-3		
	Containment Spray		≤ 82.5[2]/94[1]
6.	Containment Pressure--High-2		
	Steam Line Isolation		≤ 10[3]
7.	Steam Line Pressure — Negative Rate—High		
	Steam Line Isolation		≤ 10[3]
8.	Steam Generator Water Level--High-High		
	a.	Turbine Trip	N/A
	b.	Feedwater Isolation	≤ 7

TABLE 7.3.1-6 (SHEET 4 OF 6)

INITIATION SIGNAL AND FUNCTION	RESPONSE TIME IN SECONDS
9. Steam Generator Water Level--Low-Low	
a. Motor-Driven Auxiliary Feedwater Pumps	≤ 60
b. Turbine-Driven Auxiliary Feedwater Pump	≤ 60
10. Loss of or Degraded 4.16 kV ESF Bus Voltage	
Auxiliary Feedwater	≤ 60(14)
11. Trip of All Main Feedwater Pumps	
Motor Driven Pumps	N/A
Auxiliary Feedwater	N/A
12. RWST Level—Low-Low Coincident with Safety Injection	N/A
a. Semi-Automatic Switchover to Containment Emergency Sump	≤ 60
13. Loss of Power	
a. 4.16 kV ESF Bus Undervoltage- Loss of Voltage; Start Signal to Diesel Generator	≤ 2.0(9)
b. 4.16 kV ESF Bus Undervoltage – Grid Degraded Voltage; Start Signal to Diesel Generator	≤ 21.2(10)
14. Control Room Intake Radiogas	
Control Room Ventilation Emergency Mode Actuation	≤ 72.0(15)/102.0(16)
15. Containment Radioactivity	
a. Area Radiation Low Range- Containment Ventilation Isolation	≤ 5(8, 11)
b. Containment Ventilation Radiation- Containment Ventilation Isolation	≤ 5(8, 11)

TABLE 7.3.1-6 (SHEET 5 OF 6)

INITIATION SIGNAL AND FUNCTION	RESPONSE TIME IN SECONDS
16. Fuel Handling Building Exhaust Duct Radiation	
a. Fuel Handling Building Post Accident Ventilation Actuation	N/A

TABLE 7.3.1-6 (SHEET 6 OF 6)

TABLE NOTATIONS

(1) Time to full ECCS flow. Signal sensing, diesel generator starting, and sequencer loading delays included.

(2) Diesel generator starting delay not included. Offsite power available.

(3) Electrohydraulic valves.

(4) Deleted.

(5) Time to full ECCS flow. Diesel generator starting delay not included.

(6) Does not include valve closure time.

(7) Signal sensing, diesel generator starting and diesel generator breaker delay included.

(8) Does not include valve closure time and relates to post-accident radiation sources as specified in FSAR subsection 15.7.4.

(9) The response time shall include the time delay associated with the loss of voltage relays plus the delay associated with operation of the respective SF sequencer output relays.

(10) The response time shall include the time delay associated with the undervoltage relays plus the delay associated with operation of the respective SF sequencer output relays.

(11) Radiation detectors time response not included.

(12) Signal sensing, sequencer loading, and flow establishment delays included for the train B lead filter unit.

(13) Signal sensing, sequencer loading, train B lead fan failure, and flow establishment delays included for the train A lag filter unit.

(14) For loss of voltage, the response time begins when the loss of voltage trip setpoint of Surveillance Requirement (SR) 3.3.5.2.A has been exceeded. For degraded voltage, the response time begins when the degraded voltage trip setpoint of SR 3.3.5.2.B has been exceeded continuously for the time delay specified in SR 3.3.5.2.B.

(15) Signal sensing and flow establishment delays included for the train B lead filter unit. Response time criteria permit detection of degradation; however, analysis of record allows up to 138 s.

(16) Signal sensing, train B lead fan failure and flow establishment delays included for the train A lag filter unit. Response time permits detection of degradation; however, analysis of record allows up to 138 s.

TABLE 7.3.1-7 (SHEET 1 OF 2)

ENGINEERED SAFETY FEATURES ALLOCATION TIMES

Function	Sensor	Time	7300/NIS String	Time	SSPS Relays	Time
CNMT PRESS HI-1	Barton 764/351	1.0 s	NLP+NAL	65 ms	Input+Master+Slave	88 ms
CNMT PRESS HI-2	Barton 764/351	1.0 s	NLP+NAL	65 ms	Input+Master+Slave	88 ms
CNMT PRESS HI-3	Barton 764/351	1.0 s	NLP+NAL	65 ms	Input+Master+Slave	88 ms
STEAMLINE PRESS LO	Tobar 32PA	200 ms	NLP+NAL	65 ms	Input+Master+Slave	88 ms
	Veritrak 76PG	200 ms				
	Rosemount 1154SH9	200 ms				
	Rosemount 1153GB9	200 ms				
STEAMLINE HI NEG RATE	Tobar 32PA	200 ms	NLP+NAL	65 ms	Input+Master+Slave	88 ms
	Veritrak 76PG	200 ms				
	Rosemount 1154SH9	200 ms				
	Rosemount 1153GB9	200 ms				
PZR PRESS LO SI	Tobar 32PG	200 ms	NLP+NAL	65 ms	Input+Master+Slave	88 ms
	Veritrak 76PH	200 ms				
	Rosemount 1154SH9	200 ms				
RWST LEVEL LO-LO	Tobar 32DP	400 ms	NLP+NAL	65 ms	Input+Master+Slave	88 ms
	Veritrak 76DP	400 ms				
	Rosemount 1153DB5	200 ms				
SG LEVEL LO-LO	Tobar 32DP	400 ms	NLP+NAL	65 ms	Input+Master+Slave	88 ms
	Veritrak 76DP	400 ms				
	Rosemount 1154DH5	200 ms				

TABLE 7.3.1-7 (SHEET 2 OF 2)

Function	Sensor	Time	7300/NIS String	Time	SSPS Relays	Time
SG LEVEL HI-HI	Tobar 32DP	400 ms	NLP+NAL	65 ms	Input+Master+Slave	88 ms
	Veritrak 76DP	400 ms				
	Rosemount 1154DH5	200 ms				
CNMT AREA RADIATION LEVEL HI	Westinghouse	(1)	N/A	N/A	Input+Master+Slave	88 ms
CNMT VENT RADIATION LEVEL HI	Westinghouse	(1)	N/A	N/A	Input+Master+Slave+Slave	124 ms

Note 1: Allocated sensor times are not used for these variables. These components will continue to be tested as required.

Allocated sensor times are derived from method (3), section (9), WCAP-13632, revision 2 (Vendor Engineering Specifications). Tobar, Veritrak, and Barton times were provided in Table 9-1. Rosemount times are from Rosemount manuals 4302 and 4631. The Rosemount response time specifications may also be found in NUREG/CR-5383. Transmitter FMEAs are based upon EPRI report NP-7243, revision 1.

Values for 7300 cards are from tables 4-7 through 4-12 of WCAP-14036, revision 1. Cards installed are 4NCH, 4NRA, 6NLP 4NSA, and 9NAL or older artwork levels.

SSPS input and master relays are Potter & Brumfield KH series relays. SSPS slave relays are Potter & Brumfield MDR relays. Values are tabulated from section 4.8, Westinghouse SSPS FMEA.

TABLE 7.3.3-1 (SHEET 1 OF 2)

CONTAINMENT COMBUSTIBLE GAS CONTROL SYSTEM
ACTUATED EQUIPMENT LIST[(a)]

Description	Actuating Train A	Actuating Train B
Post-LOCA purge exhaust inside valve	X	
Post-LOCA purge exhaust inside valve		X
Containment hydrogen monitor A supply inside valves	X	
Containment hydrogen monitor B supply inside valves		X
Containment hydrogen monitor A supply outside valve		X
Containment hydrogen monitor B supply outside valve	X	
Containment hydrogen monitor A supply return valve		X
Containment hydrogen monitor B supply return valve	X	
Containment cooler fan 1	X	
Containment cooler fan 2	X	
Containment cooler fan 3		X
Containment cooler fan 4		X
Containment cooler fan 5	X	
Containment cooler fan 6	X	
Containment cooler fan 7		X
Containment cooler fan 8		X
Containment hydrogen thermal recombiner 1	X	

TABLE 7.3.3-1 (SHEET 2 OF 2)

Description	Actuating Train A	Actuating Train B
Containment hydrogen thermal recombiner 2		X
Post-LOCA cavity purge system fan 1	X	
Post-LOCA cavity purge system fan 2		X

a. Additional details are provided on the electrical schematic diagrams and the control logic diagrams referenced in section 1.7.

TABLE 7.3.4-1

CONTAINMENT PURGE ISOLATION SYSTEM ACTUATED DEVICES[a]

Containment post-loss-of-coolant accident (post-LOCA) purge isolation valve, Train A, inside reactor containment (IRC)	HV-2624A
Containment post-LOCA purge isolation valve, Train B, IRC	HV-2624B
Containment preaccess purge supply valve, Train A, IRC	HV-2626A
Containment minipurge supply valve, Train A, IRC	HV-2626B
Containment preaccess purge supply valve, Train B, outside reactor containment (ORC)	HV-2627A
Containment minipurge supply valve, Train B, ORC	HV-2627B
Containment preaccess purge exhaust valve, Train A, IRC	HV-2628A
Containment minipurge exhaust valve, Train A, IRC	HV-2628B
Containment preaccess purge exhaust valve, Train B, ORC	HV-2629A
Containment minipurge exhaust valve, Train B, ORC	HV-2629B

a. Refer to appropriate logic diagrams for additional actuated devices.

TABLE 7.3.5-1

FUEL HANDLING BUILDING VENTILATION POST-ACCIDENT ACTUATED EQUIPMENT[a]

Description	Actuating Train
Emergency HVAC system fan motor A-1542-N7-001-M01	A
Emergency HVAC system fan motor A-1542-N7-002-M01	B
Emergency HVAC system inlet damper HV-12510	A
Emergency HVAC system inlet damper HV-12511	B
Emergency HVAC system discharge damper HV-12512	A
Emergency HVAC system discharge damper HV-12513	B
Normal HVAC system supply isolation HV-2528	B
Normal HVAC system supply isolation HV-2529	A
Normal HVAC system supply isolation HV-2534	B
Normal HVAC system supply isolation HV-2535	A
Normal HVAC system exhaust isolation HV-12479	A
Normal HVAC system exhaust isolation HV-12480	B
Normal HVAC system to equipment building isolation HV-12481	A
Normal HVAC system to equipment building isolation HV-12482	B

a. Refer to appropriate logic diagrams for additional actuated devices.

TABLE 7.3.6-1

CONTROL ROOM VENTILATION ISOLATION CONTROL SYSTEM
MONITOR SENSITIVITIES AND RESPONSE TIMES

Type	Concentration Setpoint for Isolation		Limiting Isotope	Response Time
	$\mu mCi/cm^3$	ppm		
Gaseous Radioactivity	$3x10^{-6}$	-	Kr 85	(a)
Smoke	-	-	-	Manual actuation

a. Response time is radiation-level dependent.

TABLE 7.3.6-2

CONTROL ROOM VENTILATION ISOLATION CONTROL SYSTEM
ACTUATED EQUIPMENT LIST[a]

Description	Actuation Channel or Train
Control room (CR) filter unit fan motor (1-1531-N7-001-M01)	A
CR filter unit fan motor (1-1531-N7-002-M01)	B
Emergency supply outlet damper HV-12128	A
Emergency supply outlet damper HV-12129	B
Emergency return fan motor (1-1531-B7-005-M01)[b]	A
Emergency return fan motor (1-1531-B7-006-M01)[b]	B
Emergency return air damper HV-12130	A
Emergency return air damper HV-12131	B
Outside air isolation damper HV-12114[c]	A
Outside air isolation damper HV-12115[c]	B
Normal CR A/C unit fan motor A-1531-A7-001-M01	Nontrain
Normal CR A/C unit fan motor A-1531-A7-002-M01	Nontrain
Normal CR A/C unit inlet damper HV-12143	Nontrain
Normal CR A/C unit inlet damper HV-12144	Nontrain
Normal CR A/C unit discharge damper 1HV-12146	A
Normal CR A/C unit discharge damper 1HV-12147	B
Normal CR A/C unit return air 1HV-12148	B
Normal CR A/C unit return air 1HV-12149	A
CR kitchen, toilet, and conference room exhaust fan motor A-1531-B7-008-M01	Nontrain
CR kitchen, toilet, etc., fan inlet damper HV-12162	A
CR kitchen, toilet, etc., fan inlet damper HV-12163	B
Normal CR A/C unit discharge damper 2HV-12146	A
Normal CR A/C unit discharge damper 2HV-12147	B
Normal CR A/C unit return damper 2HV-12148	B
Normal CR A/C unit return damper 2HV-12149	A
Normal CR A/C return and exhaust fan motor A-1531-B7-009-M01	Nontrain
Normal CR A/C return and exhaust fan motor A-1531-B7-010-M01	Nontrain

a. Refer to appropriate logic diagrams for additional actuated devices.
b. Return air fans are disabled and abandoned in place as their function is not required.
c. Manual actuation only.

TABLE 7.3.8-1 (SHEET 1 OF 2)

MAIN STEAM, MAIN STEAM BYPASS, AND MAIN FEEDWATER AND FEEDWATER BYPASS ISOLATION ACTUATED DEVICES[(a)]

	Train
Main steam isolation valve HV-3006A SG 001	A
Main steam isolation valve bypass isolation valve HV-13005A SG 001	A
Main steam isolation valve HV-3006B SG 001	B
Main steam isolation valve bypass isolation valve HV-13005B SG 001	B
Main steam isolation valve HV-3016A SG 002	A
Main steam isolation valve bypass isolation valve HV-13007A SG 002	A
Main steam isolation valve HV-3016B SG 002	B
Main steam isolation valve bypass isolation valve HV-13007B SG 002	B
Main steam isolation valve HV-3026A SG 003	A
Main steam isolation valve bypass isolation valve HV-13008A SG 003	A
Main steam isolation valve HV-3026B SG 003	B
Main steam isolation valve bypass isolation valve HV-13008B SG 003	B
Main steam isolation valve HV-3036A SG 004	A
Main steam isolation valve bypass isolation valve HV-13006A SG 004	A
Main steam isolation valve HV-3036B SG 004	B
Main steam isolation valve bypass isolation valve HV-13006B SG 004	B
Main feedwater isolation valve HV-5227 SG 001	A and B
Main feedwater isolation valve HV-5228 SG 002	A and B

TABLE 7.3.8-1 (SHEET 2 OF 2)

	Train
Main feedwater isolation valve HV-5229 SG 003	A and B
Main feedwater isolation valve HV-5230 SG 004	A and B
Main feedwater isolation bypass valve HV-15196 SG 001	A and B
Main feedwater isolation bypass valve HV-15197 SG 002	A and B
Main feedwater isolation bypass valve HV-15198 SG 003	A and B
Main feedwater isolation bypass valve HV-15199 SG 003	A and B
Main feedwater regulating valve FV-0510 SG 001	A and B
Main feedwater regulating valve FV-0520 SG 002	A and B
Main feedwater regulating valve FV-0530 SG 003	A and B
Main feedwater regulating valve FV-0540 SG 004	A and B
Bypass feedwater regulating valve LV-5243 SG 001	A and B
Bypass feedwater regulating valve LV-5244 SG 002	A and B
Bypass feedwater regulating valve LV-5245 SG 003	A and B
Bypass feedwater regulating valve LV-5242 SG 004	A and B

a. Refer to appropriate logic diagrams for additional actuated devices.

TABLE 7.3.9-1

NSCW ACTUATED DEVICES[a]

		Train
1.	NSCW pump 1-1202-P4-001-M01	A
2.	NSCW pump 1-1202-P4-002-M01	B
3.	NSCW pump 1-1202-P4-003-M01	A
4.	NSCW pump 1-1202-P4-004-M01	B
5.	NSCW pump 1-1202-P4-005-M01	A
6.	NSCW pump 1-1202-P4-006-M01	B
7.	NSCW pump 1-1202-P4-007-M01	B
8.	NSCW pump 1-1202-P4-008-M01	A
9.	NSCW fan 1-1202-W4-001-F01	A
10.	NSCW fan 1-1202-W4-002-F01	B
11.	NSCW fan 1-1202-W4-001-F02	A
12.	NSCW fan 1-1202-W4-002-F02	B
13.	NSCW fan 1-1202-W4-001-F03	A
14.	NSCW fan 1-1202-W4-002-F03	B
15.	NSCW fan 1-1202-W4-001-F04	A
16.	NSCW fan 1-1202-W4-002-F04	B

a. Refer to appropriate logic diagrams for additional actuated devices.

TABLE 7.3.10-1

COMPONENT COOLING WATER SYSTEM
ACTUATED DEVICES[a]

	Train
Component cooling water pump 1-1203-04-001	A
Component cooling water pump 1-1203-04-003	A
Component cooling water pump 1-1203-04-005	A
Component cooling water pump 1-1203-04-002	B
Component cooling water pump 1-1203-04-004	B
Component cooling water pump 1-1203-04-006	B

a. Refer to appropriate logic diagrams for additional actuated devices.

TABLE 7.3.11-1

CONTAINMENT HEAT REMOVAL SYSTEM ACTUATED DEVICES[a]

Component	Train
Containment building cooling unit fan 1-1501-A7-001-M001	A
Containment building cooling unit fan 1-1501-A7-002-M01	A
Containment building cooling unit fan 1-1501-A7-003-M01	B
Containment building cooling unit fan 1-1501-A7-004-M01	B
Containment building cooling unit fan 1-1501-A7-005-M01	A
Containment building cooling unit fan 1-1501-A7-006-M01	A
Containment building cooling unit fan 1-1501-A7-007-M01	B
Containment building cooling unit fan 1-1501-A7-008-M01	B

a. Refer to appropriate logic diagrams for additional actuated devices.

TABLE 7.3.12-1 (SHEET 1 OF 2)

CONTROL BUILDING ESF HVAC SYSTEM
ACTUATED EQUIPMENT LIST[a]

Description	Actuation Channel or Train
Control building safety feature electrical equipment ac unit fan motor (1-1532-A7-001-M01)	A
Control building safety feature electrical equipment ac unit fan motor (1-1532-A7-002-M01	B
Control building safety feature battery room exhaust fan motor (1-1532-B7-001-M01)	A
Control building safety feature battery room exhaust fan motor (1-1532-B7-003-M01)	A
Control building safety feature battery room exhaust fan discharge dampers (HV-12742)	A
Control building safety feature battery room exhaust fan discharge damper (HV-12748)	A
Control building safety feature battery room exhaust fan motor (1-1532-B7-002-M01)	B
Control building safety feature battery room exhaust fan motor (1-1532-B7-004-M01)	B
Control building safety feature battery room exhaust fan discharge damper (HV-12727)	B
Control building safety feature battery room exhaust fan discharge damper (HV-12749)	B

TABLE 7.3.12-1 (SHEET 2 OF 2)

Description	Actuation Channel or Train
Control building control room ESF chiller room exhaust fan motor (1-1531-B7-002-M01)	A
Control building control room ESF chiller room exhaust fan motor (1-1531-B7-004-M01)	B
Control building electrical penetration filter unit heater (1-1562-N7-001-H01)	A
Control building electrical penetration filter unit heater (1-1562-N7-002-H01)	B
Control building auxiliary relay room ESF air-conditioning unit fan motor (1-1539-A7-001-M01)	A
Control building auxiliary relay room ESF air-conditioning unit fan motor (1-1539-A7-002-M01)	B

a. Refer to appropriate logic diagrams for additional actuated devices.

TABLE 7.3.13-1 (SHEET 1 OF 2)

AUXILIARY BUILDING ESF HVAC SYSTEM
ACTUATED EQUIPMENT LIST[(a)]

Description	Actuation Channel or Train
Electrical switchgear and MCC room cooler motors:	
1-1555-A7-001-M01	A
1-1555-A7-002-M01	B
1-1555-A7-003-M01	A
1-1555-A7-004-M01	B
1-1555-A7-005-M01	A
1-1555-A7-006-M01	B
Residual heat removal pump room cooler motors:	
1-1555-A7-007-M01	A
1-1555-A7-008-M01	B
Containment spray pump room cooler motors:	
1-1555-A7-009-M01	A
1-1555-A7-010-M01	B
Component cooling water pump room cooler motors:	
1-1555-A7-011-M01	A
1-1555-A7-012-M01	B
Charging pump room cooler motors:	
1-1555-A7-013-M01	A
1-1555-A7-014-M01	B
Safety injection pump room cooler reactors:	
1-1555-A7-015-M01	A
1-1555-A7-016-M01	B
Spent fuel pool heat exchanger and pump room cooler motors:	

TABLE 7.3.13-1 (SHEET 2 OF 2)

Description	Actuation Channel or Train
1-1555-A7-017-M01	A
1-1555-A7-018-M01	B
Piping penetration room filtration and exhaust unit motors:	
1-1561-N7-001-M01	A
1-1561-N7-002-M01	B
Piping penetration room filtration and exhaust unit heaters:	
1-1561-N7-001-H01	A
1-1561-N7-002-H01	B
Piping penetration room filtration and exhaust unit dampers:	
HV-12614 and PV-2550A and PV-2550B	A
HV-12616 and PV-2551A and PV-2551B	B
Piping penetration area isolation dampers:	
HV-12605 (controlled by SO-V HY-12605)	A
HV-12606 (controlled by SO-V HY-12606)	B
HV-12604 (controlled by SO-V HY-12604)	A
HV-12607 (controlled by SO-V HY-12607)	B

a. Refer to appropriate logic diagrams for additional actuated devices.

TABLE 7.3.14-1

AUXILIARY FEEDWATER PUMPHOUSE ESF HVAC SYSTEM
ACTUATED EQUIPMENT LIST[a]

Description	Actuation Channel or Train
Motor-driven auxiliary feedwater pump A room air supply fan motor (1-1593-B7-001-M01)	A
Motor-driven auxiliary feedwater pump A room air shutoff damper (HV-12006)	A
Motor-driven auxiliary feedwater pump B room air supply fan motor (1-1593-B7-002-M01)	B
Motor-driven auxiliary feedwater pump B room air shutoff damper (HV-12005)	B
Turbine-driven auxiliary feedwater pump room air intake damper (HV-12010 controlled by HY-12010 solenoid valve)	C

a. Refer to appropriate logic diagrams for additional actuated devices.

TABLE 7.3.15-1 (Sheet 1 of 2)

DIESEL GENERATOR BUILDING ESF HVAC SYSTEM
ACTUATED EQUIPMENT LIST[a]

Description	Actuations Train or Channel
ESF supply fan motors:	
1-1566-B7-001-M01	A
1-1566-B7-002-M01	B
1-1566-B7-003-M01	A
1-1566-B7-004-M01	B
Recirculation dampers:	
TV-12100A (controlled by I/P converter TY-12100A)	A
TV-12100 (controlled by I/P converter TY-12100A)	A
TV-12101A (controlled by I/P converter TY-12101A)	B
TV-12101 (controlled by I/P converter TY-12101A)	B
Air intake dampers:	
TV-12094C (controlled by I/P converter TY-12094A)	A
TV-12094A (controlled by I/P converter TY-12094A)	A
TV-12094B (controlled by I/P converter TY-12094B)	A
TV-12094D (controlled by I/P converter TY-12094B)	A
TV-12095A (controlled by I/P converter TY-12095A)	B
TV-12095C (controlled by I/P converter TY-12095A)	B
TV-12095B (controlled by I/P converter TY-12095B)	B
TV-12095D (controlled by I/P converter TY-12095B)	B
Exhaust dampers:	
TV-12086 (controlled by SOL valve TY-12086)	A
TV-12086A (controlled by SOL valve TY-12086)	A
TV-12099 (controlled by SOL valve TY-12099)	B
TV-12099A (controlled by SOL valve TY-12099)	B
TV-12096 (controlled by SOL valve TY-12096)	A
TV-12096A (controlled by SOL valve TY-12096)	A
TV-12097 (controlled by SOL valve TY-12096)	A
TV-12097A (controlled by SOL valve TY-12096)	A
TV-12098 (controlled by SOL valve TY-12098)	B
TV-12098A (controlled by SOL valve TY-12098)	B
TV-12085 (controlled by SOL valve TY-12098)	B
TV-12085A (controlled by SOL valve TY-12098)	B

TABLE 7.3.15-1 (Sheet 2 of 2)

DIESEL GENERATOR BUILDING ESF HVAC SYSTEM
ACTUATED EQUIPMENT LIST[a]

Description	Actuations Train or Channel
Non-ESF fan unit isolation dampers:	
HV-12052	A
HV-12055	B
ESF supply fan discharge dampers:	
HV-12050	A
HV-12051	A
HV-12053	B
HV-12054	B

a. Refer to appropriate logic diagrams for additional actuated devices.

TABLE 7.3.16-1

ELECTRICAL TUNNEL ESF HVAC SYSTEM
ACTUATED EQUIPMENT LIST[a]

Description	Actuation Train or Channel
Diesel power cable tunnel exhaust fan motors:	
1-1540-B7-001-M01	A
1-1540-B7-002-M01	B
NSCW tower cable tunnel exhaust fan motors:	
1-1540-B7-003-M01	A
1-1540-B7-004-M01	B
Turbine building and auxiliary building exhaust fan motor:	
2-1540-B7-005-M01 (Unit 2 only)	A
1-1540-B7-005-M01 (Unit 1 only)	A

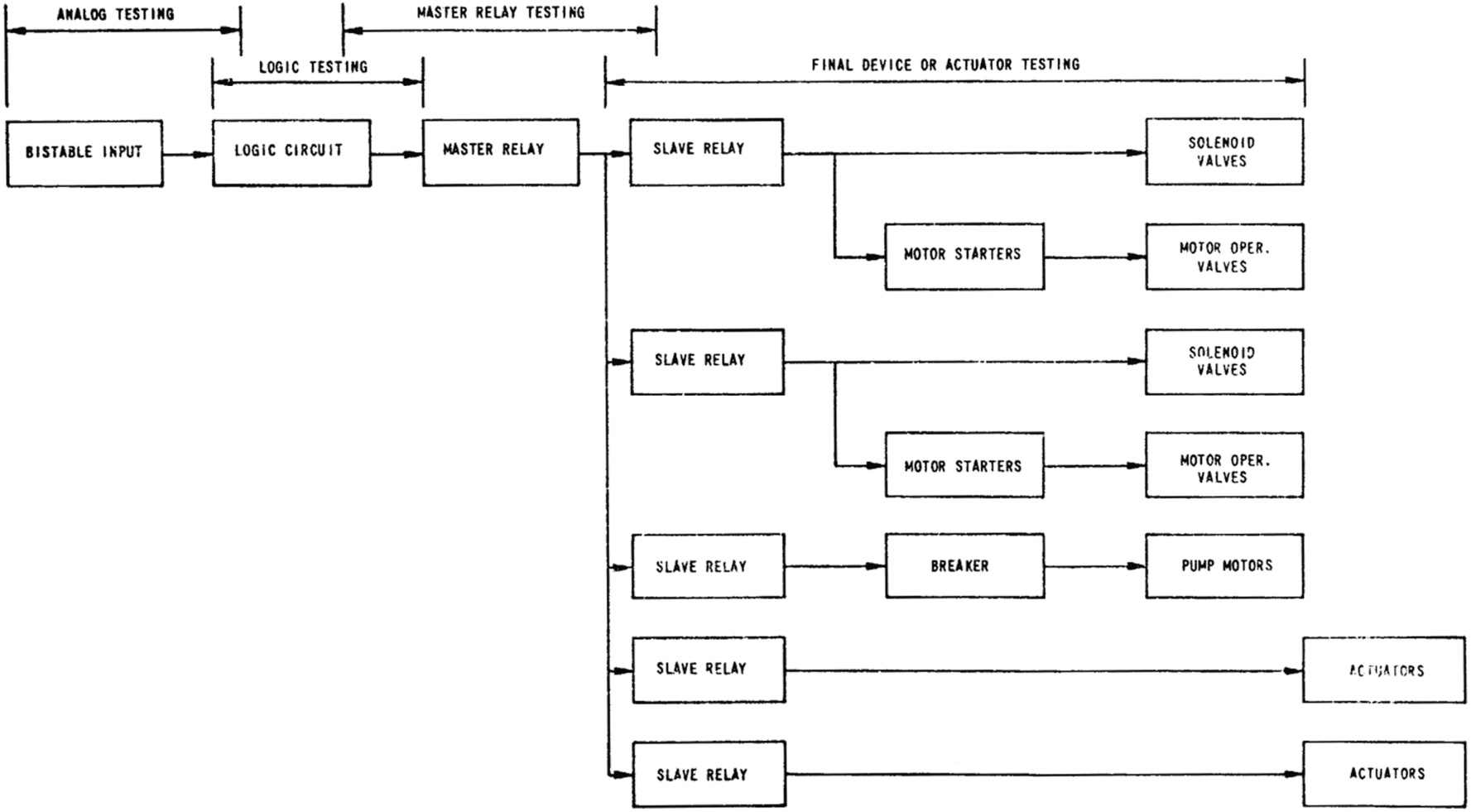
a. Refer to appropriate logic diagrams for additional actuated devices.

TABLE 7.3.17-1

DIESEL GENERATOR FUEL OIL SYSTEM
ACTUATED EQUIPMENT LIST[a]

Description	Actuation Train or Logic
Diesel fuel oil storage tank pumps:	
1-2403-P4-001	A
1-2403-P4-002	A
1-2403-P4-003	B
1-2403-P4-004	B

a. Refer to appropriate logic diagrams for additional actuated devices.



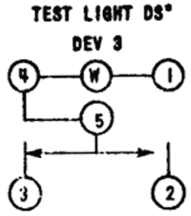
REV 14 10/07

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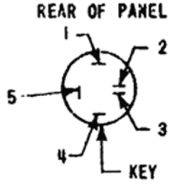
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

TYPICAL ENGINEERED SAFETY FEATURES TEST CIRCUITS

FIGURE 7.3.1–1



ILLUMINATED PUSHBUTTON SWITCH
WITH 28V LAMP NO. 327
(EXCEPT AS NOTED)



CONTACT LOCATION SCHEME

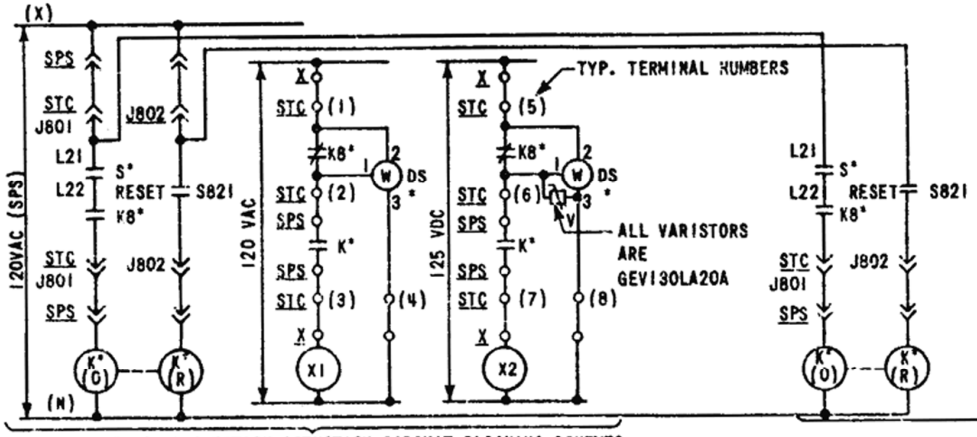
GENERAL NOTES: *

1. CIRCUITRY AND HARDWARE FOR REDUNDANT PROTECTION TRAINS "A" AND "B" TEST CABINETS ARE DUPLICATE EXCEPT AS NOTED
   - A - TRAIN "A" ONLY
   - B - TRAIN "B" ONLY
2. IN DETAILS A & B THE SYMBOL * REPRESENTS THE SUFFIX NUMBERS OF THE DEVICE REFERENCED.

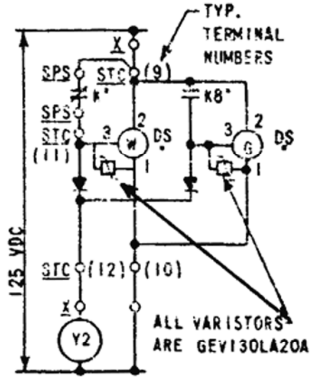
   EXAMPLE:
   - K* - SPS RELAY, K601, K602, ETC.
   - K(O) - OPERATING COIL
   - K(R) - RESET COIL
   - S* - STC TEST SWITCH, S802, S834 ETC.
   - K8* - STC RELAY, K811, K817, ETC.
   - DS* - STC LIGHT, DS8C09, DS8077, ETC.
3. "DETAIL A" & "B" TYPE CIRCUITS ARE DETAILED ON THE SCHEMATICS. "DETAIL B" CIRCUITS WILL BE SUBSTITUTED FOR "DETAIL A" CIRCUITS WHERE REQUIRED.

LOCATION LEGEND

- SPS - SOLID STATE PROTECTION SYSTEM
- STC - SAFEGUARDS TEST CABINET
- X - SWGR, MCC, AUXILIARY RELAY RACK, ETC.



DETAIL A TYPICAL PROTECTION ACTUATION CIRCUIT BLOCKING SCHEMES
(CONTACT CLOSURE FOR ACTUATION)



DETAIL B TYPICAL PROTECTION ACTUATION CIRCUIT BLOCKING SCHEMES
(CONTACT OPENING FOR ACTUATION)

*DETAILS A AND B OF THIS FIGURE ARE NOT TO BE CONFUSED WITH ALPHA DESIGNATION OF LOGIC TRAINS A AND B.

REV 14 10/07

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VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

ENGINEERED SAFEGUARDS TEST CABINET
(INDEX, NOTES, AND LEGEND)

FIGURE 7.3.1–2

7.4 SYSTEMS REQUIRED FOR SAFE SHUTDOWN

The functions necessary for safe shutdown are available through instrumentation and control channels associated with the major systems in both the primary and secondary plant. These channels are normally aligned to serve a variety of operational functions, including startup and shutdown, as well as protective functions. There are no systems specifically and solely dedicated as safe shutdown systems. However, procedures for securing and maintaining the plant in a safe condition can be instituted by appropriate alignment of selected plant systems. The discussion of these systems, together with the applicable codes, criteria, and guidelines, is found in other sections of this safety analysis report. In addition, the alignment of systems associated with the engineered safety features, which are invoked under postulated limiting fault situations, is discussed in section 6.3, paragraphs 7.3.1.1.1.1 and 7.3.1.1.1.5, and tables 7.3.1-1, 7.3.1-2, and 7.3.1-3. In the event of a turbine or reactor trip, the plant is placed in a hot standby condition. During the hot standby condition, an adequate heat sink is provided to remove reactor core residual heat. Boration capability is provided to compensate for xenon decay and to maintain the required core shutdown margin. Redundancy of systems and components is provided to enable continued maintenance of the hot standby condition. Redundant systems and components exist for taking the plant to the cold shutdown condition, if required.

The instrumentation and control systems required to be aligned for maintaining safe shutdown of the reactor, which are discussed in this section, are the minimum needed under nonaccident conditions. These systems permit the necessary operations that:

- Prevent the reactor from returning to criticality.
- Provide an adequate heat sink so that design and safety limits on reactor coolant system (RCS) temperature and pressure are not exceeded.

The designation of systems required for safe shutdown depends on identifying those systems which provide the following capabilities for maintaining a safe shutdown:

- Coolant circulation.
- Boration.
- Heat removal.
- Depressurization.

The specific systems, together with the necessary associated instrumentation and controls, are identified for both hot standby and cold shutdown in subsections 7.4.1 and 7.4.2. Table 7.4.1-1 tabulates systems available for safe shutdown.

Maintenance of a shutdown with these systems and associated instrumentation and controls has included consideration of the accident consequences that might jeopardize safe shutdown conditions. The accident consequences that are germane are those that would tend to degrade the capabilities for coolant circulation, boration, heat removal, and depressurization.

The results of the accident analysis are presented in chapter 15. Of these, the following produce consequences that might jeopardize safe shutdown conditions:

- Chemical and volume control system malfunction that results in a decrease in the boron concentration in the reactor coolant (uncontrolled boron dilution) (15.4.6).
- Loss of normal feedwater flow (15.2.7).
- Loss of external electrical load and/or turbine trip (15.2.2 and 15.2.3).
- Loss of offsite ac power to the station auxiliaries (15.2.6).

These analyses show that safety is not adversely affected by these incidents, with the associated assumptions being that one train of the instrumentation and controls discussed in subsections 7.4.1 and 7.4.2 are available to control and/or monitor shutdown. These required systems will allow the maintenance of safe shutdown even under the accident conditions listed above, which would tend toward a return to criticality or a loss of heat sink.

In addition to the operation of systems required for safe shutdown, as described below, the following are assumed:

- The turbine is tripped. (Note that this can be accomplished at the turbine as well as inside the control room.)
- The reactor is tripped. (Note that this can be accomplished at the shutdown panels as well as inside the control room.)
- All automatic systems continue functioning as long as offsite power and the plant compressed air system are available. With a loss of either, components relying on these systems are assumed to take their fail-safe position unless failure to do so is the most limiting single failure.

7.4.1 HOT STANDBY

To effect a unit shutdown, the unit is initially brought to a hot standby condition under control from the main control room or the shutdown panels. Hot standby is defined as the condition in which the reactor is subcritical (K_{eff} <0.99) and the RCS temperature and pressure are in the normal operating range. To accomplish a hot standby, the following functions are required: coolant circulation, boration, and heat removal. The portions of the reactor trip system required to achieve the shutdown condition are described in section 7.2. The system and component controls and monitoring indicators provided on the shutdown panels are listed in subsection 7.4.3. The minimum controls and monitoring indicators required to maintain a hot standby under a nonaccident condition are tabulated and discussed below. Table 7.4.2-1 lists the instrumentation and controls available for hot standby and hot or cold shutdown and provides the location of controls and indication.

A. Required Systems and Component Controls

1. Auxiliary feedwater system.
2. Condensate storage facility.
3. Main steam power-operated atmospheric relief valves.
4. Centrifugal charging pumps.
5. Nuclear service cooling water (NSCW) pumps.

6. NSCW fans.
7. Component cooling water pumps.
8. Containment fan coolers.
9. Emergency diesel generators (and associated onsite electrical distribution system).
10. Control room ventilation.
11. Emergency ventilation systems for those areas housing equipment required for safe shutdown.
12. Essential chilled water.

B. Required Monitoring Indicators

1. Steam generators
   - Water level for each steam generator.
2. Pressurizer
   - Water level.
3. Reactor Coolant System
   - Hot leg temperatures.
   - Cold leg temperatures.
   - Pressure.

7.4.1.1 Auxiliary Feedwater Control

The auxiliary feedwater pumps start automatically as described in subsection 7.3.7 or can be started manually. Start/stop motor controls are located at the shutdown panels (trains A and B) and the turbine-driven pump auxiliary feedwater local panel (train C), as well as at the main control board. Control of the motor-operated valves in the auxiliary feedwater system is provided at the shutdown panels, auxiliary feedwater local panel, and the main control board.

7.4.1.2 Power-Operated Atmospheric Steam Relief Valves

7.4.1.2.1 Description

The instrumentation and controls for the atmospheric steam relief system consist of controls, transmitters, and indicators to provide automatic or manual actuation of the power-operated atmospheric steam relief valves to remove decay heat from the RCS.

Both the main steam safety valves and the power-operated atmospheric steam relief valves are located upstream of the main steam isolation valves, outside of the containment; and both provide a means of removing decay heat in a hot standby condition. The safety valves are full-capacity, spring-loaded valves actuated by high main steam line pressure. They are described more fully in chapter 10. The power-operated atmospheric steam relief valves, however, are the preferred mode of steam relief to avoid prolonged operation of the safety valves. The power-

operated portion of the relief system is safety related, except as specifically noted otherwise in paragraph I below.

A pressure transmitter and pressure controller are provided for each of the steam generators to actuate the power-operated atmospheric steam relief valve and control the steam pressure at a predetermined setting. Manual control capability is provided both in the control room and on the shutdown panels for power-operated atmospheric steam relief valve regulation. The status of the power-operated atmospheric steam relief valves is indicated by open and closed indicating lights and by the controller output indication provided in the main control room and on the shutdown panels.

A. Initiating Circuits

No initiating circuits are required for the self-actuated, spring-loaded safety valves. Each power-operated atmospheric steam relief valve is automatically actuated to regulate the steam generator pressure via the pressure controller and can be manually actuated by selecting the manual control mode. The required instrumentation readout for manual system control is described in section 7.5.

B. Logic

No logic is required for the spring-loaded safety valves. Each power-operated atmospheric steam relief valve is individually controlled by its own pressure control loop. Normally atmospheric steam relief valve operation is automatic, but it may be operated manually.

C. Bypass

No bypass is provided. Placement of the power- operated atmospheric steam relief valve controller in the manual mode does not preclude the steam relief functional requirement, since the spring-loaded safety valves provide the code-required relief capability.

D. Interlock

No interlock is provided for the power-operated atmospheric steam relief valve system.

E. Redundancy

Any one of the four power-operated atmospheric steam relief valves provides sufficient steam relief for hot shutdown requirements. Redundancy is accomplished on a system basis, since any one of the four associated steam generators is adequate for the heat removal requirements.

F. Diversity

Diversity is accomplished by the spring-loaded safety valves operating as backup to the power-operated atmospheric steam relief valves.

G. Actuated Devices

The safety valves are self-actuated.

The power-operated atmospheric steam relief valves are electrohydraulic valves designed to fail closed.

H. Supporting Systems

The controls and power for the power-operated atmospheric steam relief valves are powered from the Class 1E power system (chapter 8).

I. Portion of System Not Required for Safety

The alarms to the station annunciator and computer are not required for safety.

J. Design Bases Information

The design bases of the power-operated atmospheric steam relief system (in accordance with Section 3 of Institute of Electric and Electronic Engineers (IEEE) Standard 279-1971) are:

1. The plant condition which requires protective action is hot standby heat removal at controlled steam generator pressure, with or without loss of offsite power.
2. The equipment is located outside the containment and is designed to withstand the temperature range, relative humidity, and atmospheric pressure for that location. (Refer to table 3.11.B.1-1 for specific values.)
3. The power-operated atmospheric steam relief system is designed to withstand the effects of earthquake without loss of function. The system is designed and its components are physically located to prevent loss of function from missile damage.
4. The power-operated atmospheric steam relief controls are analog in nature, and the response of conventional process control equipment adjusted for stable pressure controlling operation is adequate. The power-operated atmospheric steam relief valves are not intended to prevent safety valve operation when the turbine bypass system is not available. The requirement is for the power-operated steam relief valves to relieve the safety valves from a sustained pressure-controlling function in the hot standby mode. Thus, response time and accuracy are not critical for the required performance. The steam generator pressure will be relatively constant (no load steam pressure), with no rapid change required in the mass flowrate from the atmospheric steam relief valves.

7.4.1.2.2 Analysis

A. Conformance to Nuclear Regulatory Commission (NRC) General Design Criteria (GDC)

1. GDC 13 and 19

Instrumentation necessary to monitor station variables associated with hot standby is provided with adequate indication in the main control room and on the shutdown panels. Controls for the power-operated atmospheric steam relief are provided at each location. A description of the surveillance instrumentation is provided in section 7.5.

2. GDC 34

The power-operated atmospheric steam relief valves provide an adequate means of venting the steam generators to remove reactor decay heat following reactor

trip. Modulation of the power-operated atmospheric steam relief valves provides the desired rate of heat removal from the RCS to maintain the hot standby condition. The power-operated atmospheric steam relief system has sufficient redundancy to ensure its intended function, assuming a single failure.

B. Conformance to NRC Regulatory Guides

1. Regulatory Guide 1.22

   The power-operated atmospheric steam relief controls can be tested periodically.

2. Regulatory Guide 1.29

   The power-operated atmospheric steam relief controls are designed to withstand the effects of a safe shutdown earthquake (SSE) without loss of function. The power-operated atmospheric steam relief controls are classified Seismic Category 1, in accordance with the guide.

C. Conformance to IEEE Standard 279-1971

The controls for the power-operated atmospheric steam relief system conform to the applicable requirements of IEEE Standard 279-1971. The control circuits are designed so that any single failure will not prevent proper protective action (removal of reactor decay heat) when required. This is accomplished by redundant steam relief systems in that only one of the four valves is needed to provide sufficient capacity. The power-operated atmospheric relief valves utilize control power from independent Class 1E power systems. The controllers for the valves are powered from separate independent Class 1E control channels. In order to prevent interaction between the redundant systems, the control channels are wired independently and separated with no electrical connections between them.

D. Conformance to Other Criteria and Standards

Conformance to other criteria and standards is given in table 7.1.1-1.

7.4.1.3 Centrifugal Charging System Controls

7.4.1.3.1 Description

If the unit is maintained in a hot standby condition for a prolonged time, a centrifugal charging pump is required to maintain the reactor coolant inventory so that the level in the pressurizer is maintained above the heaters and to borate to compensate for xenon decay. At the time the charging pump is brought into operation to replenish the RCS, the boron concentration of the RCS may be increased, if desired. Normal operation of the charging system is automatic, as described in paragraph 7.7.1.6. Manual control is also provided both at the main control board and the shutdown panels. Control of major power-operated valves associated with establishing a charging path to the RCS is provided in the main control room and at the shutdown panels.

The following discussion is limited to the manual centrifugal charging pump controls. A detailed description of the charging system, its operation, and safety evaluation is provided in section 6.3 and subsection 9.3.4.

A. Initiating Circuits

The charging pumps can be controlled manually by the plant operator for hot standby service. For other initiating circuits, see section 7.3.

B. Logic

The control logic provides for both automatic and manual control features. The pumps can be started under manual control at any time. Refer to section 1.7 for a list of elementary and logic diagrams.

C. Bypass

No bypass of the manual controls, other than maintenance provisions, is provided.

D. Interlocks

When the shutdown panel transfer switch is in the local position, automatic start of the pump on a safety signal is defeated.

E. Redundancy

Two independent centrifugal charging pumps and control circuits are provided, either of which can provide the necessary input to the primary system for the hot standby condition.

F. Diversity

There is no diversity in the manual control circuits or power supplies for the two centrifugal charging pumps.

G. Actuated Devices

The charging pumps and associated valves are the actuated devices.

H. Supporting Systems

The charging pump controls are powered from the Class 1E power system.

I. Portion of System Not Required for Safety

The instrumentation used to monitor the charging pump operation (other than indicating lights for hand switches as an integral part of the control circuit), alarms on the station annunciator and computer, and automatic charging pump control via the pressurizer level control channels are not required for safety.

J. Design Bases Information

The design bases of the charging pump manual controls (in accordance with Section 3 of IEEE Standard 279-1971) are as follows:

1. The generating station condition that requires protective action is low pressurizer level following a reactor trip with or without loss of offsite power.
2. The equipment is located outside the containment and is designed to withstand the temperature range, relative humidity, and atmospheric pressure for its location. Refer to table 3.11.B.1-1 for specific values.

3. The charging pump manual control system is designed to withstand the effects of an SSE without loss of function. The system is designed and its components located to prevent loss of function from missile damage.

7.4.1.3.2 Analysis

A. Conformance to NRC GDC

1. GDC 13 and 19

   Instrumentation necessary to monitor station variables associated with hot standby is provided with adequate indication in the main control room and on the shutdown control panels. Manual controls for the centrifugal charging pumps are provided both inside and outside of the control room. A description of the surveillance instrumentation is provided in section 7.5.

2. GDC 33

   The centrifugal charging pump manual controls provide adequate control of the pressurizer level to preclude use of the pressurizer heaters below low-low level. One centrifugal charging pump is sufficient to provide the necessary makeup to the RCS to maintain the hot standby condition.

B. Conformance to NRC Regulatory Guides

1. Regulatory Guide 1.22

   The centrifugal charging pump manual controls can be tested periodically during operation, since the charging pumps are used during normal operation.

2. Regulatory Guide 1.29

   The centrifugal charging pump manual controls are designed to withstand the effects of an SSE without loss of function or physical damage. The centrifugal charging pump manual controls are classified Seismic Category 1, in accordance with the guide.

C. Conformance to IEEE Standard 279-1971

The centrifugal charging pump manual controls are designed to meet the portions of IEEE Standard 279-1971 applicable to manual controls. The manual control circuits are designed so that any single failure will not prevent protective action (makeup to the RCS) when required. This is accomplished by two redundant centrifugal charging pumps. The control circuit of each charging pump utilizes controls powered from an independent Class 1E power system. To prevent interaction between the redundant systems, the manual control channels are wired independently and separated with no electrical connections between them.

The normal automatic control circuits are electrically isolated from the manual controls to ensure manual control system independence.

D. Conformance to Other Criteria and Standards

Conformance to other criteria and standards is given in table 7.1.1-1.

7.4.1.4 Coolant Circulation

The preferred method of coolant circulation is forced circulation with the reactor coolant pumps supplying the driving head. With loss of offsite power, the pumps are not available; however, the RCS is designed to provide sufficient natural circulation to reach and maintain hot standby. Natural circulation flow is verified by noting the various RCS temperatures.

7.4.1.5 Other Systems Required for Hot Standby

The other major equipment and systems required to maintain the unit in the hot standby condition are listed below. For a more comprehensive and detailed listing, refer to tables 7.4.1-1 and 7.4.2-1.

A. NSCW system (subsection 9.2.1).

B. Component cooling water system (subsection 9.2.2).

C. Containment fan coolers (subsection 6.2.2).

D. Emergency diesel generators (and associated onsite electrical distribution system) (subsection 8.3.1).

E. Control room ventilation system (subsection 9.4.1).

F. Emergency ventilation system (for those areas housing equipment required for safe shutdown) (section 9.4)

G. Essential chilled water (subsection 6.2.2).

Systems A through F above are either normally operating continuously or start automatically when required. The instrumentation and controls for these systems are described in the particular section of this document where each system is described. (See A through F above.) Further discussion of the actuation and controls for the engineered safety feature systems is provided in section 7.3.

7.4.2 COLD SHUTDOWN

To perform a unit cold shutdown, the unit is brought from hot standby conditions to nearly ambient conditions from the main control room or shutdown panels. The ability to reach cold shutdown under control from the main control room utilizing safety-related components is further discussed in this section; the ability to reach cold shutdown under control from the shutdown panels is discussed in detail in subsection 7.4.3.

Cold shutdown is defined by the Technical Specifications as the condition in which the reactor is subcritical, the reactor coolant system (RCS) temperature is $\leq 200^{\circ}F$, and the RCS is depressurized. To accomplish a cold shutdown, the following functions are required:

- Coolant circulation.
- Boration.
- Heat removal.
- Depressurization.

The systems required for hot standby are also required for cold shutdown. In addition, the following systems, components, and indication are required:

A. Required Systems and Component Controls

1. Vessel head letdown and vent system.
2. Pressurizer power-operated relief valve complex.
3. Residual heat removal system.
4. Boric acid storage tank.
5. Boric acid transfer pumps.
6. Accumulator vent system.
7. Manual block of safety injection signal.
8. Pressurizer backup heaters.

The components in these systems are fully qualified and safety grade with power from a Class 1E electrical bus except the pressurizer heaters. (Two groups of pressurizer backup heaters can be administratively loaded on the non-1E emergency bus.) All control switches for these items are safety grade.

B. Required Monitoring Indicators

1. Boric acid storage tank level.
2. Boric acid charging flowrate.

Table 7.4.2-1 lists the instrumentation and controls available for hot standby and hot or cold shutdown and provides the locations of controls and indication.

Hot standby is a stable plant condition for a reactor plant that incorporates a Westinghouse nuclear steam supply system. Examination of Condition II, III, or IV events for the Westinghouse nuclear steam supply system reveals no events that require cooldown to cold shutdown conditions for safety reasons. Eventual achievement of cold shutdown conditions may be required for long-term recovery. However, there is no safety reason why this must be accomplished in some limited period of time. While the plant is in the hot standby condition, the auxiliary feedwater system and the power-operated atmospheric steam relief valves are used to remove residual heat to meet all safety requirements. The long-term safety grade supply of auxiliary feedwater allows extended operation at hot standby conditions. Additionally, the plant design includes provisions for achieving cold shutdown, even assuming a safe shutdown earthquake, a loss of offsite power, and the most limiting single failure with limited operator action outside the control room.

7.4.3 SAFE SHUTDOWN FROM OUTSIDE THE CONTROL ROOM

7.4.3.1 Description

If temporary evacuation of the control room is required because of some abnormal plant condition, the operators can establish and maintain the plant in a hot standby and hot or cold shutdown condition from outside the control room through the use of controls located at the shutdown panels. Hot standby is a stable plant condition which can be maintained safely for an extended period of time. In the event that access to the control room is restricted, the plant can be safely kept at hot standby until the control room can be reentered, by the use of the monitoring indicators and the controls listed in subsection 7.4.1. Although the prime intent of the shutdown panels is to maintain hot standby from outside the control room, the panels are also used for implementing cold shutdown from outside the control room.

7.4.3.1.1 Shutdown Panels

Hot standby and hot or cold shutdown can be established and maintained from the shutdown panels. The panels are designed to allow control of a shutdown following an evacuation of the control room, coincident with the loss of offsite power and a single active failure. No other design basis event is postulated. (See 9.5.1 for a discussion of shutdown in the event of a fire.) Remote control station equipment is designed to the same standards as the corresponding equipment in the main control room.

Two shutdown panels are provided for each unit. [redacted]. Access to each is administratively controlled. Each panel contains the control switches for the associated train's safety-grade equipment required to accomplish hot standby and hot or cold shutdown. Additionally, control of selected nonsafety-grade components is provided to accomplish a shutdown if offsite power remains available. Table 7.4.2-1 lists the instrumentation and controls available for hot and/or cold shutdown and provides the location for the various control switches and indicators.

The shutdown panels are provided for use following an evacuation of the control room only. No actions are anticipated from the shutdown panels during normal, routine shutdown, refueling, or maintenance operations. [redacted]

- [redacted]
- [redacted]
- [redacted]
- [redacted]

Because automatic features are defeated, manual action is required if a safety signal is generated while performing a shutdown from these panels.

The shutdown panels are provided with sufficient communication circuits to ensure that the operator can safely perform a hot standby and hot or cold shutdown. Communication is available between the following stations (see subsection 9.5.2):

- Control room.
- Shutdown panels.
- Onsite technical support center.
- Diesel generator local control station.
- Auxiliary feedwater local control station.

7.4.3.1.2 Controls at Other Locations

In addition to the controls and indicators provided on the shutdown panels, the following controls are provided outside the control room:

A. [redacted]

B. [redacted]

C. [redacted]

7.4.3.1.3 Design Bases Information

In accordance with Nuclear Regulatory Commission (NRC) General Design Criterion (GDC) 19, the capability of establishing a shutdown condition and maintaining the station in a safe status in that mode is considered an essential function. The controls and indications essential to this function are identified in subsection 7.4.1. To ensure the availability of the shutdown panels after control room evacuation, the following design features have been utilized:

A. The shutdown panels, including all safety-grade controls and indications mounted on them, are designed to withstand the safe shutdown earthquake (SSE) with no loss of essential functions.

B. The shutdown panels are designed to conform with the applicable portions of Institute of Electrical and Electronics Engineers (IEEE) Standard 279-1971.

C. The shutdown panels are designed to achieve and maintain hot standby conditions and to achieve cold shutdown from full power conditions and maintain cold shutdown conditions thereafter.

D. The shutdown panels are designed to achieve cold shutdown where offsite power is available and where offsite power is not available for 72 h.

E. The shutdown panels are designed to utilize only safety-related systems independent from the main control room except for indications and operator-interface modules (OIMs) for train A head vent throttle valve, centrifugal charging pump discharge throttle valves, power-operated atmospheric steam relief valves (steam generators 2 and 3 can be controlled independent of the control room at shutdown panel B by a portable plug-in signal generating device), train A accumulator tank gas vent valve and residual heat removal heat exchanger outlet and bypass valves (refer to paragraph 7.4.3.3 for a discussion on alternate shutdown indication).

F. The shutdown panels are designed for a coincident single failure. When a random event such as a fire or allowable technical specification maintenance results in one safety grade train being unavailable, a single failure in the redundant train is not postulated. When a random event other than fire causes a control room evacuation, a coincident single failure in the systems controlled from the local shutdown panel is considered.

G. The shutdown panel design allows credit for remedial operator action as discussed in paragraph 7.4.3.1.2 but minimizes the need for operator action outside of the shutdown panel area.

H. Design basis events do not occur simultaneous with nor as a consequence of any event which initiates control room evacuation.

I. The shutdown panel design minimizes the potential for inadvertent use of or sabotage of the plant from the shutdown panel.

J. Access to the shutdown panels is under administrative controls.

K. An alarm is provided in the control room to provide an indication when a component or components on the shutdown panel is/are bypassed from the main control boards to the shutdown panel.

L. Controls, switches, and indications on the shutdown panels are designed to be consistent with the design requirement for similar devices located in the main control room.

7.4.3.2 Analysis

The analysis of the control systems required for safe shutdown is found in subsection 7.4.1. The discussion below is limited to the shutdown panels.

A. Conformance to NRC GDC

1. GDC 19

   The shutdown panels provide adequate controls and indications located outside the main control room to establish and maintain the reactor and the reactor coolant system in the hot standby and hot or cold shutdown condition in the event that the main control room must be evacuated.

B. Conformance to NRC Regulatory Guides

1. Regulatory Guide 1.22

   The shutdown panels are designed to be tested periodically during station operation.

2. Regulatory Guide 1.29

   The shutdown panels are designed to withstand the effects of an SSE without loss of function or physical damage. The shutdown panels are classified Seismic Category 1. Selected instrumentation and control devices are not safety related but are qualified for seismic integrity to prevent compromising the function of safety- related devices during or after an SSE.

C. Conformance to IEEE Standard 279-1971

The shutdown panels are designed to conform to applicable portions of IEEE Standard 279-1971. The control circuits at the shutdown panels are designed so that any single failure will not prevent maintaining safe shutdown when required. This is accomplished by fully redundant controls for the systems required for hot standby and hot or cold shutdown, utilizing independent Class 1E power systems.

To prevent interaction between the redundant systems, the redundant control channels are wired independently and separated with no electrical connections between them. Non-Class 1E circuits available for safe shutdown are electrically isolated from Class 1E circuits.

D. Conformance to Other Guides, Criteria, and Standards

The additional guides, criteria, and standards listed in table 7.1.1-1 apply only to the essential instrumentation and control required for safe shutdown from outside the control room.

7.4.3.3 Alternate Shutdown Indication System

7.4.3.3.1 Description

The alternate shutdown indication system is designed to provide indication and controllers; i.e., OIMs, necessary for cold shutdown that are independent from the control room in the event of a control room fire. No other events are postulated to occur either during or after a control room fire; consequently, the design is exempted from the single failure criteria, Seismic Category 1 criteria, and the other design basis accident criteria, except where required for other reasons (e.g., due to interfacing with or impacting on existing systems).

The plant safety monitoring system (PSMS) (refer to paragraph 7.5.3.6) and the alternate shutdown indication system cabinet are used to process and output isolated signals which go to the control room and the train B shutdown panel. The PSMS provides isolated signals for the alternate shutdown indication parameters for which the PSMS performs data acquisition and display. The alternate shutdown indication system cabinet isolates signals which are required in the process cabinets and the OIM control loops.

The reactor may be tripped from the main control board before leaving the control room or tripped from either of the shutdown panels immediately after entering the shutdown panel rooms. Both shutdown panels are fully equipped panels that may act as the point of control for performing a shutdown and cooldown of the plant given that the control room is inaccessible. However, only the train B shutdown panel is provided with electrically isolated instrumentation and controls for use as the alternate shutdown point of control following a control room fire.

7.4.3.3.2 Design Bases Information

The alternate shutdown indication system is designed to meet Branch Technical Position CMEB 9.5-1 requirement C.5.C (see appendix 9B).

7.4.3.3.2.1 Safety Design Bases. The alternate shutdown indication system shall not compromise safety-related systems and associated inputs nor prevent safe shutdown.

7.4.3.3.2.2 Power Generation Design Basis. The alternate shutdown indication system provides electrically isolated signals into the control room during power generation. It is designed to function during and after a control room fire.

A. The alternate shutdown indication system controls, in conjunction with remote shutdown panel B controls, are used to achieve and maintain hot standby condition and achieve cold shutdown from full power conditions in 72 h following a control room fire and maintain cold shutdown conditions thereafter.

B. The alternate shutdown indication system instrumentation, in conjunction with remote shutdown panel B instrumentation, provides direct readings and controls to monitor the process variables necessary to perform and control the following shutdown functions:

   1. Reactivity control.
   2. Reactor coolant makeup/inventory.
   3. Reactor heat removal.

C. The alternate shutdown indication system consists of the following required parameters and OIMs (see table 7.4.2-1):

1. Neutron flux.
2. Reactor coolant system wide range T_{cold} - (loops 2 and 3).
3. Incore thermocouples in the quadrants corresponding to loops 2 and 3 (Unit 1) and loops 1 and 4 (Unit 2).
4. Reactor coolant system wide range pressure.
5. Steam generator wide range level (loops 2 and 3).
6. Pressurizer level.
7. Head vent throttle valve (OIM).
8. Accumulator tank gas vent valve (OIM).

D. The alternate shutdown indication system accommodates post-fire conditions where offsite power is available and where offsite power is not available for 72 h.

E. The alternate shutdown indication system is not damaged by a control room fire.

F. The alternate shutdown indication system and associated circuits design are exempted from Seismic Category I criteria, single failure criteria, or other design basis accident criteria, except where required for other reasons (e.g., because of interface with or impact on existing safety systems).

G. The alternate shutdown indication system is electrically isolated from the control room so that a fire-induced, hot short, open circuit, or short to ground in the alternate shutdown control room indication circuits will not prevent operation of the alternate shutdown indication at the shutdown panel.

H. Access to the alternate shutdown indication system is under administrative control.

I. The alternate shutdown indication system OIMs are activated manually following evacuation of the control room. This actuation does not disturb control, process, protection, or nuclear instrumentation circuits except those associated with the alternate shutdown indication system.

J. The alternate shutdown indication system is electrically isolated from the control room so that a fire-induced hot short, open circuit, or short to ground in any of the Class 1E circuits will not prevent operation of the alternate shutdown equipment from the shutdown panel.

K. An alarm is provided in the control room to provide an indication in the event that the alternate shutdown OIMs are bypassed from the main control board to the shutdown panel.

7.4.3.3.2.3 Guides, Criteria, and Standards. The alternate shutdown indication system conforms to GDC 19, the applicable portions of IEEE Standards 279-1971, 323-1974, and 344-1975, Regulatory Guide 1.22, and Branch Technical Position CMEB 9.5-1.

TABLE 7.4.1-1

SYSTEMS AVAILABLE FOR SAFE SHUTDOWN

Auxiliary feedwater system

Condensate storage facility

Chemical and volume control system (boration and makeup functions)

Pressurizer power-operated relief valve complex

Reactor vessel head letdown system

Residual heat removal system

Main steam power-operated atmospheric relief valve complex

Component cooling water system

NSCW system

Onsite standby power supply (diesel generators and associated outside electrical distribution system)

Ventilation systems (control room and engineered safety features rooms)

Associated instrumentation and controls

Safety injection system (accumulator vents systems)

Containment fan coolers

Essential chilled water

ACCW

TABLE 7.4.2-1 (SHEET 9 OF 9)

a.

AC	air conditioner
AFW	auxiliary feedwater
BAST	boric acid storage tank
BAT	boric acid transfer
BIT	boron injection tank
CB	control building
CBSF	control building safety feature
CCW	component cooling water
CRDM	control rod drive mechanism
CST	condensate storage tank
CTB	containment building
DG	diesel generator
ESF	engineered safety features
HX	heat exchanger
NSCW	nuclear service cooling water
PORV	power-operated relief valve
PRT	pressurizer relief tank
RCP	reactor coolant pump
RCS	reactor coolant system
RHR	residual heat removal
RWST	refueling water storage tank
SG	steam generator
SI	safety injection
VCT	volume control tank

b.

CR	control room
PSDA	shutdown panel A
PSDB	shutdown panel B
ASI	alternate shutdown indication

c. The items identified in this column are provided with control room circuitry isolation, via the alternate shutdown indication panel, to provide the ability for shutdown from remote shutdown panel B in the event of a control room fire.

d. Unit 1 only.

e. Available only on plant computer in control room.

7.5 INFORMATION SYSTEMS IMPORTANT TO SAFETY

7.5.1 SAFETY-RELATED DISPLAY INSTRUMENTATION INTRODUCTION

An analysis was conducted to identify the appropriate variables and to establish the appropriate design bases and qualification criteria for instrumentation employed by the operator for monitoring conditions in the reactor coolant system, the secondary heat removal system, and the containment, including engineered safety functions and the systems employed for attaining a safe shutdown condition.

The instrumentation is used by the operator to monitor the VEGP throughout all operating conditions including anticipated operational occurrences and accident and post-accident conditions.

The emergency response facilities and support systems consisting of the onsite technical support center, emergency operations facility, operational support center, safety parameter display system, and the emergency response data system (ERDS) are discussed in the VEGP Emergency Plan, section H. Table 7.5.2-1 indicates the specific plant parameters which are associated with ERDS. Modifications to these parameters in the plant may require NRC notification per 10 CFR 50, Appendix E, Part VI.

7.5.2 DESCRIPTION OF INFORMATION SYSTEMS

The plant safety analyses and evaluations define the design basis accident (DBA) event scenarios for which preplanned operator actions are required. Accident monitoring instrumentation is necessary to permit the operator to take required actions to address these analyzed situations. However, instrumentation is also necessary for unforeseen situations (i.e., to ensure that, should plant conditions evolve differently than predicted by the safety analyses, the control room operating staff has sufficient information to evaluate and monitor the course of the event). Additional instrumentation is also needed to indicate to the operating staff whether the integrity of the fuel cladding, the reactor coolant pressure boundary (RCPB), or the reactor containment has degraded beyond the prescribed limits defined as a result of the plant safety analyses and other evaluations.

Five classifications of variables have been identified to provide this instrumentation:

A. Those variables that provide information needed by the operator to perform manual actions identified in the operating procedures that are associated with DBA events are designated type A. These variables are restricted to preplanned actions for DBA events. The basis for selecting type A variables is given in paragraph 7.5.2.2.1.

B. Those variables needed to assess that the plant critical safety functions are being accomplished or maintained, as identified in the plant safety analysis and other evaluations, are designated type B.

C. Those variables used to monitor for the gross breach or the potential for gross breach of the fuel cladding, the RCPB, or the containment are designated type C.

D. Those variables needed to assess the operation of individual safety systems and other systems important to safety are designated type D.

E. Those variables that are required for use in determining the magnitude of the postulated releases and continually assessing any such releases of radioactive materials are designated type E.

The five classifications of variable are not mutually exclusive, in that a given variable (or instrument) may be included in one or more types. When a variable is included in one or more of the five classifications, the equipment monitoring this variable meets the requirements of the highest category identified.

Three categories of design and qualification criteria have been identified. The differentiation is made in order that an importance of information hierarchy can be recognized in specifying accident monitoring instrumentation. Category 1 instrumentation has the highest performance requirements and should be utilized for information which cannot be lost under any circumstances. Category 2 and Category 3 instruments are of lesser importance in determining the state of the plant and do not require the same level of operational assurance.

The primary differences between category requirements are in qualification, application of single failure, power supply, and display requirements. Category 1 requires seismic and environmental qualification, the application of a single failure criteria, utilization of emergency power, and an immediately accessible display. Category 2 requires environmental and seismic qualification commensurate with the required function but does not require the single failure criteria, emergency power, or an immediately accessible display. Category 2 requires, in effect, a rigorous performance verification for a single instrument channel. Category 3, which is high quality commercial grade, does not require qualification, single failure criteria, emergency power, or an immediately accessible display.

Table 7.5.2-1 summarizes the following information for each variable identified:

A. Instrument range or status.

B. Type and category.

C. Environmental qualification.

D. Seismic qualification.

E. Number of channels.

F. Display methodology.

G. Implementation date.

7.5.2.1 Definitions

7.5.2.1.1 Design Basis Accident Events

Those events, any one of which could occur during the lifetime of a particular unit, and those events not expected to occur but postulated because their consequences would include the potential for release of significant amounts of radioactive gaseous, liquid, or particulate material to the environment are DBA events. Excluded are those events (defined as normal and anticipated operational occurrences in 10 CFR 50) expected to occur more frequently than once during the lifetime of a particular unit.

The limiting accidents that were used to determine instrument functions are:

- Loss-of-coolant accident (LOCA).

- Steam line break.
- Feedwater line break.
- Steam generator tube rupture.

7.5.2.1.2 Hot Standby

Hot standby is the state of the plant in which the reactor is subcritical such that k_{eff} is less than or equal to 0.99 and the reactor coolant system (RCS) temperature is greater than or equal to 350°F.

7.5.2.1.3 Cold Shutdown

Cold shutdown is the state of the plant in which the reactor is subcritical such that k_{eff} is less than or equal to 0.99, the RCS temperature is less than 200°F, and the RCS pressure is less than or equal to 10 CFR 50, Appendix G limits.

7.5.2.1.4 Controlled Condition

A controlled condition is the state of the plant that is achieved when the "subsequent action" portion of the plant emergency procedures is implemented and the critical safety functions are being accomplished or maintained by the control room operating staff.

7.5.2.1.5 Critical Safety Functions

Critical safety functions are those safety functions that are essential to prevent a direct and immediate threat to the health and safety of the public. These are the accomplishing or maintaining of:

- Reactivity control.
- RCS pressure control.
- Reactor coolant inventory control.
- Reactor core cooling.
- Heat sink maintenance.
- Reactor containment environment.

7.5.2.1.6 Immediately Accessible Information

Immediately accessible information is information that is visually available to the control room operating staff immediately (i.e., within human response time requirements), once they have made the decision that the information is needed.

7.5.2.1.7 Primary Information

Primary information is information that is essential for the direct accomplishment of the preplanned manual actions necessary to bring the plant into a safe condition in the event of a DBA event; it does not include those variables that are associated with contingency actions.

7.5.2.1.8 Contingency Actions

Contingency actions are those manual actions that address conditions beyond the DBA events.

7.5.2.1.9 Key Variables

Key variables are those variables which provide the most direct measure of the information required.

7.5.2.1.10 Backup Information

Backup information is that information, made up of additional variables beyond those classified as key, that provide supplemental and/or confirmatory information to the control room operating staff. Backup variables do not provide indication which is as reliable or complete as that provided by primary variables and are not usually relied upon as the sole source of information.

7.5.2.2 Variable Types

These accident monitoring variables and information display channels are those that are required to enable the control room operating staff to perform the functions defined by type A, B, C, D, and E classifications as follows.

7.5.2.2.1 Type A

Type A variables provide the primary information required to permit the control room operating staff to:

A. Perform the diagnosis specified in the VEGP emergency operating instructions.

B. Take the specified, preplanned, manually controlled actions for which no automatic control is provided that are required for safety systems to accomplish their safety function in order to recover from the DBA.

C. Attain and maintain a cold shutdown condition.

The verification of the actuation of safety-related systems has been excluded from the type A definition. The variables which provide this verification are included in the definition of type D.

Type A variables are restricted to preplanned actions for DBA events. Variables used for contingency actions and additional variables which might be utilized are of types B, C, D, and E.

7.5.2.2.2 Type B

Type B variables provide to the control room operating staff information to assess the process of accomplishing or maintaining critical safety functions (i.e., reactivity control, RCS pressure control, RCS inventory control, reactor core cooling, heat sink maintenance, and reactor containment environment).

7.5.2.2.3 Type C

Type C variables provide the control room operating staff information to monitor:

A. The extent to which variables that indicate the potential for causing a gross breach of a fission product barrier have exceeded the design basis values.

B. The incore fuel cladding, the RCPB, or the primary reactor containment which may have been subject to gross breach.

   These variables include those required to initiate the early phases of the emergency plan. Excluded are those associated with monitoring of radiological release from the plant which are included in type E.

Type C variables used to monitor the potential for breach of a fission product barrier have an arbitrarily determined extended range. The extended range was chosen to minimize the probability of instrument saturation even if conditions exceed those predicted by the safety analysis.

Although variables selected to fulfill type C functions may rapidly approach the values that indicate an actual gross failure, it is the final steady-state value reached that is important. Therefore, a high degree of accuracy and a rapid response time are not necessary for type C information display channels.

7.5.2.2.4 Type D

Type D variables provide the control room operating staff sufficient information to monitor the performance of:

A. Plant safety systems employed for mitigating the consequences of an accident and subsequent plant recovery to attain a cold shutdown condition. These include verification of the automatic actuation of safety systems.

B. Other systems normally employed for attaining a cold shutdown condition.

7.5.2.2.5 Type E

Type E variables provide the control room operating staff information to:

A. Monitor the habitability of the control room.

B. Monitor the plant areas where access may be required to service equipment necessary to monitor or mitigate the consequences of an accident.

C. Estimate the magnitude of release of radioactive material through identified pathways and continually assess such releases.

D. Monitor radiation levels and radioactivity in the environment surrounding the plant.

7.5.2.3 Variable Categories

The qualification requirements of the type A, B, C, D, and E accident monitoring instrumentation are subdivided into three categories. Descriptions of the three categories are given below. Table 7.5.2-2 briefly summarizes the selection criteria for type A, B, C, D, and E variables into each of the three categories. Table 7.5.2-3 briefly summarizes the design and qualification requirements of the three designated categories.

7.5.2.3.1 Category 1

7.5.2.3.1.1 Selection Criteria for Category 1. The selection criteria for Category 1 variables have been subdivided according to the variable type. For type A, those key variables used for diagnosis or providing information for necessary operator action have been designated Category 1. For type B, those key variables used for monitoring the process of accomplishing or maintaining critical safety functions have been designated Category 1. For type C, those key variables used for monitoring the potential for breach of a fission product barrier have been designated Category 1. There are no type D or type E Category 1 variables.

7.5.2.3.1.2 Qualification Criteria for Category 1. The instrumentation is environmentally and seismically qualified in accordance with sections 3.11 and 3.10, respectively. Instrumentation shall continue to read within the required accuracy following but not necessarily during a seismic event.

At least one instrumentation channel is qualified from the sensor up to and including the display. For the other instrumentation channels, qualification as a minimum is applied up to and includes the channel isolation device. (Refer to paragraph 7.5.2.3.4 in regard to extended range instrumentation qualification.)

7.5.2.3.1.3 Design Criteria for Category 1.

A. No single failure within either the accident-monitoring instrumentation, its auxiliary supporting features, or its power sources, concurrent with the failures that are a cause of or result from a specific accident, will prevent the control room operating staff from being presented the required information. Where failure of one accident-monitoring channel results in information ambiguity (e.g., the redundant displays disagree), the additional information is provided to allow the control room operating staff to analyze the actual conditions in the plant. This may be accomplished by providing additional independent channels of information of the same variable (addition of an identical channel) or by providing independent channels which monitor different variables which bear known relationships to the channels (addition of a diverse channel(s)). Redundant or diverse channels are electrically independent and physically separated from each other with two-train separation and from equipment not classified important to

safety in accordance with Regulatory Guide 1.75, Physical Independence of Electric Systems.

If ambiguity does not result from failure of the channel, then a third redundant or diverse channel is not required.

B. The instrumentation is energized from station emergency standby power sources, battery backed where momentary interruption is not tolerable, as discussed in Regulatory Guide 1.32, Criteria for Safety-Related Electric Power Systems for Nuclear Power Plants.

C. The out-of-service interval is based on normal Technical Specification requirements for the system it serves where applicable or where specified by other requirements.

D. Servicing, testing, and calibration programs are specified to maintain the capability of the monitoring instrumentation. For those instruments where the required interval between testing is less than the normal time interval between generating station shutdowns, a capability for testing during power operation is provided.

E. Whenever means for removing channels from service are included in the design, the design facilitates administrative control of the access to such removal means.

F. The design facilitates administrative control of the access to all setpoint adjustments, module calibration adjustments, and test points.

G. The monitoring instrumentation design minimizes the development of conditions that would cause meters, annunciators, recorders, alarms, etc., to give anomalous indications that could be potentially confusing to the control room operating staff.

H. The instrumentation is designed to facilitate the recognition, location, replacement, repair, or adjustment of malfunctioning components or modules.

I. To the extent practicable, monitoring instrumentation inputs are from sensors that directly measure the desired variables. An indirect measurement is made only when it can be shown by analysis to provide unambiguous information.

J. Periodic checking, testing, calibration, and calibration verification are performed in accordance with the applicable portions of Regulatory Guide 1.118, Periodic Testing of Electric Power and Protection Systems.

K. The range selected for the instrumentation encompasses the expected operating range of the variable being monitored to the extent that saturation does not negate the required action of the instrument in accordance with the applicable portions of Regulatory Guide 1.105, Instrument Setpoints.

7.5.2.3.1.4 Information Processing and Display Interface Criteria for Category 1. The interface criteria specified here provide requirements to be implemented in the processing and displaying of the information.

A. The control room operating staff have immediate access to the information from redundant or diverse channels in units of measure familiar to the staff; i.e. for temperature readings, degrees should be used, not volts. Where two or more

instruments are needed to cover a particular range, overlapping instrument spans are provided.

B. A historical record of at least one instrumentation channel for each process variable is maintained. A recorded pre-event history for these channels is required for a minimum of 1 h, and continuous recording of these channels is required following an accident until continuous recording of such information is no longer deemed necessary. The term "continuous recording" is not intended to exclude the use of discrete time sample data storage systems. This recording is available when required and does not need to be immediately accessible.

The time period of 1 h was selected based on a representatively slow transient which bounds this time requirement. A 1/2-in.- equivalent break area LOCA was selected since the reactor trip occurs at approximately 50 min after the break. Where direct and immediate trend or transient data is essential for operator information or action, the recording is immediately accessible.

7.5.2.3.2 Category 2

7.5.2.3.2.1 Selection Criteria for Category 2. The selection criteria for Category 2 variables are subdivided according to the variable type. For types A, B, and C, those variables which provide preferred backup information are designated Category 2. For type D, those key variables that are used for monitoring the performance of safety systems have been designated Category 2. For type E, those key parameters to be monitored for use in determining the magnitude of the release of radioactive materials and for continuously assessing such releases have been designated Category 2.

7.5.2.3.2.2 Qualification Criteria for Category 2. Category 2 instrumentation is qualified from the sensor up to and including the channel isolation device for at least the environment (seismic and/or environmental) in which it must operate to serve its intended function.

7.5.2.3.2.3 Design Criteria for Category 2.

A. Category 2 instrumentation that is required to operate following a safe shutdown earthquake to mitigate a consequential plant incident is energized from a seismically qualified power source, which is battery backed where momentary interruption is not tolerable. The instrumentation required to function after a seismic event is the safety-related cold shutdown instrumentation described in section 7.4. Otherwise, the instrumentation is energized from a highly reliable onsite power source, not necessarily the emergency standby power, which is battery backed where momentary interruption is not tolerable.

B. The out-of-service interval is based on paragraph 7.5.2.4, Post Accident Monitoring Program.

C. Servicing, testing, and calibration programs are specified to maintain the capability of the monitoring instrumentation. For those instruments where the required interval between testing is less than the normal time interval between generating station shutdowns, a capability for testing during power operation is provided.

D. Whenever means for removing channels from service are included in the design, the design facilitates administrative control of the access to such removal means.

E. The design facilitates administrative control of the access to all setpoint adjustments, module calibration adjustments, and test points.

F. The monitoring instrumentation design minimizes the potential for the development of conditions that would cause meters, annunciators, recorders, and alarms, etc., to give anomalous indications that could be potentially confusing to the operator.

G. The instrumentation is designed to facilitate the recognition, location, replacement, repair, or adjustment of malfunctioning components or modules.

H. To the extent practicable, monitoring instrumentation inputs are from sensors that directly measure the desired variables. An indirect measurement is made only when it can be shown by analysis to provide unambiguous information.

I. Periodic checking, testing, calibration, and calibration verification are in accordance with applicable portions of Regulatory Guide 1.118, Periodic Testing of Electric Power and Protection Systems.

J. The range selected for the instrumentation encompasses the expected operating range of the variable being monitored to the extent that saturation does not negate the required action of the instrument in accordance with the applicable portions of Regulatory Guide 1.105, Instrument Setpoints.

7.5.2.3.2.4 Information Processing and Display Interface Criteria for Category 2. The instrumentation signal is, as a minimum, processed for display on demand. Recording requirements are variable specific and are determined on a case-by-case basis.

7.5.2.3.3 Category 3

7.5.2.3.3.1 Selection Criteria for Category 3. The selection criteria for Category 3 variables have been subdivided according to the variable type. For types B and C, those variables which provide backup information have been designated Category 3. For types D and E, those variables which provide preferred backup information have been designated Category 3. There are no Category 3 type A variables.

7.5.2.3.3.2 Qualification Criteria for Category 3. The instrumentation is high quality, commercial grade which is not required to provide information when exposed to a post-accident adverse environment.

7.5.2.3.3.3 Design Criteria for Category 3.

A. Servicing, testing, and calibration programs are specified to maintain the capability of the monitoring instrumentation. For those instruments where the

required interval between testing is less than the normal time interval between generating station shutdown, a capability for testing during power operation is provided.

B. Whenever means for removing channels from service are included in the design, the design facilitates administrative control of the access to such removal means.

C. The design facilitates administrative control of the access to all setpoint adjustments, module calibration adjustments, and test points.

D. The monitoring instrumentation design minimizes the potential for the development of conditions that would cause meters, annunciators, recorders, and alarms, etc., to give anomalous indications that could be potentially confusing to the operator.

E. The instrumentation is designed to facilitate the recognition, location, replacement, repair, or adjustment of malfunctioning components or modules.

F. To the extent practicable, monitoring instrumentation inputs are from sensors that directly measure the desired variables. An indirect measurement is made only when it can be shown by analysis to provide unambiguous information.

7.5.2.3.3.4 Information Processing and Display Interface Criteria for Category 3. The instrumentation signal is, as a minimum, processed for display on demand. Recording requirements are variable specific and are determined on a case-by-case basis.

7.5.2.3.4 Extended Range Instrumentation Qualification Criteria

The qualification environment for extended range instrumentation is based on the DBA events; the assumed maximum qualification value of the monitored variable shall be equal to the specified maximum range for the variable. The monitored variable is assumed to approach this peak by extrapolating the most severe initial ramp associated with the DBA events. The decay is considered proportional to the decay for this variable associated with the DBA events. No additional qualification margin needs to be added to the extended range variable. All environmental envelopes, except those pertaining to the variable measured by the information display channel, are those associated with the DBA events. The environmental qualification requirement for extended range instrument does not account for steady-state elevated levels that may occur in other environmental parameters associated with the extended range variable. For example, a sensor measuring containment pressure must be qualified for the measured process variable range (i.e., three times design pressure for concrete containments), but the corresponding ambient temperature is not mechanistically linked to that pressure. Rather, the ambient temperature value is the bounding value for DBA events analyzed in chapter 15. The extended range requirement is to ensure that the instrument will continue to provide information if conditions degrade beyond those postulated in the safety analysis. Since extended variable ranges are nonmechanistically determined, extension of associated parameter levels is not justifiable and is therefore not required.

7.5.2.4 Post Accident Monitoring Instrumentation Program

A program shall be maintained, the post accident monitoring instrumentation program, which ensures the capability to monitor plant variables and systems operating status during and following an accident. This program shall include those instruments provided to indicate system operating status and furnish information regarding the release of radioactive materials (Category 2 and 3 instrumentation as defined in Regulatory Guide 1.97 Revision 2) and provide the following:

A. Preventive maintenance and/or periodic surveillance of instrumentation.

B. Preplanned operating procedures and backup instrumentation to be used if one or more monitoring instruments become inoperable.

C. Administrative procedures for returning inoperable instruments to operable status as soon as practicable.

7.5.3 DESCRIPTION OF VARIABLES

7.5.3.1 Type A Variables

Type A variables are defined in paragraph 7.5.2.2.1. They are the variables which provide primary information required to permit the control room operating staff to:

A. Perform the diagnosis specified in the VEGP emergency operating procedures.

B. Take specified preplanned manually controlled actions for which no automatic control is provided that are required for safety systems to accomplish their safety function to recover from a design basis accident (DBA) event. (Verification of actuation of safety systems is excluded from type A and is included as type D.)

C. Attain and maintain a cold shutdown condition.

Key type A variables have been designated Category 1. These are the variables which provide the most direct measure of the information required. The key type A variables are:

- Reactor coolant system (RCS) wide-range (WR) pressure.
- WR hot leg reactor coolant temperature (T_{hot}).
- WR cold leg reactor coolant temperature (T_{cold}).
- WR steam generator level.
- Narrow-range (NR) steam generator level.
- Pressurizer level.
- Containment pressure.
- Steam line pressure.
- Containment water level (WR).

- Containment water level (NR).
- Condensate storage tank level.
- Refueling water storage tank level.
- Auxiliary feedwater flow.
- Containment radiation level (high range).
- Core exit temperature.
- Steam line radiation.
- RCS subcooling.

No type A variable has been designated Category 2 or 3. A summary of type A variables is provided in table 7.5.3-1.

7.5.3.2 Type B Variables

Type B variables are defined in paragraph 7.5.2.2.2. They are the variables that provide information to the control room operating staff to assess the process of accomplishing or maintaining critical safety functions, i.e.:

- Reactivity control.
- RCS pressure control.
- Reactor coolant inventory control.
- Reactor core cooling.
- Heat sink maintenance.
- Primary reactor containment environment.

Variables which provide the most direct indication (i.e., key variable) to assess each of the six critical safety functions have been designated Category 1. Preferred backup variables have been designated Category 2. These are listed in table 7.5.3-2.

7.5.3.3 Type C Variables

Type C variables are defined in paragraph 7.5.2.2.3. Basically, they are the variables that provide to the control room operating staff information to monitor the potential for breach or actual gross breach of:

- Incore fuel clad.
- RCS boundary.

- Containment boundary.

(Variables associated with monitoring of radiological release from the plant are included in type E.)

Those type C key variables which provide the most direct measure of the potential for breach of one of the three fission product boundaries have been designated Category 1. Backup information indicating potential for breach is designated Category 2. Variables which indicate actual breach and have been designated as preferred backup information are designated Category 2. All other backup variables have been designated Category 3.

Table 7.5.3-3 summarizes the selection of type C variables.

7.5.3.4 Type D Variables

Type D variables are defined in paragraph 7.5.2.2.4. They are those variables that provide sufficient information to the control room operating staff to monitor the performance of:

A. Plant safety systems employed for mitigating the consequences of an accident and subsequent plant recovery to attain a safe shutdown condition, including verification of the automatic actuation of safety systems.

B. Other systems normally employed for attaining a cold shutdown condition.

Type D key variables are designated Category 2. Preferred backup information is designated type D Category 3.

The following systems or major components have been identified as requiring type D information to be monitored:

A. Pressurizer level and pressure control (assess status of the pressurizer following return to normal pressure and level control under certain post-accident conditions).

B. Chemical and volume control system (CVCS) (employed for attaining a safe shutdown under certain post-accident conditions).

C. Secondary pressure and level control (employed for restoring/maintaining a secondary heat sink under post-accident conditions).

D. Emergency core cooling system (ECCS).

E. Auxiliary feedwater.

F. Containment systems.

G. Component cooling water (CCW).

H. Nuclear service cooling water.

I. Residual heat removal (RHR).

J. Heating, ventilation, and air-conditioning (HVAC) (if required for engineered safety features operation).

K. Electric power to vital safety systems.

L. Verification of automatic actuation of safety systems.

M. Reactor coolant system status.

N. Reactivity control.

Table 7.5.3-4 lists the key variables identified for each system listed above.

For the purpose of specifying seismic qualification for type D Category 2 variables, it is assumed that a seismic event and a break in Seismic Category 1 piping will not occur concurrently. As a result, the limiting event is an unisolated (single failure of a main steam isolation valve) break in Nuclear Safety Class 2 main steam piping. Instrumentation necessary to monitor this event and associated with the safety systems which are required to mitigate should be seismically qualified. Similarly, the environmental qualification of type D Category 2 variables depends on whether the instrumentation is subject to a high-energy line break when required to provide information.

7.5.3.5 Type E Variables

Type E variables are defined in paragraph 7.5.2.2.5. They are those variables that provide the control room operating staff with information to:

A. Monitor the habitability of control room.

B. Monitor the plant areas where access may be required to service equipment necessary to monitor or mitigate the consequences of an accident.

C. Estimate the magnitude of release of radioactive materials through identified pathways.

D. Monitor radiation levels and radioactivity in the environment surrounding the plant.

Key type E variables are qualified to Category 2 requirements. Preferred backup type E variables are qualified to Category 3 requirements.

Table 7.5.3-5 lists the key type E variables.

7.5.3.6 Plant Safety Monitoring System

The plant safety monitoring system (PSMS) is a microprocessor-based monitoring system used to process and output many of the Regulatory Guide 1.97, Revision 2 variables in proper format to internal plasma displays and external indicators, displays, cabinets and other equipment. The PSMS consists of three types of modular components: the remote processing unit, the display processing unit, and the plasma display. These components perform the data acquisition and processing, the data base consolidation and comparison, and the data selection and display, respectively.

The system is seismically and environmentally qualified and is configured to address single failure criteria. Qualification details are available in sections 3.10 and 3.11. In addition, the PSMS has the capability for online testing without affecting reactor protection and control.

The configuration of the PSMS on VEGP consists of a remote processing unit associated with each protection channel set and a remote processing unit assigned to the monitoring of non-Class 1E signals. Each remote processing unit acts independently to perform data acquisition, engineering unit conversion, and limit checking. Through this independence, the system is immune to common mode failures. The remote processing units associated with the four protection channel sets are powered by the same vital buses as the protection sets.

The remote processing units also provide an additional function: isolated data links from each remote processing unit serve as inputs to the integrated plant computer for transfer of the Regulatory Guide 1.97 data set.

The plasma display modules are redundant, qualified graphic/alphanumeric modules for displaying Category 1 and certain Category 2 variables on demand (as indicated in table 7.5.2-1). The displays have been human factor engineered in order to provide to the operator a concise display of plant conditions. Access to particular information is via functional keys integral to the PSMS. These displays will be used in conjunction with other control room instrumentation to monitor the VEGP throughout all operating conditions including anticipated operational occurrences and accident and post-accident conditions.

Additional discussions on the features of PSMS are provided in paragraphs 7.4.3.3, Alternate Shutdown Indication System, 7.7.2.7, Core Cooling Monitor, and 7.7.2.8, Reactor Vessel Level Instrumentation System.

7.5.4 ADDITIONAL INFORMATION

A cross-reference of variables and categories for each instrument identified in the VEGP survey is included in table 7.5.4-1.

Table 7.5.4-2 is included as a cross-reference to identify post-accident monitoring systems instruments utilized at VEGP which also address the recommendations of NUREG-0737. The instruments identified meet the intent of the guidance provided in NUREG-0737.

7.5.5 BYPASSED AND INOPERABLE STATUS INDICATION FOR ENGINEERED SAFETY FEATURES SYSTEMS

7.5.5.1 Description

In accordance with the guidance of Regulatory Guide 1.47, means are provided for automatic system level indication of the plant's engineered safety features (ESF) systems which are bypassed or inoperable. The system status monitoring panel (QBPS) serves the purpose of such indication and is located in the control room, next to the main control board. The QBPS panel is safety grade and seismically qualified and hence remains functional during and after a design basis event. However, no credit is taken in the accident analysis (chapter 15) for the QBPS indications being available to the operator. Most of the information displayed on the QBPS panel can be derived by the operator from other safety-grade instrumentation in the control room, such as:

- Individual ESF equipment status indicating lights and controls' position.
- ESF equipment monitor (light boxes on the main control board), light groups 1 through 5.
- Lights indicating the status and manual override of the automatic actuation signals for the control room and fuel handling building ESF heating, ventilation, and air-conditioning (HVAC) systems.

The bypasses that are applied manually to test-block the automatic safety actuation signal for either train of the two latter HVAC systems are monitored only on the QBPS panel.

Each ESF system monitored on the QBPS panel (table 7.5.5-1) has one monitoring light and an adjacent selector handswitch for each of its trains. The light-switch pairs belonging to each train are grouped together for easy train identification. Under normal circumstances, i.e., when no bypass condition has been detected, all monitoring lights are off. A detection of such condition in any monitored component in either train of an ESF system causes the corresponding light on the panel to illuminate. The engraving on the light readily identifies the bypassed system; the location of the light on the panel determines the train. Each system monitoring light can be illuminated either automatically or manually by its corresponding selector handswitch. The QBPS panel circuitry automatically detects any of the following conditions as applicable in each monitored component of the ESF systems:

- Loss of control power.
- Control handswitch in pull-to-lock position.
- Overcurrent lockout relay tripped (process-control loads).
- Circuit breaker not in operating position.
- Control transferred from the control room to a local panel.
- Manual block of the actuation of one safety train to test the other train.[(a)]
- Loss of power to the relay actuation logic.[(a)]
- Manual override of an automatic actuation signal.[(a)]
- Incorrect status of a hand-operated component, defeating the safety function of an ESF system.[(b)]

The components monitored by the system status monitoring circuits are the ESF pumps, fans, compressors, valves, dampers, and relay logic circuits.

In accordance with Regulatory Guide 1.47, the automatic system level indication of bypass and inoperable status is provided only for automatically actuated systems, including those systems that directly support the automatically initiated systems but may not be automatically initiated because they are normally in the operating mode. No automatic indication is provided for the bypasses that are expected to occur less frequently than once per year or when the system is not required to be operable. These may include such maintenance features as manual valves provided for isolation of equipment for repairs, electrical cable connections, or other manual disconnects. However, manual initiation of ESF equipment bypass indication on a system level basis is provided; each status monitoring light on the QBPS panel can be manually lit by its adjacent selector handswitch. Under administrative control, manual bypass indication can be set up or removed to further enhance the operator's awareness of the current status of the ESF systems. The automatic indication feature cannot be removed by operator action.

[a] Applies only to the control room and fuel-handling building ESF HVAC systems.
[b] Applies only to the reactor water storage tank main drain isolation valve No. 1204-207, which, if closed, disables the safety injection system.

The illumination of any monitoring light on the QBPS panel will activate the annunciator alarm on the main control board and will also be registered by the plant computer. Annunciator response procedures are discussed in subsection 13.5.2. In accordance with Institute of Electrical and Electronics Engineers (IEEE) Standard 384 and Regulatory Guide 1.75, proper isolation is provided between Class 1E circuits in different trains and between the Class 1E and non-Class 1E circuits.

The QBPS monitoring lights and circuits are powered from the same train as the ESF equipment they monitor. The operability of each train of the systems status monitoring system can be readily verified by pressing a test pushbutton for that train, which activates all status monitoring lights, annunciator, and the computer input in that train.

The availability of power to the ESF system status monitoring circuitry is indicated on the QBPS panel (one normally lit indicating light per train). Loss of power is immediately annunciated.

7.5.5.2 Conformance to Regulatory Guide 1.47

As required by Nuclear Regulatory Commission Regulatory Guide 1.47, the ESF status monitoring system comprising the QBPS panel and related circuitry provides the following functions:

A. Bypassed and inoperable status is automatically indicated at the system level for protection systems and systems actuated or controlled by protection systems (i.e. for primary ESF systems). Automatic status indication for primary ESF systems does not occur for the bypassing or inoperable condition of auxiliary or supporting systems which must be operable for the primary ESF systems to perform their safety-related functions. However, the intent of Position C.2 of RG 1.47 is met in that automatic indication is provided for bypassed and inoperable status for these auxiliary and supporting systems; and in that VEGP procedures will include steps to ensure that an operator, in responding to the annunciator alarm for inoperable indication of a support system, will manually activate the inoperable status indication for the appropriate primary ESF systems.

B. The automatic indication discussed in A is highly reliable and provided in the control room, while manual activation of the system level indicators is also provided in the control room.

C. Automatic indication is provided for all those bypasses or deliberately induced inoperable conditions that are expected to occur more frequently than once per year, have significant bearing upon the ability to perform safety functions, and are expected to occur when the affected system is normally required to be operable.

7.5.5.3 Conformance to Branch Technical Position ICSB-21

The guidelines set forth in Branch Technical Position ICSB-21, Revision 2, are complied with by the following:

A. Bypass indicators are arranged to enable the operator to determine the status of each safety system to determine if continued reactor operation is permissable.

B. The operator cannot cancel erroneous bypass indications.

C. The ESF status monitoring system is not used to perform functions that are essential to safety.

D. The ESF status monitoring system is designed and installed in a manner that precludes the possibility of adverse effects on plant safety systems and does not reduce the required independence between redundant safety systems.

E. The design of the ESF status monitoring system provides the capability of assuming its operable status during normal plant operation by means of verifying its indicating and annunciating functions.

F. The ESF status monitoring system is designed with shared system bypass condition(s) monitored in both units.

7.5.5.4 System Drawings

The logic and elementary diagrams pertaining to the ESF systems status monitoring system are included in the tables in subsection 1.7.1.

TABLE 7.5.2-1 (SHEET 1 OF 14)

POST-ACCIDENT MONITORING INSTRUMENTATION

Variable	Range/ Status	Type/ Category	Qualification Environmental	Qualification Seismic	Number of Instruments	Control Room Indication	Implementation Date	Power Supply	EOF Indication	TSC Indication	Conformance	ERDS Parameter(ae)
Reactor coolant pressure (wide range)	0 to 3000 psi	A1, B1, B2, C1, C2, D2	Yes	Yes	4 per unit	Plasma display 4 meters 1 dual recorder	Core load	1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2	Yes
RCS wide range T_{hot}	0 to 700 F	A1, B1, B2	Yes	Yes	1 per loop	Plasma display 4 meters 1 dual recorder	Complete	1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2	Yes
RCS wide range T_{cold}	0 to 700 F	A1, B1, B2	Yes	Yes	1 per loop	Plasma display 4 meters 1 dual recorder	Complete	1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2	Yes
Wide range steam generator water level	0 to 100 percent of span	A1, B1, B2, D2	Yes	Yes	1 per steam generator	Plasma display 4 meters 2 dual recorders	Complete	1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2	Yes
Narrow range steam generator water level	0 to 100 percent of span	A1, B1, D2	Yes	Yes	3 per steam generator	Plasma display 12 meters	Complete	1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2	No
Pressurizer level	0 to 100 percent of span	A1, B1, D2	Yes	Yes	3 per unit	Plasma display 3 meters 1 recorder	Complete	1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2	Yes
Containment pressure	0 to 75 psig	A1, B1, B2, C2, D2	Yes	Yes	4 per unit	Plasma display 4 meters 1 dual recorder	Complete	1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2	Yes

TABLE 7.5.2-1 (SHEET 2 OF 14)

Variable	Range/ Status	Type/ Category	Qualification Environmental	Qualification Seismic	Number of Instruments	Control Room Indication	Implementation Date	Power Supply	EOF Indication	TSC Indication	Conformance	ERDS Parameter[ae]
Steam line pressure	0 to 1300 psig	A1, B1, D2	Yes	Yes	3 per loop	Plasma display 12 meters 2 dual recorders	Complete	1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2	Yes
Refueling water storage tank level	0 to 100 percent of span	A1, D2	Yes	Yes	4 per unit	Plasma display 4 meters 1 dual recorder	Complete	1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2	Yes
Containment water level (narrow range)	0 to 48 in.[z]	A1, B1, B2, C2, D2	Yes	Yes	2 per unit	Plasma display	Core load	1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2	Yes
Containment water level (wide range)	0 to 120 in.[z]	A1, B1, B2, C2, D2	Yes	Yes	2 per unit	Plasma display 2 meters	Complete	1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2	Yes
Condensate storage tank level	0 to 100 percent of span	A1, B1, D2	Yes	Yes	2 per tank	Plasma display 4 meters 1 dual recorder	Complete	1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2	No
Auxiliary feedwater flow	0 to 600 gal/min	A1, B1, D2	Yes	Yes	2 per loop	Plasma display 4 meters	Core load	1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2	Yes
Containment radiation level (wide range)	1 to 10^8R/h	A1, B1, B2, E2	Yes	Yes	2 per unit	Safety-related display console	Complete	1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2	Yes
Steamline radiation monitor	10^{-1} to 10^3 μCi/cm^3	A1, B2, E2	Yes	Yes	1 per loop	Safety-related display console	Core load	1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2	Yes
Core exit temperature	200 to 2300 F	A1, B1, C1	Yes	Yes	2 per core quadrant per train	Plasma display	Core load	1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2	Yes
RCS subcooling	600 F subcooling to 350 F superheat	A1, B1, D2	Yes	Yes	2 per unit	Plasma display	Core load	1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2	Yes

TABLE 7.5.2-1 (SHEET 3 OF 14)

Variable	Range/ Status	Type/ Category	Qualification Environmental	Qualification Seismic	Number of Instruments	Control Room Indication	Implementation Date	Power Supply	EOF Indication	TSC Indication	Conformance	ERDS Parameter[ae]
Neutron flux (Extended range)	Lower: 10^{-1} to 10^{6} CPS Upper: 10^{-3} to 200 percent power	B1, D2	Yes	Yes	2 per unit	Plasma display	Core load	1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2[a g]	No
Reactor vessel water level	Upper range: 60-120 percent Full range: 0-120 percent Dynamic Head: 0-120 percent liquid	B1, C2, D2	Yes	Yes	2 per unit	Plasma display	Core load	1E	Yes	Yes	Conforms to Reg. Guide 1.97, Rev. 2	Yes
Containment isolation valve status	Open/ closed	C1,[ai] C2, D2	Yes	Yes	1 per valve	1 pair of[ak] lights per valve	Complete	1E	Yes	Yes	(a)	No
Containment hydrogen concentration	0 to 10 percent partial pressure	B3, C3	Yes	Yes	2 per unit	2 meters 1 recorder plasma display	Complete	1E	Yes	Yes	Conforms to Reg. Guide 1.97, Rev. 2	Yes
Containment pressure (extended range)	-5 to 160 psig	C1, C2	Yes	Yes	2 per unit	Plasma display	Core load	1E	Yes	Yes	Conforms to Reg. Guide 1.97, Rev. 2	Yes
Plant vent[t] level radiogas	10^{-6} to $10^{4}\mu Ci/cm^{3}$	C2, E2	Yes[o]	Yes[x]	1 per unit	Communication console	Core load	Non-1E[p]	Yes	Yes	Conforms to Reg. Guide 1.97, Rev. 2	Yes
Plant vent particulates and halogen (passive filters)	NA	C2, E2	Yes	Yes	NA	NA	Complete	NA	NA	NA	Conforms to Reg. Guide 1.97, Rev. 2	No

TABLE 7.5.2-1 (SHEET 4 OF 14)

Variable	Range/ Status	Type/ Category	Qualification Environmental	Qualification Seismic	Number of Instruments	Control Room Indication	Implementation Date	Power Supply	EOF Indication	TSC Indication	Conformance	ERDS Parameter[ae]
Auxiliary building radiation level (portable sampling)	NA	C3	No	No	NA	NA	Complete	NA	NA	NA	Conforms to Reg. Guide 1.97, Rev. 2	No
Site environmental radiation level (portable sampling)	NA	C3, E3	No	No	NA	NA	Complete	NA	NA	NA	Conforms to Reg. Guide 1.97, Rev. 2	No
RCS activity	N/A	C3	No	No	NA	No	Complete	NA	No	No	(b)	No
Reactor coolant Pump status	Running/ stopped	D2	Yes	Yes	1 per pump	1 pair of lights per pump plasma display	Complete	1E	Yes	Yes	(d)	No
Pressurizer pressure	1700/ 2500 psig	D2	Yes	Yes	4 per unit	4 meters	Complete	1E	Yes	Yes	(d)	No
Power-operated relief (PORV) valve status	Open/ closed	D2	Yes	Yes	1 per valve	1 pair of lights per valve	Complete	1E	Yes	Yes	Conforms to Reg. Guide 1.97, Rev. 2	No
Primary safety valve status	Open/ Closed	D2	Yes	Yes	1 per valve	Plasma display[q]	Core load	1E	Yes	Yes	Conforms to Reg. Guide 1.97, Rev. 2	No
Pressurizer heater current	0-800 amps	D2	Yes[o]	No	2 per unit	[q]	Core load	Non-1E UPS	Yes	Yes	Conforms to Reg. Guide 1.97, Rev. 2	No
Pressurizer relief tank temperature	50 to 350 F	D3	No	No	1 per unit	1 meter	8/1/86	Non-1E UPS	Yes	Yes	(w)	No
Charging system flow	0 to 200 gal/min	D2	Yes[o]	No	1 per path	1 meter	Core load	Non-1E UPS	Yes	Yes	Conforms to Reg. Guide 1.97, Rev. 2	Yes
Emergency charging flow	0 to 150 gal/min	D2	Yes	Yes	1 per path	2 meters	Core load	Non-1E UPS	Yes	Yes	Conforms to Reg. Guide 1.97, Rev. 2	No

TABLE 7.5.2-1 (SHEET 5 OF 14)

Variable	Range/ Status	Type/ Category	Qualification Environmental	Qualification Seismic	Number of Instruments	Control Room Indication	Implementation Date	Power Supply	EOF Indication	TSC Indication	Conformance	ERDS Parameter[ae]
Letdown flow	0 to 200 gal/min	D2	Yes[a]	No	1 per path	1 meter	Core load	Non-1E UPS	Yes	Yes	Conforms to Reg. Guide 1.97, Rev. 2	No
Emergency letdown	0 to 80 gal/min	D2	Yes[a]	No	1 per path	2 meters	Core load	Non-1E UPS	Yes	Yes	Conforms to Reg. Guide 1.97, Rev. 2	No
Volume control tank level	0 to 100 percent of span	D2	Yes[a]	No	1 per tank	1 meter	Complete	Non-1E UPS	Yes	Yes	Conforms to Reg. Guide 1.97, Rev. 2	No
Chemical and volume control system valve status	Open/ closed	D2	Yes	Yes	1 per valve	1 pair of[ab] lights per valve	Complete	1E/Non-1E UPS	Yes	Yes	(d)	No
Chemical and volume control system pump status	Running/ stopped	D2	Yes	Yes	1 per pump	1 pair of lights per pump	Complete	1E	Yes	Yes	(d)	No
Reactor coolant pump seal injection flow	0 to 20 gal/min	D2	Yes	Yes	1 per pump	4 meters	Complete	Non-1E UPS	Yes	Yes	(d)	No
Steam generator atmospheric PORV status	Open/ closed	D2	Yes	Yes	1 per valve	1 pair of lights per valve	Complete	Non-1E UPS	Yes	Yes	Conforms to Reg. Guide 1.97, Rev. 2	No
Main steam line isolation valve status	Open/ closed	B2, D2	Yes	Yes	1 per valve	1 pair of[ab] lights per valve	Complete	1E	Yes	Yes	(d)	No
Main steam isolation valve bypass isolation valve status	Open/ closed	B2, D2	Yes	Yes	1 per valve	1 pair of[ab] lights per valve	Complete	1E	Yes	Yes	(d)	No
Steam generator system status main steam flow	0 to 4.8 million b/h	D2	Yes	Yes	2 per steam generator	8 meters 4 recorders	Complete	Non-1E UPS	Yes	Yes	Conforms to Reg. Guide 1.97, Rev. 2	No

TABLE 7.5.2-1 (SHEET 6 OF 14)

Variable	Range/ Status	Type/ Category	Qualification Environmental	Qualification Seismic	Number of Instruments	Control Room Indication	Implementation Date	Power Supply	EOF Indication	TSC Indication	Conformance	ERDS Parameter[ae]
Main feedwater regulating valve status	Open/ closed	D2	Yes	Yes	1 per valve	1 pair of[ab] lights per valve	Complete	1E	Yes	Yes	(d)	No
Main feedwater bypass valve status	Open/ closed	D2	Yes	Yes	1 per valve	1 pair of[ab] lights per valve	Complete	1E	Yes	Yes	(d)	No
Main feedwater flow	0 to 4.8 million b/h	D2	Yes[o]	Yes[aa]	2 per steam generator	8 meters 4 recorders	Complete	Non-1E UPS	Yes	Yes	Conforms to Reg. Guide 1.97, Rev. 2	Yes
Steam generator blowdown isolation valve status	Open/ closed	D2	Yes	Yes	1 per valve	1 pair of lights per valve	Complete	1E	Yes	Yes	(d)	No
Steam generator sample line isolation valve status	Open/ closed	D2	Yes	Yes	1 per valve	1 pair of lights per valve	Complete	1E	Yes	Yes	(d)	No
High-head safety injection	0 to 1000 gal/min	D2	Yes	Yes	1 per valve	1 meter	Complete	1E	Yes	Yes	Conforms to Reg. Guide 1.97, Rev. 2	Yes
Low-head safety injection	0 to 800 gal/min	D2	Yes	Yes	1 per train	2 meters	Complete	1E	Yes	Yes	Conforms to Reg. Guide 1.97, Rev. 2	No
Main feedwater isolation valve status	Open/ closed	D2	Yes	Yes	1 per valve	2 pair of lights per valve	Complete	1E	Yes	Yes	(d)	No
Emergency core[u] cooling system valve status	Open/ closed	D2	Yes	Yes	1 per valve	1 pair of lights per valve	Complete	1E	Yes	Yes	(d)	No
Accumulator pressure	0 to 700 psig	D3	No	No	1 per tank	4 meters	Core load	Non-1E UPS	Yes	Yes	(aj)	No

TABLE 7.5.2-1 (SHEET 7 OF 14)

Variable	Range/ Status	Type/ Category	Qualification		Number of Instruments	Control Room Indication	Implementation Date	Power Supply	EOF Indication	TSC Indication	Conformance	ERDS Parameter[ae]
			Environmental	Seismic								
Safety injection pump status	Running/ stopped	D2	Yes	Yes	1 per pump	1 pair of lights per pump	Complete	1E	Yes	Yes	(d)	No
Auxiliary feedwater valve status	Open/ closed	D2	Yes	Yes	1 per valve	1 pair of lights per valve	Complete	1E	Yes	Yes	(d)	No
Feedwater isolation bypass valve status	Open/ closed	D2	Yes	Yes	1 per valve	1 pair of lights[ab] per valve	Complete	1E	Yes	Yes	(d)	No
Auxiliary feedwater pump status	Running/ stopped	D2	Yes	Yes	1 per pump	1 pair of lights per pump	Complete	1E	Yes	Yes	Conforms to Reg. Guide 1.97, Rev. 2	No
Containment spray valve status	Open/ closed	D2	Yes	Yes	1 per valve	1 pair of lights per valve	Complete	1E	Yes	Yes	(d)	No
Containment spray pump status	Running/ stopped	D2	Yes	Yes	1 per pump	1 pair of lights per pump	Complete	1E	Yes	Yes	(d)	No
Containment fan cooler damper position	Open/ closed	D2	Yes	Yes	1 per damper	1 pair of lights per damper	Complete	1E	Yes	Yes	(d)	No
Containment fan cooler breaker position	High/low speed, on/off	D2	Yes	Yes	1 per breaker	1 pair of lights per breaker	Complete	1E	Yes	Yes	(d)	No
Component cooling water header pressure	0 to 200 psig	D2	Yes[o]	Yes[aa]	1 per train	2 meters	Complete	Non-1E UPS	Yes	Yes	(d)	No
Component cooling water header temperature	0 to 300 F	D2	Yes[o]	Yes[aa]	1 per train	IPC	Complete	Non-1E UPS	Yes	Yes	Conforms to Reg. Guide 1.97, Rev 2	No

TABLE 7.5.2-1 (SHEET 8 OF 14)

Variable	Range/ Status	Type/ Category	Qualification Environmental	Qualification Seismic	Number of Instruments	Control Room Indication	Implementation Date	Power Supply	EOF Indication	TSC Indication	Conformance	ERDS Parameter[ae]
Component cooling water surge tank level	0 to 100 percent	D2	Yes[o]	Yes[aa]	1 per train	IPC	Complete	Non-1E UPS	Yes	Yes	(d)	No
Component cooling water flow to engineered safety features components	0 to 15,000 gal/min	D2	Yes[ah]	Yes[aa]	1 per train	2 meters	Complete	Non-1E UPS	Yes	Yes	Conforms to Reg. Guide 1.97, Rev 2	No
Component cooling water pump status	Running/ stopped	D2	Yes	Yes	1 per pump	1 pair of lights per pump	Complete	1E	Yes	Yes	(d)	No
Auxiliary component cooling water flow from RCP seals	0-400 gal/min	D2	Yes	Yes	2 per path	2 meters	Complete	1E	No	No	(d)	No
Nuclear service cooling water system flow	0 to 25,000 gal/min	D2	Yes[o]	Yes[aa]	1 per train	2 meters	Complete	Non-1E UPS	Yes	Yes	(d)	No
Nuclear service cooling water valve status	Open/ closed	D2	Yes	Yes	1 per valve	1 pair of lights per valve	Complete	1E	Yes	Yes	(d)	No
Nuclear service cooling water pump status	Running/ stopped	D2	Yes	Yes	1 per pump	1 pair of lights per pump	Complete	1E	Yes	Yes	(d)	No
Nuclear service cooling water fan status	On/off	D2	Yes	Yes	1 per fan	1 pair of lights per fan	Complete	1E	Yes	Yes	(d)	No

TABLE 7.5.2-1 (SHEET 9 OF 14)

Variable	Range/ Status	Type/ Category	Qualification Environmental	Qualification Seismic	Number of Instruments	Control Room Indication	Implementation Date	Power Supply	EOF Indication	TSC Indication	Conformance	ERDS Parameter[ae]
Heating, ventilation, and air-conditioning system status[v]	Open/ closed	D2	Yes	Yes	1 per damper	1 pair of lights per damper	Complete	1E	Yes[ac]	Yes[ac]	(d)	No
Engineered safety features (ESF) environment temperature	High/ normal	D2	Yes	Yes	1 per ESF component	1 alarm per channel	Complete	1E/ Non-1E	Yes[ad]	Yes[ad]	(d, n)	No
ESF environment cooler fan status	On/off	D2	Yes	Yes	1 per ESF component	1 pair of lights per channel	Complete	1E	Yes	Yes	(d)	No
ac, dc, vital instrument voltage	Bus specific	D2	Yes	Yes	1 per bus	1 meter per bus	Complete	1E	Yes	Yes	Conforms to Reg. Guide 1.97, Rev 2	No
Residual heat removal (RHR) heat exchanger discharge temperature	50 to 400 F	D2	Yes	Yes	1 per train	IPC	Core load	Non-1E UPS	Yes	Yes	Conforms to Reg. Guide 1.97, Rev 2	No
RHR flow	0 to 5000 gal/min	D2	Yes	Yes	1 per train	2 meters	Complete	1E	Yes	Yes	Conforms to Reg. Guide 1.97, Rev 2	Yes
RHR valve status	Open/ closed	D2	Yes	Yes	1 per valve	1 pair of lights per valve	Complete	1E	Yes	Yes	(d)	No
RHR pump status	Running/ stopped	D2	Yes	Yes	1 per pump	1 pair of lights per pump	Complete	1E	Yes	Yes	(d)	No
Control rod position indication	0 to 228 steps	D3	No	No	2 per control rod	Light-emitting diodes	Complete	Non-1E	No	No	Conforms to Reg. Guide 1.97, Rev 2	No
Reactor trip breaker position	Tripped/ not tripped	D2	Yes	Yes	1 per breaker	1 pair of lights per breaker	Complete	1E	Yes	Yes	(d)	No

TABLE 7.5.2-1 (SHEET 10 OF 14)

Variable	Range/ Status	Type/ Category	Qualification Environmental	Qualification Seismic	Number of Instruments	Control Room Indication	Implementation Date	Power Supply	EOF Indication	TSC Indication	Conformance	ERDS Parameter[ae]
SI actuation	Actuated/ not actuated	D2	Yes	Yes	1 per train	1 light per unit	Complete	1E	Yes	Yes	(d)	No
Turbine stop valve status	Closed/ not closed	D2	No[o]	No	1 per valve	1 light per valve	Complete	Non-1E	Yes	Yes	(d)	No
First stage turbine pressure	0 to 120% power	D2	Yes	Yes	1 per train	2 meters	Complete	1E	Yes	Yes	(d)	No
Plant vent airflow rate	Unit 1 3200 to 160,000 Sft^3/min Unit 2 2000 to 100,000 Sft^3/min	E2	Yes[o]	Yes[x]	1 per unit	(q)	Core load	Non-1E[p]		Yes	(y)	No
Condenser air ejector radiation	5 x 10 $10^5 \mu Ci/cm^3$	E3	No	No	1 per unit	Communication console	Complete	Non-1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2	Yes
Area radiation monitors												
Control room monitor	10^{-2} to 10^3 mr/h	E3	No	No	1 per plant	Communication console	Complete	Non-1E diesel-backed	Yes	Yes	(e)	No
Fuel handling building monitor	10^{-1} to 10^4 mr/h	E3	No	No	1 per plant	Communication console	Complete	Non-1E diesel-backed	Yes	Yes	(e)	No
Sampling room monitor	10^{-1} to 10^4 mr/h	E3	No	No	1 per plant	Communication console	Complete	Non-1E diesel-backed	Yes	Yes	(e)	No
Accident sampling capability												
Primary coolant and sump		E3	No	No	NA		Complete	NA			al.	No
• Grab Sample												

TABLE 7.5.2-1 (SHEET 11 OF 14)

Variable	Range/ Status	Type/ Category	Qualification Environmental	Qualification Seismic	Number of Instruments	Control Room Indication	Implementation Date	Power Supply	EOF Indication	TSC Indication	Conformance	ERDS Parameter[ae]
Containment air		E3	No	No	NA		Complete	NA			al.	No
• Grab Sample												
High level radioactive liquid tank	0 to 100 percent of span	D3	No	No	1 per tank	High level alarm	Complete	Non-1E	No	No	(r)	No
Radioactive gas holdup tank pressure	0 to 150 psig	D3	No	No	1 per tank	High pressure alarm	Complete	Non-1E	No	No	(s)	No
Meteorological parameters	NA	E3	No	No	20 per plant	[q]	Complete	Non-1E	Yes	Yes	Conforms to Reg Guide 1.97, Rev 2	Yes[af]
Containment sump radiation	NA	E3	No	No	NA	NA	Complete	NA	NA	NA	(i)	No
Containment atmospheric temperature											(f)	Yes
Containment sump water temperature	0-400 F	D2	Yes	Yes	1 per unit	1 meter	Core load	Non-1E	No	No	Conforms to Reg Guide 1.97, Rev 2	No
Heat removal by the containment fan system											(h)	No
Containment sump and containment atmospheric sampling											(i)	No
Boric acid charging flow											(j)	No
RCS extended range pressure											(k)	No

TABLE 7.5.2-1 (SHEET 12 OF 14)

Variable	Range/ Status	Type/ Category	Qualification Environmental	Qualification Seismic	Number of Instruments	Control Room Indication	Implementation Date	Power Supply	EOF Indication	TSC Indication	Conformance	ERDS Parameter[ae]
Accumulator tank level											(m)	No

a. Containment Isolation Valve Status - Containment isolation valves in series are provided with a single indication on each valve. The indications for the two valves in series are powered from different trains.

b. RCS Activity - Normal sampling system can be used to sample the RCS for later activity analysis.

c. Deleted

d. The plant specific study performed on VEGP indicated that these parameters were included in the minimum set of parameters necessary to monitor the performance of:

1. Plant safety systems employed for mitigating the consequences of an accident and subsequent plant recovery to attain a cold shutdown condition, including verification of the automatic actuation of safety systems.
2. Other systems normally employed for attaining a cold shutdown condition.

e. Area Radiation - VEGP will use health physics plant survey monitoring to determine radiation levels in the areas or locations where post accident access may be required or which may contain a significant post-accident radiation source.

f. Containment Atmospheric Temperature - The VEGP emergency response guidelines do not require the operator to take an action that would result in adverse consequences if the containment temperature was indicating an erroneous value. As such, the containment temperature indication is considered a Category 3 parameter.

g. Deleted

h. Heat Removal by the Containment Fan/Heat Removal System - Other parameters were designated as VEGP type D variables to demonstrate that the containment heat removal systems are operating properly. These include the following:

- Containment spray flow.
- Containment spray system valve status.
- Containment pressure.
- Containment water level.
- Containment spray pump status.
- Fan cooler damper position.
- Fan cooler breaker position.

i. Containment Sump and Atmospheric Sampling - Data obtained from containment sump and atmospheric sampling are not used by the operator to take any manual action to mitigate the consequences of an accident. Hence, these parameters are considered backup variables (Category 3) and the capability to obtain grab samples is maintained. See Note al.

j. Boric Acid Charging Flow - For monitoring the performance of the emergency core cooling system (ECCS), VEGP has designated refueling water storage tank level, high-head safety injection flow, low-head safety injection flow, containment water level, and ECCS valve status. Since the ECCS does not normally take suction from the boric acid tank (BAT), the boric acid charging flow was not designated a key variable. If the operator uses the BAT for boration following an accident, normal charging flow and reactor coolant system (RCS) sampling is used to demonstrate that the RCS is being borated.

k. RCS Extended Range Pressure - RCS wide-range pressure will be used.

l. Deleted

m. Reactor coolant system pressure indication and valve position indication for the accumulator discharge isolation and accumulator vent valves provide adequate status of the accumulators.

n. The environment/temperature for the diesel generator building is monitored by instrumentation that is designed for a mild environment, nonseismic, and powered from a non-Class 1E UPS.

o. These devices are backup verification to qualified system status parameters. These devices are purchased to perform in their anticipated service environments for the plant conditions for which they must function.

p. The alternate (auxiliary) power supply will be either battery-backed or powered by the diesel generator.

q. These parameters may be displayed on the integrated plant computer monitor display in the control room.

r. The liquid radwaste system is not required following an event. The containment normal sumps and the reactor coolant drain tank discharge lines are isolated by a containment isolation signal. These lines will be isolated subsequent to any LOCA. ECCS pumps are located in flood retaining rooms that prevent the spread of post-LOCA recirculation fluid if a pump were to fail. The recycle holdup tank level is alarmed (at 95 percent volume) in the control room as part of the liquid radwaste system trouble alarm. An operator would be dispatched to the local control panel to clear the annunciated signal.

s. The design pressure of each of the waste gas decay tanks (WGDT) is 150 psig. There are seven WGDTs per unit and two shared shutdown decay tanks. Each tank is provided with a pressure transmitter and a high pressure alarm in the control room as part of the gaseous waste processing system trouble alarm. An operator would be dispatched to the local control panel to clear the annunciated signal. The alarms for the WGDTs are set at 100 psig. All of the WDGTs are provided with relief valves set at or below the tanks design pressure. The relief valves for the WGDTs discharge to the shutdown decay tanks which are normally at low pressure. Should an extended discharge to the shutdown decay tank occur a high alarm would be received prior to lifting of the shutdown decay tank relief valve. The relief valves for the shutdown decay tanks discharge to the plant vent which is monitored by the plant vent monitor. Failure of one of these tanks was analyzed in subsection 15.7.1. Based upon the protection afforded by the installed tank relief valves and the potential eventual release to the plant vent, the gaseous waste processing system trouble alarm in the control room is adequate for providing information concerning the status of the WGDTs.

t. The containment effluent radioactivity is monitored by the plant vent monitor. The plant vent receives the discharges from the containment purge system, auxiliary building, control building, and the fuel handling building.

u. Accumulator isolation valve status is a part of the ECCS valve status.

v. Emergency ventilation damper position is part of the heating, ventilation, and air conditioning system status.

w. The upper range of pressurizer relief tank temperature indication was chosen to envelope the maximum expected temperature in the tank. The rupture disk on the relief tank has a blowout pressure of 92 psig, corresponding to a saturation temperature of 331 F.

x. Seismic qualification is limited to the isokinetic nozzles, flow transmitter, monitoring skid, and associated data processing module.

y. Additional flow instrumentation is available that provides a range of 3,000-190,000 Sft^3/min for Unit 1 and 1,200-120,000Sft^3/min for Unit 2.

z. Containment narrow range water level has a 0-48 in. instrument span starting at 6 in. above the sump bottom and ending at 6 in. above the top of the sump. Containment wide range water level has a 0-120 in. instrument span starting at 9 in. above the containment floor and ending at 10 ft.-9 in. above the containment floor.

aa. Applies to the seismic integrity of the pressure boundary only.

ab. Other position indication (via the system status monitoring panel and/or monitoring light boxes) is available if the pair of lights associated with the valve hand switch which is powered from the same source as the solenoid valve is lost due to intrusion of a harsh environment on the unsealed solenoid.

ac. The auxiliary feedwater turbine driven pump room pneumatic damper position is not available in the TSC and EOF. This damper receives an open signal on auxiliary feedwater turbine driven pump start and fails open in the event of loss of offsite power to provide ventilation via natural circulation. Damper position indication may be obtained from the control room.

ad. Indications of ESF environmental temperatures in the TSC and EOF are high alarms. High temperatures are of concern for equipment operability in these environments following an accident; therefore, alarms of low temperatures are not needed.

ae. Changes to these parameters may require NRC notification per 10 CFR 50, Appendix E, Section VI. Additional parameters transmitted, but not shown in the table, are NIS Source, Intermediate, and Power Range Detectors; RCS flows; liquid effluent radiation; CVCS letdown radiation; and Steam Generator Blowdown Radiation.

af. Meteorological parameters transmitted through ERDS are 10-meter wind speed, 10-meter wind direction, and 60- to 10-meter delta temperature.

ag. Due to post LOCA flooding of the reactor cavity/incore instrument tunnel, the extended range neutron flux monitors could possibly be submerged and thus become inoperable. These detectors are required from the time of event initiation, until a stabilized subcritical condition is established, return to critical is no longer credible, and manual sampling is established. This is consistent with Regulatory Guide 1.97, Revision 2 which indicates the function of the excore neutron flux detectors is "function detection; accomplishment of mitigation," and the function of boron concentration measurement is "verification"; and with NUREG-0737 (item II.B.3) which requires the capability to obtain and analyze boron samples within approximately 1 hour, i.e., well before the excore neutron detectors may become inoperable.

ah. Procured for mild environment.

ai. Only one channel of positive indication is provided per valve, but the NRC determined that with this one exception, containment isolation valve position indication meets all other category 1 criteria. See NUREG-1137.

aj. Per NRC letter from C. E. Carpenter, Jr. to C. K. McCoy dated November 19, 1993, the accumulator pressure variable is Category 3 not Category 2.

ak. Several containment isolation valves are operated by both the A-Train and the B-Train. In this case, the control room indication for these valves is 2 pair of lights per valve.

al. With License Amendments 123 and 101, the NRC staff approved elimination of the post-accident sampling system (PASS). However, pursuant to the NRC safety evaluation for WCAP-14986 which provided the basis for NRC approval of PASS elimination, contingency plans have been developed for obtaining and analyzing highly-radioactive samples of reactor coolant, containment sump liquid, and containment atmosphere.

TABLE 7.5.2-2

SUMMARY OF SELECTION OF CRITERIA

Type	Category 1	Category 2	Category 3
A	Key variables that are used for diagnosis or providing information necessary for operator action	Variables which provide preferred backup information	None
B	Key variables that are used for monitoring the process of accomplishing or maintaining critical safety functions	Variables which provide preferred backup information	Variables which provide backup information
C	Key variables that are used for monitoring the potential for breach of a fission product barrier	Variables which provide preferred backup information	Variables which provide backup information
D	None	Key variables which are used for monitoring the performance of plant systems	Variables which provide preferred backup information which are used for monitoring the performance of plant systems
E	None	Key variables to be monitored for use in determining the magnitude of the release of radioactive materials and for continuously assessing such releases.	Variables to be monitored which provide preferred backup information for use in determining the magnitude of the release of radioactive materials and for continuously assessing such releases.

TABLE 7.5.2-3

SUMMARY OF DESIGN, QUALIFICATION, AND INTERFACE REQUIREMENTS

Qualification	Category 1	Category 2	Category 3
Environmental	Yes	As appropriate (See paragraph 7.5.2.3.2.2.)	No
Seismic	Yes	As appropriate (See paragraph 7.5.2.3.2.2.)	No
Design			
Single failure	Yes	No	No
Power supply	Emergency diesel generator	Onsite	As required
Channel out of service	Technical Specifications	Technical LAM Specifications	As required
Testability	Yes	Yes	As required
Interface			
Minimum indication	Immediately access ble	Demand	Demand
Recording	Yes	As required (See paragraph 7.5.2.3.2.4.)	As required (See paragraph 7.5.2.3.3.4.)

(a) As defined in paragraph 7.5.2.4, Post Accident Monitoring Instrumentation Program.

TABLE 7.5.3-1

SUMMARY OF TYPE A VARIABLES

Variable	Variable Function	Type/ Category
RCS pressure (wide range (WR))	Key	A1
T_{hot} (WR)	Key	A1
T_{cold} (WR)	Key	A1
Steam generator level (WR)	Key	A1
Steam generator level (narrow range (NR))	Key	A1
Pressurizer level	Key	A1
Containment pressure	Key	A1
Steam line pressure	Key	A1
Containment water level (WR)	Key	A1
Containment water level (NR)	Key	A1
Condensate storage tank level	Key	A1
Refueling water storage tank level	Key	A1
Auxiliary feedwater flow	Key	A1
Containment area radiation level (WR)	Key	A1
Core exit temperature	Key	A1
Steam line radiation monitor	Key	A1
RCS subcooling	Key	A1

TABLE 7.5.3-2 (SHEET 1 OF 2)

SUMMARY OF TYPE B VARIABLES

Function Monitored	Variable	Variable Function	Type/ Category
Reactivity control	Extended range Neutron Flux	Key	B1
	WR T_{hot}	Backup (P)	B2
	WR T_{cold}	Backup (P)	B2
RCS pressure control	RCS pressure (WR)	Key	B1
	WR T_{hot}	Key	B1
	WR T_{cold}	Key	B1
	Containment pressure	Backup (P)	B2
	Containment area radiation (WR)	Backup (P)	B2
	Stream line radiation	Backup (P)	B2
Reactor coolant inventory control	Pressurizer level	Key	B1
	Reactor vessel water level	Key	B1
	Containment water level (NR)	Backup (P)	B2
	Containment water level (WR)	Backup (P)	B2
	WR steam generator level	Backup (P)	B2
Reactor core cooling	Core exit temperature	Key	B1
	RCS subcooling	Key	B1
	CST level	Key	B1
	Reactor vessel water level	Key	B1
	WR T_{hot}	Backup (P)	B2
	WR T_{cold}	Backup (P)	B2
	RCS pressure (WR)	Backup (P)	B2
Heat sink maintenance	NR steam generator level	Key	B1
	WR steam generator level	Key	B1
	Auxiliary feedwater flow	Key	B1
	Core exit temperature	Key	B1
	Steam line pressure	Key	B1
	Main steam line isolation and bypass valve status	Backup (P)	B2

TABLE 7.5.3-2 (SHEET 2 OF 2)

Function Monitored	Variable	Variable Function	Type/ Category
Containment environment	Containment pressure	Key	B1
	Containment area radiation	Key	B1
	Containment water level (NR)	Key	B1
	Containment water level (WR)	Key	B1
	Containment hydrogen concentration	Key	B1

TABLE 7.5.3-3

SUMMARY OF TYPE C VARIABLES

Function Monitored	Variable	Condition	Variable Function	Type/ Category
Incore fuel clad	Core exit temperature	Potential for breach	Key	C1
	Reactor vessel water level	Potential for breach	Backup (P)	C2
	RCS activity	Actual breach	Backup	C3
RCS boundary	RCS pressure (WR)	Potential for breach	Key	C1
	RCS pressure (WR)	Actual breach	Backup (P)	C2
	Containment pressure	Actual breach	Backup (P)	C2
	Containment water level (NR)	Actual breach	Backup (P)	C2
	Containment water level (WR)	Actual breach	Backup (P)	C2
Containment boundary	Containment pressure (extended range)	Potential for breach	Key	C1
	Containment hydrogen concentration	Potential for breach	Key	C1
	Plant vent radiation level	Actual breach	Backup (P)	C2
	Containment isolation valve status	Actual breach	Key	C2
	Containment pressure (extended range)	Actual breach	Backup (P)	C2
	Site environmental radiation	Actual breach	Backup	C3
	Auxiliary building radiation	Actual breach	Backup	C3

TABLE 7.5.3-4 (SHEET 1 OF 4)

SUMMARY OF TYPE D VARIABLES

System	Variable	Variable Function	Type/ Category
Pressurizer level and pressure control	Power-operated relief valve (PORV) status	Key	D2
	Safety valve status	Key	D2
	Pressurizer level	Key	D2
	RCS pressure (WR)	Key	D2
	Pressurizer heater power availability	Key	D2
	Pressurizer pressure	Key	D2
CVCS	Charging system flow	Key	D2
	Letdown flow	Key	D2
	Volume control tank level	Key	D2
	Seal injection flow	Key	D2
	CVCS valve status	Key	D2
Secondary pressure and level control	Steam generator atmospheric steam dump valve status	Key	D2
	Main steam flow	Key	D2
	Main steam isolation valve and bypass valve status	Key	D2
	Steam generator blowdown isolation valve status	Key	D2
	Steam line pressure	Key	D2
	Auxiliary feedwater flow	Key	D2
	Steam generator level (WR)	Key	D2
	Steam generator level (NR)	Key	D2
	Main feedwater control and bypass valve status	Key	D2
	Main feedwater isolation valve and bypass valve status	Key	D2
	Main feedwater flow	Key	D2
	SG sample line isolation valve status	Key	D2

TABLE 7.5.3-4 (SHEET 2 OF 4)

System	Variable	Variable Function	Type/ Category
Reactor coolant	RCS subcooling	Key	D2
System status	Reactor coolant pump status	Key	D2
	Reactor vessel water level	Key	D2
ECCS	Refueling water storage tank level	Key	D2
	HHSI and LHSI flow	Key	D2
	Containment water level (NR)	Key	D2
	Containment water level (WR)	Key	D2
	ECCS valve status	Key	D2
	Accumulator pressure	Backup	D3
Auxiliary feedwater	Auxiliary feedwater flow	Key	D2
	Auxiliary feedwater valve status	Key	D2
	Condensate storage tank level	Key	D2
Containment	Containment spray flow	Key	D2
	Containment water level (WR and NR)	Key	D2
	Containment spray valve status	Key	D2
	Containment spray pump status	Key	D2
	Containment pressure	Key	D2
	Containment fan cooler damper position	Key	D2
	Containment fan cooler breaker position	Key	D2
	Containment isolation valve status	Key	D2
	Containment sump water temperature	Key	D2
CCW	Header pressure	Key	D2
	Header temperature	Key	D2
	Surge tank level	Key	D2
	CCW flow	Key	D2
	CCW pump status	Key	D2
	A flow from RCP seals	Key	D2

TABLE 7.5.3-4 (SHEET 3 OF 4)

System	Variable	Variable Function	Type/ Category
Nuclear service cooling water system	Valve status	Key	D2
	System flow	Key	D2
	Fan status	Key	D2
	Pump status	Key	D2
Reactor coolant	RCS subcooling	Key	D2
System status	Reactor coolant pump status	Key	D2
	Reactor vessel water level	Key	D2
RHR	Heat exchanger discharge temperature	Key	D2
	Flow	Key	D2
	Valve status	Key	D2
	RCS pressure (WR)	Key	D2
	Pump status	Key	D2
HVAC	Environment for ESF components	Key	D2
	System status	Key	D2
	ESF environment cooler status	Key	D2
Electric power	ac/dc vital instrument voltage	Key	D2
Verification of automatic actuation of safety systems	Reactor trip breaker position	Key	D2
	Reactor trip bypass breaker position	Key	D2
	Rod position indication	Backup	D3
	SI activation	Key	D2
	Turbine stop valve position	Key	D2
	First-stage turbine pressure	Key	D2
	Main feedwater control bypass valve status	Key	D2
	Main feedwater isolation valve status	Key	D2
	Auxiliary feedwater pump status	Key	D2

TABLE 7.5.3-4 (SHEET 4 OF 4)

System	Variable	Variable Function	Type/ Category
	Safety injection pump status	Key	D2
	Nuclear service cooling water pump status	Key	D2
	CCW pump status	Key	D2
	Containment isolation valve status	Key	D2
	Containment fan cooler status	Key	D2
	RHR pump status	Key	D2
	Containment spray pump status	Key	D2
	CVCS pump status	Key	D2
Reactivity control system	Extended range neutron flux	Key	D2
	Control rod position indication	Backup	D3

TABLE 7.5.3-5

SUMMARY OF TYPE E VARIABLES

Variable	Variable Function	Type/ Category
Containment area radiation (WR)	Key	E2
Plant vent radiation level	Key	E2
Steam line radiation	Key	E2
Plant vent air flow rate	Key	E2
Condenser air ejector radiation	Backup (P)	E3
Area radiation monitors		
Control room monitor	Backup (P)	E3
Radiochemistry lab monitor	Backup (P)	E3
Fuel handling building monitor	Backup (P)	E3
Sampling room monitor	Backup (P)	E3
Decontamination station (large parts)	Backup (P)	E3
Decontamination station (small parts)	Backup (P)	E3
Instrument decontamination station	Backup (P)	E3
Site environmental radiation level	Backup (P)	E3
Meteorological parameters	Backup (P)	E3
Containment sump radiation	Backup (P)	E3

TABLE 7.5.4-1 (SHEET 1 OF 5)

SUMMARY OF VARIABLES AND CATEGORIES

	Type and Category				
Variable	Type A	Type B	Type C	Type D	Type E
Reactor coolant system (RCS) pressure (wide range (WR))	1	1,2	1,2	2	
WR T_{hot}	1	1,2			
WR T_{cold}	1	1,2			
WR steam generator level	1	1,2		2	
Narrow range (NR) steam generator level	1	1		2	
Pressurizer level	1	1		2	
Containment pressure	1	1,2	2	2	
Steam line pressure	1	1		2	
Refueling water storage tank level	1			2	
Containment water level (WR and NR)	1	1,2	2	2	
Condensate storage tank level	1	1		2	
Auxiliary feedwater flow	1	1			2
Containment radiation level (high range)	1	1,2			2
Steam line radiation monitor	1	2		2	
Core exit temperature	1	1		1	
RCS subcooling	1	1		2	
Condenser air ejector					3
Extended range neutron flux		1		2	

TABLE 7.5.4-1 (SHEET 2 OF 5)

Variable	Type and Category				
	Type A	Type B	Type C	Type D	Type E
Reactor vessel water level		1	2	2	
Containment isolation valve status			1,2	2	
Control rod position				3	
Containment hydrogen concentration		1	1		
Containment pressure (extended range)			1,2		
RCS activity			3		
Plant vent radiogas level			2		2
Auxiliary building radiation level (portable sample)			3		
Site environmental radiation level			3		3
Reactor coolant pump status				2	
Pressurizer pressure				2	
Power-operated relief valve (PORV) status				2	
Primary safety valve status				2	
Pressurizer heater current				2	
Pressurizer relief tank temperature				3	
Charging system flow				2	
Emergency charging flow				2	
Letdown flow				2	
Emergency letdown				2	
Volume control tank level				2	
Chemical and volume control system (CVCS) valve status				2	
CVCS pump status				2	

TABLE 7.5.4-1 (SHEET 3 OF 5)

Variable	Type and Category				
	Type A	Type B	Type C	Type D	Type E
Reactor coolant pump seal injection flow				2	
Steam generator atmospheric PORV status				2	
Main steam line isolation valve status		2		2	
Main steam line isolation valve bypass isolation valve status		2		2	
Steam generator system status-main steamflow				2	
Main feedwater control valve status				2	
Main feedwater control bypass valve status				2	
Main feedwater isolation bypass valve status				2	
Main feedwater isolation valve status				2	
Main feedwater flow				2	
Steam generator blowdown isolation valve status				2	
Steam generator sample line isolation valve status				2	
High-head safety injection flow				2	
Low-head safety injection flow				2	
Emergency core cooling system valve status				2	
Accumulator pressure				3	
Auxiliary feedwater valve status				2	
Containment spray valve status				2	
Containment spray pump status				2	

TABLE 7.5.4-1 (SHEET 4 OF 5)

Variable	Type and Category				
	Type A	Type B	Type C	Type D	Type E
Containment fan cooler damper position				2	
Containment fan cooler breaker position				2	
Component cooling water (CCW) header pressure				2	
CCW header temperature				2	
CCW surge tank level				2	
CCW flow				2	
Auxiliary component cooling water from RCP seals				2	
Nuclear service cooling water system flow				2	
Nuclear service cooling water system valve status				2	
Residual heat removal (RHR) heat exchanger discharge temperature				2	
RHR flow				2	
RHR valve status				2	
RHR pump status				2	
Engineered safety features (ESF) environment temperature				2	
ESF environment cooler status				2	
Heating, ventilation, and air-conditioning system status				2	
ac and dc vital instrument voltage				2	
SI actuation				2	
Reactor trip breaker position				2	

TABLE 7.5.4-1 (SHEET 5 OF 5)

Variable	Type and Category				
	Type A	Type B	Type C	Type D	Type E
Turbine stop valve position				2	
First-stage turbine pressure				2	
Auxiliary feedwater pump status				2	
Safety injection pump status				2	
Nuclear service cooling water pump status				2	
Nuclear service cooling water fan status				2	
CCW pump status				2	
Area radiation					
Control room monitor					3
Fuel handling building area radiation					3
Sampling room monitor					3
Plant vent airflow rate					2
Meteorological parameters					3
Containment sump radiation					3
Accident sampling capability					3
Containment sump water temperature				2	

TABLE 7.5.4-2

NUREG-0737 CONFORMANCE

Applicable Section of NUREG-0737	Variable
II.D.3	Pressurizer PORV status
II.F.1, Attachment 4	Containment pressure (extended range)
II.F.1, Attachment 5	Containment water level (NR and WR)
II.F.1, Attachment 6(2)	Containment H@ concentration
II.F.2	Core exit temperature
	Reactor vessel level RCS subcooling
I.D.2	Safety parameter display system
II.E.1.2	Auxiliary feedwater flow
II.F.1, Attachment 3(1)	Containment area radiation (high range)
II.F.1, Attachment 2	Sampling and analysis of plant effluent (See section 11.5.)
II.F.1, Attachment 1	Noble gas effluent monitors (See section 11.5.)
II.K.1.5	ECCS and other system valve status

1. Calibration of high-range monitors is performed in accordance with the manufacturer's recommendation.

2. Accurate indication of containment hydrogen concentration is available to the operators within 90 minutes of initiating safety injection following a LOCA.

TABLE 7.5.5-1

ESF SYSTEMS MONITORED ON THE SYSTEM STATUS MONITORING PANEL

System Name	System No.	Monitored Train
Nuclear service cooling water system	1202	A, B
Component cooling water system	1203	A, B
Spent fuel pit cooling system	1213	A, B
Auxiliary component cooling water system	1217	A, B
Safety injection system	1204	A, B
Chemical and volume control system	1208	A, B
Auxiliary feedwater system (motor driven)	1302	A, B
Auxiliary feedwater system (turbine driven)	1302	C
Containment spray system	1206	A, B
Residual heat removal system	1205	A, B
Containment building air cooling system	1501	A, B
Essential chilled water system	1592	A, B
Auxiliary building ESF equipment room coolers and auxiliary feedwater pumphouse HVAC system	1555, 1593	A, B
Control building ESF electrical equipment room HVAC system	1532	A, B
Control building control room HVAC system	1531	A, B
Fuel handling building ESF HVAC system	1542	A, B
Piping penetration filtration and exhaust system	1561	A, B
Electrical tunnel ventilation system	1540	A, B
Diesel generator standby power system and diesel generator, fuel oil, air start and diesel generator building HVAC systems	1566, 1821, 2403	A, B
Containment hydrogen recombiner system and CTB post LOCA cavity purge system	1513, 1516	A, B

7.6 INTERLOCK SYSTEMS IMPORTANT TO SAFETY

7.6.1 INSTRUMENTATION AND CONTROL POWER SUPPLY SYSTEM

7.6.1.1 Description

The following is a description of the vital instrumentation and control power supply system. Further information is provided in paragraph 8.3.1.1.5.

A. Refer to drawing 1X3D-AA-G01A for a single-line diagram of the vital instrumentation and control power supply system.

B. There are six Class 1E inverters, six distribution panels, four manual transfer switches, and four Class 1E batteries, each with two battery chargers. The manual transfer switches installed for one of the two battery chargers per train allow for incoming power connection from a FLEX diesel generator following BDBEE (Beyond Desgin Basis External Event) but are normally aligned to their Class 1E power sources. Each inverter is connected independently to one distribution panel. Channels I and II have two inverters each; channels III and IV each have only one inverter.

C. The six Class 1E inverters provide a source of 120-V, 60-Hz power for the reactor protection system, the engineered safety feature actuation system, and the nuclear steam supply system control and instrumentation, the post-accident monitoring system, and the safety-related radiation monitoring system. The power for the channel I, II, III, and IV inverters is from the Class 1E 125-V dc Train A, B, C, and D station batteries, respectively. The station batteries ensure continued operation of instrumentation systems in the event of a station blackout.

D. Each distribution panel may be connected to a backup source of 120-V ac power. The tie is through a local, manually operated breaker, which is mechanically interlocked with the breaker connecting the inverter to the distribution panel such that the distribution panel cannot be connected to both sources simultaneously. The backup 120-V ac power is derived from the train A and B 480-V ac distributing system via 480-120-V regulating transformers that are qualified as isolation devices.

7.6.1.2 Analysis

Since the inverters for each of the four channels are connected to independent battery systems, a loss of a single dc bus can only affect the dc power supply to one of the four channels.

Each inverter is independently connected to its respective instrument distribution panel so that the loss of an inverter cannot affect more than one of the six distribution panels.

Therefore no single failure in the instrumentation and control power supply system or its associated power supplies can cause a loss of power to more than one of the redundant loads.

The inverters are designed to maintain their outputs within acceptable limits. The loss of the dc input is alarmed in the control room, as is the loss of an inverter's output.

A more complete description of the instrumentation and control power supply system, including physical separation and provisions to protect against fire, is discussed in chapter 8.

Based on the scope definitions presented in references 1 through 3, the criteria which are applicable to the instrumentation and control power supply system is Institute of Electrical and Electronics Engineers (IEEE) Standard 308-1974. The design is in compliance with IEEE Standard 308-1974 and Regulatory Guide 1.6. Availability of this system is continuously indicated by the operational status of the systems it serves (drawing 1X3D-AA-G01A) and is verified by periodic testing performed on the served systems. The inverters have been seismically and environmentally qualified as discussed in sections 3.10 and 3.11.

7.6.1.3 References

1. The Institute of Electrical and Electronic Engineers, Inc., "IEEE Criteria for Class 1E Electric Systems for Nuclear Power Generating Stations," IEEE Standard 308-1974.
2. The Institute of Electrical and Electronic Engineers, Inc., "IEEE Standard: Criteria for Protection Systems for Nuclear Power Generating Stations," IEEE Standard 279-1971.
3. The Institute of Electrical and Electronic Engineers, Inc., "IEEE Trial-Use Criteria for the Periodic Testing of Nuclear Power Generating Station Protection System," IEEE Standard 338-1975.

7.6.2 RESIDUAL HEAT REMOVAL ISOLATION VALVES

7.6.2.1 Description

The residual heat removal system (RHRS) isolation valves are normally closed and are only opened for residual heat removal (RHR) after system pressure is reduced to approximately 365 psig and system temperature has been reduced to approximately 350°F.

There are two motor-operated valves in series in each of the two RHR pump suction lines from the reactor coolant system (RCS) hot legs. The two valves nearest the RCS (valves HV8701B and HV8702B) are designated as the inner isolation valves, while the two valves nearest the RHR pumps (valves HV8701A and HV8702A) are designated as the outer isolation valves. The interlock logic provided for the isolation valves is shown in figure 7.6.2-1. Logic for the outer valves is identical to that provided for the inner isolation valves, except that equipment diversity is employed by virtue of the fact that the pressure transmitter set used for valve interlocks on the inner valves is manufactured differently from the pressure transmitter set used for the outer valve interlocks.

Each valve is interlocked so that it cannot be opened at the main control board unless the RCS pressure is below a preset pressure. This interlock prevents the valve from being opened at the main control board when the RCS pressure plus the RHR pump pressure would be above the RHRS design pressure. A second pressure interlock is provided which initiates a control room alarm to alert the operators if one or both of the valves are not fully closed and the RCS pressure is greater than 420 psig.

The interlock table for the inner and outer isolation valves is shown in table 7.6.2-1. Inner isolation valve HV8701B (in train C) and outer isolation valve HV8702A (in train D) are interlocked by valve position signals derived from stem-mounted switches on valves in the opposite train. The valves themselves are identified in table 7.6.2-1. These switches

(designated limit switches No. 2) are operated by the position of the valve stem and are separate from the switches supplied with the valve motor operator (designated No. 1). The motor-operated limit switch on each valve is connected to the same electrical train as the valve motor with which it is interlocked.

Reactor coolant system pressure control during low temperature operation is discussed in paragraph 5.2.2.10.

7.6.2.2 Analysis

Based on the scope definitions presented in references 1 and 2, these criteria do not apply to the RHR isolation valve interlocks; however, in order to meet Nuclear Regulatory Commission (NRC) requirements and because of the possible severity of the consequences of loss of function, the requirements of Institute of Electrical and Electronic Engineers (IEEE) Standard 279-1971 will be applied with the following comments:

A. For the purpose of applying IEEE Standard 279-1971 to this circuit, the following definitions will be used:

   1. Protection System

      The two valves in series in each line and all components of their interlocking and closure circuits.

   2. Protective Action

      The initiation of an alarm in the main control room when RCS pressures are above the preset value and one or both valves are open.

B. IEEE Standard 279-1971, Section 4.10

   The above mentioned pressure interlock signals and logic will be tested online to the maximum extent possible without adversely affecting safety. This test will include the analog signal through to the train signal which activates the slave relays. (The slave relays provide the final output signal to the valve control circuit.) This is done in the best interests of safety, since an actual actuation to permit opening the valve or to test the automatic closing of an open valve could potentially leave only one remaining valve to isolate the low-pressure RHRS from the RCS.

C. IEEE Standard 279-1971, Section 4.15

   This requirement does not apply, as the setpoints are independent of mode of operation and are not changed.

D. IEEE Standard 279-1971, Section 4.12

   This requirement does not apply to these interlocks because bypass of the interlocks at the local station is under strict administrative control with the valves power removed (breaker locked in the open position) to prevent unauthorized opening of the valves.

Environmental qualification of the valves and wiring is discussed in section 3.11.

In those cases where nuclear steam supply system-furnished valves for engineered safety features (table 7.6.2-1) are in one train with interlocking provisions by valve position signals derived from stem-mounted switches on valves in the opposite train, the field wiring as well as

motor control center wiring are in the balance of plant scope. They are wired and installed to ensure the following conditions:

A. Electrical independence of and separation of the switch(es) designated No. 2 (stem mounted) from the control circuit of the valve whose position is being sensed.

B. Separation of the field wiring of the valves switch(es) designated No. 2 (stem mounted) from the field wiring of the valves control circuit in accordance with IEEE Standard 384-1974.

Under the above conditions, a single failure analysis of either the controlling valves or the controlled valves train-oriented control circuitry will show that a single failure in the train-oriented control circuitry will not defeat the protective action of the redundant train-oriented control circuitry. Other than the cases where a stem-mounted limit switch No. 2 is used for position signals between valves assigned to the redundant train, there are no cases where interlocks mounted on equipment of one train are used in the circuitry of the redundant train.

7.6.2.3 References

1. The Institute of Electrical and Electronic Engineers, Inc., "IEEE Standard: Criteria for Protection Systems for Nuclear Power Generating Stations," IEEE Standard 279-1971.

2. The Institute of Electrical and Electronic Engineers, Inc., "IEEE Trial-Use Criteria for the Periodic Testing of Nuclear Power Generating Station Protection System," IEEE Standard 338-1975.

3. The Institute of Electrical and Electronics Engineers, Inc., "Trial-Use Standard Criteria for Separation of Class 1E Equipment and Circuits," IEEE Standard 384-1974.

7.6.3 REFUELING INTERLOCKS

Electrical interlocks (i.e., limit switches) as discussed in subsection 9.1.4 are provided for minimizing the possibility of damage to the fuel during fuel handling operations.

7.6.4 ACCUMULATOR MOTOR-OPERATED VALVES

The design of the interconnection of these signals to the accumulator isolation valve meets the following criteria established in previous Nuclear Regulatory Commission (NRC) positions on this matter:

A. Automatic opening of the accumulator valves when:

1. The primary coolant system pressure exceeds a preselected value (specified in the Technical Specifications).

2. A safety injection (SI) signal has been initiated.

Both signals shall be provided to the valves.

B. Utilization of an SI signal to automatically remove (override) any bypass features that are provided to allow an isolation valve to be closed for short periods of time when the reactor coolant system (RCS) is at pressure (in accordance with the

provisions of the Technical Specifications). As a result of the confirmatory SI signal, isolation of an accumulator with the reactor at pressure is acceptable.

The control circuit for these valves is shown in figure 7.6.4-1. The valves and control circuits are further discussed in subsections 6.3.2 and 6.3.5.

The SI system accumulator discharge isolation valves are motor- operated, normally open valves, which are controlled from the main control board. Technical Specifications require these valves to be open with power removed in mode 3 when RCS pressure is above 1000 psig.

These valves are also interlocked such that:

A. They open automatically on receipt of an SI signal with the main control board switch in either the "auto" or "close" position.

B. They open automatically whenever the RCS pressure is above the SI unblock pressure (P-11) specified in the Technical Specifications only when the main control board switch is in the "auto" position.

C. They cannot be closed as long as an SI signal is present.

The four main control board position switches for these valves provide a "spring return to auto" when placed in the open position and a "maintain position" when placed in the closed position.

The "maintain closed" position is required to provide an administratively controlled manual block of the automatic opening of the valve at pressure above the SI unblock pressure (P-11). The manual block or "maintain closed" position is required when performing periodic check valve leakage tests when the reactor is at pressure. The maximum permissible time that an accumulator valve can be closed when the reactor is at pressure is specified in the Technical Specifications.

Administrative control is required to ensure that any accumulator valve, which has been closed at pressures above the SI unblocking pressure, is returned to the "auto" position. Verification that the valve automatically returns to its normal full open position is also required.

During plant shutdown, the accumulator valves are closed. To prevent an inadvertent opening of these valves during that period, the accumulator valve motor circuit breakers are opened or withdrawn. Administrative control is again required to ensure that these valve circuit breakers are closed during the prestartup procedures.

Power lockout is provided for these valves as discussed in paragraph 8.3.1.1.11.

These normally open motor-operated valves have alarms, indicating a malpositioning (with regard to their emergency core cooling system function during the injection phase). The alarms sound in the main control room.

An alarm will sound for any accumulator isolation valve under the following conditions when the RCS pressure is above the SI unblocking pressure.

A. Valve motor-operated limit switch indicates valve not open.

B. Valve stem-operated limit switch indicates valve not open. The alarm on this switch will repeat itself at given intervals.

7.6.5 SWITCHOVER FROM INJECTION TO RECIRCULATION

The details of achieving cold leg recirculation following safety injection (SI) and after a loss-of-coolant accident (LOCA) are given in paragraph 6.3.2.8.

7.6.5.1 Description of Instrumentation Used for Switchover

Protection logic is provided to automatically open the two residual heat removal (RHR) containment emergency sump isolation valves (HV8811A in train A and HV8811B in train B) when two of four refueling water storage tank (RWST) level sensors indicate less than the low-low level setpoint in conjunction with the maintained SI signal. The SI signal is maintained by the contact of a slave relay in the solid-state protection system (SSPS) output cabinet that closes on SI and remains closed until manually reset from the control board. This manual reset switch is separate from the main SI reset switch, which is not associated with this circuit. The sump valve automatic open circuit reset switch permits the operator to remove the actuation signal if the corresponding sump isolation valve must be closed and retained in a closed position following a LOCA, e.g., for maintenance purposes.

7.6.5.2 Initiation Circuit

The two out of four low-low RWST level is the trip signal which, in coincidence with the SI signal, provides the initiation function to automatically open the containment sump isolation valves.

7.6.5.3 Logic

The logic function derived from the RWST level sensors and the SI signal is depicted on sheet 1, figure 7.6.5-1.

7.6.5.4 Bypass

The manual reset logic function is shown on sheet 2, figure 7.6.5-1; its purpose and action are described in paragraph 7.6.5.1. As noted, the SI signal is retained by latching it; it is not removed by action of the main SI reset used by the operator per emergency procedures to remove the SI signal to other equipment prior to realignment for switchover to the recirculation mode following a postulated LOCA.

7.6.5.5 Interlocks

The trip signal logic consists of four RWST water level transmitters, each of which provides a level signal to one of the four RWST level channel bistables. The RWST level channel bistables are:

- Normally deenergized.
- Deenergized on loss of power.
- Energized on low-low setpoint.

Each level channel bistable is assigned to a separate instrumentation and control power supply. A trip signal is provided from both train A and train B SSPS cabinets to the corresponding sump

isolation valves logic, should two of the four water level channel bistables receive a RWST level signal lower than the low-low level setpoint, following the generation of an SI signal.

The following valve in the A electrical train (HV8811A SI recirculation sump to RHR pump No. 1 isolation valve) will be interlocked with the following valve in the C electrical train (HV-8701B loop 1 RHR suction line from reactor coolant system isolation valve).

The following valve in the B electrical train (HV8811B SI recirculation sump to RHR pump No. 2 isolation valve) will be interlocked with the following valve in the D electrical train (HV-8702A loop 2 RHR suction line from RCS isolation valve).

Note that in the case of these RHR containment emergency sump isolation valves, the protective action is independent of the position of the interlocking valves. The sump valve is opened automatically by the two out of four low-low RWST water level signal and the control from the inner and outer RHR pump suction isolation valves, and RWST/RHR suction valve positions are bypassed. The valve position interlocking is used only during online testing of the SI recirculation sump isolation valves.

7.6.5.6 Sequence

This circuit is energized directly from the SSPS output cabinet and is not sequenced following an accident that requires its functioning. The SSPS output cabinet is powered from the Class 1E 120-V vital ac distribution panels as discussed in subsections 7.6.1 and 8.3.1. Power is available for operation of all valves on the first step of the diesel generator sequencing.

7.6.5.7 Redundancy

The function of this semiautomatic switchover is available from both train A and train B down to the actuated equipment. The function including the actuated equipment is therefore redundant, and train separation and independence is maintained from sensor to actuated equipment.

7.6.5.8 Diversity

Diversity of components and equipment between the redundant trains is not required to protect against systematic failures, such as multiple failures resulting from a credible single event. The associated components are environmentally and seismically qualified in accordance with the procedures described in sections 3.10 and 3.11. Function diversity is provided in that manual operation is available as a backup to the semiautomatic mode.

7.6.5.9 Actuated Devices

The actuated devices are the two motor control center starters, one for each of the motor-operated sump valves, HV8811A and HV8811B. These valves are the only equipment actuated by the semiautomatic switchover from injection to recirculation following a LOCA. The remaining sequence of switchover operations is manual and includes alignment of the appropriate miniflow valves. For the sequence of these manual operations, refer to table 6.3.2-7.

7.6.6 INTERLOCKS ISOLATING SAFETY SYSTEMS FROM NONSAFETY SYSTEMS

7.6.6.1 This paragraph has been deleted.

7.6.6.2 Refueling Water Storage Tank Isolation

7.6.6.2.1 Description

As described in detail in section 6.3, the refueling water storage tank (RWST) provides a source of water for the emergency core cooling operations. The RWST is a nuclear safety class, Seismic Category 1 structure. However, the sludge mixing pump and the electric circulation heater connected to the tank do not meet these qualification requirements; therefore, an isolation capability is provided to prevent a loss of the RWST water volume. Two train-oriented, air-operated, seismically qualified valves mounted in series on the suction line to the sludge mixing pump provide this capability. When closed, they isolate the safety-related portion of the line (connecting to the RWST) from its nonsafety-related, nonseismic portion connected to the sludge mixing pump. Both valves fail closed upon the loss of instrument air and/or control power. Each valve is automatically actuated to close upon a RWST low-level signal from a redundant level switch in its respective safety train. The isolation valves will remain closed until individually opened by the operator's manual action. Such action will result in the opening of the valve only if the RWST low-level signal is no longer present.

The capability for remote-manual isolation from the control room is also provided. The status of each isolation valve is displayed by the indicating lights in the control room.

The closure of either isolation valve will cause an automatic trip of the sludge mixing pump and the electric circulation heater due to low flow.

7.6.6.2.2 Initiating Circuits

Automatic initiation of the RWST isolation takes place when a low-level signal is received from the redundant level switches. For manual isolation, each isolation valve has a corresponding handswitch on the miscellaneous equipment panel in the control room.

7.6.6.2.3 Logic

The isolation logic is shown in drawing 1X5DN114-4.

7.6.6.2.4 Bypass

Neither bypass indication nor manual override of the automatic actuation are provided.

7.6.6.2.5 Interlocks

The isolation interlocks are shown in drawing 1X5DN114-4.

7.6.6.2.6 Sequencing

The solenoid valves controlling the RWST isolation valves are powered from a safety-related, 125-V dc, battery-backed power supply and are not sequenced. (See section 8.3.)

7.6.6.2.7 Redundancy

The RWST isolation equipment is fully redundant; two completely independent trains (A and B) are provided.

7.6.6.2.8 Diversity

Functional diversity is provided in that manual operation is available as a backup to the automatic mode.

7.6.6.2.9 Actuated Devices

The actuated devices are two solenoid valves (HY-10957 and HY-10958) controlling the air supply to the pneumatic actuators of the gate-type isolation valves (HV-10957 and HV-10958, respectively).

7.6.6.2.10 Supporting Systems

The following systems support the RWST isolation equipment:

- Class 1E, 125-V dc power supply.
- Instrument air systems.

Failure of either or both systems will cause immediate closure of the isolation valves and will not degrade the isolation function.

7.6.6.2.11 Analysis

Analysis is provided in paragraph 7.6.6.9.

7.6.6.2.12 Periodic Testing

Provisions for the periodic go testing of the actuation system at full power are discussed in the Technical Specifications.

7.6.6.3 This paragraph had been deleted.

7.6.6.4 Isolation of Reactor Coolant Pump Thermal Barrier Cooling Water

7.6.6.4.1 Description

As described in detail in subsection 9.2.8, the auxiliary component cooling water (ACCW) system cools the thermal barriers of the reactor coolant pumps. The portion of the ACCW system related to this function is safety-related and Seismic Category 1; it interconnects with the nonsafety-related and nonseismic portion of the system. An isolation valve (HV-2041) is provided separating the safety from nonsafety class, nonseismic portions of the thermal barriers ACCW discharge line. The valve is a safety-related, motor-operated gate valve interlocked in a manner that prevents a spill of the reactor coolant from the postulated, breached thermal barrier should a break occur in the nonsafety-related piping downstream of that valve. The closure of the valve is either automatic or manual. The high-pressure and high-flow signals derived from safety-related transmitters cause the actuation, which enhances the reliability of the isolation. The high-pressure signal is indicative of a reactor coolant pump thermal barrier breach. The high-flow signal can result from the same cause or from a pipe break in the nonsafety-related portion of the piping, downstream of the isolation valve. The capability for remote-manual isolation from the control room is also provided. The isolation valve can be manually opened at any time, but it will automatically close when the handswitch handle is released, unless both actuating signals have cleared. The valve status is displayed in the control room by the indicating lights.

7.6.6.4.2 Initiating Circuits

The automatic closure of the reactor coolant pumps' thermal barrier ACCW isolation valve will occur upon high line pressure or high line flow signal. Each signal is derived from a safety-related transmitter. Both transmitters belong to instrumentation channel 2 and are powered from train B. The valve's control circuit and motor are powered from the same train. The handswitch for manual control is located on the main control board in the control room.

7.6.6.4.3 Logic

The isolation logic is shown in drawing 1X5DN094-3.

7.6.6.4.4 Bypass

Bypass of the isolation valve resulting from a loss of the control power is indicated in the control room. To manually override the automatic actuation, the handswitch handle is maintained in the open position, keeping the valve open. When released, the handle returns to the neutral position; and the valve closes if either of the actuating signals is still present.

7.6.6.4.5 Interlocks

The isolation interlocks are shown in drawing 1X5DN094-3.

7.6.6.4.6 Sequencing

The isolation valve motor is sequenced on the first (0.5-s) step of load sequencing.

7.6.6.4.7 Redundancy

Valve HV-2041 is not redundant; however, upstream from that valve, each of the ACCW discharge lines from the individual reactor coolant pumps have isolation valves powered and actuated by safety train A. Since valve HV-2041 is powered and actuated by train B, the functional redundance is maintained.

7.6.6.4.8 Diversity

The reactor coolant pumps' thermal barrier ACCW isolation valve closes automatically upon receiving either of the high-pressure or high-flow signals, providing actuation diversity. Functional diversity exists in the capability for manual actuation provided as a backup for the automatic mode.

7.6.6.4.9 Actuated Devices

The actuated device is the valve's 480-V motor starter.

7.6.6.4.10 Supporting Systems

The isolation function relies on the operability of the safety-related, 480-V ac power supply system, train B.

7.6.6.4.11 Analysis

Analysis is provided in paragraph 7.6.6.9.

7.6.6.4.12 Instrumentation

Annunciator windows are provided in the control room to alert the operator on a high flow or high pressure signal from the reactor coolant thermal barrier and auxiliary component cooling water interface system. Separate annunciator windows for high flow are provided to distinguish between which reactor coolant pump thermal barrier has failed. Isolation valve status is provided in the control room by indicating lights.

7.6.6.4.13 Periodic Testing

Provisions for the periodic go testing of the actuation system at full power are discussed in the Technical Requirements Manual.

7.6.6.5 Electric Steam Boiler Isolation

The electric steam boilers and all associated equipment located in the electric steam boiler building have been removed. All other equipment associated with the electric steam boiler that is located in areas other than the electric steam boiler building has been abandoned in place. The steam lines from the electric steam boiler going to the boric acid batching tank, which is the only safety-related interface with this system, have been cut and capped near the tank.

7.6.6.6 Steam Generator Blowdown Isolation

7.6.6.6.1 Description

As discussed in paragraph 10.4.8.1, the steam generator blowdown system (SGBS) is designed to maintain optimum secondary side water chemistry during normal operation and during anticipated operational occurrences of main condenser inleakage or primary to secondary leakage. Because portions of the SGBS are classifed as high energy lines, design features are provided to rapidly isolate the blowdown path should a rupture occur in the system piping outside of the containment.

The two isolation valves (in series) per blowdown line are located inside containment, powered from different safety trains and are automatically closed by the signals generated by separate instrumentation channels. The isolation valves (HV 15212A through D and 15216A through D) are air-operated, globe valves and fail closed upon loss of instrument air and/or control power. The capability for remote-manual isolation is provided in the control room as a backup for the automatic mode. Attempts to manually open either valve will not succeed, unless all actuating signals have cleared. The status of the isolation valves is displayed in the control room by the indicating lights.

7.6.6.6.2 Initiating Circuits

Each blowdown isolation valve in any given steam generator blowdown line will automatically close when a high-flow signal or any one of the high-temperature signals are received. The high temperature signals are generated by sensors located in four different safety-related equipment rooms within the auxiliary building. There are four temperature sensors and one flow transmitter associated with each valve. The automatic actuation signals are trained, channelized, and derived from safety-related flow transmitters and resistance temperature detectors. The signals generated are indicative of a steam generator blowdown line break outside of containment. Each valve can be manually actuated from the control room by the handswitch mounted on the miscellaneous system/equipment panel.

7.6.6.6.3 Logic

The isolation logic is shown in drawings 1X5DN152-1 and 1X5DN152-2.

7.6.6.6.4 Bypass

Neither bypass indication nor manual override of the automatic actuation are provided for the steam generator blowdown isolation valves.

7.6.6.6.5 Interlocks

The interlocks are shown in drawings 1X5DN152-1 and 1X5DN152-2.

7.6.6.6.6 Sequencing

The solenoid valves controlling the pneumatic isolation valves are powered from a safety-related, 125-V-dc, battery-backed power supply and are not sequenced. (See section 8.3).

7.6.6.6.7 Redundancy

All insulation valves, logic, instrumentation, controls, and power supplies are fully redundant and arranged in two completely independent trains, A and B. Each train is capable of providing 100-percent isolation of the appropriate steam generator blowdown line. The signals actuating train A and B isolation valves are generated by the redundant sensors and processed by redundant circuitry in two independent channels (3 and 4, respectively) in the balance of plant process instrumentation.

7.6.6.6.8 Diversity

Isolation can be initiated on either high line flow or high temperature, providing actuation diversity function. Manual isolation capability is provided as a backup for the automatic mode, which provides functional diversity.

7.6.6.6.9 Actuated Devices

The actuated devices are two solenoid valves per steam generator blowdown line (HY-15212A through D and HY-15216A through D) controlling the air supply to the pneumatic actuators of the globe-type isolation valves (HV-15212A through D and HV-15216A through D, respectively).

7.6.6.6.10 Supporting Systems

The following systems support the steam generator blowdown line isolation equipment:

- Class 1E, 125-V dc power supply.

- Instrument air system.

Failure of either or both of those systems will cause immediate closure of the isolation valves and will not degrade the isolation function.

7.6.6.6.11 Analysis

Analysis is provided in paragraph 7.6.6.9.

7.6.6.6.12 Instrumentation

A common annunciator window is provided in the control room to alert the operator of high temperature in any of four equipment rooms. The specific room may be identified by means of redundant temperature indicators with a selector switch in the control room. Isolation valve status is provided in the control room by indicating lights.

7.6.6.6.13 Periodic Testing

Provisions for the periodic go testing of the actuation system at full power are discussed in the Technical Requirements Manual.

7.6.6.7 CVCS Letdown Line Isolation

7.6.6.7.1 Description

As described in paragraph 9.3.4.1.2.1, the CVCS letdown line functions to maintain a programmed water level in the RCS pressurizer, thus maintaining proper reactor coolant inventory. Because the CVCS letdown line is classified as high energy, design features are provided to rapidly isolate the letdown path should a rupture occur in the system piping outside of the containment.

The two isolation valves (in series) are located inside containment, powered from different safety trains and are automatically closed by the signals generated by separate instrumentation channels. The isolation valves (HV-15214 and HV-8160) are air-operated globe valves and fail closed upon loss of instrument air and/or control power. The capability for remote-manual isolation is provided in the control room as a backup for the automatic mode. Attempts to manually open either valve will not succeed, unless all actuating signals have cleared. The status of the isolation valves is displayed in the control room by the indicating lights.

7.6.6.7.2 Initiating Circuits

Each isolation valve in the CVCS letdown will automatically close when any one of the high temperature signals is received. Isolation valve HV-8160 will also automatically close upon receipt of a containment isolation signal (see FSAR Section 6.2.4). The high temperature signals are generated by sensors located in three different safety-related equipment rooms within the auxiliary building. There are three temperature sensors associated with each valve. The automatic actuation signals are trained, channelized, and derived from safety-related

resistance temperature detectors. The signals generated are indicative of a CVCS letdown line break outside of containment. Each valve can be manually actuated from the control room by the handswitch mounted on the main control board.

7.6.6.7.3 Logic

The isolation logic is shown in drawings 1X5DN151-1 and 1X5DN151-2.

7.6.6.7.4 Bypass

Neither bypass indication nor manual override of the automatic actuation are provided for the CVCS letdown line isolation valves.

7.6.6.7.5 Interlocks

The interlocks are shown in drawings 1X5DN151-1 and 1X5DN151-2.

7.6.6.7.6 Sequencing

The solenoid valves controlling the pneumatic isolation valves are powered from a safety-related, 125-V dc, battery-backed power supply and are not sequenced. (See section 8.3.)

7.6.6.7.7 Redundancy

All isolation valves, logic, instrumentation, controls, and power supplies are fully redundant and arranged in two completely independent trains, A and B. Each train is capable of providing 100-percent isolation of the CVCS letdown line. The signals actuating train A and B isolation valves are generated by the redundant sensors and processed by redundant circuitry in two independent channels (3 and 4, respectively) in the balance of plant process instrumentation.

7.6.6.7.8 Diversity

Isolation is initiated on high temperature for valve HV-15214 and either high temperature or an LI signal for HV-8160. Manual isolation capability is provided as a backup for the automatic mode, which provides functional diversity.

7.6.6.7.9 Actuated Devices

The actuated device is solenoid valve HY-15214 controlling the air supply to the pneumatic actuators of the globe-type isolation valve HV-15214 and solenoid valves HY-8160 and HY-15215 for valve HV-8160.

7.6.6.7.10 Supporting Systems

The following systems support the CVCS letdown isolation equipment:

- Class 1E, 125-V dc power supply.
- Instrument air system.

Failure of either or both of those systems will cause immediate closure of the isolation valves and will not degrade the isolation function.

7.6.6.7.11 Analysis

Analysis is provided in paragraph 7.6.6.9.

7.6.6.7.12 Instrumentation

A common annunciator window is provided in the control room to alert the operator of high temperature in any of three equipment rooms. The specific room may be identified by means of redundant temperature indicators with a selector switch in the control room. Isolation valve status is provided in the control room by indicating lights.

7.6.6.7.13 Periodic Testing

Provisions for the periodic go testing of the actuation system at full power are discussed in the Technical Requirements Manual.

7.6.6.8 Analysis

A. Conformance to Nuclear Regulatory Commission Regulatory Guides

1. Regulatory Guide 1.22

   Isolation controls and interlocks can be tested periodically.

2. Regulatory Guide 1.29

   Isolation equipment, circuitry, and instrumentation are designed to withstand the effects of an earthquake without loss of function.

   All related components are classified as Seismic Category 1, in accordance with the guide.

3. Regulatory Guide 1.62

   Isolation controls discussed previously provide manual actuation capability, in accordance with the applicable portions of the guide.

4. Regulatory Guide 1.75 and Institute of Electrical and Electronics Engineers (IEEE) Standard 384

   Those isolation circuits discussed above that are classified safety-related conform to the requirements of both the guide and the standard.

B. Conformance to IEEE Standard 279-1971

Those isolation circuits and devices discussed above that are classified safety-related conform to the applicable portions of the standard.

C. Conformance to Other Criteria and Standards

Conformance to other criteria and standards is indicated in table 7.1.1-1.

7.6.7 INTERLOCKS FOR REACTOR COOLANT SYSTEM PRESSURE CONTROL DURING LOW-TEMPERATURE OPERATIONS

The basic function of the reactor coolant system (RCS) pressure control during low-temperature operation is discussed in paragraph 5.2.2.1. As noted in paragraph 5.2.2.10, this pressure control includes semiautomatic actuation logic for two pressurizer power-operated relief valves (PORVs). The function of this actuation logic is to continuously monitor RCS temperature and pressure conditions, with actuation logic armed by operator action by means of an arm/block main control board switch, which is placed in the block position when the plant is at operating pressure. The monitored system temperature signals are auctioneered low; this auctioneered signal, when below the setpoint, is annunciated to direct the operator to arm the circuit. The auctioneered temperature is processed to generate monitored RCS pressure. This comparison provides an actuation signal to an actuation device which, if manually armed, causes the PORV to automatically open if necessary to prevent pressure conditions from exceeding allowable limits. (See drawings 1X6AA02-494 and 1X6AA02-495, for the logic diagram showing the basic elements used to process the generating station variables for this low-temperature RCS overpressurization preventive interlock. These two sheets present the logic diagram for the pressurizer pressure relief system for trains A and B that is part of the safety-grade cold shutdown system.)

The wide range temperature signals are used as input to generate the reference pressure limit program considering the plant's allowable pressure and temperature limits. This reference pressure is then compared to the actual RCS pressure monitored by the wide range pressure channel. The error signals derived from the difference between the reference pressure and the measured pressure first annunciates a main board alarm whenever the measured pressure approaches, within a predetermined amount, the reference pressure. On a further increase in measured pressure and after the operator has armed the circuit, the error signal generates an annunciated actuation signal channel; train independence between protection sets and between trains A and B is maintained from sensors to the PORVs.

Upon receipt of the actuation signal, the actuation device automatically causes the PORV to open. Upon sufficient RCS inventory letdown, the operating RCS pressure decreases, clearing the actuation signal. Removal of this signal from the actuation device causes the PORV to close.

7.6.7.1 Analysis of Interlock

The logic functions and actuation signals shown in drawings 1X6AA02-494 and 1X6AA02-495, are implemented in nuclear steam supply system protection equipment. For the criteria to which the protection system was designed and which apply equally well to the interlocks that are part of this protection system, see sections 7.2 and 7.3. The primary purpose of these interlocks is automatic transient mitigation. These interlocks do not perform a primary protective function but rather provide semiautomatic pressure control at low temperature as backup to the operator. However, to ensure a well-engineered design and improved operability, the instrumentation and control portions of the interlocks for RCS pressure control during low-temperature operation

satisfy applicable sections of Nuclear Regulatory Commission (NRC) Branch Technical Position RSB 5-2 that addresses instrumentation and control. Institute of Electrical and Electronics Engineers (IEEE) Standard 279-1971 is applied with the following comments:

A. For the purpose of applying IEEE Standard 279-1971 to this circuit, the following definitions are used:

1. Safety-Grade System

   The safety-grade system is the block valve and the PORV in series in each of the redundant lines and in all components of the interlocks for RCS pressure control during low-temperature operation. The instrumentation and control equipment for one redundant line is defined as the train A system; the instrumentation and control equipment for the other redundant line is defined as the train B system.

2. Safety Function

   The safety function is defined as the automatic control of the RCS pressure during low-temperature operation to prevent the actual pressure from exceeding the calculated reference pressure limit. This safety function is satisfied by either train of the redundant system, the train A system or the train B system.

B. IEEE Standard 279-1971, Paragraph 4.2

Any single random failure within the train A system or the train B system will not prevent protective action at the system level when required.

C. IEEE Standard 279-1971, Paragraph 4.10

The above mentioned pressure interlock signals and logic is tested online while the system's manually operated main control board arm/block switches are in the block position. This online at power testing will be done to the maximum extent possible without adversely affecting safety. This test will include the analog signal through to the associated output bistables in the process equipment. There is no practical design which will permit an integrated online test through to the final simultaneous openings of the PORV and the PORV block valves. Furthermore, the valves themselves are testable when the reactor is shut down.

D. IEEE Standard 279-1971, Paragraph 4.12

The safety function is manually blocked by operator action of the main control board arm/block switch which places it in the block position when the plant is at temperatures greater than the range of concern for RCS low-temperature operation. The annunciator initiated by the low-temperature auctioneered circuit will alarm to warn the operator that the arm/block switch should be placed in the arm position. Whether or not the system should be armed and actually is not armed will be indicated to the operator when this annunciator is initiated and the switch is positioned to the maintained block position. In addition, if the system is armed and the PORV block valve is not fully open, this condition is also annunciated.

7.6.7.2 Pressurizer Pressure Relief System

The interlocks described in paragraph 7.6.7.1, together with pressurizer pressure control and the pressurizer pressure control interlock shown in drawings 1X6AA02-235 and 1X6AA02-230, respectively, and the interlocks for the pressurizer block valves, 8000 A and B, shown on drawings 1X6AA02-494 and 1X6AA02-495, are referred to as the pressurizer pressure relief system.

The pressurizer pressure relief system provides the following:

A. Capability for RCS overpressure mitigation during cold shutdown, heatup, and cooldown operations to minimize the potential for impairing reactor vessel integrity when operating at or near the vessel ductility limits.

B. Capability for RCS depressurization following Condition II, III, and IV events.

C. Interlock that, with the pressurizer PORVs and PORV block valves in auto control, closes the PORV block valves and prevents signals from the pressurizer pressure control signal from opening the PORVs when the pressurizer pressure is low and the system is not manually armed.

Interlocks for the pressurizer pressure relief system control the valve positions of the pressurizer PORVs and the PORV block valves. The interlocks provide the following functions:

- Pressurizer pressure control.
- RCS pressure control during low-temperature operation.
- RCS pressure control to achieve and maintain a cold shutdown.

The interlock functions that provide pressurizer pressure control are derived from process parameters as shown in drawings 1X6AA02-230, 1X6AA02-235, 1X6AA02-494, and 1X6AA02-495. The functions shown in drawings 1X6AA02-494 and 1X6AA02-495 include those needed for the PORV block valves as well as the pressurizer PORVs to meet both interlock logic and manual operation requirements where manual operation can be either at the main control board or on the local shutdown panel.

TABLE 7.6.2-1

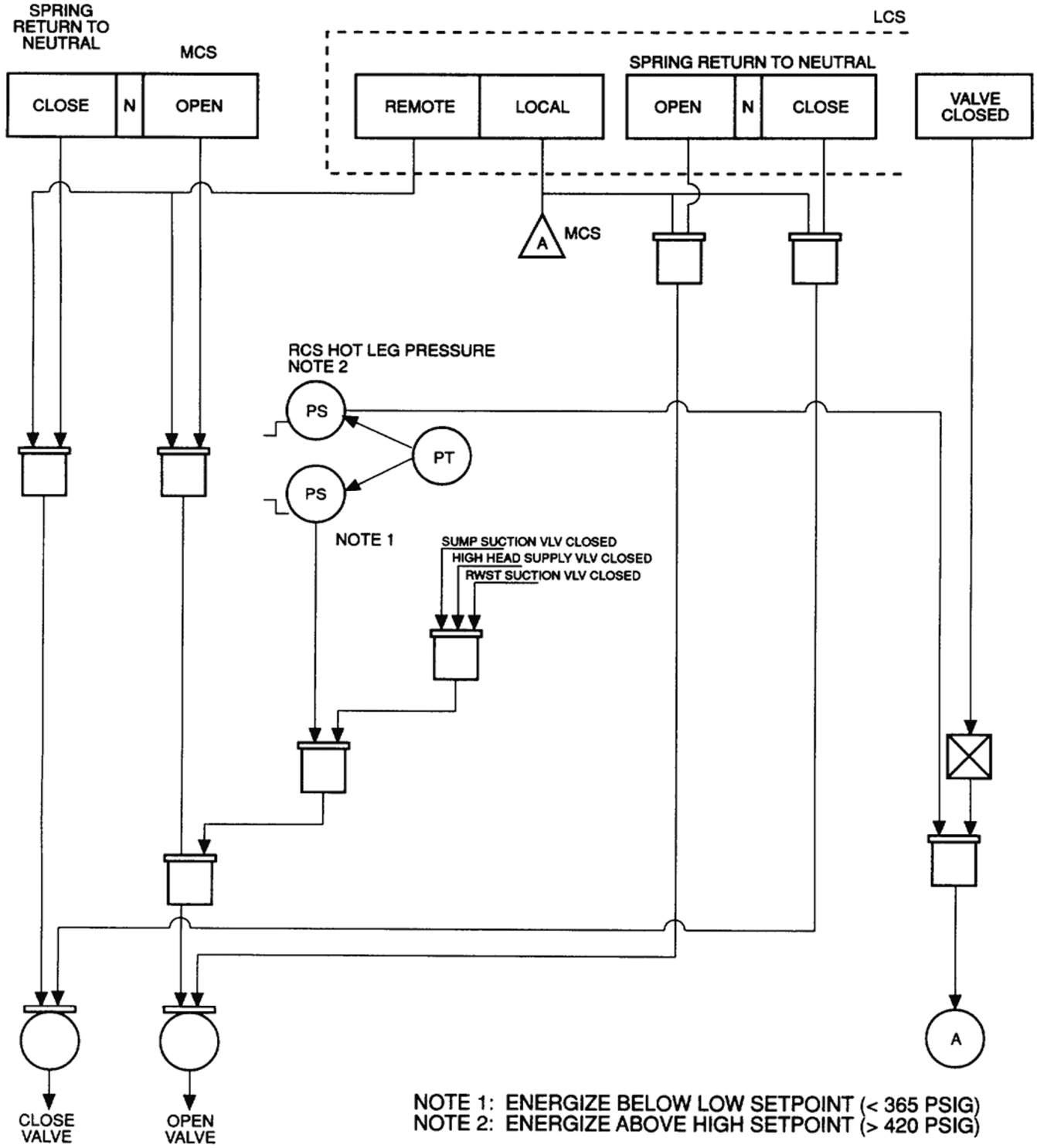
INTERLOCK TABLE FOR OUTER ISOLATION VALVES
(Refer to figure 7.6.2-1)

Outer Isolation Valve			HV8701A (Train A)	HV8702A (Train D)
Interlock	↓	Pressure transmitter →	PT 438	PT 418
Recirculation line valve	/	Limit switch[a]	HV8804A/No. 1[a]	HV8804B/No. 2[a]
Refueling water storage tank (RWST) isolation valve	/	Limit switch[a]	HV8812A/No. 1[a]	HV8812B/No. 2[a]
Sump line isolation valve	/	Limit switch[a]	HV8811A/No. 1[a]	HV8811B/No. 2[a]

INTERLOCK TABLE FOR INNER ISOLATION VALVES
(Refer to figure 7.6.2-1)

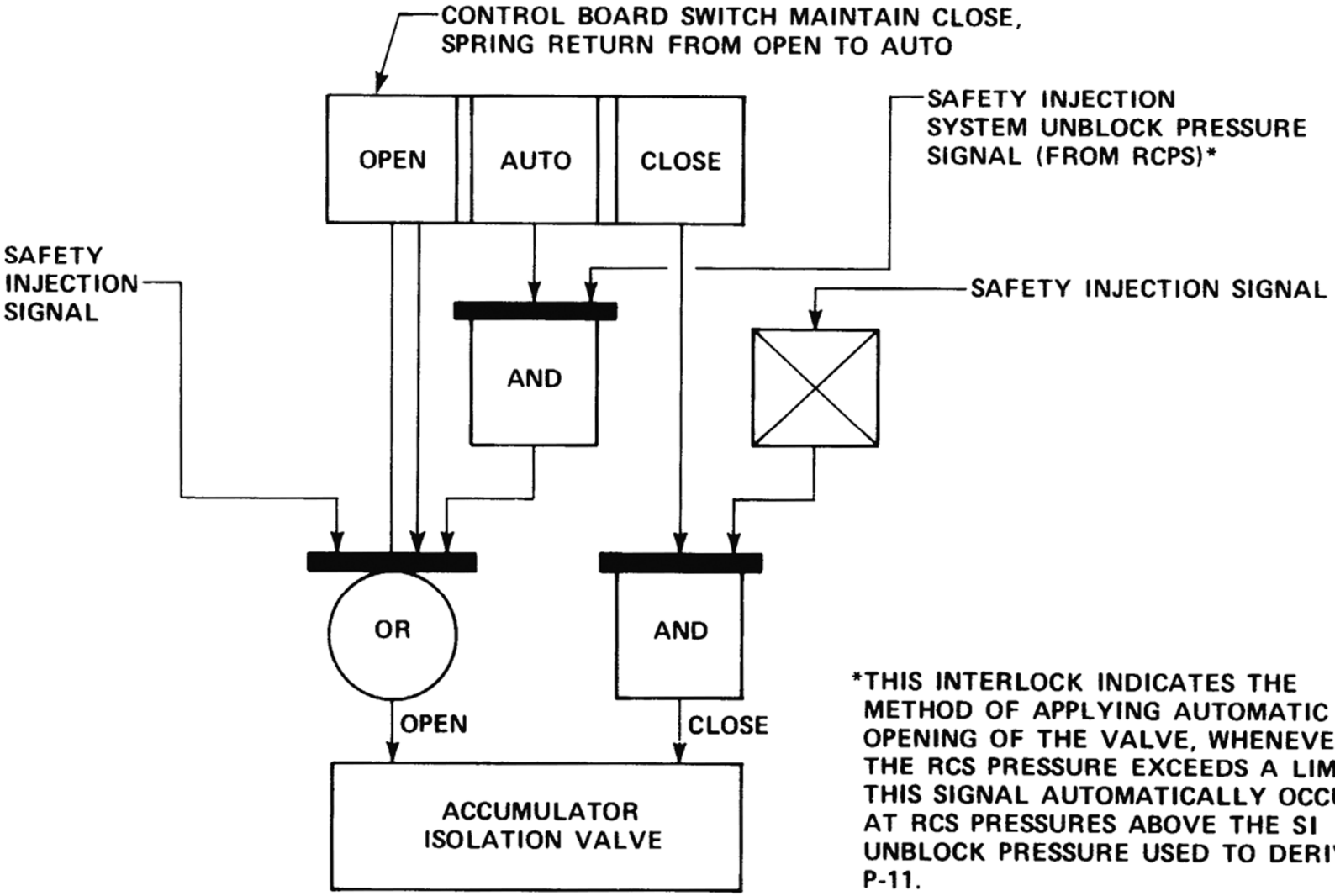
RCS-RHRS Inner Isolation Valve			HV8701B (Train C)	HV8702B (Train B)
Interlock	↓	Pressure transmitter →	PT 408	PT 428
Recirculation line valve	/	Limit switch[a]	HV8804A/No. 2[a]	HV8804B/No. 1[a]
RWST isolation valve	/	Limit switch[a]	HV8812A/No. 2[a]	HV8812B/No. 1[a]
Sump line isolation valve	/	Limit switch[a]	HV8811A/No. 2[a]	HV8811B/No. 1[a]

a. Limit switch No. 1 is a gear-driven limit switch supplied with a valve. Limit switch No. 2 is an added stem-mounted limit switch.



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SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	LOGIC DIAGRAM FOR THE RHRS ISOLATION VALVES
	FIGURE 7.6.2–1



REV 14 10/07

SOUTHERN COMPANY
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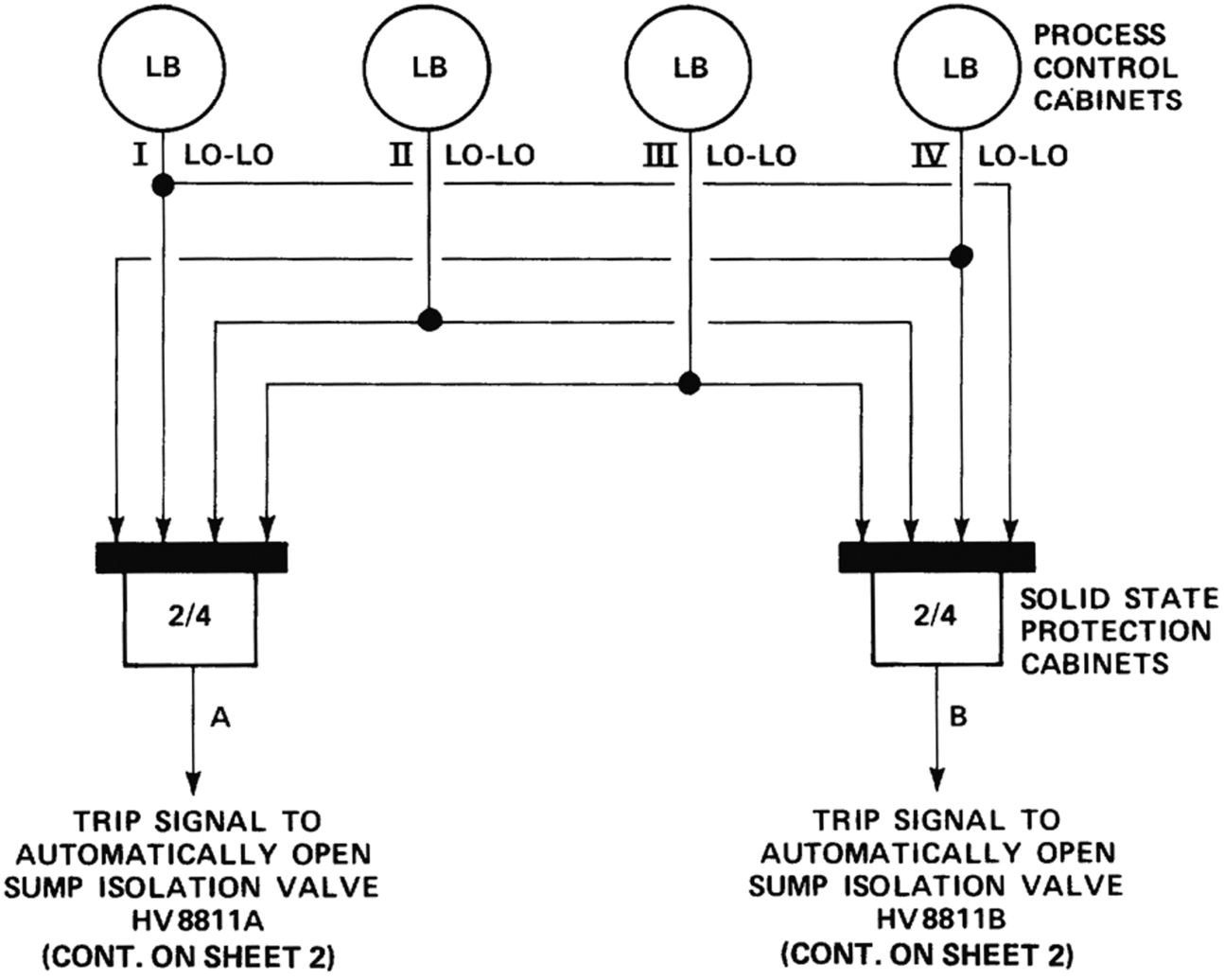
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

FUNCTIONAL BLOCK DIAGRAM
OF ACCUMULATOR ISOLATION VALVE

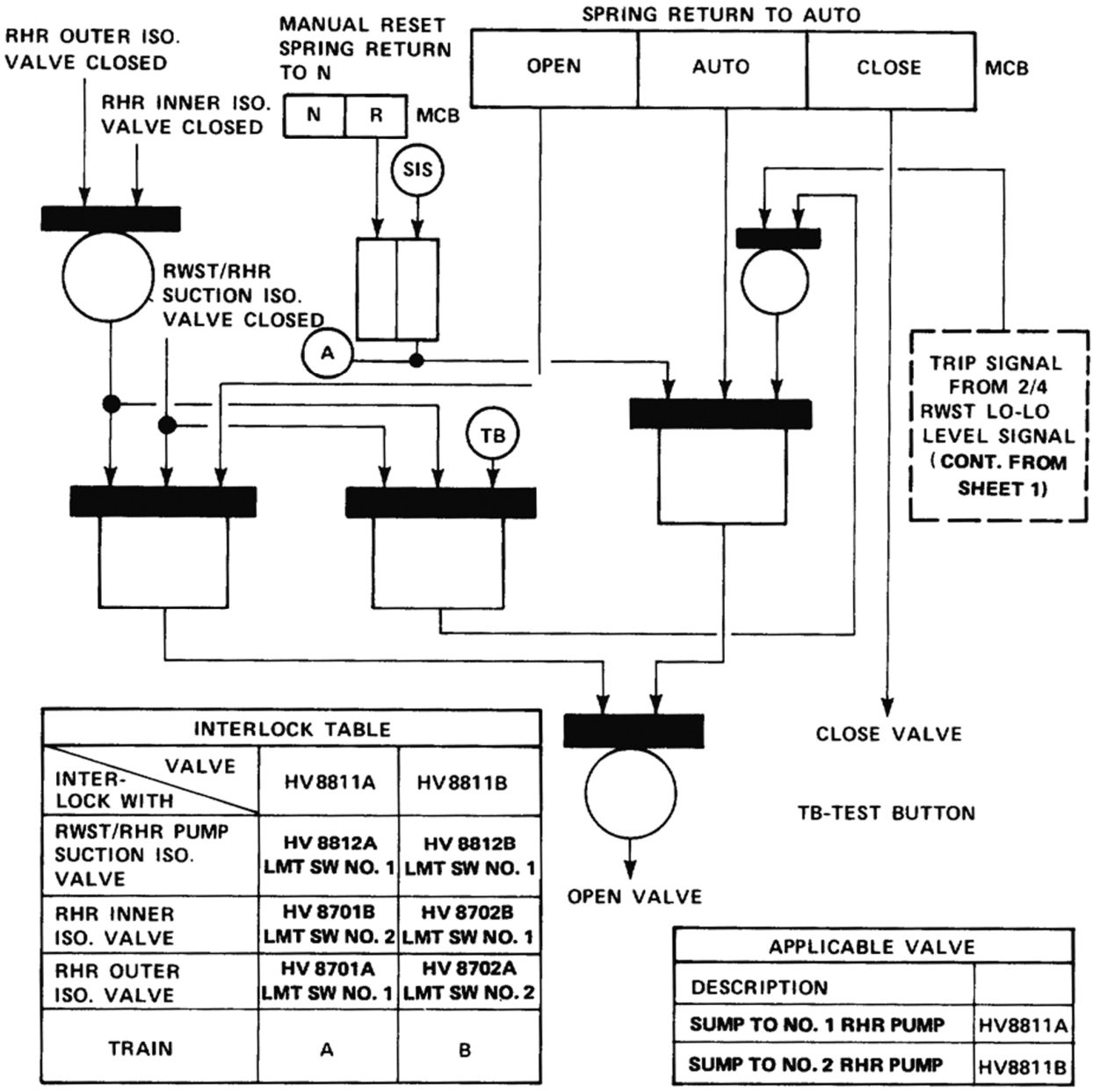
FIGURE 7.6.4–1

RWST LEVEL CHANNEL BISTABLES

1) NORMALLY DEENERGIZED
2) DEENERGIZED ON LOSS OF POWER
3) TRIP SIGNAL PROVIDED WHEN ENERGIZED
4) ENERGIZED ON LO-LO SETPOINT



REV 14 10/07



INTERLOCK TABLE		
VALVE / INTER-LOCK WITH	HV8811A	HV8811B
RWST/RHR PUMP SUCTION ISO. VALVE	HV 8812A LMT SW NO. 1	HV 8812B LMT SW NO. 1
RHR INNER ISO. VALVE	HV 8701B LMT SW NO. 2	HV 8702B LMT SW NO. 1
RHR OUTER ISO. VALVE	HV 8701A LMT SW NO. 1	HV 8702A LMT SW NO. 2
TRAIN	A	B

APPLICABLE VALVE	
DESCRIPTION	
SUMP TO NO. 1 RHR PUMP	HV8811A
SUMP TO NO. 2 RHR PUMP	HV8811B

LIMIT SWITCH NO. 1 IS THE NORMAL POSITION SIGNAL AND IS USED FOR POSITION SIGNALS BETWEEN VALVES ASSIGNED TO THE SAME TRAIN.
LIMIT SWITCH NO. 2 IS THE STEM MOUNTED POSITION SWITCH AND IT IS USED FOR POSITION SIGNALS BETWEEN VALVES ASSIGNED TO OPPOSITE TRAINS.

(A) ACTUATION SIGNAL LAMP

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SOUTHERN COMPANY
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VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

SAFETY INJECTION SYSTEM
RECIRCULATION SUMP AND RHR
SUCTION ISOLATION VALVES

FIGURE 7.6.5–1 (SHEET 2 OF 2)

7.7 CONTROL SYSTEMS NOT REQUIRED FOR SAFETY

The general design objectives of the plant control systems are:

A. To establish and maintain power equilibrium between the primary and secondary system during steady-state unit operation.

B. To constrain operational transients so as to preclude unit trip and to reestablish steady-state unit operation.

C. To provide the reactor operator with monitoring instrumentation that indicates all required input and output control parameters of the systems and provides the operator with the capability of assuming manual control of the system.

7.7.1 DESCRIPTION OF CONTROL SYSTEMS NOT REQUIRED FOR SAFETY

The plant control systems described in this section perform the following functions:

A. Reactor Control System

1. Enables the nuclear plant to accept a step-load increase or decrease of 10 percent and a ramp increase or decrease of 5 percent/min within the load range of 15 to 100 percent, without reactor trip, steam dump, or pressurizer relief actuation, subject to possible xenon limitations.

2. Maintains reactor coolant average temperature (T_{avg}) within prescribed limits by creating the bank demand signals for moving groups of full-length rod cluster control assemblies during normal operation and operational transients. The T_{avg} control also supplies a signal to pressurizer water level control and steam dump control.

B. Rod Control System

1. Provides for reactor power modulation by manual or automatic control (automatic rod insertion only) of full-length control rod banks in a preselected sequence and for manual operation of individual banks.

2. Includes systems for monitoring and indicating which:

- Provide alarms to alert the operator if the required core reactivity shutdown margin is not available because of excessive control rod insertion.

- Display control rod position.

- Provide alarms to alert the operator in the event of control rod deviation exceeding a preset limit.

C. Plant Control System Interlocks

1. Prevent further withdrawal of the control banks when signal limits are approached that indicate the approach to a departure from nucleate boiling ratio limit or kW/ft limit.

2. Limit automatic turbine load increase to values for which the nuclear steam supply system has been designed.

D. Pressurizer Pressure Control

1. Maintains or restores the pressurizer pressure to the design pressure ±35 psi (which is within reactor trip and relief and safety valve actuation setpoint limits) following normal operational transients that induce pressure changes by control (manual or automatic) of heaters and spray in the pressurizer.

2. Provides steam relief by controlling the power relief valves.

E. Pressurizer Water Level Control

Establishes, maintains, and restores pressurizer water level within specified limits as a function of the average coolant temperature. Changes in level are caused by coolant density changes induced by loading, operational, and unloading transients. Level changes are produced by means of charging flow control (manual or automatic), as well as by manual selection of letdown orifices. Maintaining coolant level in the pressurizer within prescribed limits by actuating the charging and letdown system thus provides control of the reactor coolant water inventory.

F. Steam Generator Water Level Control

1. Establishes and maintains the steam generator water level within predetermined limits during normal operating transients.

2. Maintains the steam generator water level within predetermined limits and unit trip conditions. It regulates the feedwater flowrate such that under operational transients the water level for the reactor coolant system does not decrease below a minimum value. Steam generator water inventory control is manual or automatic through the use of feedwater regulating valves.

G. Steam Dump Control

1. Permits the nuclear plant to accept a sudden loss of load without incurring reactor trip. Steam is dumped to the condenser as necessary to accommodate excess power generation in the reactor during turbine load reduction transients.

2. Ensures that stored energy and residual heat are removed following a reactor trip to bring the plant to equilibrium no-load conditions without actuation of the steam generator safety valves.

3. Maintains the plant at no-load conditions to permit a manually controlled cooldown of the plant.

H. Incore Instrumentation

Provides information on the neutron flux distribution and on the core outlet temperatures at selected core locations.

7.7.1.1 Reactor Control System

The reactor control system enables the nuclear plant to follow load changes automatically, including the acceptance of step-load increases or decreases of 10 percent and ramp increases or decreases of 5 percent/min within the load range of 15 to 100 percent without reactor trip, steam dump, or pressure relief (subject to possible xenon limitations). The system is also

capable of restoring coolant average temperature to within the programmed temperature dead band following a change in load. Automatic control rod insertion may be used for temperature control. However, rod withdrawal can only be performed manually. Manual control rod operation may be performed at any time within the range of defined insertion limits.

The reactor control system controls the reactor coolant average temperature by regulating control rod bank position. The reactor coolant loop average temperatures are determined from hot leg and cold leg measurements in each reactor coolant loop. There is an average coolant temperature (T_{avg}) computed for each loop, where:

$$T_{avg} = \frac{T_{hot} + T_{cold}}{2}$$

The error between the programmed reference temperature (based on turbine impulse chamber pressure) and the highest of the T_{avg} measured temperatures (which is processed through a lead-lag compensation unit) from each of the reactor coolant loops constitutes the primary control signal as shown in general in figure 7.7.1-1 and in more detail on the functional diagrams shown in drawing 1X6AA02-233. The system is capable of automatically restoring coolant average temperature to the programmed value following a decrease in load. Manual rod control may be needed to restore the coolant average temperature to the programmed value following an increase in load. The programmed coolant temperature increases linearly with turbine load from zero power to the full power condition. The T_{avg} also supplies a signal to pressurizer level control and steam dump control and to rod insertion limit monitoring.

The temperature channels needed to derive the temperature input signals for the reactor control system are fed from protection channels via isolation amplifiers.

An additional control input signal is derived from the reactor power versus turbine load mismatch signal. This additional control input signal improves system performance by enhancing response and reducing transient peaks.

The core axial power distribution is controlled during load follow maneuvers by changing (a manual operator action) the boron concentration in the reactor coolant system. The control board $\Delta\phi$ displays (paragraph 7.7.1.3.1) indicate the need for an adjustment in the axial power distribution. Adding boron to the reactor coolant will reduce T_{avg} requiring the control rods to be withdrawn manually. This action will reduce power peaks in the bottom of the core. Removing boron from the reactor coolant will automatically move the rods further into the core to control power peaks in the top of the core.

7.7.1.2 Rod Control System

The rod control system receives rod speed and direction signals from the T_{avg} control system. Control rods may be inserted automatically; however, rod withdrawal can only be performed manually. The rod speed demand signal varies over the corresponding range of 5 to 45 in./min (8 to 72 steps/min), depending on the magnitude of the input signal. Manual control is provided to move control banks in or out at prescribed fixed speeds.

The shutdown banks are always in the fully withdrawn position during normal operation and are moved to this position at a constant speed by manual control prior to criticality. A reactor trip signal causes them to fall by gravity into the core. There are five shutdown banks.

The control banks are the only rods that can be manipulated under automatic control. Each control bank is divided into two groups to obtain smaller incremental reactivity changes per step.

All rod control cluster assemblies in a group are electrically paralleled to move simultaneously. There is individual position indication for each rod cluster control assembly.

Power to rod drive mechanisms is supplied by two motor generator sets operating from two separate 480-V, 3-phase buses. Each generator is the synchronous type and is driven by a 200-h induction motor. The ac power is distributed to the rod control power cabinets through the two series-connected reactor trip breakers.

The variable speed rod drive programmer affords the ability to insert small amounts of reactivity at low speed to accomplish fine control of reactor coolant average temperature about a small temperature dead band, as well as furnishing control at high speed. A summary of the rod cluster control assembly sequencing characteristics is given below:

A. Two groups within the same bank are stepped such that the relative position of the groups will not differ by more than one step.

B. The control banks are programmed such that withdrawal of the banks is sequenced in the following order: control bank A, control bank B, control bank C, and control bank D. The programmed insertion sequence is the opposite of the withdrawal sequence; i.e., the last control bank withdrawn (D) is the first control bank inserted.

C. The control bank withdrawals are programmed such that when the first bank reaches a preset position, the second bank begins to move out simultaneously with the first bank. This preset position is determined by the maximum allowable overlap between banks (typically 100 to 115 steps). When the second bank reaches a preset position, the third bank begins to move out. The first bank will be fully withdrawn before the third bank starts to move. Therefore, it is possible for only two banks to be withdrawn at any one time. This withdrawal sequence continues until the unit reaches the desired power level. The control bank insertion sequence is the opposite.

D. Overlap between successive control banks is adjustable between 0 to 50 percent (0 and 115 steps), with an accuracy of $\pm$1 step.

E. Rod speeds for either the shutdown banks or manual operation of the control banks are capable of being controlled between a minimum of 6 steps/min and a maximum of 72 (+0, -10) steps/min.

7.7.1.3 Plant Control Signals for Monitoring and Indicating

7.7.1.3.1 Monitoring Functions Provided by the Nuclear Instrumentation System

The power range channels are important because of their use in monitoring power distribution in the core within specified safe limits. They are used to measure power level, axial flux imbalance, and radial flux imbalance. Suitable alarms are derived from these signals as described below.

Basic power range signals are:

A. Total current from a power range detector (four signals from separate detectors); these detectors are vertical and have a total active length of 10 ft.

B. Current from the upper half of each power range detector (four signals).

C. Current from the lower half of each power range detector (four signals).

The following (including standard signal processing for calibration) are derived from these basic signals:

A. Indicated nuclear power (four signals).

B. Indicated axial flux imbalance ($\Delta\phi$), derived from upper half flux minus lower half flux (four signals).

Alarm functions derived are as follows:

A. Deviation (maximum minus minimum of four) in indicated nuclear power.

B. Upper radial tilt (maximum to average of four) on upper half currents.

C. Lower radial tilt (maximum to average of four) on lower half currents.

Provision is made to continuously record, on the control board, the eight ion chamber signals; i.e., upper and lower currents for each detector. Nuclear power and axial imbalance are selectable for recording as well. Indicators are provided on the control board for nuclear power and for axial flux imbalance.

The axial flux difference imbalance deviation $\Delta\phi$ alarms are derived from the plant process computer which determines the 1-min averages of the excore detector outputs to monitor $\Delta\phi$ in the reactor core and alerts the operator where $\Delta\phi$ alarm conditions exist. Alarm messages are output immediately when $\Delta\phi$ exceeds the RAOC limits above a preset (50 percent) power level.

Additional background information on the nuclear instrumentation system can be found in reference 1.

7.7.1.3.2 Rod Position Monitoring

Two separate systems are provided to sense and display control rod position as described below:

A. Digital Rod Position Indication System

The digital rod position indication system measures the actual position of each control and shutdown rod using a detector which consists of discrete coils mounted concentrically with the rod drive pressure housing. The coils are located axially along the pressure housing and magnetically sense the entry and presence of the rod drive shaft through its center line. For each detector, the coils are interlaced into two data channels and are connected to the containment electronics (data A and B) by separate multiconductor cables. By employing two separate channels of information, the digital rod position indication system can continue to function (at reduced accuracy) when one channel fails. Multiplexing is used to transmit the digital position signals from the containment electronics to the control board display unit.

The control board display unit contains a column of light-emitting diodes (LEDs) for each rod. At any given time, the one LED illuminated in each column shows the position for that particular rod. Each rod has its position displayed to ± 4 steps throughout its range of travel.

Included in the system is a rod at bottom signal for each rod that operates a local alarm. Also a control room annunciator is actuated when any shutdown rod or control bank A rod is at bottom.

B. Demand Position System

The demand position system counts pulses generated in the rod drive control system to provide a digital readout of the demanded bank position.

The demand position and digital rod position indication systems are separate systems, but safety criteria were not involved in the separation, which was a result only of operational requirements. Operating procedures require the reactor operator to compare the demand and indicated (actual) readings from the rod position indication system to verify operation of the rod control system.

7.7.1.3.3 Control Bank Rod Insertion Monitoring

When the reactor is critical, the normal indication of reactivity status in the core is the position of the control bank in relation to reactor power (as indicated by the reactor coolant system loop ΔT) and coolant average temperature. Insertion limits for the control banks are defined as a function of reactor power. Two alarms are provided for each control bank.

A. The low alarm alerts the operator of rod insertion approaching the control bank insertion limits.

B. The low-low alarm alerts the operator of rod insertion at or below the insertion bank limits. If actual control bank position is below the insertion limits, the operator is to initiate action per the Technical Specifications.

The purpose of the control bank rod insertion monitor is to give warning to the operator of excessive rod insertion. The insertion limit maintains sufficient core reactivity shutdown margin following reactor trip, provides a limit on the maximum inserted rod worth in the unlikely event of a hypothetical rod ejection, and limits rod insertion such that acceptable nuclear peaking factors are maintained. Since the amount of shutdown reactivity required for the design shutdown margin following a reactor trip increases with increasing power, the allowable rod insertion limits must be decreased (the rods must be withdrawn further) with increasing power. Two parameters which are proportional to power are used as inputs to the insertion monitor. These are the ΔT between the hot leg and the cold leg, which is a direct function of reactor power, and T_{avg}, which is programmed as a function of power. The rod insertion monitor uses parameters for each control rod bank as follows:

$$Z_{LL} + A(\Delta T)_{auct} + B(T_{avg})_{auct} + C$$

where:

Z_{LL}	=	maximum permissible insertion limit for affected control bank.
$(\Delta T)_{auct}$	=	highest ΔT of all loops.
$(T_{avg})_{auct}$	=	highest T_{avg} of all loops.
A, B, C	=	constants chosen to maintain $Z_{LL} \geq$ actual limit based on physics calculations.

The control rod bank demand position (Z) is compared to Z_{LL} as follows:

If $Z - Z_{LL} \leq D$, a low alarm is actuated.

If $Z - Z_{LL} \leq E$, a low-low alarm is actuated.

Since the highest values of T_{avg} and ΔT are chosen by auctioneering, a conservatively high representation of power is used in the insertion limit calculation.

Actuation of the low alarm alerts the operator of rod insertion approaching the control bank insertion limits. Actuation of the low-low alarm alerts the operator of rod insertion to at or below the control bank insertion limits. If actual control bank position is below the insertion limits, the operator is to initiate action per the Technical Specifications. The value of D is chosen such that the low alarm actuates above the low-low alarm. The value of E is chosen such that the low-low alarm actuates at the insertion limit except near the top of rod motion. Figure 7.7.1-2 shows a block diagram representation of the control rod bank insertion monitor. The monitor is shown in more detail on the functional diagrams shown in drawing 1X6AA02-233. In addition to the rod insertion monitor for the control banks, the plant computer, which monitors individual rod positions, provides an alarm that is associated with the rod deviation alarm (discussed in paragraph 7.7.1.3.4) which warns the operator if any shutdown rod cluster control assembly leaves the fully withdrawn position.

Rod insertion limits are established by:

A. Establishing the allowed rod reactivity insertion at full power consistent with the purposes given above.

B. Establishing the differential reactivity worth of the control rods when moved in normal sequence.

C. Establishing the change in reactivity with power level by relating power level to rod position.

D. Linearizing the resultant limit curve. All key nuclear parameters in this procedure are measured as part of the initial and periodic physics testing program.

Any unexpected change in the position of the control bank under automatic control (insertion only), or a change in coolant temperature under manual control, provides a direct and immediate indication of a change in the reactivity status of the reactor. In addition, samples are taken periodically of coolant boron concentration. Variations in concentration during core life provide an additional check on the reactivity status of the reactor, including core depletion.

7.7.1.3.4 Rod Deviation Alarm

The position of any control rod is compared to the position of other rods in the bank.

The demanded and measured rod position signals are monitored by the plant computer, which drives an input to the main control board annunciator system whenever an individual rod position signal deviates from the other rods in the bank by a preset limit. The alarm can be set with appropriate allowance for instrument error and within sufficiently narrow limits to preclude exceeding core design hot channel factors.

Figure 7.7.1-3 is a block diagram of the rod deviation comparator and alarm system implemented by the plant computer.

7.7.1.3.5 Rod Bottom Alarm

A rod bottom signal for the control and shutdown rods in the digital rod position indication system is used to operate a control relay which generates the "rod bottom rod drop" alarm.

7.7.1.4 Plant Control System Interlocks

The listing of the plant control system interlocks and the description of their derivations and functions are presented in table 7.7.1-1. The designation numbers for these interlocks are preceded by "C." The development of these logic functions is shown in the functional diagrams. (See drawings 1X6AA02-233, 1X6AA02-234, 1X6AA02-235, 1X6AA02-236, 1X6AA02-237, 1X6AA02-238, 1X6AA02-239, and 1X6AA02-240.)

7.7.1.4.1 Rod Stops

Rod stops are provided to prevent abnormal power conditions which could result from excessive control rod withdrawal initiated by either a control system malfunction or operator violation of administrative procedures.

Rod stops are the C-1 through C-4 control interlocks identified in table 7.7.1-1. The C-3 rod stop derived from overtemperature ΔT and the C-4 rod stop derived from overpower ΔT are also used for turbine runback, which is discussed below.

7.7.1.4.2 Automatic Turbine Load Runback

Automatic turbine load runback is initiated by an approach to an overpower or overtemperature condition. This will prevent high power operation that might lead to an undesirable condition which, if reached, will be protected by reactor trip.

Turbine load reference reduction is initiated by either an overtemperature or overpower ΔT signal. Two-out-of-four coincidence logic is used.

A rod stop and turbine runback are initiated for both the overtemperature and the overpower condition when:

$$\Delta T > \Delta T_{rod\ stop}$$

for either condition in general:

$$\Delta T_{rod\ stop} = \Delta T_{setpoint} - B_p$$

where:

$\Delta T_{setpoint}$ = overtemperature and overpower ΔT reactor trip value.

B_p = a setpoint bias.

The turbine runback is continued until ΔT is equal to or less than $\Delta T_{rod\ stop}$.

This function serves to maintain an essentially constant margin to trip.

7.7.1.4.3 Turbine Loading Stop

An interlock (C-16) is provided to limit turbine loading during a rapid return to power transient when a reduction in reactor coolant temperature is used to increase reactor power (through the negative moderator coefficient). This interlock limits the reduction in coolant temperature so that it does not reach cooldown accident limits and preserves satisfactory steam generator operating conditions. Subsequent automatic turbine loading can begin after the interlock has

been cleared by anincrease in coolant temperature, which is accomplished by reducing the boron concentration in the coolant.

7.7.1.5 Pressurizer Pressure Control

The reactor coolant system pressure is controlled by using either the heaters (in the water region) or the spray (in the steam region) of the pressurizer plus steam relief for large transients. The electrical immersion heaters are located near the bottom of the pressurizer. A portion of the heater group is proportionally controlled to correct small pressure variations. These variations are caused by heat losses, including heat losses resulting from a small continuous spray. The remaining (backup) heaters are turned on when the pressurizer pressure controlled signal demands approximately 100-percent proportional heater power.

The spray nozzles are located on the top of the pressurizer. Spray is initiated when the pressure controller spray demand signal is above a given setpoint. The spray rate increases proportionally with the increasing spray demand signal until it reaches a maximum value. For additional information on the pressurizer heaters and their power supply, see subsection 5.4.10.

Steam condensed by the spray reduces the pressurizer pressure. A small continuous spray is normally maintained to reduce thermal stresses and thermal shock and to help maintain uniform water chemistry and temperature in the pressurizer.

Power relief valves limit system pressure for large positive pressure transients. In the event of a large load reduction, not exceeding the design plant load rejection capability, the pressurizer power-operated relief valves might be actuated for the most adverse conditions; e.g., the most negative Doppler coefficient and the minimum incremental rod worth. The relief capacity of the power-operated relief valves is sized large enough to limit the system pressure to prevent actuation of high-pressure reactor trip for the above condition.

A block diagram of the pressurizer pressure control system is shown on figure 7.7.1-4.

7.7.1.6 Pressurizer Water Level Control

The pressurizer operates by maintaining a steam cushion over the reactor coolant. As the density of the reactor coolant varies with temperature, the steam water interface is adjusted to compensate for cooling density variations with relatively small pressure disturbances.

The water inventory in the reactor coolant system is maintained by the chemical and volume control system. During normal plant operation, the charging flow varies to produce the flow demanded by the pressurizer water level controller. The pressurizer water level is programmed as a function of coolant average temperature, with the highest average temperature (auctioneered) being used. The pressurizer water level decreases as the load is reduced from full load. This is a result of coolant contraction following programmed coolant temperature reduction from full power to low power. The programmed level is designed to match as nearly as possible the level changes resulting from the coolant temperature changes.

To control pressurizer water level during startup and shutdown operations, the charging flow is manually regulated from the main control room. Low level in the pressurizer initiates a control-grade process protection signal to close the letdown isolation valves and deenergize the pressurizer heaters. This action is intended to maintain pressurizer level above the heaters. Failure of the letdown isolation valves to close and terminate letdown has been evaluated and has been found to have no impact on plant safety.

A block diagram of the pressurizer water level control system is shown on figure 7.7.1-5.

7.7.1.7 Steam Generator Water Level Control

Each steam generator is equipped with a three-element feedwater flow controller that maintains a programmed water level, which is capable of being controlled as a function of turbine load, but is really a fixed value. The three-element feedwater controller regulates the feedwater valve by continuously comparing the feedwater flow signal, the water level signal, the programmed level, and the pressure compensated steamflow signal.

The feedwater pump speed is varied to maintain sufficient discharge pressure to support feedwater flow demand. The speed controller receives a demand signal, which is a programmed function of the highest feedwater flow demand across all loops. Continued delivery of feedwater to the steam generators is required as a sink for the heat stored and generated in the reactor following a reactor trip and turbine trip. An override signal closes all feedwater valves when the average coolant temperature is below a given temperature and the reactor has tripped. Manual override of the feedwater control system is available at all times.

When the nuclear plant is operating at very low power levels (as during startup), the steam and feedwater flow signals will not be usable for control. Therefore, a secondary automatic control system is provided for operation at low power. This system uses the narrow range steam generator water level and wide range steam generator water level in a feed forward control scheme to position a bypass valve which is parallel to the main feedwater regulating valve. Transition from the bypass feedwater control system (low power) to parallel use of the main feedwater control valve is performed automatically by the control system at approximately 20-percent power. The feedwater control system automatically closes the bypass valve at approximately 40-percent power.

Block diagrams of the steam generator water level control system and the main feedwater pump speed control system are shown in figures 7.7.1-6 and 7.7.1-7.

7.7.1.8 Steam Dump Control

The plant is designed to accept a 50-percent loss of net load without tripping the reactor.

The automatic steam dump system is able to accommodate this abnormal load rejection and to reduce the effects of the transient imposed upon the reactor coolant system. By bypassing main steam directly to the condenser, an artificial load is thereby maintained on the primary system. The rod control system can then reduce the reactor temperature to a new equilibrium value without causing overtemperature and/or overpressure conditions. The steam dump steamflow capacity is 40 percent of full load steamflow at full load steam pressure.

If the difference between the reference T_{avg} (T_{ref}) based on turbine impulse chamber pressure and the lead-lag compensated auctioneered T_{avg} exceeds a predetermined amount, and if the interlock mentioned below is satisfied, a demand signal will actuate the steam dump to maintain the reactor coolant system temperature within control range until a new equilibrium condition is reached.

To prevent actuation of steam dump on small load perturbations, an independent load rejection sensing circuit is provided. This circuit senses the rate of decrease in the turbine load as detected by the turbine impulse chamber pressure. It is provided to unblock the dump valves when the rate of load rejection exceeds a preset value corresponding to a 10-percent step-load decrease or a sustained ramp-load decrease of 5 percent/min.

Steam dump solenoid valves and circuits in the turbine building are designed to criteria similar to the reactor trip on turbine trip circuits as described in paragraph 7.2.1.1.2.F. A block diagram of the steam dump control system is shown on figure 7.7.1-8.

7.7.1.8.1 Load Rejection Steam Dump Controller

This circuit prevents a large increase in reactor coolant temperature following a large, sudden load decrease. The error signal is a difference between the lead-lag compensated auctioneered T_{avg} and the reference T_{avg} is based on turbine impulse chamber pressure.

The T_{avg} signal is the same as that used in the reactor coolant system. The lead-lag compensation for the T_{avg} signal is to compensate for lags in the plant thermal response and in valve positioning. Following a sudden load decrease, T_{ref} is immediately decreased and T_{avg} tends to increase, thus generating an immediate demand signal for steam dump. Since control rods are available in this situation, steam dump terminates as the error comes within the maneuvering capability of the control rods.

7.7.1.8.2 Plant Trip Steam Dump Controller

Following a reactor trip, the load rejection steam dump controller is defeated and the plant trip steam dump controller becomes active. Since control rods are not available in this situation, the demand signal is the error signal between the lead-lag compensated auctioneered T_{avg} and the no-load reference T_{avg}. When the error signal exceeds a predetermined setpoint, the dump valves are tripped open in a prescribed sequence. As the error signal reduces in magnitude, indicating that the reactor coolant system T_{avg} is being reduced toward the reference no-load value, the dump valves are modulated by the plant trip controller to regulate the rate of removal of decay heat and thus gradually establish the equilibrium hot shutdown condition.

7.7.1.8.3 Steam Header Pressure Controller

Residual heat removal at operating temperature is maintained by the steam generator pressure controller (manually selected), which controls the amount of steamflow to the condensers. This controller operates a portion of the same steam dump valves to the condensers which are used during the initial transient following turbine or reactor trip on load rejection.

7.7.1.9 Incore Instrumentation

The incore instrumentation system consists of chromel-alumel thermocouples at fixed core outlet positions and movable miniature neutron detectors which can be positioned at the center of selected fuel assemblies anywhere along the length of the fuel assembly vertical axis. The basic system for insertion of these detectors is shown in figure 7.7.1-9.

7.7.1.9.1 Thermocouples

Chromel-alumel thermocouples are threaded into guide tubes that penetrate the reactor vessel head through seal assemblies and terminate at the exit flow end of the fuel assemblies. The thermocouples are provided with two primary seals, a core exit thermocouple nozzle assembly

(CETNA) and a swage-type seal from conduit to head. Thermocouple readings are monitored by the plant safety monitoring system.

7.7.1.9.2 Movable Neutron Flux Detector Drive System

Miniature fission chamber detectors can be remotely positioned in retractable guide thimbles to provide flux mapping of the core. The stainless steel detector shell is welded to the leading end of helical wrap drive cable and to stainless steel sheathed coaxial cable. The retractable thimbles, into which the miniature detectors are driven, are pushed into the reactor core through conduits which extend from the bottom of the reactor vessel down through the concrete shield area and then up to a thimble seal table. Their distribution over the core is nearly uniform, with about the same number of thimbles located in each quadrant.

The thimbles are closed at the leading ends, are dry inside, and serve as the pressure barrier between the reactor water pressure and the atmosphere. Mechanical seals between the retractable thimbles and the conduits are provided at the seal table. During reactor operation, the retractable thimbles are stationary. They are extracted downward from the core during refueling to avoid interference within the core. A space above the seal table is provided for the retraction operation.

The drive system for the insertion of the miniature detectors consists basically of drive assemblies, 5-path transfer assemblies, and 10-path transfer assemblies, as shown in figure 7.7.1-9. The drive system pushes hollow helical wrap drive cables into the core with the miniature detectors attached to the leading ends of the cables and small diameter sheathed coaxial cables threaded through the hollow centers back to the ends of the drive cables. Each drive assembly consists of a gear motor which pushes a helical wrap drive cable and a detector through a selective thimble path by means of a special drive box and includes a storage device that accommodates the total drive cable length. Each thimble location can be accessed by at least two detectors controlled from different drive units.

7.7.1.9.3 Control and Readout Description

The control and readout system provides means for inserting the miniature neutron detectors into the reactor core and withdrawing the detectors while plotting neutron flux versus detector position. The control system is located in the control room. Limit switches in each transfer device provide feedback of path selection operation. Each gearbox drives a resolver for position feedback. One five-path transfer selector is provided for each drive unit to insert the detector in one of five functional modes of operation. One 10-path transfer is also provided for each drive unit that is then used to route a detector into any one of up to 10 selectable paths. A common path is provided to permit cross-calibration of the detectors.

The control room contains the necessary equipment for control, position indication, and flux recording for each detector.

A flux mapping consists, briefly, of selecting flux thimbles in given fuel assemblies at various core quadrant locations. The detectors are driven to the top of the core and stopped automatically. An x-y plot (position versus flux level) is initiated with the slow withdrawal of the detectors through the core from top to a point below the bottom. In a similar manner other core locations are selected and plotted. Each detector provides axial flux distribution data along the center of a fuel assembly.

Various radial positions of detectors are then compared to obtain a flux map for a region of the core.

The number and location of these thimbles have been chosen to permit measurement of local to average peaking factors (F_Q) to an accuracy of ±5 percent (95-percent confidence). Measured nuclear peaking factors (F_Q) will be increased by 5 percent to allow for this accuracy, when using ≥ 44 detector thimbles. When using ≥ 29 and < 44 detector thimbles, the measurement uncertainty is 1.05 + [2.0 {3-T / (14.5)}] / 100, where T equals the number of thimbles (reference 45). If the measured power peaking is larger than acceptable, reduced power capability will be indicated.

Operating plant experience has demonstrated the adequacy of the incore instrumentation in meeting the design bases stated.

7.7.1.10 Boron Concentration Measurement System

The boron concentration measurement system utilizes a sample measurement unit which contains a neutron source and neutron detector located in a shield tank. Piping within the shield tank is arranged to maintain coolant sample flow between the neutron source and the neutron detector. Neutron absorption by the boron in the coolant sample flow reduces the number of neutrons which contact the detector per unit time. Therefore, the time required to count a fixed number of neutron contacts is variable and dependent upon the concentration of boron solution.

The sample tank subassembly of the BCMS consists of a 100 gallon stainless steel tank with a cover assembly which includes the annular sample chamber, heat exchange coil, source and detector holders, temperature monitoring and control hardware, and all necessary plumbing. The sample tank is filled with water which provides for coolant sample temperature control and shielding for the neutron source (Am-Be). The sample tank subassembly is designed so that all tank connections are at the top or near the top to preclude the possibility of loss of water shielding due to accidental leakage.

The boron concentration measurement system is designed for use as an advisory system. It is not designed as a safeguards system or component of a safeguards system. The boron concentration measurement system is not part of a control element or control system nor is it designed for this use. No credit is taken for this system in any accident analysis. Therefore, redundancies of measurement components, self-checking subsystems, malfunction annunciations, and diagnostic circuitry are not included in this system. As a general operating aid it provides information as to when additional check analyses are warranted rather than as a basis for fundamental operating decisions.

7.7.1.11 ATWS Mitigation System Actuation Circuitry (AMSAC) Description

7.7.1.11.1 System Description

The ATWS (anticipated transient without scram) mitigation system actuation circuitry (AMSAC) provides a backup to the reactor trip system (RTS) and engineered safety features (ESF) actuation system (ESFAS) for initiating turbine trip and auxiliary feedwater flow in the event of an anticipated transient (e.g., in the event of complete loss of main feedwater). The AMSAC is independent of and diverse from the RTS and the ESFAS with the exception of the final actuation devices and is classified as non-Class 1E. It is a highly-reliable, microprocessor-based, single-train system powered by a non-Class 1E source. The AMSAC meets the applicable requirements of Part 50.62 of Title 10 of the Code of Federal Regulations and the

quality assurance requirements of NRC Generic Letter 85-06. No other standards apply to the AMSAC.

The AMSAC continuously monitors main feedwater flow, which is an anticipatory indication of a loss of heat sink, and initiates certain functions when the flow drops below a predetermined setpoint in three of the four main feedwater lines for a delayed amount of time (dependent on turbine load). These initiated functions are the tripping of the turbine, the initiation of auxiliary feedwater, and isolation of the steam generator blowdown and sample lines.

The AMSAC is designed to be highly reliable, resistant to inadvertent actuation, and easily maintained. Reliability is assured through the use of internal redundancy and continual self-testing by the system. Inadvertent actuations are minimized through the use of internal redundancy and majority voting at the output stage of the system. The time delay on the low main feedwater flow and the coincidence logic used also minimize inadvertent actuations.

The AMSAC automatically performs its actuations when above a preselected power level, which is determined using turbine impulse chamber pressure, and remains armed sufficiently long after that pressure drops below the setpoint to ensure that its function will be performed in the event of a turbine trip.

7.7.1.11.2 Equipment Description

The AMSAC consists of a single train of equipment located in a seismically qualified cabinet.

The design of the AMSAC is based on the industry standard Intel multibus format, which permits the use of various readily available, widely used microprocessor cards on a common data bus for various functions.

The AMSAC consists of the following:

1. System Hardware

   The system hardware consists of two primary systems: the actuation logic system (ALS) and the test/ maintenance system (T/MS).

   A. Actuation Logic System

      The ALS monitors the analog and digital inputs, performs the functional logic required, provides actuation outputs to trip the turbine and initiate auxiliary feedwater flow, and provides status information to the test/maintenance system. The ALS consists of three groups of input/output (I/O) modules, three actuation logic processors (ALPs), two majority voting modules, and two output relay panels. The I/O modules provide signal conditioning, isolation, and test features for interfacing the ALS and T/MS. Conditioned signals are sent to three identical ALPs for analog-to-digital conversion, setpoint comparison, and coincidence logic performance. Each of the ALPs perform identical logic calculations using the same inputs and derive component actuation demands, which are then sent to the majority voting modules. The majority voting modules perform a two-out-of-three vote on the ALP demand signals. These modules drive the relays providing outputs to the existing turbine trip and auxiliary feedwater initiation circuits. A simplified block diagram of the AMSAC ALS architecture is presented in figure 7.7.1-14.

B. Test/Maintenance System

The test/maintenance system provides the AMSAC with automated and manual testing as well as a maintenance mode. Automated testing is the continuously performed self-checking done by the system during normal operation. ALS status is monitored by the T/MS and sent to the plant computer and the main control board. Manual testing of the system can be performed on-line to provide assurance that the ALS system is fully operational. The maintenance mode permits, under administrative control, modification of channel setpoints, channel status and timer values, and initiation of channel calibration.

The T/MS consists of a test/maintenance processor, a digital-to-analog conversion board, a memory board, expansion boards, a self-health board, digital output modules, a test/maintenance panel, and a portable terminal/printer.

2. Feedwater Flow Sensing

The AMSAC utilizes the feedwater flow signal as measured with the four differential pressure-type flow transmitters, one for each of the main feedwater lines shown in drawing 1X6AA02-231.

3. Turbine Impulse Pressure

The AMSAC also utilizes the turbine impulse pressure signal as measured with two pressure transmitters located in the steam supply line near the turbine shown in drawing 1X6AA02-240.

4. Equipment Actuation

The output relay panels provide component actuation signals through isolation relays, which then drive the final actuation circuitry shown in drawings 1X6AA02-239 and 1X6AA02-240 for initiation of auxiliary feedwater and for turbine trip.

7.7.1.11.3 Functional Performance Requirements

Analyses have shown that the most limiting ATWS event is a loss of feedwater event without a reactor trip. AMSAC performs the mitigative actuations of automatically initiating auxiliary feedwater, tripping the turbine, and isolating the steam generator blowdown and sampling lines. These are initiated in order to ensure a secondary heat sink following an anticipated transient (ANS Condition II) without a reactor trip, in order to limit core damage following an anticipated transient without a reactor trip, and to ensure that the energy generated in the core is compatible with the design limits to protect the reactor coolant pressure boundary by maintaining the reactor coolant pressure to within ASME Stress Level C.

7.7.1.11.4 AMSAC Interlocks

A single interlock, designated as C-20, is provided to allow for the automatic arming and blocking of the AMSAC (see table 7.7.1-1). The system is blocked at sufficiently low reactor power levels when the actions taken by the AMSAC following an ATWS need not be automatically initiated. Turbine impulse chamber pressure in a two-out-of-two logic scheme is

used for this permissive. Turbine impulse chamber pressure above the setpoint will automatically defeat any block, i.e., will arm the AMSAC. Dropping below this setpoint will automatically block the AMSAC. Removal of the C-20 permissive is automatically delayed for a predetermined time. The operating status of the AMSAC is displayed on the main control board.

7.7.1.11.5 Trip System

The feedwater flow and turbine impulse chamber pressure inputs are used by the AMSAC to determine trip demand. Signal conditioning is performed on the transmitter output and used by each of the ALPs to derive a component actuation demand. If three of the four feedwater lines have a low flow at a power level greater than the C-20 permissive, then a trip demand signal is generated. This signal drives output relays for performing the necessary mitigative actions.

7.7.1.11.6 Isolation Devices

AMSAC is independent of the reactor trip and engineered safety features actuation systems. The AMSAC feedwater flow inputs are non-Class 1E signals from the process control cabinets. No isolation into AMSAC is needed for these inputs. The AMSAC turbine impulse chamber pressure inputs are made downstream of Class 1E isolation devices, which are located within the process protection cabinets. These isolation devices ensure that the existing protection system continues to meet all applicable safety criteria by providing isolation. Buffering of the AMSAC outputs from the safety-related final actuation device circuits is achieved through qualified relays. A credible fault occurring in the nonsafety-related AMSAC will not propagate through and degrade the RTS and ESFAS.

7.7.1.11.7 AMSAC Diversity from the Reactor Protection Systems

Equipment diverse from the RTS and ESFAS is used in the AMSAC to prevent common mode failures that might affect the AMSAC and the RTS or ESFAS. The AMSAC is a digital, microprocessor-based system with the exception of the analog feedwater flow and turbine impulse pressure transmitter inputs, whereas the reactor trip system utilizes an analog-based protection system. Also where similar components are utilized for the same function in both AMSAC and the reactor trip system, the components used in AMSAC are provided from a different manufacturer.

Common mode failure of identical components in the analog portion of the RTS that results in the inability to generate a reactor trip signal will not impact the ability of the digital AMSAC to generate the necessary mitigative actuations. Similarly, a postulated common mode failure affecting analog components in ESFAS, affecting its ability to initiate auxiliary feedwater, will not impact the ability of the digital-based AMSAC to automatically initiate auxiliary feedwater.

7.7.1.11.8 Power Supply

The AMSAC power supply is a non-Class 1E vital bus, which is independent from the RTS power supplies, and is backed by batteries which are independent from the existing batteries which supply the RTS.

7.7.1.11.9 Environmental Variations

AMSAC equipment is not designed as safety-related equipment; therefore, it is not required to be qualified as safety-related equipment. The AMSAC equipment is located in a controlled environment such that variations in the ambient conditions are minimized. No AMSAC equipment is located inside containment. The transmitters (feedwater flow and turbine impulse chamber pressure) that supply the input into AMSAC are located outside containment and the turbine building, respectively, and are qualified for the environment in which they are located.

7.7.1.11.10 Setpoints

The AMSAC makes use of two setpoints in the coincidence logic in order to determine whether mitigative functions are required. Feedwater flow in each main feedwater line is sensed to determine a loss of secondary heat sink is imminent. The low-flow setpoint is selected in such a manner that a true lowering of the flow will be detected by the system. The normal small variations in feedwater flow will not result in a spurious AMSAC signal.

The C-20 permissive setpoint is selected in order to be consistent with ATWS investigations showing that the mitigative actions performed by the AMSAC need not be automatically actuated below a certain power level. The maximum allowable value of the C-20 permissive setpoint is defined by these investigations.

To avoid inadvertent AMSAC actuation on the loss of one main feedwater pump, a time delay unit is required on the low main feedwater flow channels to adjust the AMSAC actuation response time. The delay unit is designed such that the delay time is dependent on turbine power. This will ensure the reactor protection system will provide the first trip signals.

To ensure that the AMSAC remains armed sufficiently long to permit its function in the event of a turbine trip, the C-20 permissive is maintained for a preset time delay, after the turbine impulse chamber pressure drops below the setpoint.

The setpoints and the capability for their modification in the AMSAC are under administrative control.

7.7.1.12 REFERENCE

1. Lipchak, J. B., "Nuclear Instrumentation System," WCAP-8255, January 1974 (for additional background information only).

7.7.2 ANALYSIS OF CONTROL SYSTEMS NOT REQUIRED FOR SAFETY

The plant control systems are designed to ensure high reliability in any anticipated operational occurrences. Equipment used in these systems is designed and constructed with a high level of reliability.

Proper positioning of the control rods is monitored in the control room by bank arrangements of the individual position columns for each rod cluster control assembly (RCCA). A rod deviation alarm alerts the operator of a deviation of one RCCA from the other rods in that bank position. There are also insertion limit monitors with visual and audible annunciation. A rod bottom alarm signal is provided to the control room for each full-length RCCA. Four excore long ion chambers also detect asymmetrical flux distribution indicative of rod misalignment.

Overall reactivity control is achieved by the combination of soluble boron and RCCAs. Long-term regulation of core reactivity is accomplished by adjusting the concentration of boric acid in the reactor coolant. Short-term reactivity control for power changes is accomplished by the plant control system, which automatically moves RCCAs. This system uses input signals including neutron flux, coolant temperature, and turbine load.

The axial core power distribution is controlled by moving the control rods through changes in reactor coolant system boron concentration. Adding boron causes the rods to move out, thereby reducing the amount of power in the bottom of the core; this allows power to redistribute toward the top of the core. Reducing the boron concentration causes the rods to move into the core, thereby reducing the power in the top of the core; the result redistributes power toward the bottom of the core.

The plant control systems will prevent an undesirable condition in the operation of the plant that, if reached, will be protected by reactor trip. The description and analysis of this protection is covered in section 7.2. Worst-case failure modes of the plant control systems are postulated in the analysis of off-design operational transients and accidents covered in chapter 15, such as the following:

- Uncontrolled RCCA bank withdrawal from a subcritical or low power startup condition.
- Uncontrolled RCCA bank withdrawal at power.
- RCCA misalignment.
- Loss of external electrical load and/or turbine trip. Loss of all nonemergency ac power to the station auxiliaries (station blackout).
- Feedwater system malfunctions that result in a decrease in feedwater temperature.
- Excessive increase in secondary steamflow.
- Inadvertent opening of a steam generator relief or safety valve.

These analyses show that a reactor trip setpoint is reached in time to protect the health and safety of the public under those postulated incidents and that the resulting coolant temperatures produce a departure from nucleate boiling ratio well above the limiting value. Thus, there will be no cladding damage and no release of fission products to the reactor coolant system under the assumption of these postulated worst-case failure modes of the plant control system.

7.7.2.1 Separation of Protection and Control System

In some cases, it is advantageous to employ control signals derived from individual protection channels through isolation amplifiers contained in the protection channel. As such, a failure in the control circuitry does not adversely affect the protection channel. Test results have shown that a short circuit or the application (credible fault voltage from within the cabinets) of 118-V ac or 140-V dc on the isolated output portion of the circuit (nonprotection side of the circuit) will not affect the input (protection) side of the circuit.

Where a single random failure can cause a control system action that results in a generating station condition requiring protective action and can also prevent proper action of a protection system channel designed to protect against the condition, the remaining redundant protection

channels are capable of providing the protective action even when degraded by a second random failure. This meets the applicable requirements of section 4.7 of Institute of Electrical and Electronic Engineers (IEEE) Standard 279-1971.

The pressurizer pressure channels needed to derive the control signals are electrically isolated from control.

7.7.2.2 Response Considerations of Reactivity

Reactor shutdown with control rods is completely independent of the control functions, since the trip breakers interrupt power to the rod drive mechanisms regardless of existing control signals. The design is such that the system can withstand accidental withdrawal of control groups or unplanned dilution of soluble boron without exceeding acceptable fuel design limits. The design meets the requirements of General Design Criterion 25.

No single electrical or mechanical failure in the rod control system could cause the accidental withdrawal of a single RCCA from the partially inserted bank at full power operation. The operator could deliberately withdraw a single RCCA in the control bank; this feature is necessary in order to retrieve a rod, should one be accidentally dropped. In the extremely unlikely event of simultaneous electrical failures which could result in single RCCA withdrawal, rod deviation would be displayed on the plant annunciator, and the individual rod position readouts would indicate the relative positions of the rods in the bank. Withdrawal of a single RCCA by operator action, whether deliberate or by a combination of errors, would result in activation of the same alarm and the same visual indications.

Each bank of control and shutdown rods in the system is divided into two groups of up to four or five mechanisms each. The rods comprising a group operate in parallel through multiplexing thyristors. The two groups in a bank move sequentially, such that the first group is always within one step of the second group in the bank. The group 1 and group 2 power circuits are installed in different cabinets, as shown in figure 7.7.2-1, which also shows that one group is always within one step (5/8 in.) of the other group. A definite schedule of actuation or deactuation of the stationary gripper, movable gripper, and lift coils of a mechanism is required to withdraw the RCCA attached to the mechanism. Since the four stationary gripper, movable gripper, and lift coils associated with the RCCAs of a rod group are driven in parallel, any single failure which could cause rod withdrawal would affect a minimum of one group of RCCAs. As a result of Generic Letter 93-04, "Rod Control System Failure and Withdrawal of Rod Control Cluster Assemblies, 10 CFR 50.54(f)," a test is performed following each refueling outage which ensures that the proper current order timing for the grippers and lift coils is maintained. Mechanical failures are in the direction of insertion or immobility.

Figure 7.7.2-2 illustrates the design features that ensure no single electrical failure could cause the accidental withdrawal of a single RCCA from the partially inserted bank at full power operation.

Figure 7.7.2-2 shows the typical parallel connections on the lift, movable, and stationary coils for a group of rods. Since single failures in the stationary or movable circuits will result in dropping or preventing rod (or rods) motion, the discussion of single failure will be addressed to the lift coil circuits:

A. Because of the method of wiring the pulse transformers which fire the lift coil multiplex thyristors, three of the four thyristors in a rod group could remain turned off when required to fire, if for example the gate signal lead failed open at point X_1. Upon "up" demand, one rod in group 1 and four rods in group 2 would

withdraw. A second failure at point X_2 in the group 2 circuit is required to withdraw one RCCA.

B. Timing circuit failures will affect the four mechanisms of a group or the eight mechanisms of the bank and will not cause a single rod withdrawal.

C. More than two simultaneous component failures are required (other than the open wire failures) to allow withdrawal of a single rod.

The identified multiple failures involving the least number of components consists of open circuit failure of the proper 2 out of 16 wires connected to the gate of the lift coil thyristors. The probability of open wire (or terminal) failure is 0.016×10^{-6} per hour by MIL-HDB217A. These wire failures would have to be accompanied by failure or disregard of the indications mentioned above. The probability of this occurrence is therefore too low to have any significance.

Concerning the human element, to erroneously withdraw a single rod cluster control assembly, the operator would have to improperly set the bank selector switch, the lift coil disconnect switches, and the in-hold-out switch. In addition, the three indications would have to be disregarded or ineffective. Such series of errors would require a complete lack of understanding and administrative control. A probability number cannot be assigned to a series of errors such as these.

The rod position indication system provides direct visual displays of each control rod assembly position. The plant computer alarms for deviation of rods from their banks. In addition, a rod insertion limit monitor provides an audible and visual alarm to warn the operator of an approach to an abnormal condition resulting from dilution. The low-low insertion limit alarm alerts the operator to follow emergency boration procedures. The facility reactivity control systems are such that acceptable fuel damage limits will not be exceeded even in the event of a single malfunction of either system.

An important feature of the control rod system is that insertion is provided by gravity fall of the rods.

In all analyses involving reactor trip, the single highest worth rod cluster control assembly is postulated to remain untripped in its full out position.

One means of detecting a stuck control rod assembly is available from the actual rod position information displayed on the control board. The control board position readouts, one for each rod, give the plant operator the actual position of the rod in steps. The indications are grouped by banks (e.g., control bank A, control bank B, etc.) to indicate to the operator the deviation of one rod with respect to other rods in a bank. This serves as a means to identify rod deviation.

The plant computer monitors the actual position of all rods. Should a rod be misaligned from the other rods in that bank by more than 12 steps, the rod deviation alarm is actuated.

Misaligned RCCAs are also detected and alarmed in the control room via the flux tilt monitoring system, which is independent of the plant computer.

Isolated signals derived from the nuclear instrumentation system are compared with one another to determine whether a preset amount of deviation of average power level has occurred. Should such a deviation occur, the comparator output will operate a bistable unit to actuate a control board annunciator. This alarm will alert the operator to a power imbalance caused by a misaligned rod. By use of individual rod position readouts, the operator can determine the deviating control rod and take corrective action. The design of the plant control systems meets the requirements of General Design Criterion 23.

Refer to section 4.3 for additional information on response considerations resulting from reactivity.

7.7.2.3 Step-Load Changes Without Steam Dump

The plant control system is capable of restoring equilibrium conditions, without a trip, following a ±10-percent step change in load demand, over the 15- to 100-percent power range. Automatic control allows control rod insertion only. Control rod withdrawal can only be performed manually. Steam dump is blocked for load decrease less than or equal to 10 percent. A load demand greater than full power is prohibited by the turbine control load valve position limit on Unit 1 and load limit devices on Unit 2.

The plant control system minimizes the reactor coolant average temperature deviation during the transient within a given value and restores average temperature to the programmed setpoint.

Excessive pressurizer pressure variations are prevented by using spray and heaters and power relief valves in the pressurizer.

The control system must limit nuclear power overshoot to acceptable values following a 10- to 100-percent increase in load.

7.7.2.4 Loading and Unloading

Ramp loading and unloading of 5 percent/min can be accepted over the 15- to 100-percent power range under manual control for loading and automatic control for unloading without tripping the plant. Control rod insertion may be performed automatically. However, control rod withdrawal can only be performed manually. The function of the control system is to maintain the coolant average temperature as a function of turbine-generator load.

The coolant average temperature increases during loading and causes a continuous insurge to the pressurizer as a result of coolant expansion. The sprays limit the resulting pressure increase. Conversely, as the coolant average temperature is decreasing during unloading, there is a continuous outsurge from the pressurizer resulting from coolant contraction. The pressurizer heaters limit the resulting system pressure decrease. The pressurizer water level is programmed such that the water level is above the setpoint for heater cutout during the loading and unloading transients. The primary concern during loading is to limit the overshoot in nuclear power and to provide sufficient margin in the overtemperature ΔT setpoint.

The automatic load controls are designed to adjust the unit generation to match load requirements within the limits of the unit capability and licensed rating.

During rapid loading transients, a drop in reactor coolant temperature is sometimes used to increase core power. (Refer to paragraph 3.9.N.1.1.1.5.) This mode of operation is applied when the control rods are not inserted deep enough into the core to supply all the reactivity requirements of the rapid load increase. (The boron control system is relatively ineffective for rapid power changes.) The reduction in temperature is initiated by continued turbine loading past the point where the control rods are completely withdrawn from the core. The temperature drop is recovered and nominal conditions restored by a boron dilution operation.

Excessive drops in coolant temperature are prevented by interlock C-16. This interlock circuit monitors the auctioneered low coolant temperature indications and the programmed reference temperature, which is a function of turbine impulse pressure, and stops the turbine loading sequence when the decreased temperature reaches the setpoints.

7.7.2.5 Load Rejection Furnished by Steam Dump System

When a load rejection occurs and if the difference between the required temperature setpoint of the reactor coolant system and the actual average temperature exceeds a predetermined amount, a signal will actuate the steam dump to maintain the reactor coolant system temperature within control range until a new equilibrium condition is reached.

The reactor power is reduced at a rate consistent with the capability of the rod control system. Reduction of the reactor power is automatic. The steam dump flow reduction is as fast as RCCAs are capable of inserting negative reactivity.

The rod control system can then reduce the reactor temperature to a new equilibrium value without causing overtemperature and/or overpressure conditions. The steam dump steamflow capacity is 40 percent of full load steamflow at full load steam pressure.

The steam dump flow decreases proportionally as the control rods act to reduce the average coolant temperature. The artificial load is therefore removed as the coolant average temperature is restored to its programmed equilibrium value.

The dump valves are modulated by the reactor coolant average temperature signal. The required number of steam dump valves can be tripped quickly to stroke full open or modulate, depending upon the magnitude of the temperature error signal resulting from loss of load.

7.7.2.6 Turbine-Generator Trip with Reactor Trip

Whenever the turbine-generator unit trips at an operating power level above 40-percent power on Unit 1 and 50-percent power on Unit 2, the reactor also trips. The unit is operated with a programmed average temperature as a function of load, with the full load average temperature significantly greater than the temperature corresponding to saturation pressure at the steam generator safety valve setpoint. The thermal capacity of the reactor coolant system is greater than that of the secondary system, and because the full load average temperature is greater than the no-load temperature, a heat sink is required to remove heat stored in the reactor coolant to prevent actuation of steam generator safety valves for a trip from full power. This heat sink is provided by the combination of controlled release of steam to the condenser and by makeup of feedwater to the steam generators.

The steam dump system is controlled from the reactor coolant average temperature signal, whose setpoint values are programmed as a function of turbine load. Actuation of the steam dump is rapid to prevent actuation of the steam generator safety valves. With the dump valves open, the average coolant temperature starts to reduce quickly to the no-load setpoint. A direct feedback of temperature acts to proportionally close the valves to minimize the total amount of steam which is bypassed.

The feedwater flow is cut off following a reactor trip when the average coolant temperature decreases below a given temperature or when the steam generator water level reaches a given high level.

Additional feedwater makeup is then controlled manually to restore and maintain steam generator water level, while ensuring that the reactor coolant temperature is at the desired value. Residual heat removal is maintained by the steam header pressure controller (manually selected) which controls the amount of steamflow to the condensers. This controller operates a portion of the same steam dump valves to the condensers which are used during the initial transient following turbine and reactor trip.

The pressurizer pressure and water level fall rapidly during the transient because of coolant contraction. The pressurizer water level is programmed so that the level following the turbine and reactor trip is above the heaters. However, if the heaters become uncovered following the trip, the chemical and volume control system will provide full charging flow to restore water level in the pressurizer. Heaters are then turned on to restore pressurizer pressure to normal.

The steam dump and feedwater control systems are designed to prevent the average coolant temperature from falling below the programmed no-load temperature following the trip to ensure adequate reactivity shutdown margin.

7.7.2.7 Core Cooling Monitor

A core cooling monitor (T saturation meter) is provided to meet the requirements of item II.F.2 of NUREG-0737 to provide instrumentation for the detection of inadequate core cooling. For information on the system see the "Summary Report, Westinghouse Reactor Vessel Level Instrumentation System for Monitoring Inadequate Core Cooling."[1]

The core cooling monitor utilizes inputs from the existing hot leg resistance temperature detectors, selected incore thermocouples, and the reactor coolant system pressure sensors. The plant safety monitoring system (PSMS) is employed to calculate the reactor coolant system saturation temperature for the existing system pressure, compares this value to the measured reactor coolant system temperature, and continuously indicates the margin-to-saturation on the PSMS plasma display. Margin-to-saturation is available on the PSMS plasma display based on both auctioneered high hot leg temperature and on auctioneered high incore thermocouples. In addition to the plasma display, alarms are provided to indicate first the development of off-normal conditions and then the approach to loss of normal core cooling.

Details of seismic and environmental qualification can be found in sections 3.10 and 3.11.

Procedures for safety injection and natural circulation require the operator to monitor reactor coolant system temperature for subcooling conditions during the initial phase of the incident and during followup conditions when natural circulation is being verified.

7.7.2.8 Reactor Vessel Level Instrumentation System

A reactor vessel level instrumentation system (RVLIS) is provided to meet the requirement of item II.F.2 of NUREG-0737 to provide instrumentation for the detection of inadequate core cooling. For information on the system see the "Summary Report, Westinghouse Reactor Vessel Level Instrumentation System for Monitoring Inadequate Core Cooling."[1]

Reactor vessel level is also utilized to indicate the need to vent noncondensable gases from the reactor vessel head. The RVLIS utilizes two sets of differential pressure cells to measure reactor vessel level.

The narrow range RVLIS instrument provides an indication of reactor vessel water level from the bottom of the reactor vessel to the top of the reactor vessel when one or no reactor coolant pump is operating. The narrow range instrument also measures the reactor core and internals pressure drop and therefore provides an indication of the relative void content or density of the circulating fluid, when only one reactor coolant pump is operating. When more than one reactor coolant pump is operating, the narrow range instrument reading will be off scale.

The wide range RVLIS instrument provides an indication of reactor core, internals, and outlet nozzle pressure drop for any combination of operating reactor coolant pumps. Comparison of the measured pressure drop with the normal, single-phase pressure drop provides an

approximate indication of the relative void content or density of the circulating fluid. The wide range instrument monitors vessel level on a continuous basis.

Details of the seismic and environmental qualification can be found in sections 3.10 and 3.11.

7.7.2.9 Control Systems Failure Analysis

An analysis has been performed on VEGP to confirm that the consequences of a random initiating failure in a control system or its supporting systems will not cause plant conditions more severe than those bounded by the chapter 15 analysis (multiple independent failures are excluded). The analysis addressed the consequences of control system failures due to the following:

- Loss of any single instrument.
- Break of any single instrument line.
- Loss of power to a single inverter.
- Loss of power to a protection set.
- Loss of power to a control group.

7.7.2.10 Anticipated Transient Without Scram (ATWS) Mitigation System Actuation Circuitry (AMSAC) Analysis

7.7.2.10.1 Safety Classification/Safety-Related Interface

The AMSAC is not safety-related and therefore need not meet the requirements of IEEE 279-1971. The AMSAC has been implemented such that the reactor trip system (RTS) and the engineered safety features (ESF) actuation system (ESFAS) continue to meet all applicable safety-related criteria. The AMSAC is independent of the RTS and ESFAS. The isolation provided between the RTS and the AMSAC and between the ESFAS and the AMSAC by the isolator modules and the isolation relays ensures that the applicable safety-related criteria are met for the RTS and the ESFAS.

7.7.2.10.2 Redundancy

System redundancy has not been provided. Since AMSAC is a backup nonsafety-related system to the redundant RTS, redundancy is not required. To ensure high system reliability, portions of the AMSAC have been implemented as internally redundant, such that a single failure of an input channel or actuation logic processor (ALP) will neither actuate nor prevent actuation of the AMSAC.

7.7.2.10.3 Diversity

Diverse equipment has been selected in order that common cause failures affecting both the RTS and the AMSAC or both the ESFAS and the AMSAC will not render these systems inoperable simultaneously. A more detailed discussion of the diversity between the RTS and the AMSAC and between the ESFAS and the AMSAC is presented in paragraph 7.7.1.11.7.

7.7.2.10.4 Electrical Independence

The AMSAC is electrically independent of the RTS and ESFAS from the process control cabinets up to the final actuation devices. Isolation devices are provided to isolate the nonsafety AMSAC circuitry from the safety-related actuation circuits of the auxiliary feedwater system as discussed in 7.7.1.11.6.

7.7.2.10.5 Physical Separation from the RTS and ESFAS

AMSAC needs to be and is physically separated from the existing protection system hardware. The AMSAC outputs are provided from separate relay panels within the cabinets. The two trains are separated (in accordance with Regulatory Guide 1.75, Rev. 2) within the AMSAC cabinet by a combination of metal barriers, conduit, and distance.

7.7.2.10.6 Environmental Qualification

Equipment related to the AMSAC is qualified to operate under conditions resulting from anticipated operational occurrences for the respective equipment location. The AMSAC equipment, with the exception of the isolation devices located outside containment in a mild environment, is not designated as safety-related equipment and therefore is not required to be qualified as safety related per the requirements of IEEE Standard 279-1971, "IEEE Standard for Criteria for Protection Systems for Nuclear Power Generating Stations."

7.7.2.10.7 Seismic Qualification

It is required that only the isolation devices comply with seismic qualification. The AMSAC output isolation device is qualified in accordance with a program that was developed to implement the requirements of IEEE Standard 344-1975, "IEEE Standard for Seismic Qualification of Class 1E Electrical Equipment for Nuclear Power Generating Stations."

7.7.2.10.8 Test, Maintenance, and Surveillance Quality Assurance

NRC Generic Letter 85-06, "Quality Assurance Guidance for ATWS Equipment that is not Safety Related," requires quality assurance procedures commensurate with the nonsafety-related classification of the AMSAC. The quality controls for the AMSAC are, at a minimum, consistent with existing plant procedures or practices for nonsafety-related equipment.

Design of the AMSAC followed procedures relating to equipment procurement, document control, and specification of system components, materials, and services. In addition, specifications also define quality assurance practices for inspections, examinations, storage, shipping, and tests as appropriate to a specific item or service.

A computer software verification program and a firmware validation program have been implemented commensurate with the nonsafety-related classification of the AMSAC to ensure that the system design requirements implemented with the use of software have been properly implemented and to ensure compliance with the system functional, performance, and interface requirements.

System testing is completed prior to the installation and operation of the AMSAC, as part of the normal factory acceptance testing and the validation program. Periodic testing is performed both automatically through use of the system automatic self-checking capability, and manually, under administrative control via the AMSAC test/maintenance panel.

7.7.2.10.9 Power Supply

Power to the AMSAC is from a battery-backed, non-Class 1E vital bus independent of the power supplies for the RTS and ESFAS. The station battery supplying power to the AMSAC is independent of those used for the RTS and ESFAS. The AMSAC is an energize-to-actuate system capable of performing its mitigative functions with a loss of offsite power.

7.7.2.10.10 Testability at Power

The AMSAC is testable at power. This testing is done via the system test/maintenance panel. The capability of the AMSAC to perform its mitigative actuations is bypassed at a system level while in the test mode. Total system testing is performed as a set of three sequential, partial, and overlapping tests. The first of the tests checks the analog input portions of the AMSAC in order to verify accuracy. Each of the analog input modules is checked separately. The second test checks each of the ALPs to verify that the appropriate coincidence logic is sent to the majority voter. Each ALP is tested separately. The last test exercises the majority voter and the integrity of the associated output relays. The majority voter and associated output relays are tested by exercising all possible input combinations to the majority voter. The integrity of each of the output relays is checked by confirming continuity of the relay coils without operating the relays. The capability to individually operate the output relays, confirm integrity of the associated field wiring, and operate the corresponding isolation relays and final actuation devices at plant shutdown is provided.

7.7.2.10.11 Inadvertent Actuation

The AMSAC has been designed such that the frequency of inadvertent actuations is minimized. This high reliability is ensured through use of three redundant ALPs and a majority voting module. A single failure in any of these modules will not result in a spurious AMSAC actuation. In addition, a three-out-of-four low feedwater flow coincidence logic and a time delay (dependent on turbine load) have been selected to further minimize the potential for inadvertent actuations.

7.7.2.10.12 AMSAC Bypass

7.7.2.10.12.1 Maintenance Bypasses. The AMSAC is blocked at the system level during maintenance, repair, calibration, or test. While the system is blocked, the bypass condition is continuously indicated in the main control room.

7.7.2.10.12.2 Operating Bypasses. The AMSAC has been designed to allow for operational bypasses with the inclusion of the C-20 permissive. Above the C-20 setpoint, the AMSAC is automatically unblocked (i.e., armed); below the setpoint, the system is automatically blocked. The operating status of the AMSAC is continuously indicated in the main control room via an annunciator window.

7.7.2.10.12.3 Indication of Bypasses. Whenever the mitigative capabilities of the AMSAC are bypassed or deliberately rendered inoperable, this condition is continuously indicated in the main control room. In addition to the operating bypass, any manual maintenance bypass is indicated via the AMSAC general warning sent to the main control room.

7.7.2.10.12.4 Means for Bypassing. A permanently installed system bypass selector switch is provided to bypass the system. This is a two-position selector switch with "NORMAL" and "BYPASS" positions. At no time is it necessary to use any temporary means, such as installing jumpers or pulling fuses, to bypass the system.

7.7.2.10.13 Completion of Mitigative Actions Once Initiated

The AMSAC mitigative actions go to completion as long as the coincidence logic is satisfied and the time delay requirements are met. If the flow in the feedwater lines is reinitiated before the timer expires and increases to above the low-flow setpoint, then the coincidence logic will no longer be satisfied and the actuation signal disappears. If the coincidence logic conditions are maintained for the duration of the time delay, then the mitigative actions go to completion. The auxiliary feedwater initiation signal is latched in at the component actuating devices and the turbine trip is latched at the turbine digital electrohydraulic control system (DEHC) (Unit 1) or the electrohydraulic control system (Unit 2). Deliberate operator action is then necessary to terminate auxiliary feedwater flow, clear the turbine trip signal using the DEHC main control board human machine interfaces (HMIs) (Unit 1) or the main control board turbine trip reset switch (Unit 2), and proceed with the reopening of the turbine stop valves.

7.7.2.10.14 Manual Initiation

Manual initiation of the AMSAC is not provided. The capability to initiate the AMSAC mitigative functions manually, i.e., initiate auxiliary feedwater, trip the turbine, and isolate steam generator blowdown and sampling lines, exists at the main control board independently of AMSAC.

7.7.2.10.15 Information Readout

The AMSAC has been designed such that the operating and maintenance staffs have accurate, complete, and timely information pertinent to the status of the AMSAC. A system-level general warning alarm is indicated in the control room. Diagnostic capability exists from the test/maintenance panel to determine the cause of any unanticipated inoperability or deviation.

7.7.2.11 REFERENCE

1. Anderson, T. M., (Westinghouse), to Eisenhut, D. G., (NRC), "Summary Report, Westinghouse Reactor Vessel Level Instrumentation System for Monitoring Inadequate Core Cooling," December 23, 1980.

TABLE 7.7.1-1 (SHEET 1 OF 2)

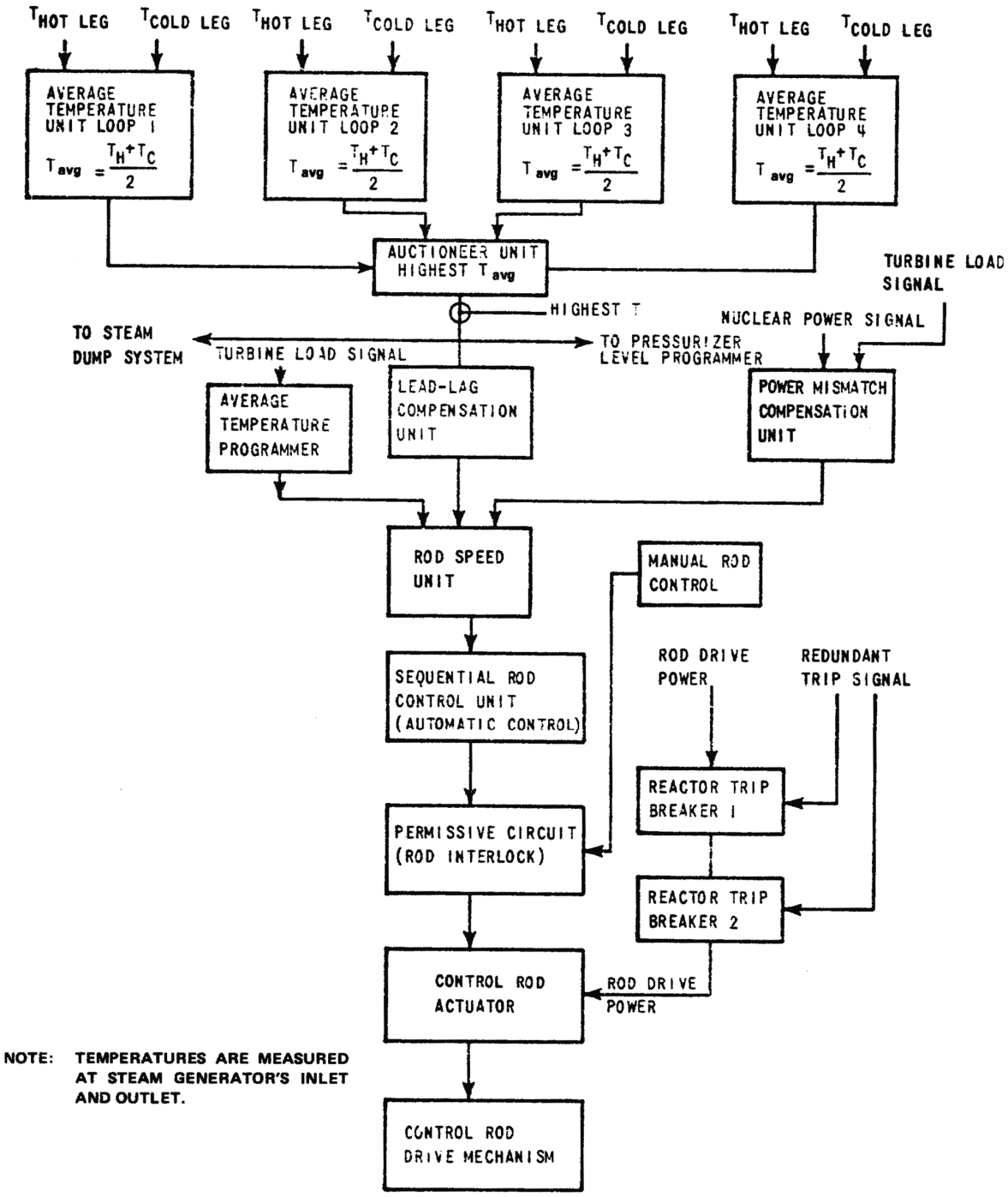
PLANT CONTROL SYSTEM INTERLOCKS

Designation	Derivation	Function
C-1	1/2 neutron flux (intermediate range) above setpoint	Blocks control rod withdrawal
C-2	1/4 neutron flux (power range) above setpoint	Blocks control rod withdrawal
C-3	2/4 overtemperature ΔT above setpoint	Blocks control rod withdrawal Actuates turbine runback via load reference
C-4	2/4 overpower ΔT above setpoint	Blocks control rod withdrawal Actuates turbine runback via load reference
C-5	1/1 turbine impulse chamber pressure below setpoint	Indication only

TABLE 7.7.1-1 (SHEET 2 OF 2)

Designation	Derivation	Function
C-7	1/1 time derivative (absolute value) of turbine impulse chamber pressure (decrease only) above setpoint	Makes steam dump valves available for either tripping or modulation
C-9	Any condenser pressure above setpoint or no circulating water pumps running	Blocks steam dump to condenser
C-11	Not used.	Not used.
C-16	Reduce limit in coolant temperature above normal setpoint	Stops automatic turbine loading until condition clears
C-20[(a)]	Two-of-two turbine impulse chamber pressure above setpoint	Arms AMSAC; below setpoint blocks AMSAC (generated in AMSAC; see section 7.7)
P-4	Reactor trip	Blocks steam dump control via load T_{avg} controller
		Makes steam dump valves available for either tripping or modulation
	Absence of P-4	Blocks steam dump control via plant trip T_{avg} controller

a. Not part of control system (non-Class 1E).



NOTE: TEMPERATURES ARE MEASURED AT STEAM GENERATOR'S INLET AND OUTLET.

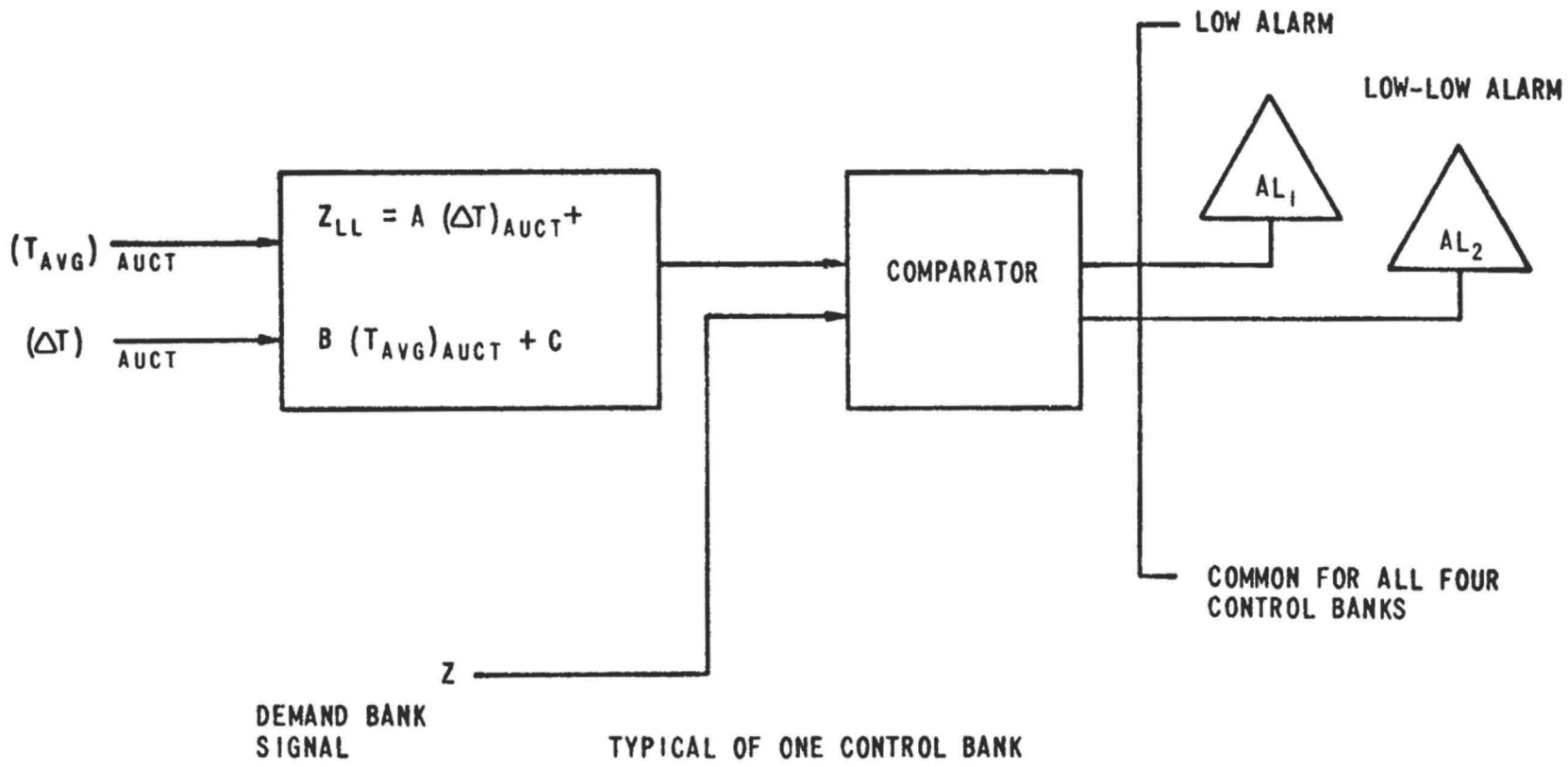
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SIMPLIFIED BLOCK DIAGRAM OF REACTOR CONTROL SYSTEM

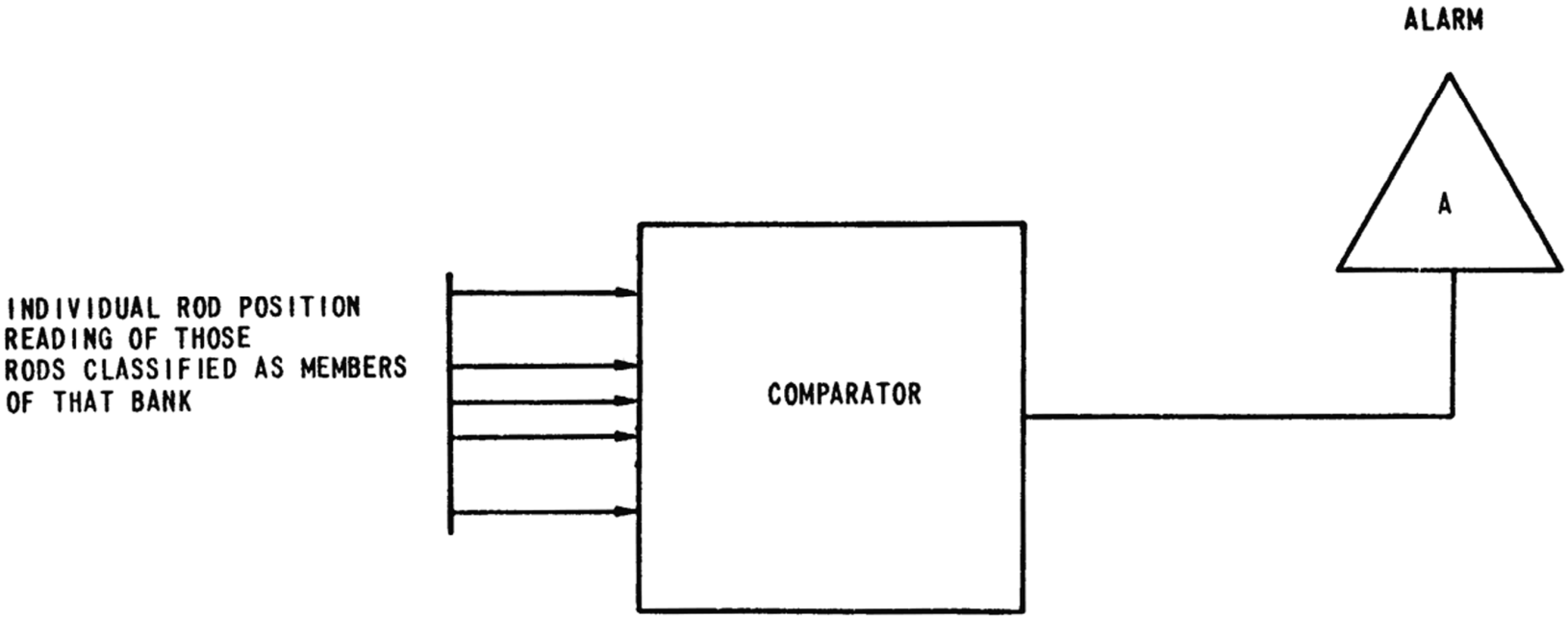
FIGURE 7.7.1–1



NOTE: 1. COMPUTER ALGORITHM IS USED FOR THE COMPARATOR NETWORK
2. COMPARISON IS DONE FOR ALL CONTROL BANKS

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SOUTHERN COMPANY VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	CONTROL BANK ROD INSERTION MONITOR
	FIGURE 7.7.1–2



NOTE:
1. DIGITAL OR ANALOG SIGNALS MAY BE USED FOR THE COMPARATOR COMPUTER INPUTS.
2. THE COMPARATOR WILL ENERGIZE THE ALARM IF THERE EXISTS A POSITION DIFFERENCE GREATER THAN A PRESET LIMIT BETWEEN ANY INDIVIDUAL ROD AND THE DEMAND BANK SIGNAL.
3. COMPARISON IS INDIVIDUALLY DONE FOR ALL CONTROL BANKS.

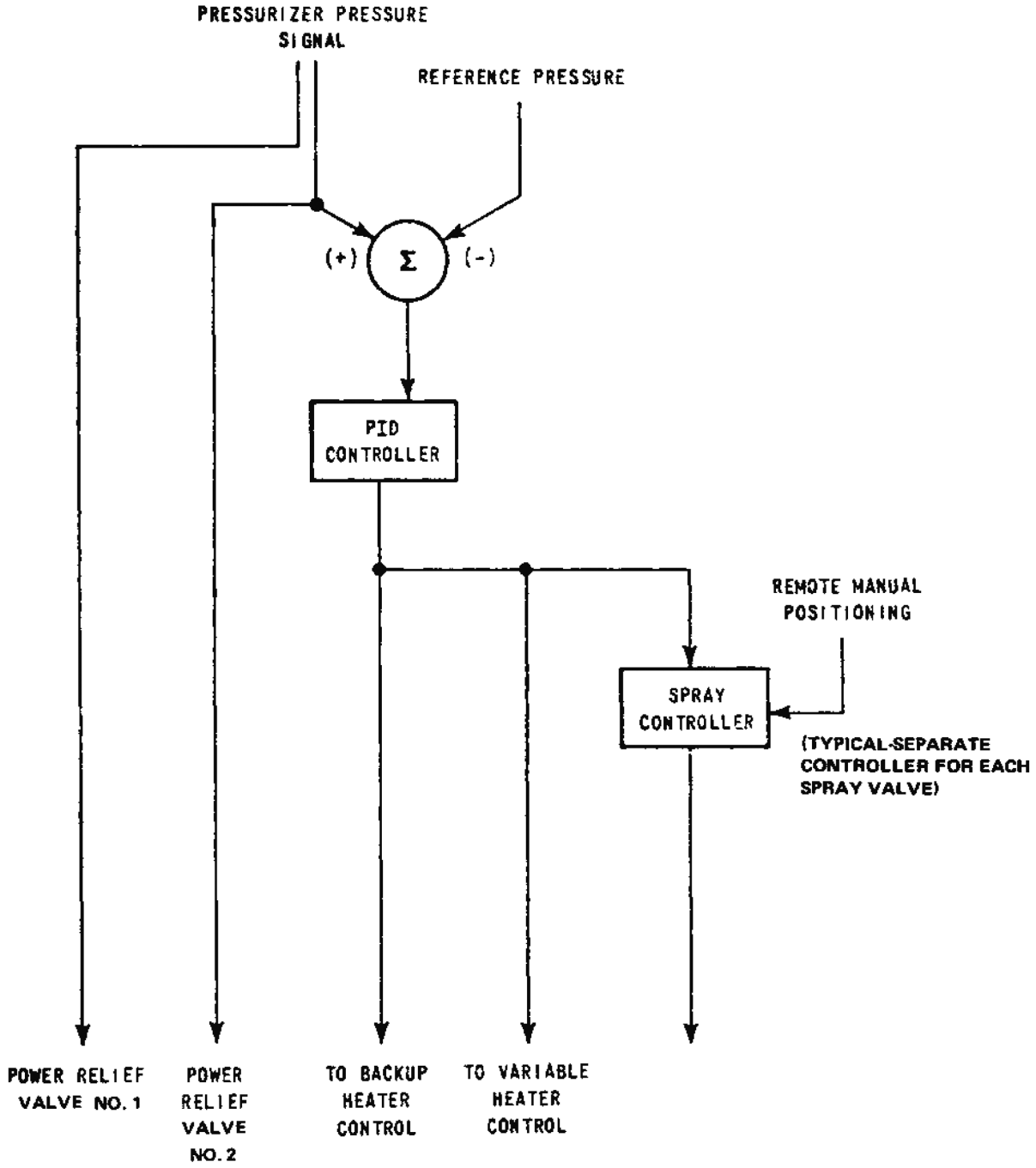
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ROD DEVIATION COMPARATOR

FIGURE 7.7.1–3



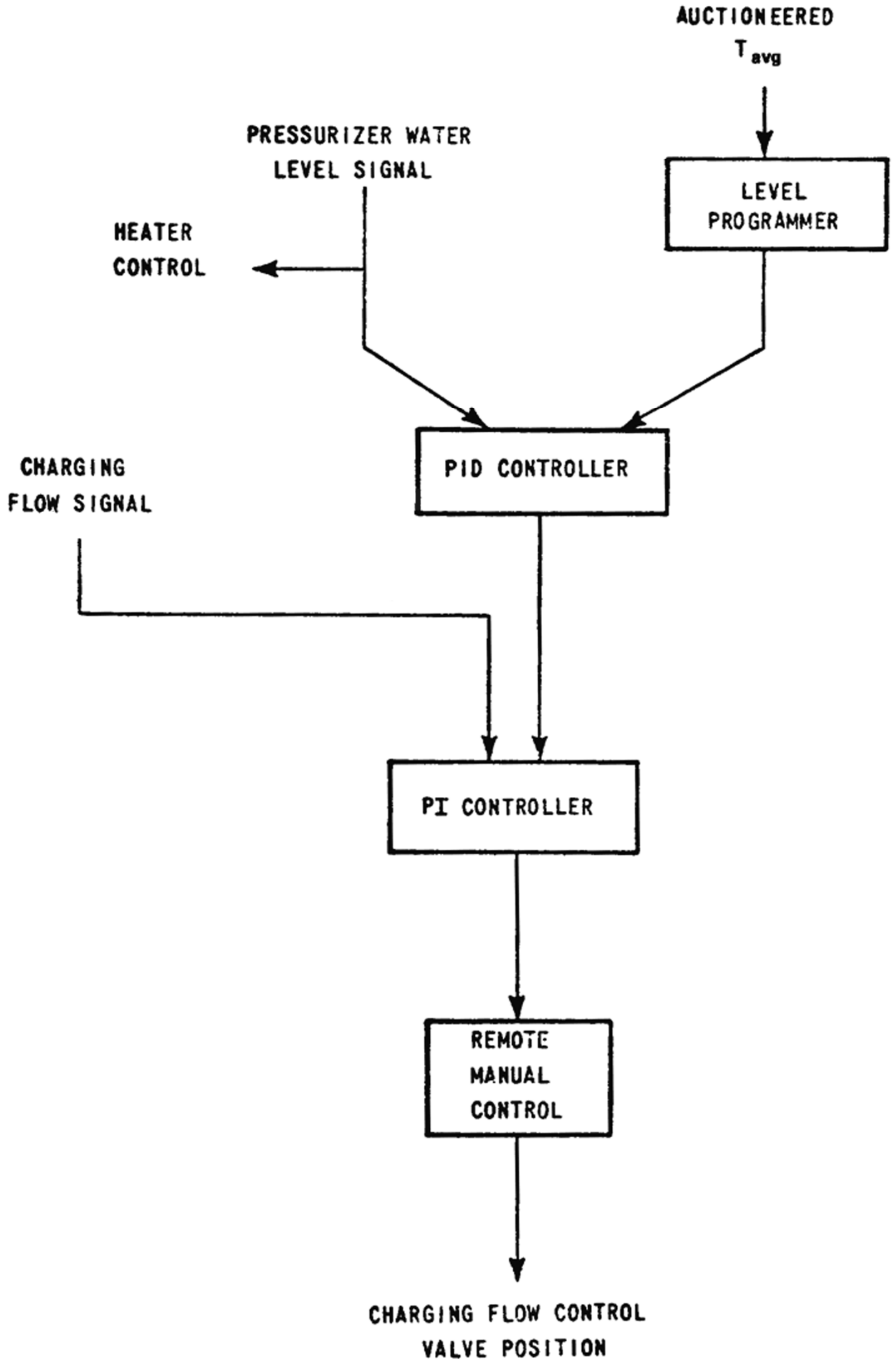
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BLOCK DIAGRAM OF PRESSURIZER PRESSURE CONTROL SYSTEM

FIGURE 7.7.1–4



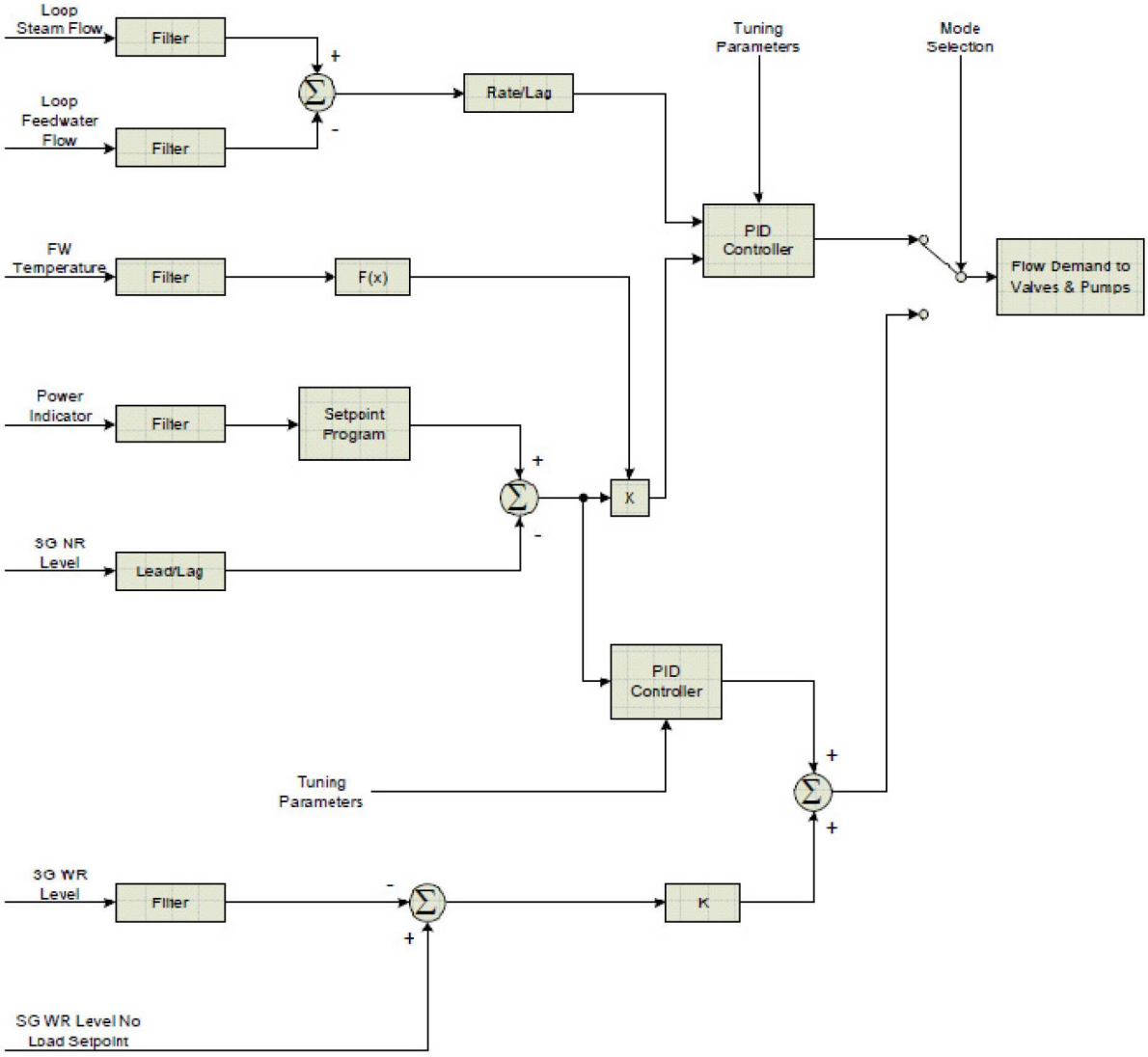
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BLOCK DIAGRAM OF PRESSURIZER LEVEL CONTROL SYSTEM

FIGURE 7.7.1–5



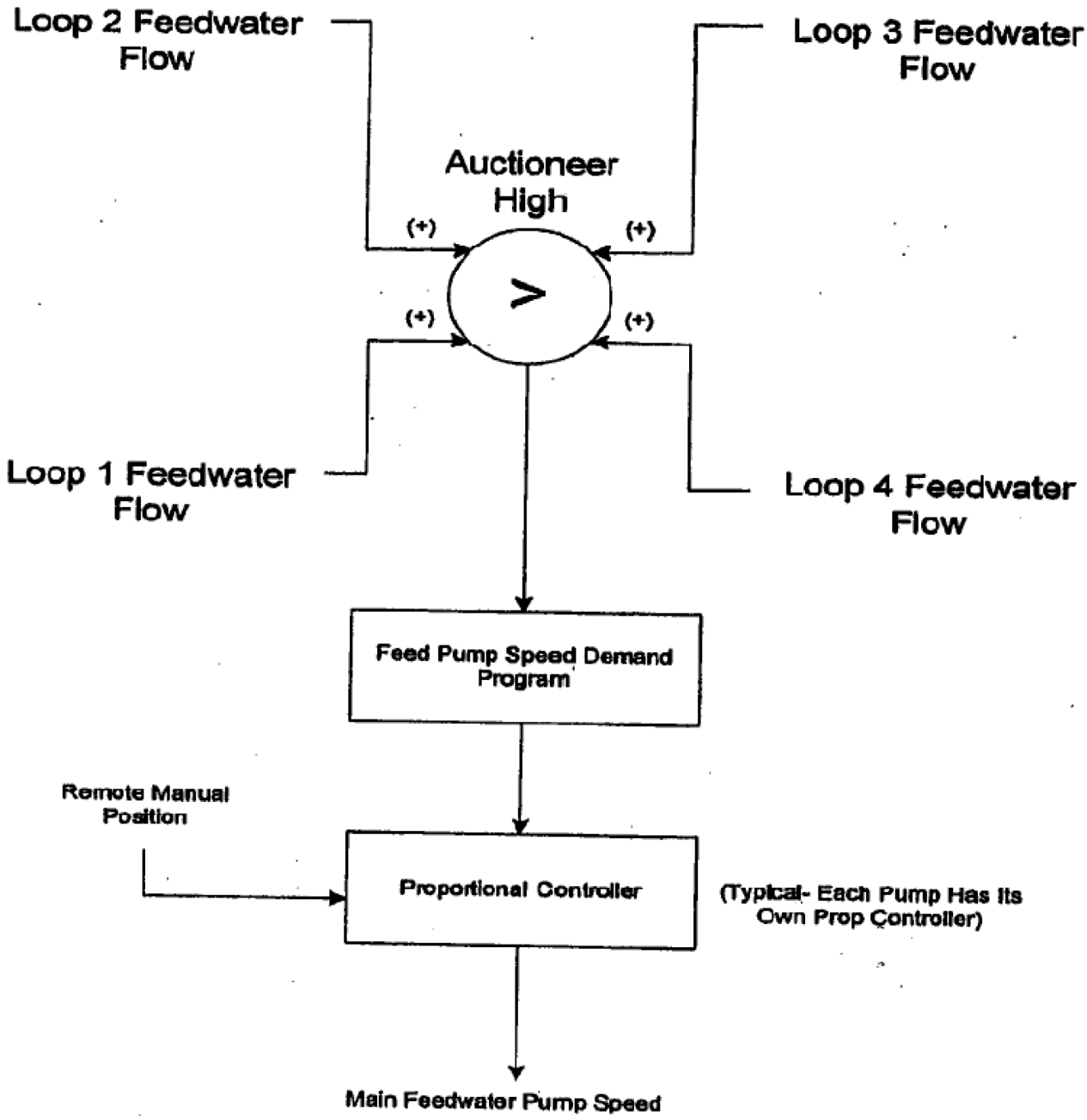
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BLOCK DIAGRAM OF STEAM GENERATOR WATER LEVEL CONTROL SYSTEM

FIGURE 7.7.1-6



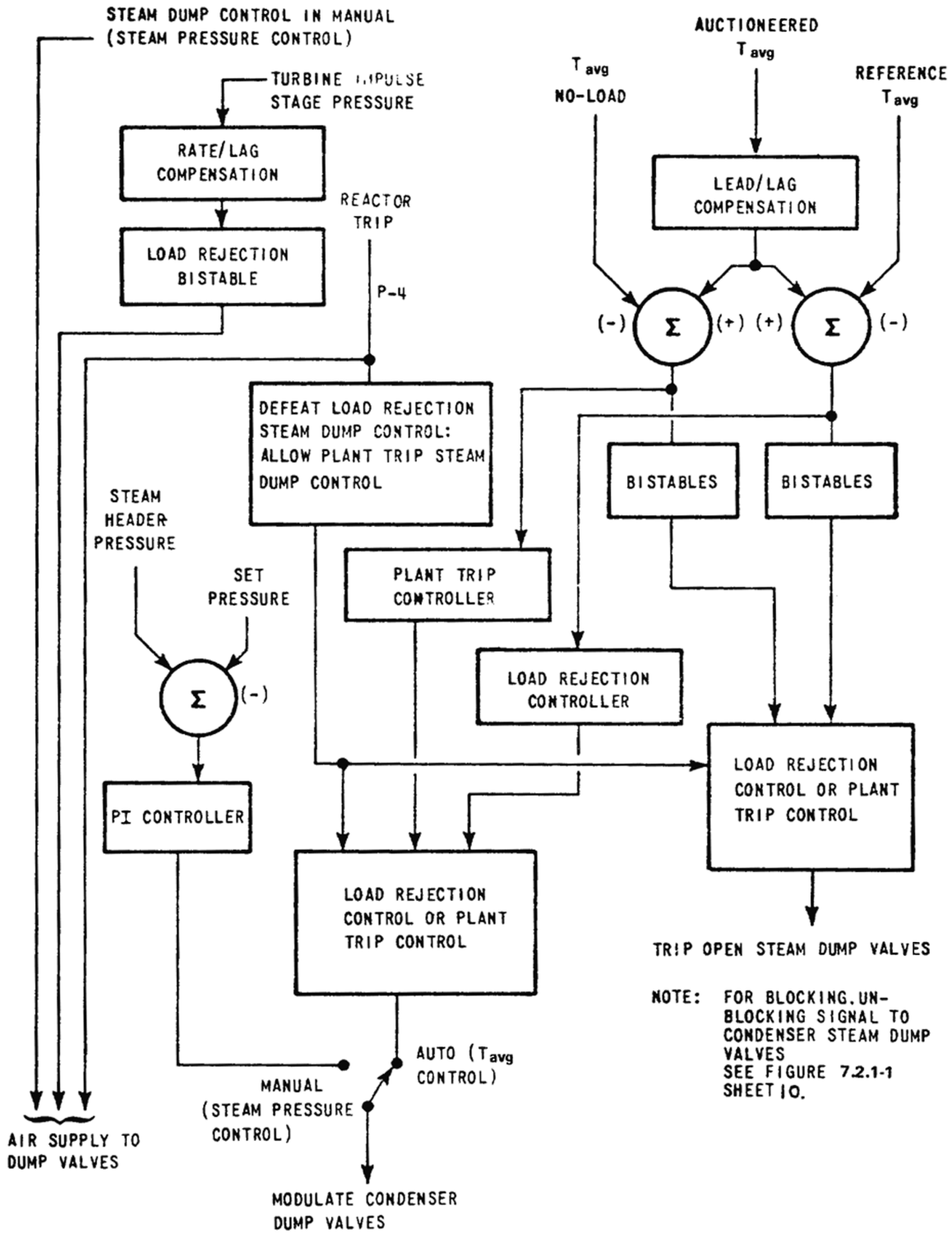
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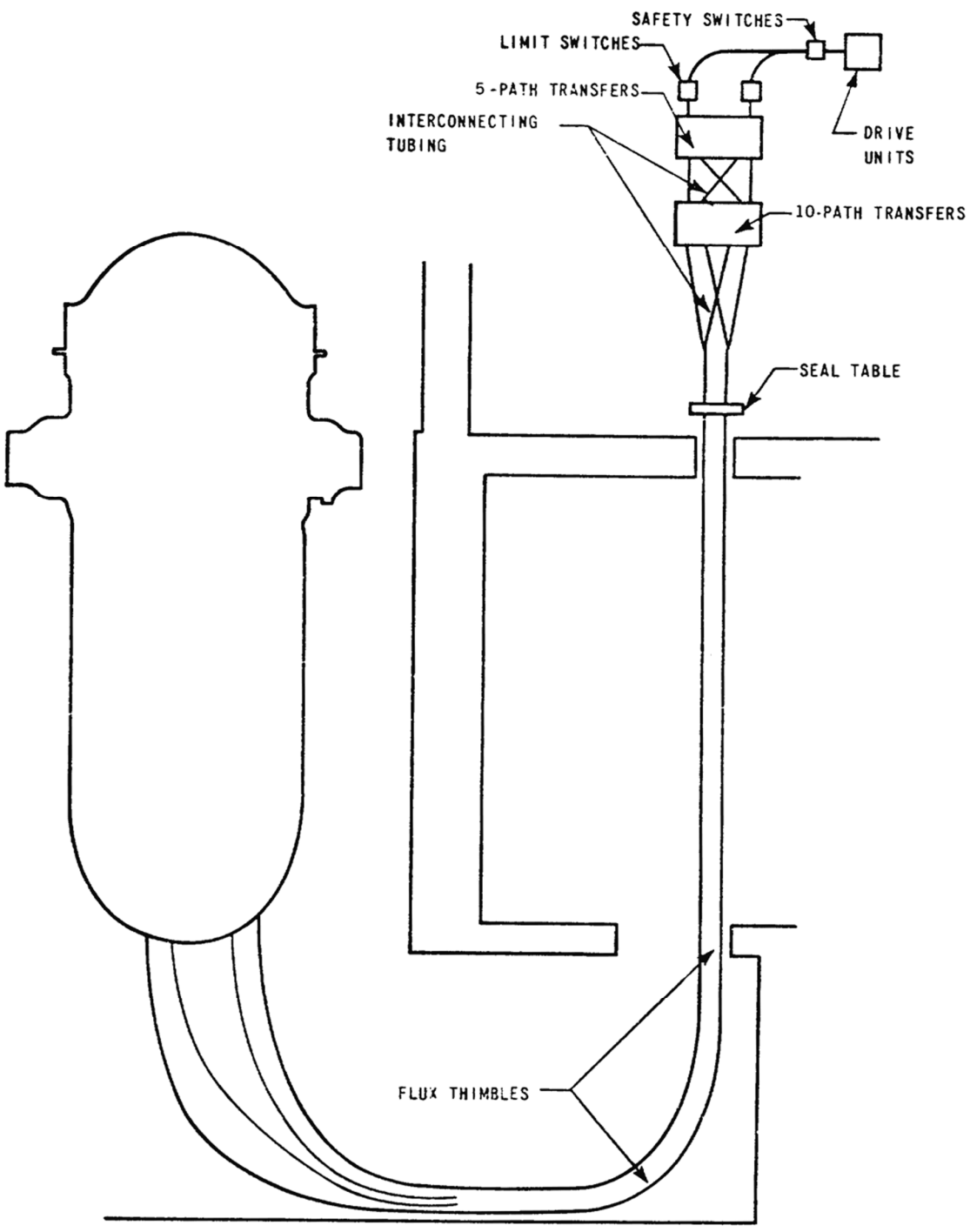
BLOCK DIAGRAM OF MAIN FEEDWATER PUMP SPEED CONTROL SYSTEM

FIGURE 7.7.1-7



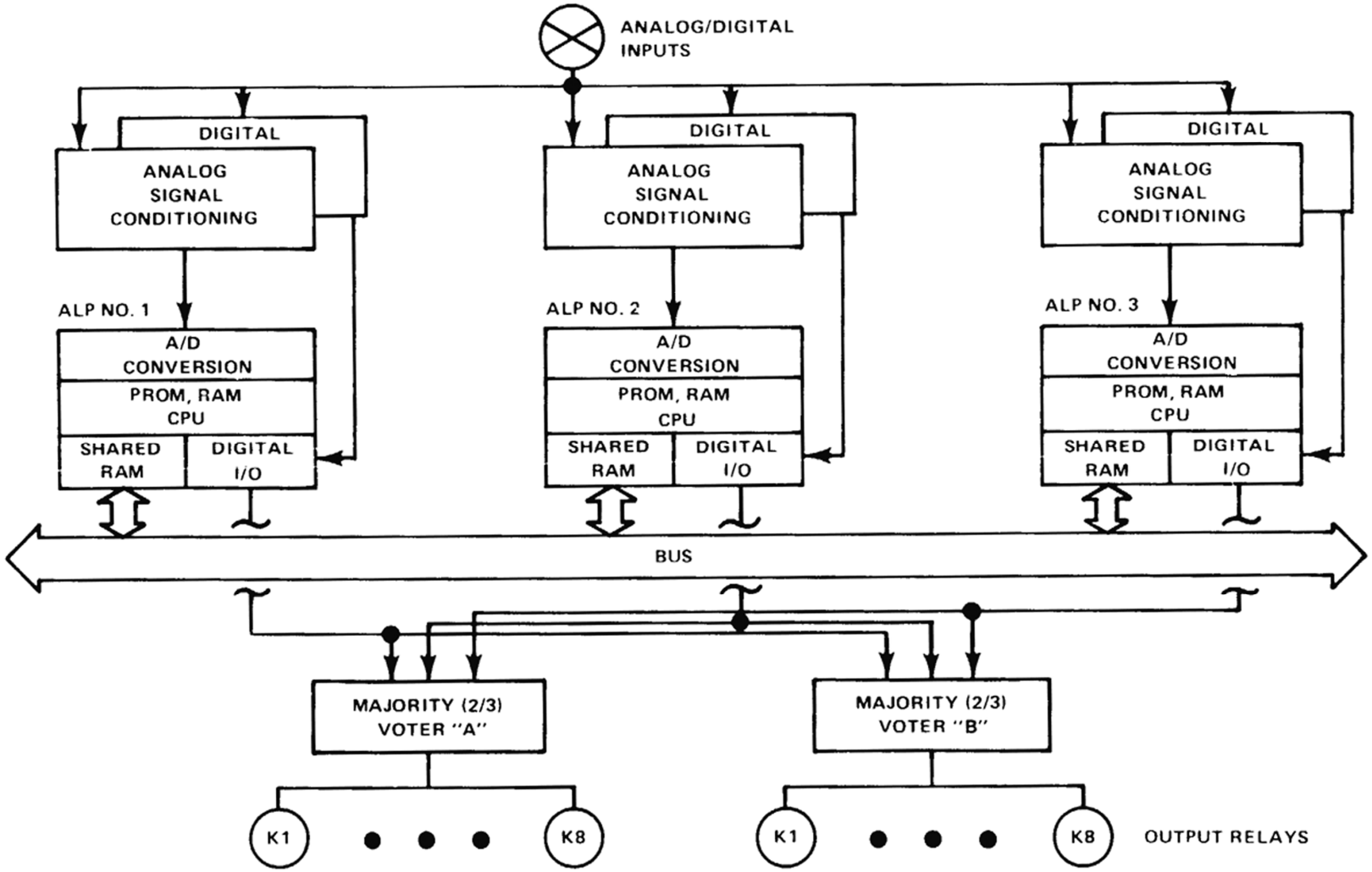
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SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	BLOCK DIAGRAM OF STEAM DUMP CONTROL SYSTEM
	FIGURE 7.7.1–8



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SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	BASIC FLUX–MAPPING SYSTEM
	FIGURE 7.7.1–9



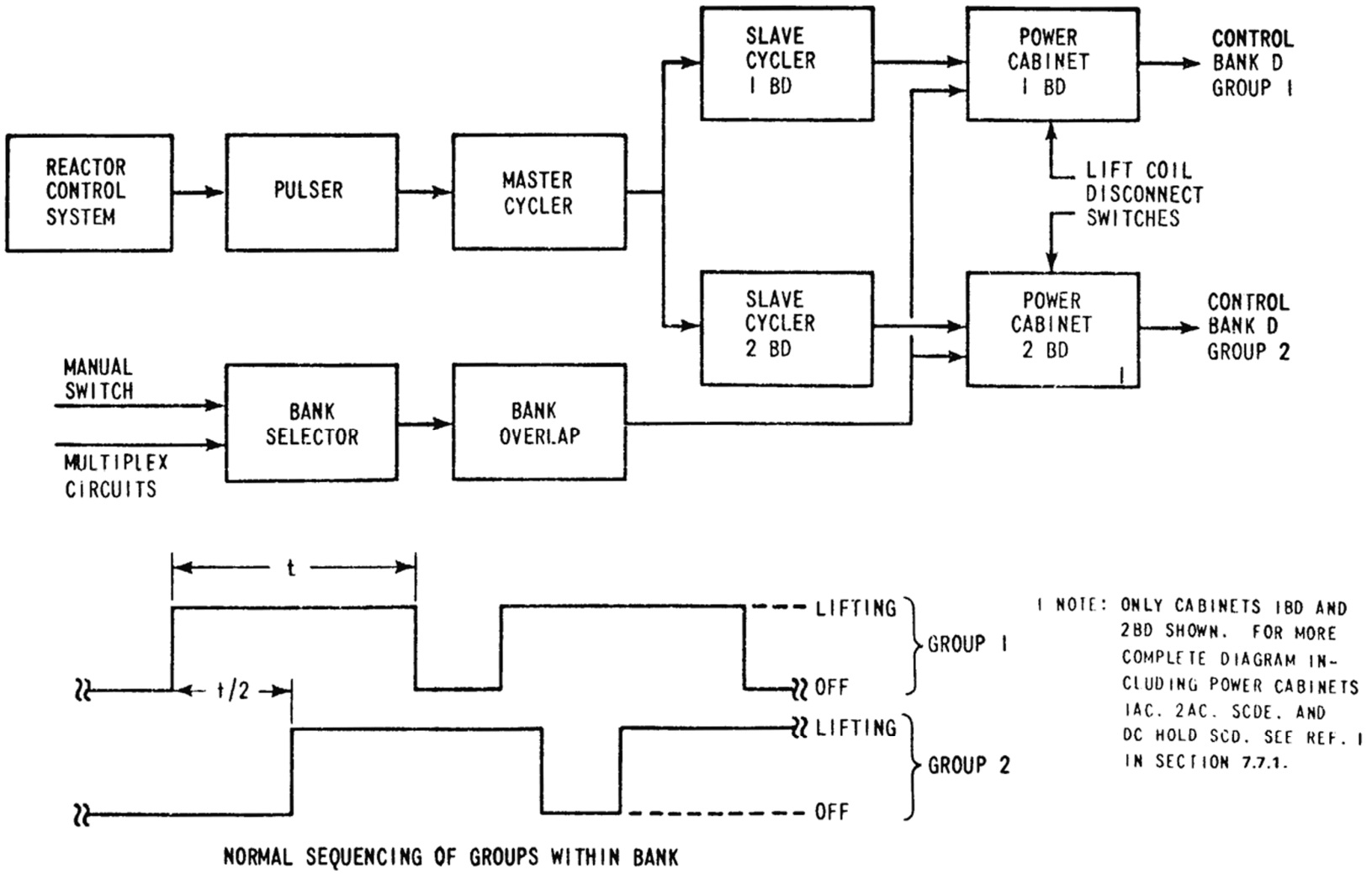
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ACTUATION LOGIC SYSTEM ARCHITECTURE

FIGURE 7.7.1-10



NORMAL SEQUENCING OF GROUPS WITHIN BANK

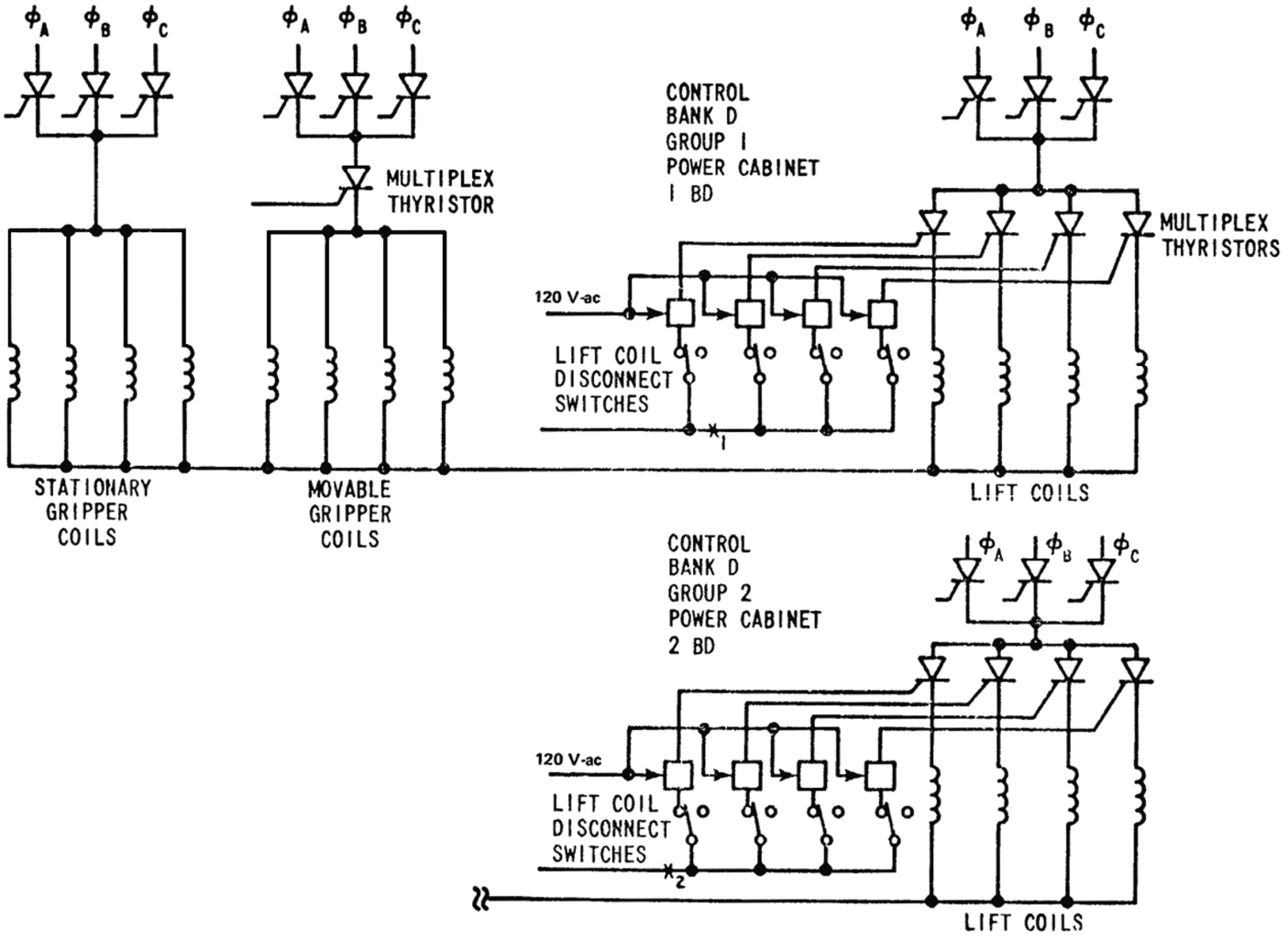
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SIMPLIFIED BLOCK DIAGRAM
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FIGURE 7.7.2–1



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CONTROL BANK D
PARTIAL SIMPLIFIED SCHEMATIC
DIAGRAM POWER CABINETS 1BD AND 2BD

FIGURE 7.7.2–2